Analog Electronics II: Op-amp & 555 Timer

Course instructor:

Dr. Mohsin Sajjad

Assistant Professor

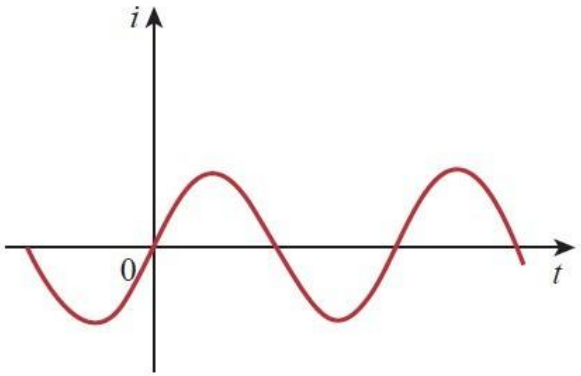
North South University

EEE 311

Lecture # 1

DC vs AC

A direct current (dc) flows only in one direction and can be constant or time varying.

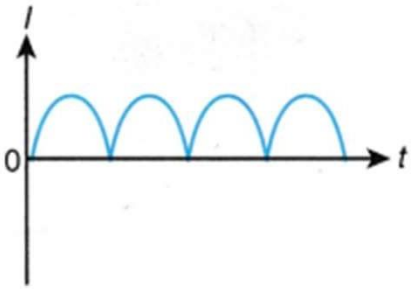


Source: Alexander and Sadiku

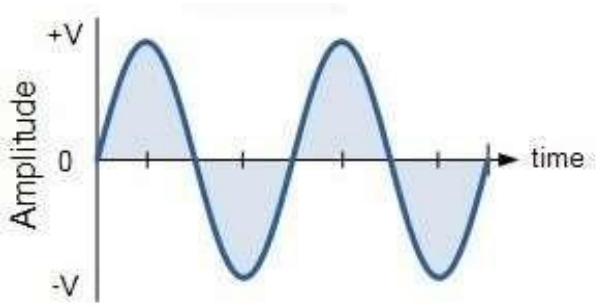
DC or AC?

1

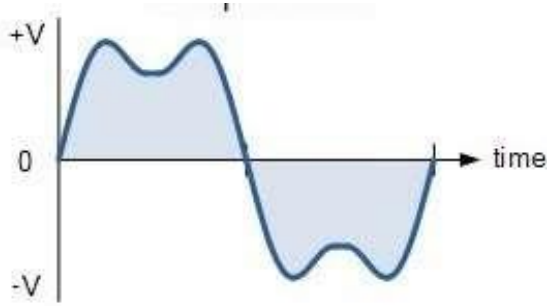
2



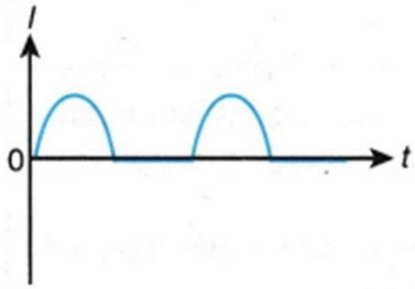
5



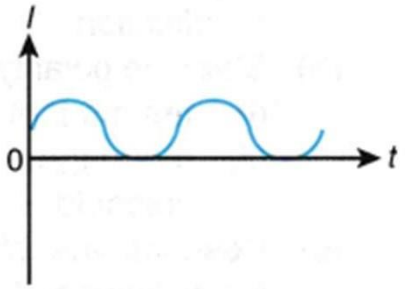
6



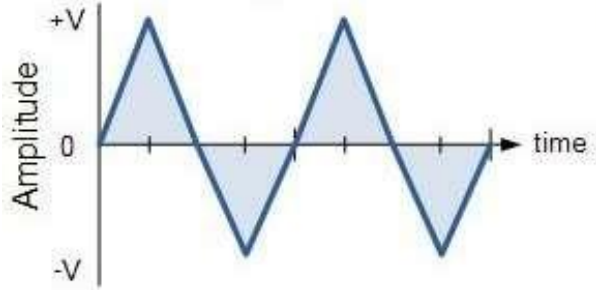
3



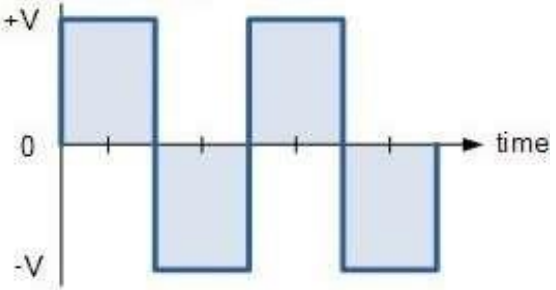
4



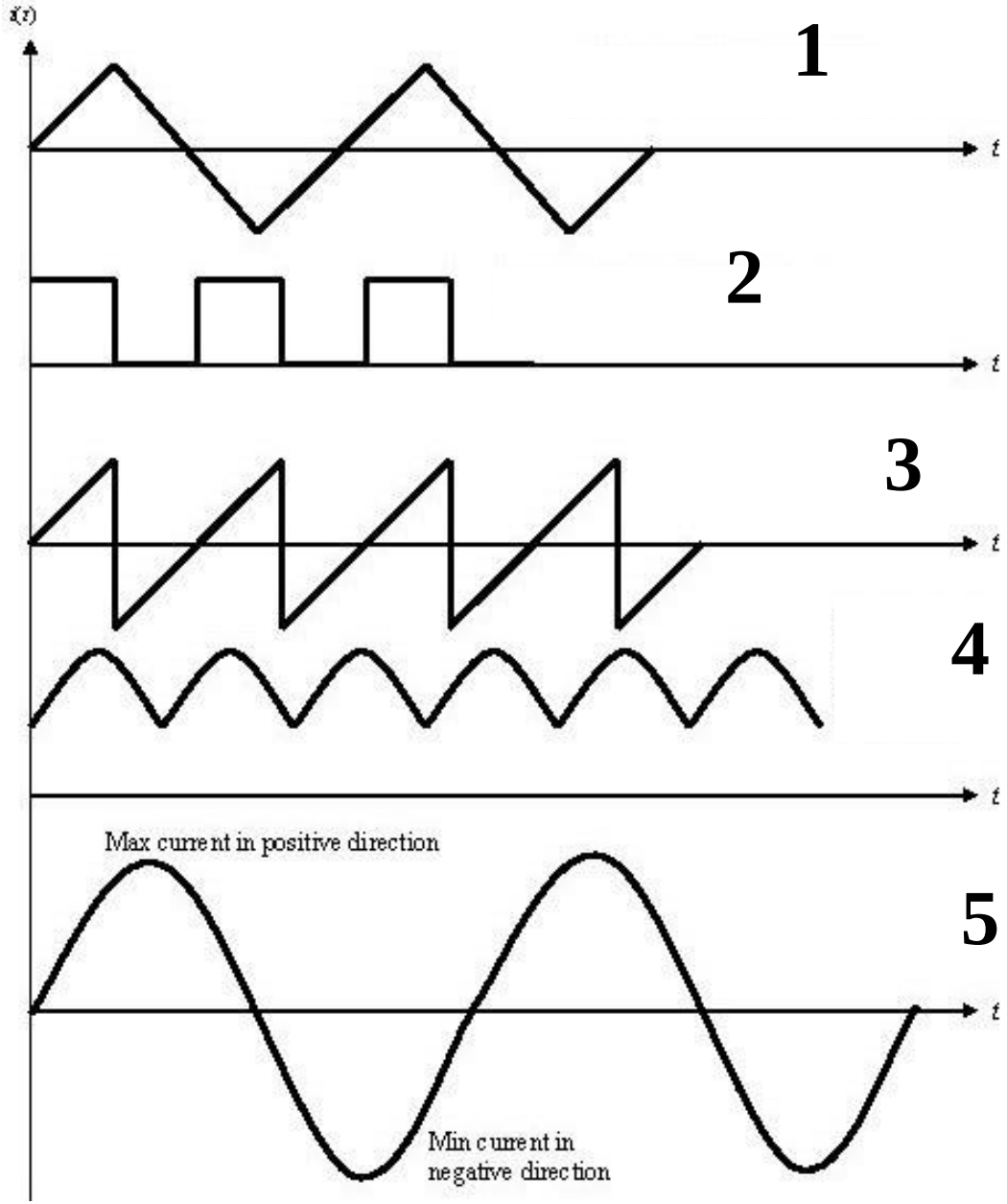
7



8



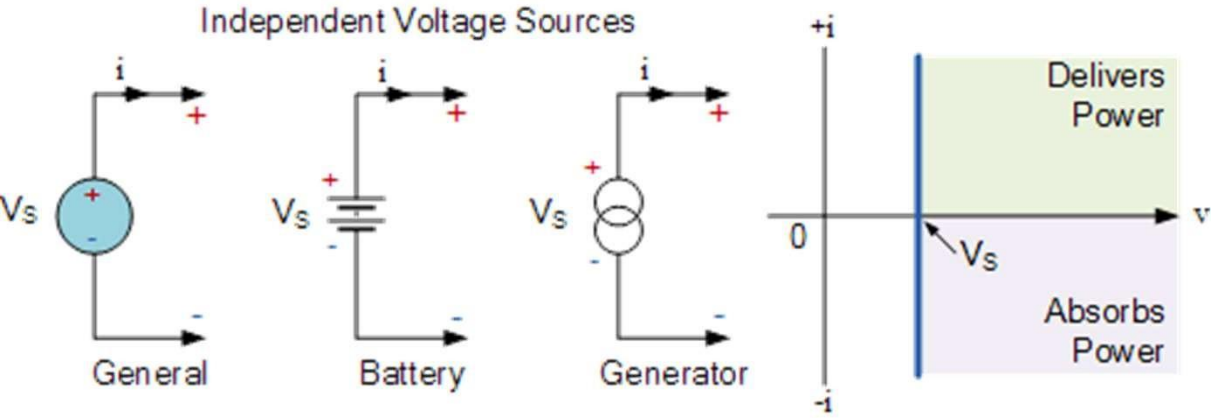
DC or AC?


1
2
3
4
5
t
Max current in positive direction
Min current in
negative direction

Active circuit elements - independent

Generators, Batteries, Voltage and Current sources

An **ideal independent source** is an active element that provides a specified voltage or current that is completely independent of other circuit elements.



ElectronicsTutorials

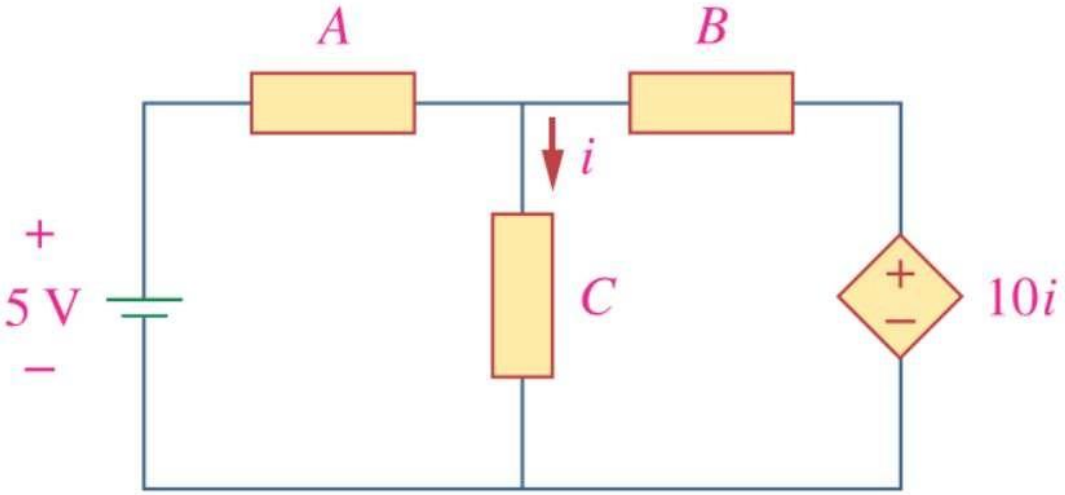
Source: Alexander and Sadiku

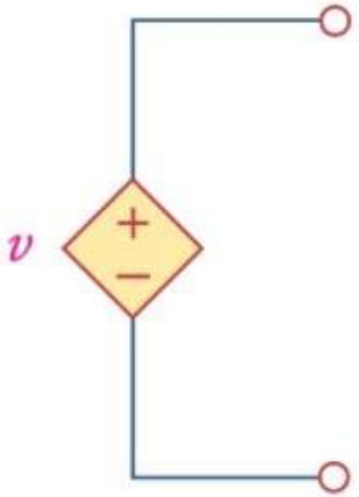
Active circuit elements - dependent

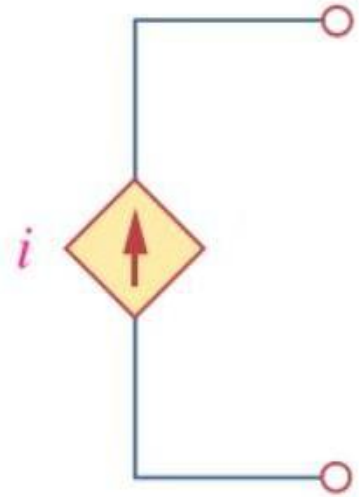
Voltage and Current sources

An **ideal dependent** (or **controlled**) **source** is an active element in which the source quantity is controlled by another voltage or current.

1. A voltage-controlled voltage source (VCVS).
2. A current-controlled voltage source (CCVS).
3. A voltage-controlled current source (VCCS).
4. A current-controlled current source (CCCS).

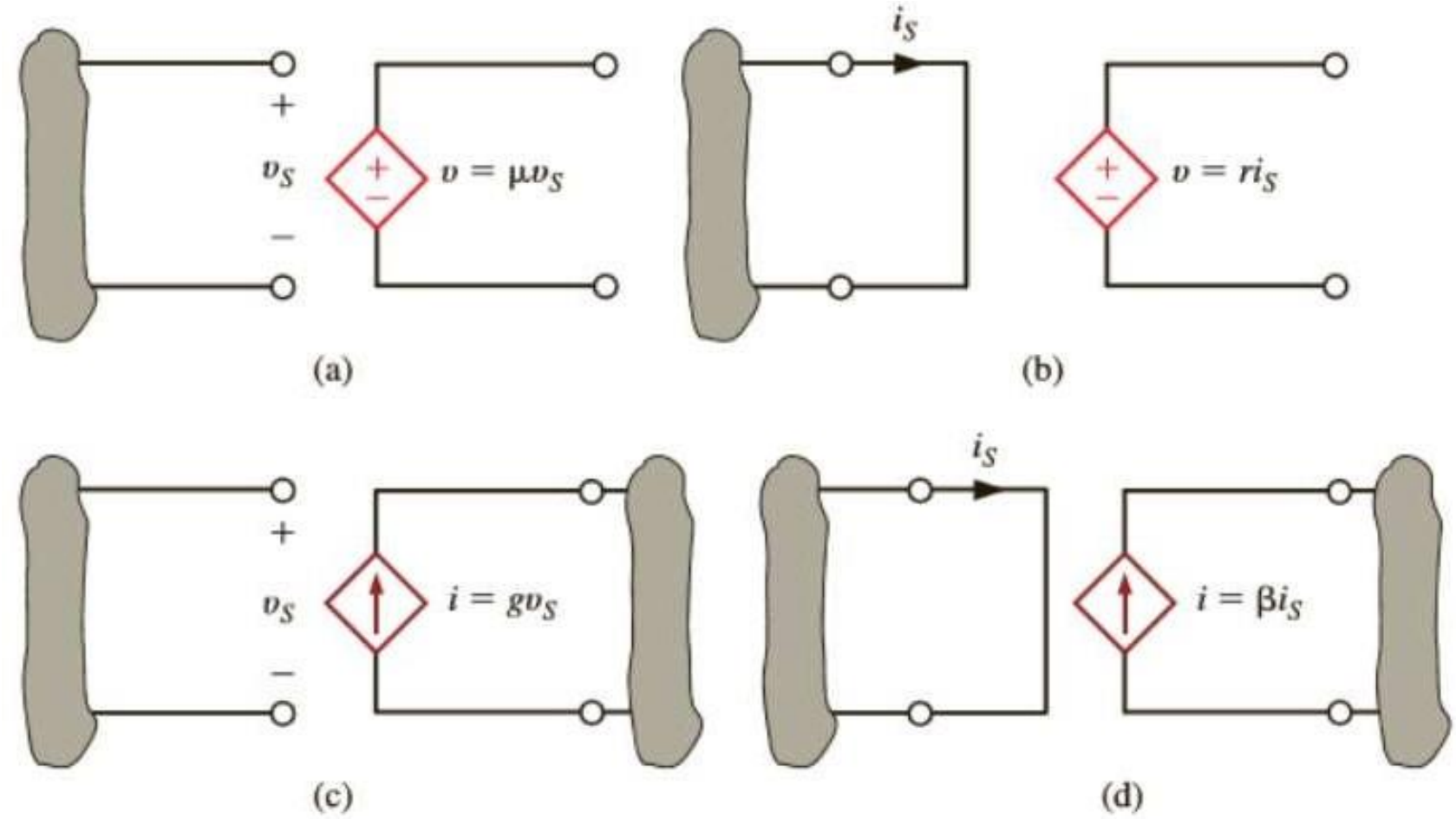






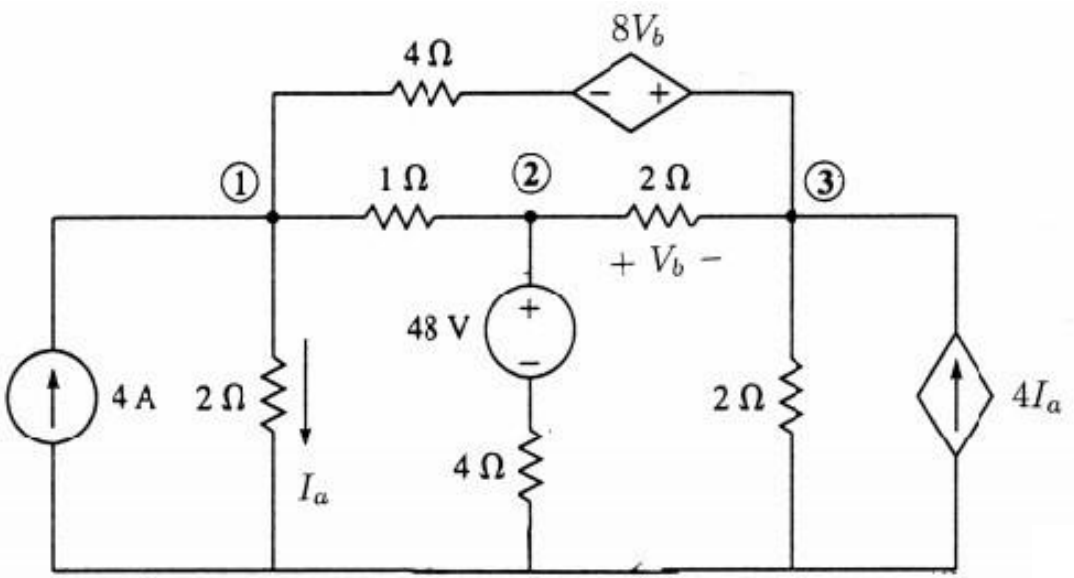
Source: Alexander and Sadiku

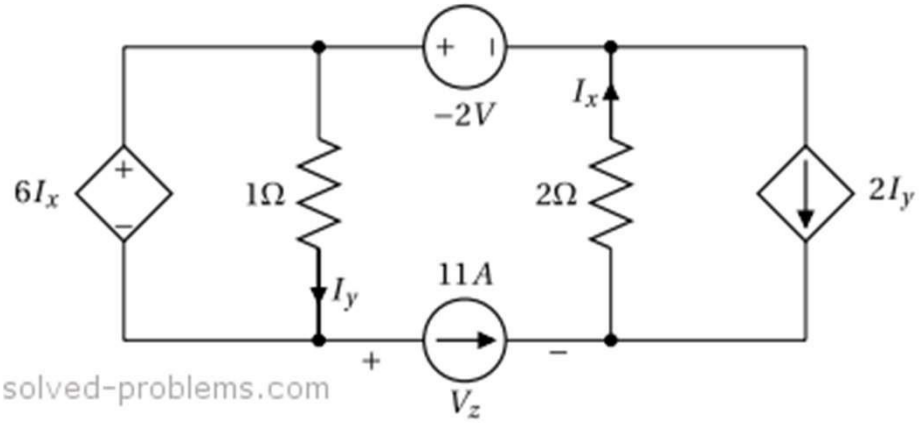
Active circuit elements - dependent

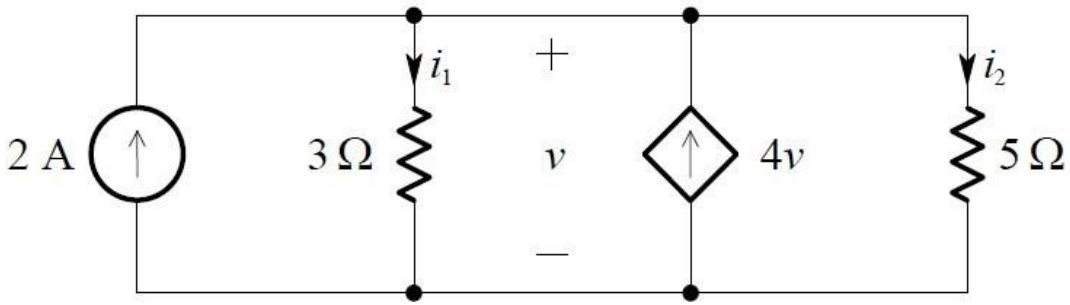


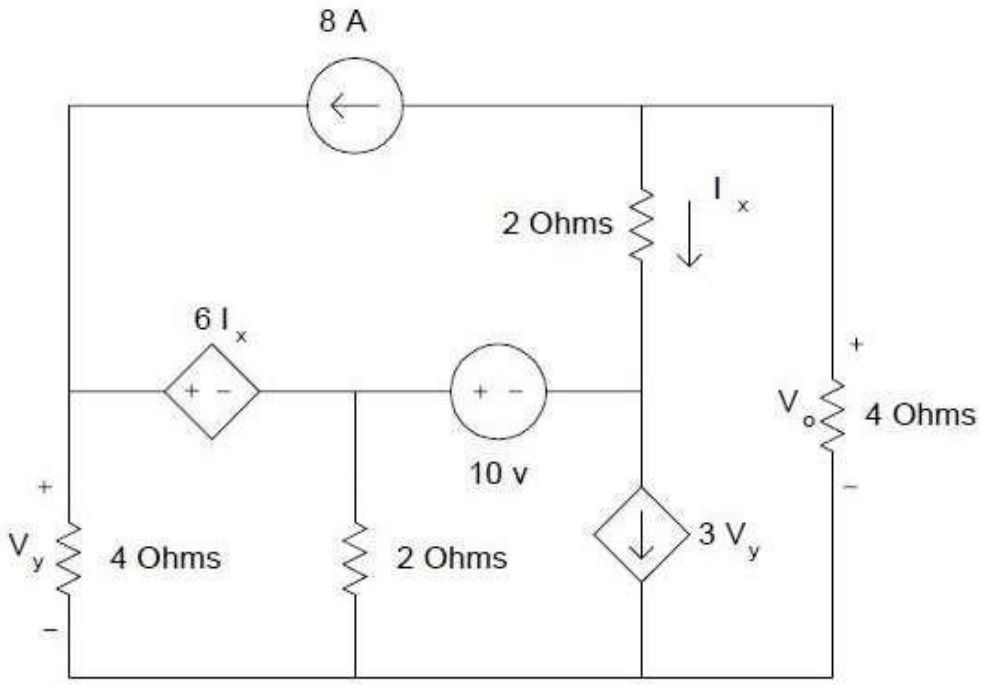
Source: Irwin and Nelms

Which type of Dependent source?



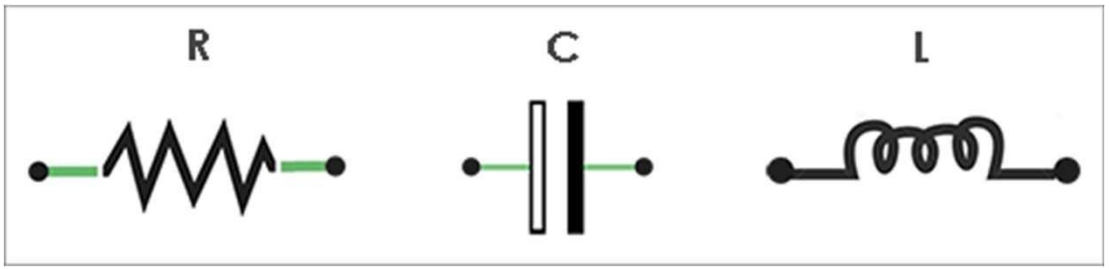






Source: Irwin and Nelms

Passive circuit elements

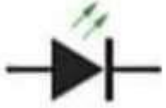
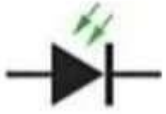
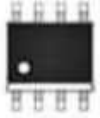
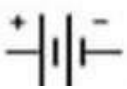


Resistor, Capacitor, Inductor

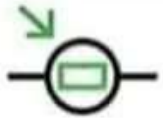
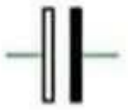
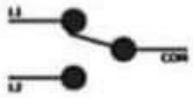
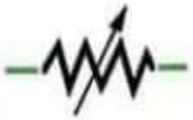
Passive elements are not capable to provide energy on their own

Source: https://www.codrey.com/dc-circuits/active-and-passive-elements/

ACTIVE

Transistor		
Diode		
LED		
Photodiode		
Integrated Circuit		-
Operational Amplifier		
Seven Segment Display		
Battery		

PASSIVE

Resistor		
LDR		
Thermistor		
Capacitor		
Inductor		
Switch		
Variable Resistor		
Transformer		

HACKATRONIC

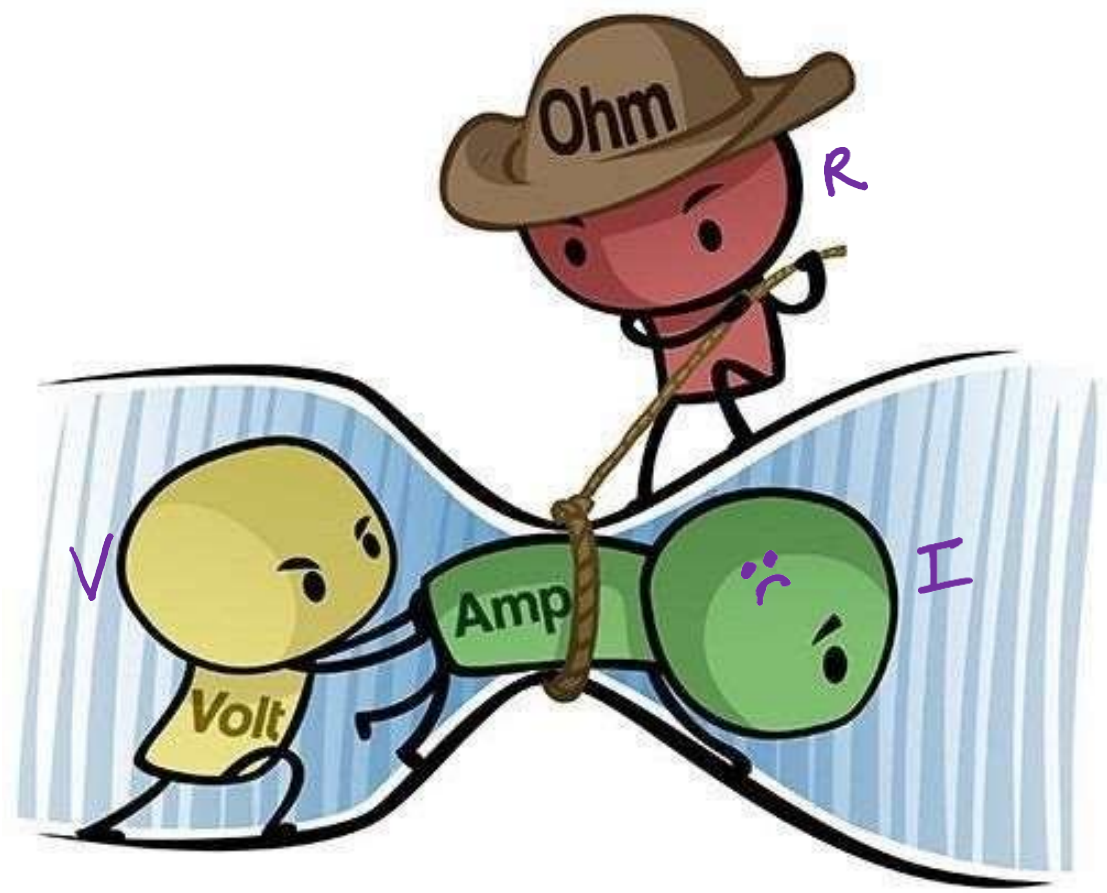
BASIS	ACTIVE COMPONENTS	PASSIVE COMPONENT
Nature of source	Active components deliver power or energy to the circuit.	Passive elements utilizes power or energy in the circuit.
Examples	Diodes, Transistors, SCR, Integrated circuits etc.	Resistor, Capacitor, Inductor etc.
Function of the component	Devices which produce energy in the form of voltage or current.	Devices which stores energy in the form of voltage or current.
Power Gain	They are capable of providing power gain.	They are incapable of providing power gain.

BASIS	ACTIVE COMPONENTS	PASSIVE COMPONENT
Flow of current	Active components can control the flow of current.	Passive components cannot control the flow of the current.
Requirement of external source	They require an external source for the operations.	They do not require any external source for the operations.
Nature of energy	Active components are energy donor.	Passive components are energy acceptor.

HACKATRONIC

Current, voltage, resistance

Ohm's law



$I \propto \frac{1}{R}$ (V const)

$I \propto V$ (R const)

Source: Amino

Ohm's law

Ohm's law states that the voltage v across a resistor is directly proportional to the current i flowing through the resistor.

That is,

$$v \propto i \tag{2.2}$$

$$\boxed{v = iR}$$

The *resistance* R of an element denotes its ability to resist the flow of electric current; it is measured in ohms (Ω).

We may deduce from Eq. (2.3) that

$$R = \frac{v}{i} \tag{2.4}$$

so that

$$1\ \Omega = 1\ \text{V/A}$$

Source: Alexander and Sadiku

Example 2.1: An electric iron draws 2 A at 120 V. Find its resistance.

v = 120V

i = 2A

$R = \frac{v}{i} = \frac{120}{2} = 60\ \Omega$

Source: Alexander and Sadiku

2.2 Short Circuit

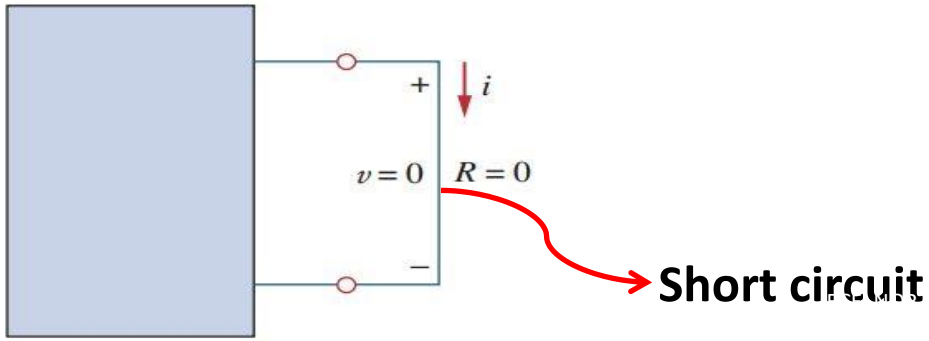
Short circuit: An element with $\mathrm{R} = 0\Omega$

Short circuit: circuit element with resistance approaching zero.

$$v = iR = i \times 0 = 0V$$

Voltage is zero, current is not zero

Usually, a connecting wire assumed to be a perfect conductor



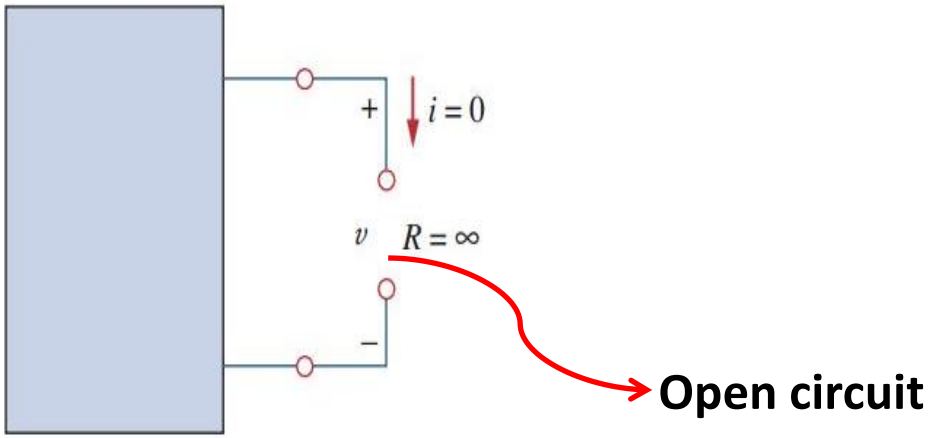
2.2 Open Circuit

Open circuit: An element with $\mathrm{R} = \infty\Omega$

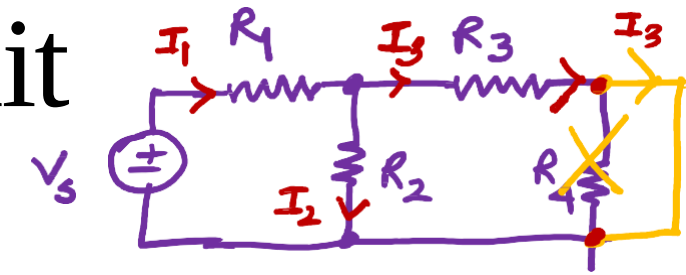
Open circuit: circuit element with resistance approaching infinity.

$$i = \frac{v}{R} = \frac{v}{\infty} = v \times 0 = 0A$$

Current is zero, voltage is not zero

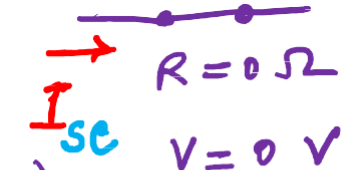


Short circuit and open circuit



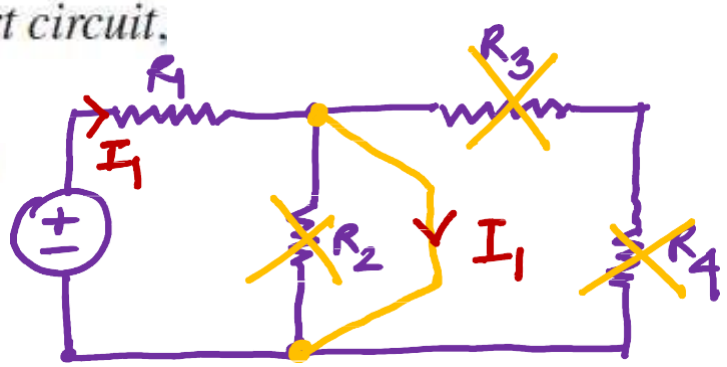
A short circuit is a circuit element with resistance approaching zero.

$R = 0$ is called a *short circuit*.



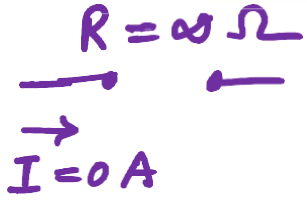
depends on the rest of network

$$v = iR = 0$$



An open circuit is a circuit element with resistance approaching infinity.

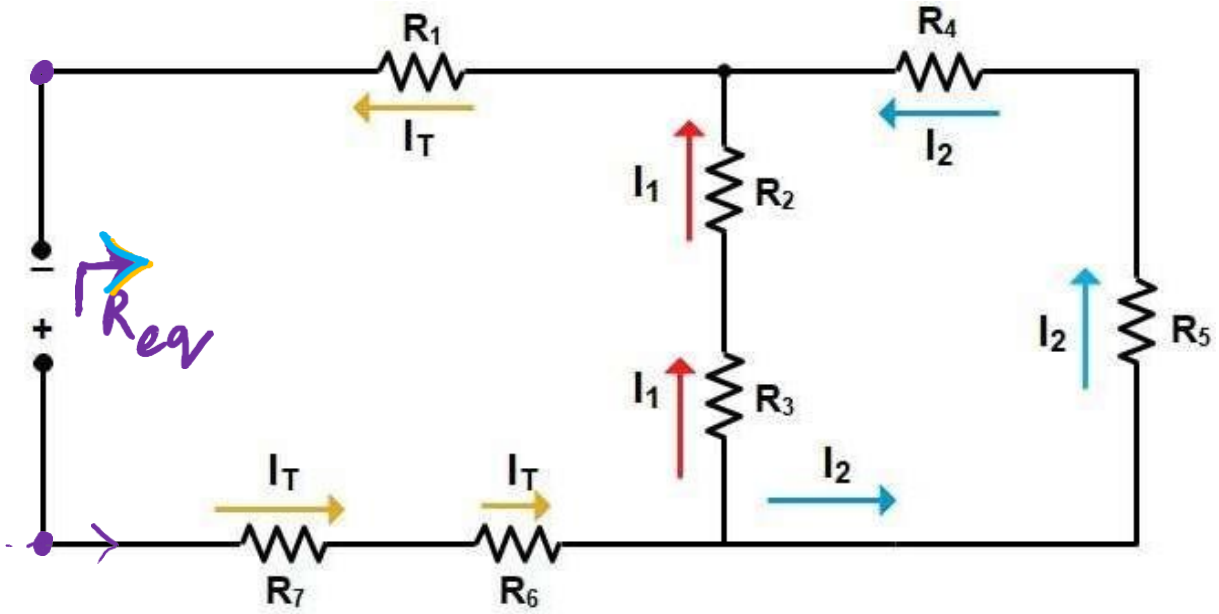
$R = \infty$ is known as an *open circuit*.



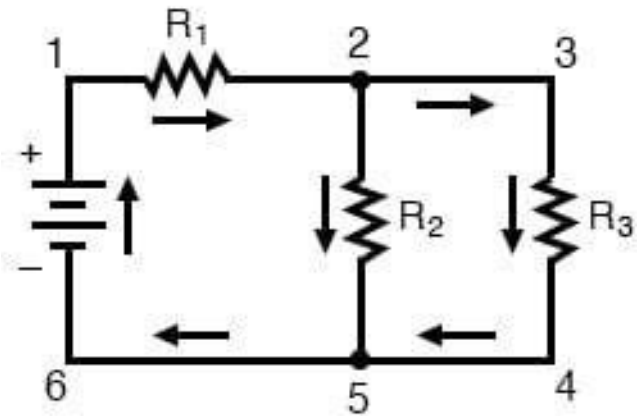
$$i = \lim_{R \to \infty} \frac{v}{R} = 0$$

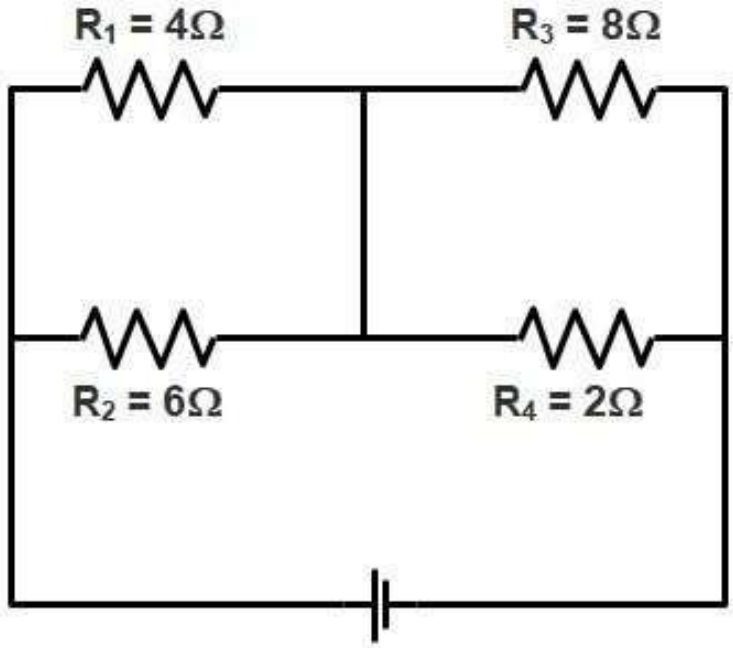
Source: Alexander and Sadiku

Series or parallel or none?



Tracing common current paths to identify series circuit





$R_{p_1} + R_{p_2}$

Source: Alexander and Sadiku

Example 2.9: Find R_{eq} for the circuit

Always start from the opposite side of R_{eq}

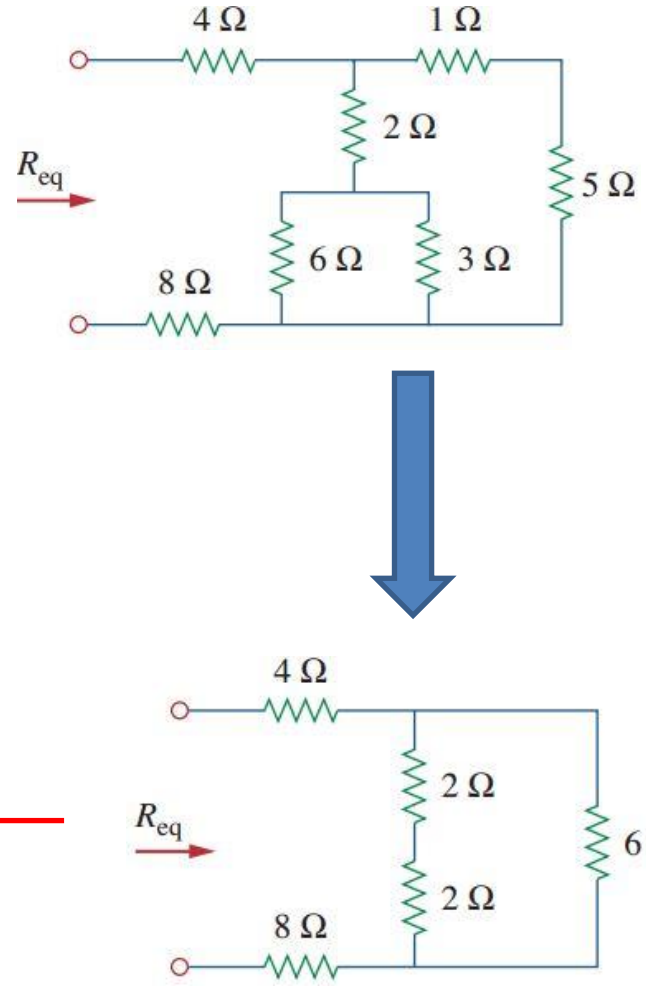
$1\ \Omega$ & $5\ \Omega$ series = $6\ \Omega$

$3\ \Omega$ & $6\ \Omega$ parallel = $\frac{3\times6}{3+6} = 2\ \Omega$

$2\ \Omega$ & $2\ \Omega$ series = $4\ \Omega$

$4\ \Omega$ & $6\ \Omega$ parallel = $\frac{4\times6}{4+6} = 2.4\ \Omega$

$4\ \Omega, 8\Omega$ & $2.4\ \Omega$ series = $14.4\ \Omega = R_{eq}$



2.4 Kirchhoff's current law (KCL)

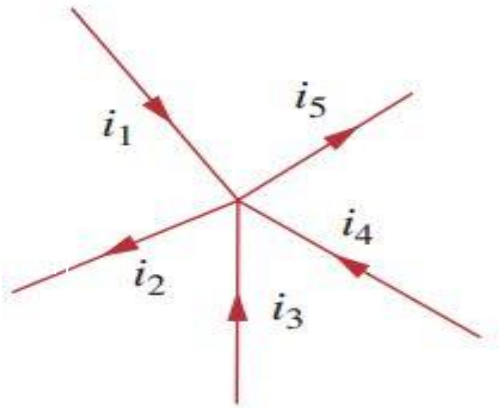
KCL: algebraic sum of currents entering a node (or a closed boundary) is zero.

Sum of the currents entering a node is equal to the sum of the currents leaving the node.

$$\sum_{node} i = 0A \rightarrow \sum_{node} i_{entering} = \sum_{node} i_{leaving}$$

$$+i_1 + (-i_2) + i_3 + i_4 + (-i_5) = 0$$

$$\rightarrow \boldsymbol{i_1 + i_3 + i_4 = i_2 + i_5}$$



2.4 Kirchhoff's voltage law (KVL)

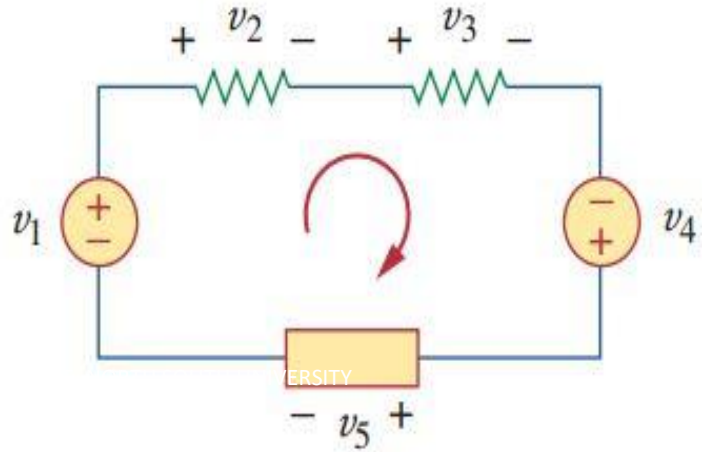
KVL: algebraic sum of all volt-ages around a closed path (or loop) is zero

Sum of voltage drops = Sum of voltage rises

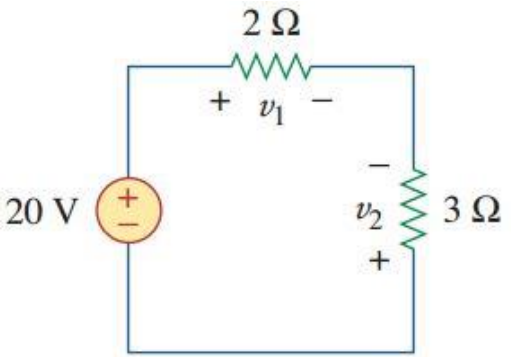
$$\sum_{loop} v = 0V \rightarrow \sum_{loop} v_{rise} = \sum_{loop} v_{drop}$$

$$-v_1 + v_2 + v_3 + v_5 - v_4 = 0$$

$$v_1 + v_4 = v_2 + v_3 + v_5$$



Example 2.5: Find v_1 and v_2

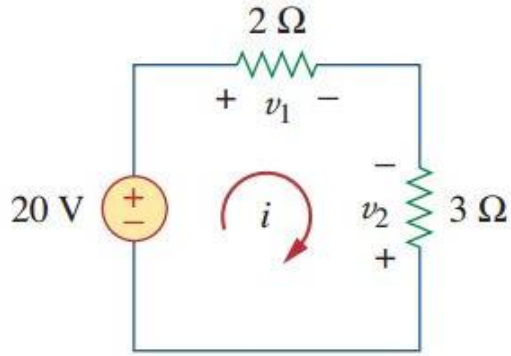


From Ohm's law,

$$v_1 = 2i, \qquad v_2 = -3i \tag{2}$$

Applying KVL around the loop gives

$$+20 + v_1 - v_2 = 0 \tag{2}$$



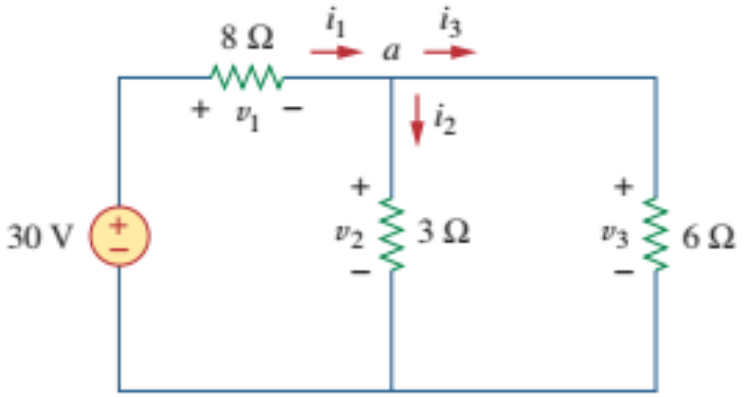
Substituting Eq. (2.5.1) into Eq. (2.5.2), we obtain

$$-20 + 2i + 3i = 0 \quad \text{or} \quad 5i = 20 \quad \Rightarrow \quad i = 4\ \text{A}$$

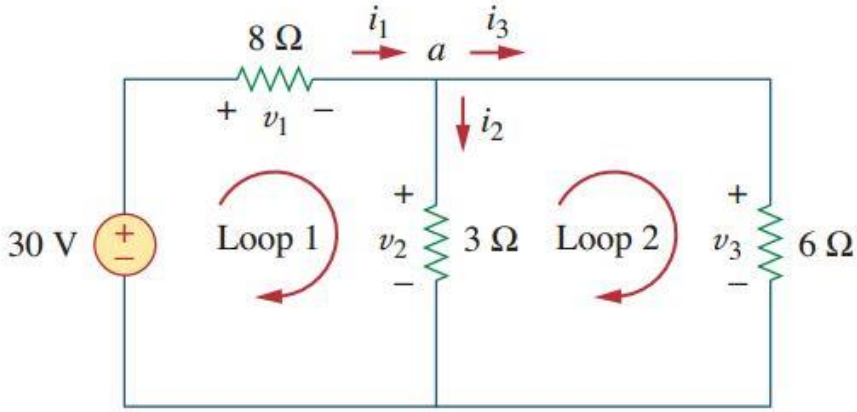
Substituting i in Eq. (2.5.1) finally gives

$$v_1 = 8\ \text{V}, \qquad v_2 = -12\ \text{V}$$

Example 2.8: Find currents and voltages in the circuit



- 6 unknowns. Need 6 equations.
- Ohm's Law: $v_1 = 8i_1; v_2 = 3i_2; v_3 = 6i_3$
- 3Ω and 6Ω: parallel
- KVL Loop 2: $v_2 = 3i_2 = v_3 = 6i_3$
- KCL at node a: $i_1 = i_2 + i_3$
- KVL Loop 1: $-30V + v_1 + v_2 = 0$
- $i_1 = 3A; i_2 = 2A; i_3 = 1A$
- $v_1 = 24V; v_2 = 6V; v_3 = 6V$



Analog Electronics II: Op-amp & 555 Timer

Course instructor:

Dr. Mohsin Sajjad

Assistant Professor

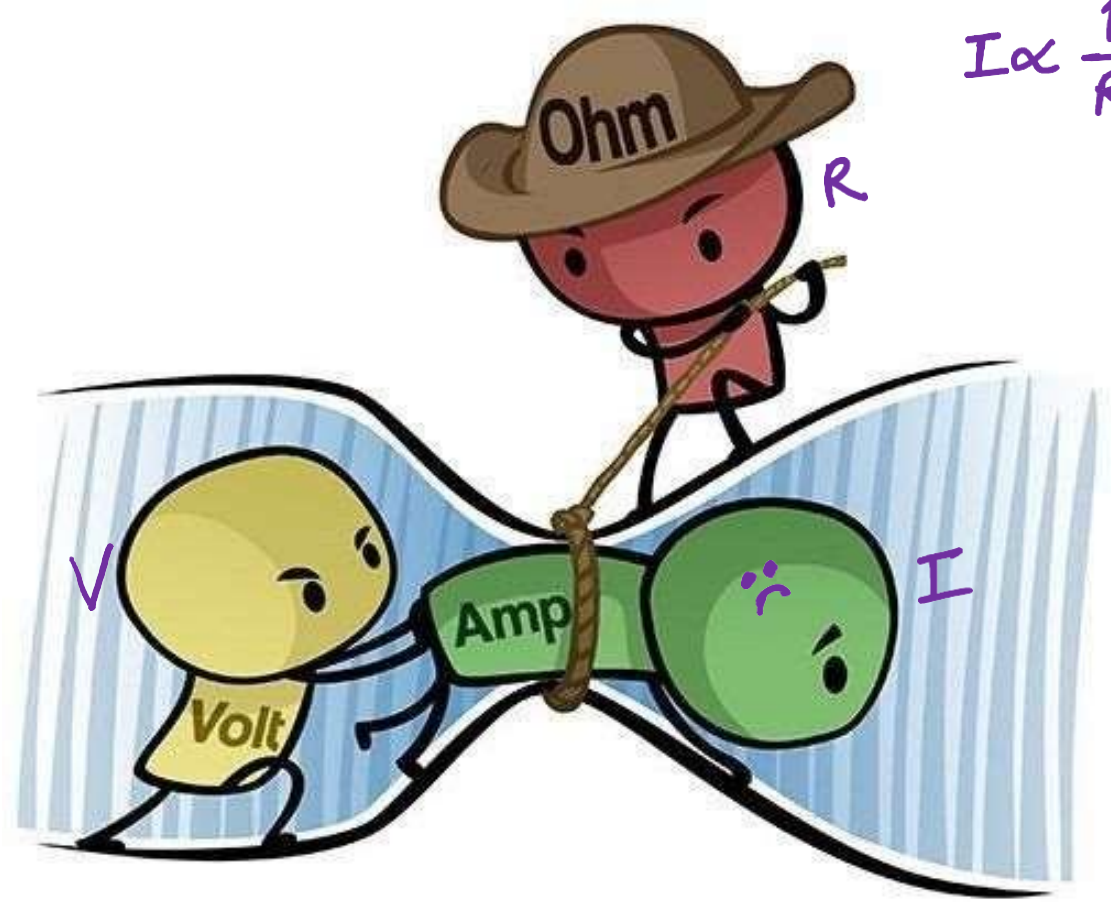
North South University

EEE 311

Lecture # 2

Current, voltage, resistance

Ohm's law



$I \propto \frac{1}{R}$ (V const)

$I \propto V$ (R const)

Source: Amino

Ohm's law

Ohm's law states that the voltage v across a resistor is directly proportional to the current i flowing through the resistor.

That is,

$$v \propto i \tag{2.2}$$

$$\boxed{v = iR}$$

The *resistance* R of an element denotes its ability to resist the flow of electric current; it is measured in ohms (Ω).

We may deduce from Eq. (2.3) that

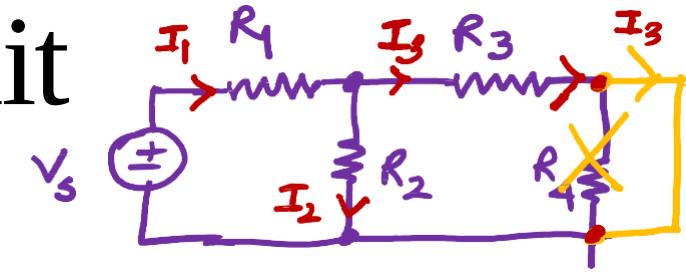
$$R = \frac{v}{i} \tag{2.4}$$

so that

$$1\ \Omega = 1\ \text{V/A}$$

Source: Alexander and Sadiku

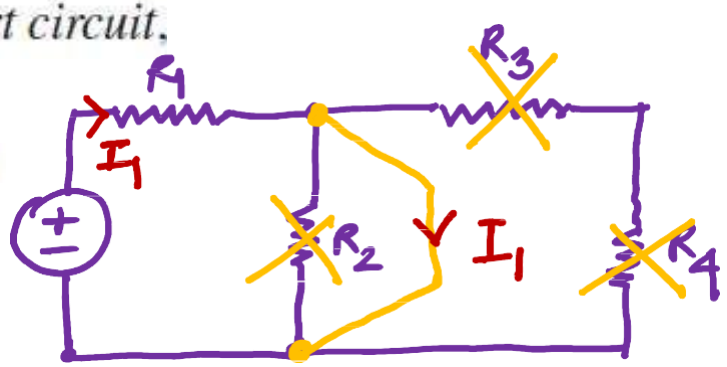
Short circuit and open circuit



A short circuit is a circuit element with resistance approaching zero.

$R = 0$ is called a *short circuit*.

$$v = iR = 0$$



An open circuit is a circuit element with resistance approaching infinity.

$R = \infty$ is known as an *open circuit*.

$$i = \lim_{R \to \infty} \frac{v}{R} = 0$$

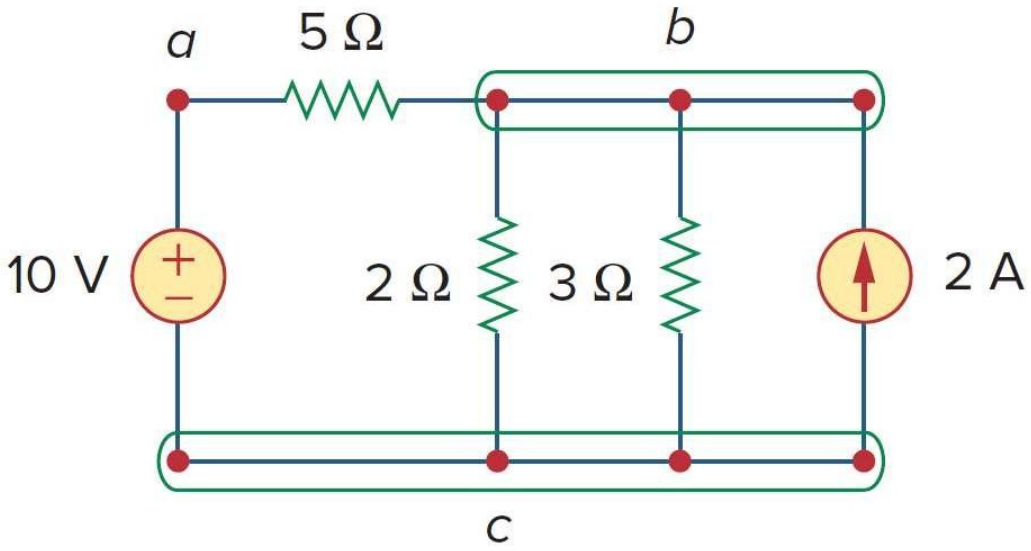
Source: Alexander and Sadiku

Nodes, Branches, and Loops

A branch represents a single element such as a voltage source or a resistor.

A node is the point of connection between two or more branches.

A loop is any closed path in a circuit.



Source: Alexander and Sadiku

Kirchhoff ’s Laws

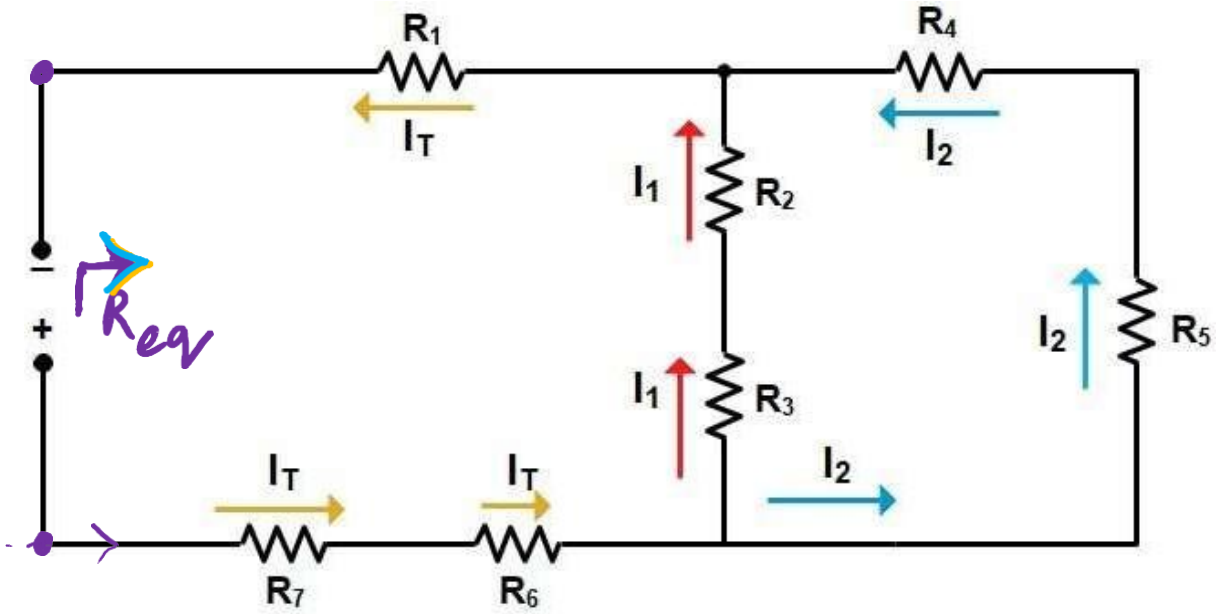
Kirchhoff's current law (KCL) states that the algebraic sum of currents entering a node (or a closed boundary) is zero.

The sum of the currents entering a node is equal to the sum of the currents leaving the node.

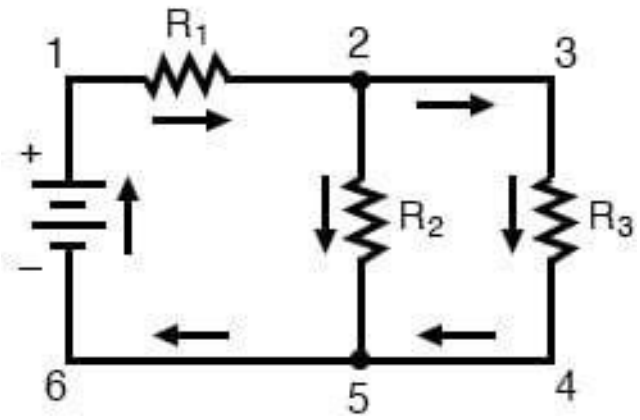
Kirchhoff's voltage law (KVL) states that the algebraic sum of all voltages around a closed path (or loop) is zero.

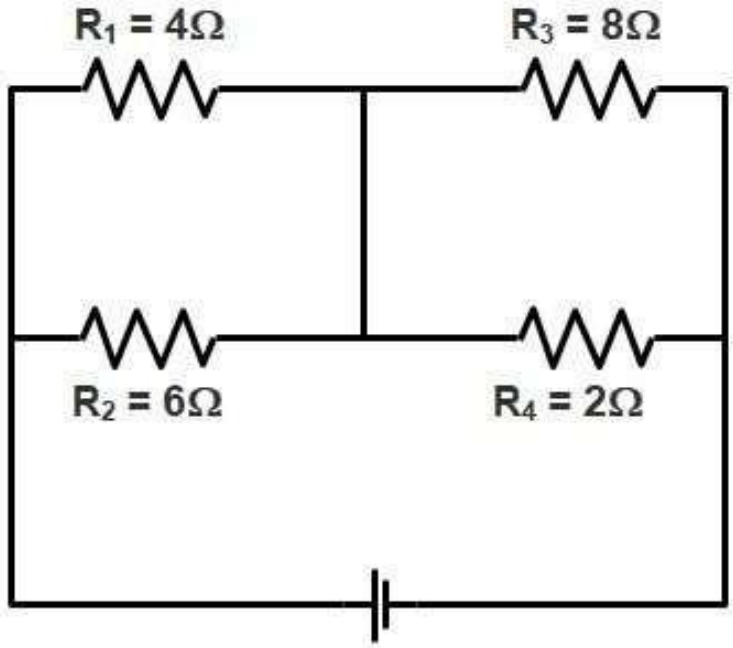
Source: Alexander and Sadiku

Series or parallel or none?



Tracing common current paths to identify series circuit

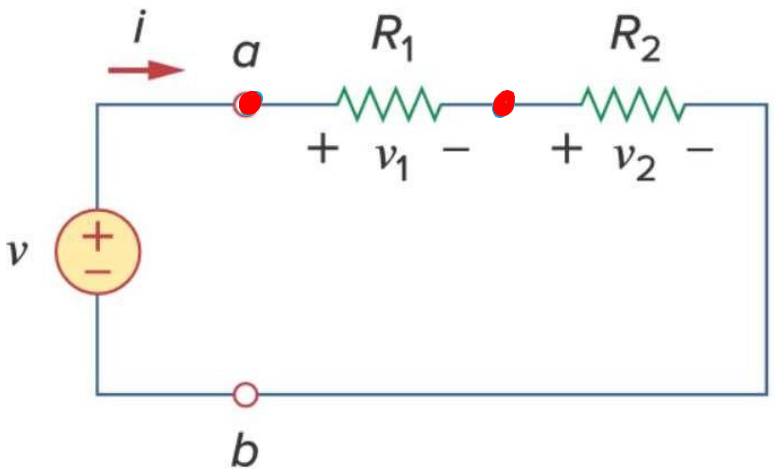


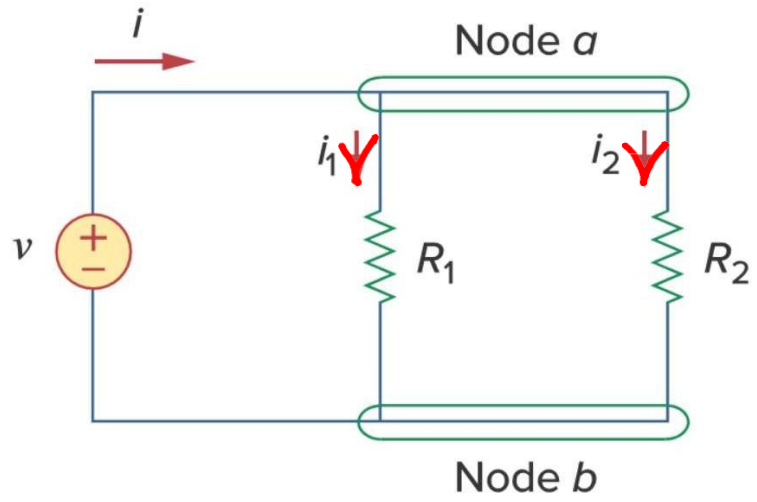


$R_{p_1} + R_{p_2}$

Source: Alexander and Sadiku

Voltage divider rule and Current Divider rule





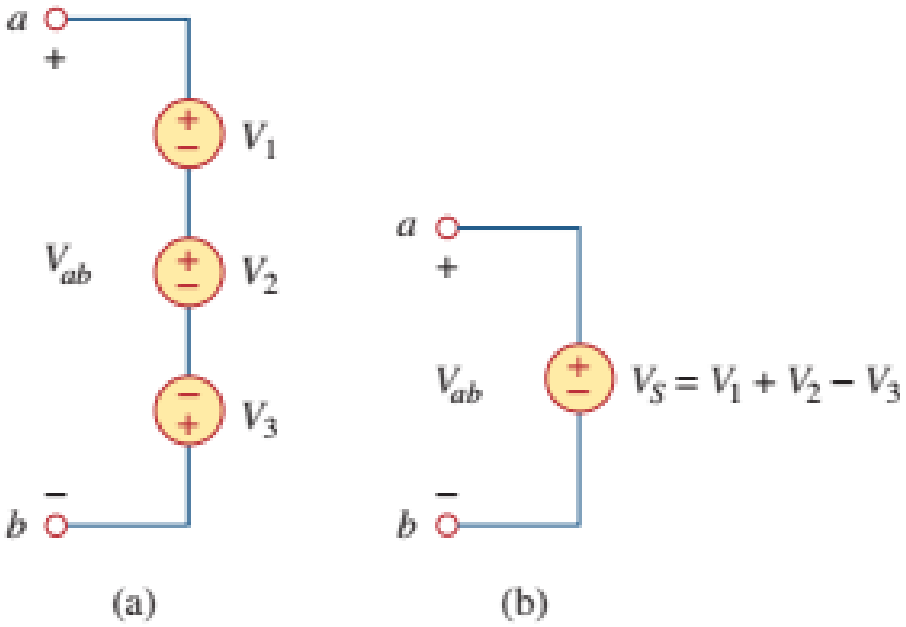
$$v_1 = \frac{R_1}{R_1 + R_2} v$$

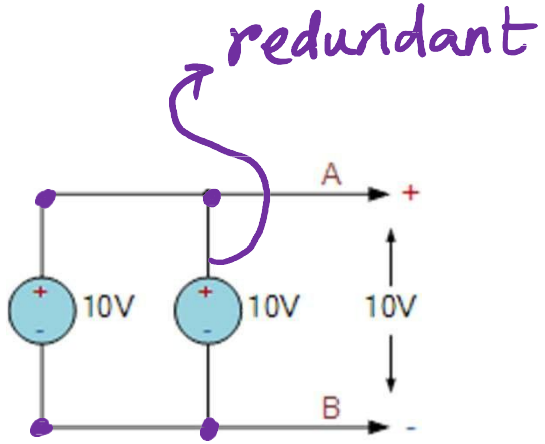
$$v_2 = \frac{R_2}{R_1 + R_2} v$$

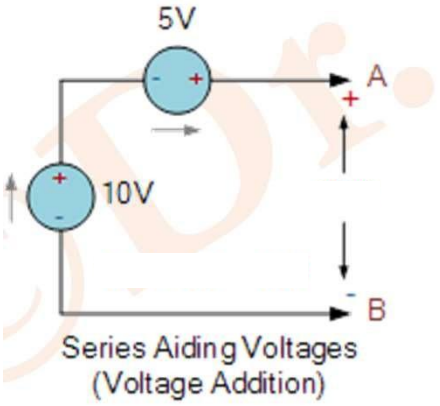
$$i_1 = \frac{R_2\, i}{R_1 + R_2}$$

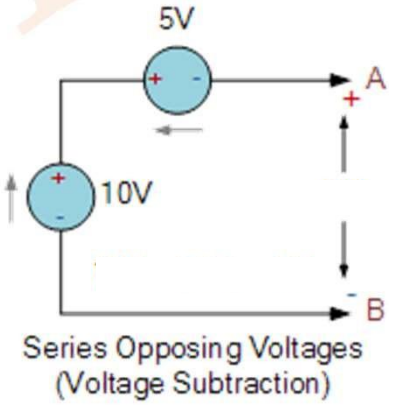
$$i_2 = \frac{R_1\, i}{R_1 + R_2}$$

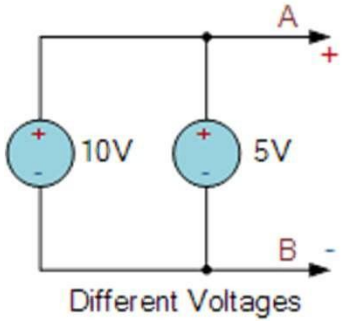
Voltage sources in series and parallel

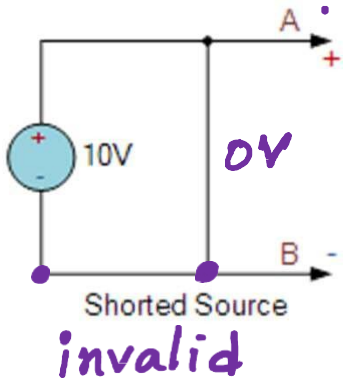






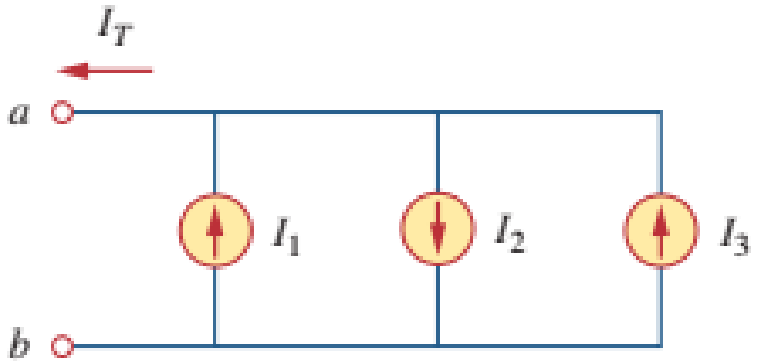




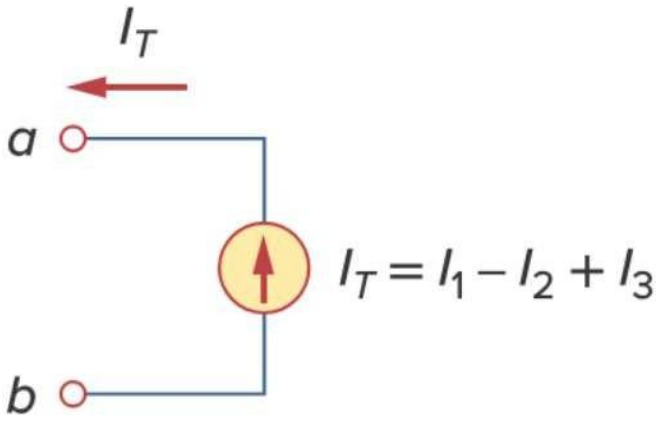


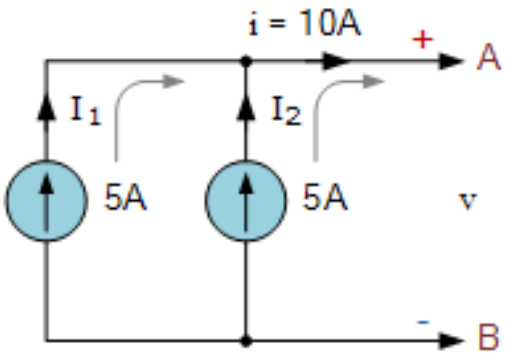
Source: Alexander and Sadiku

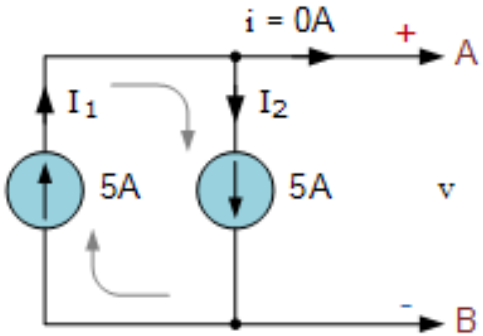
Current sources in series and parallel

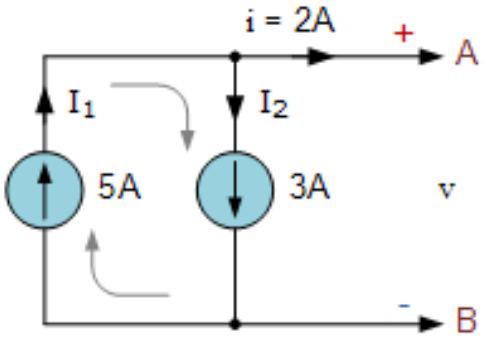


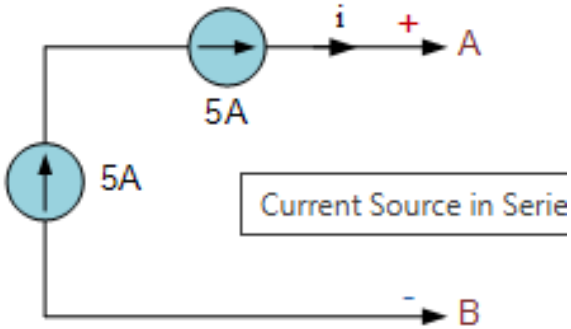
(a)

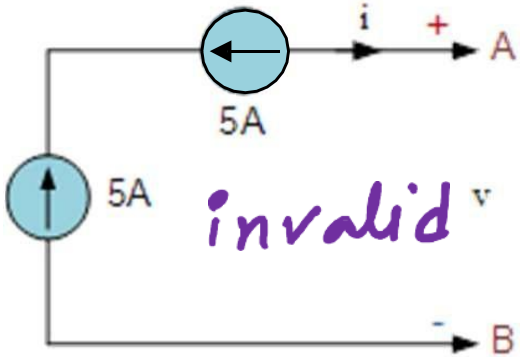










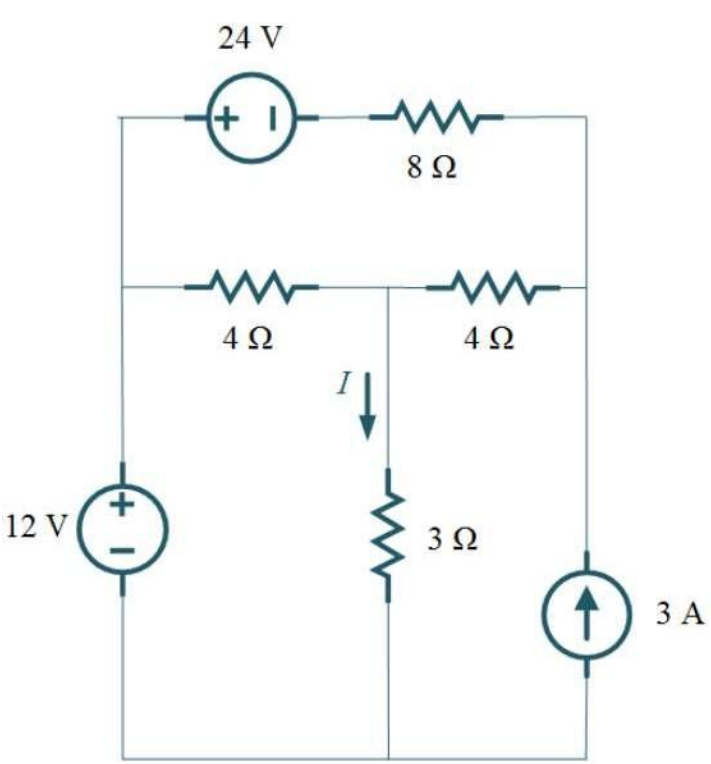


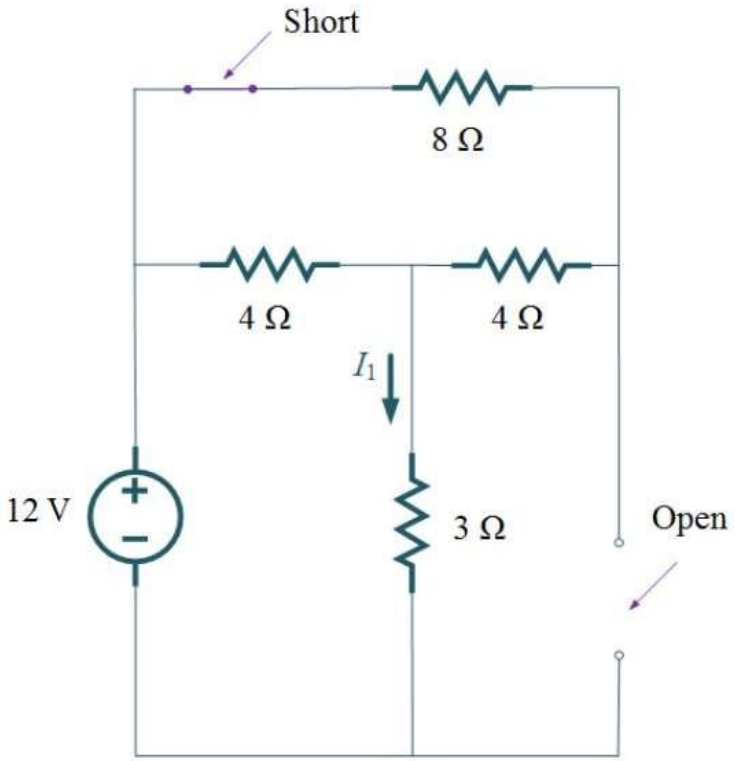
Source: Alexander and Sadiku

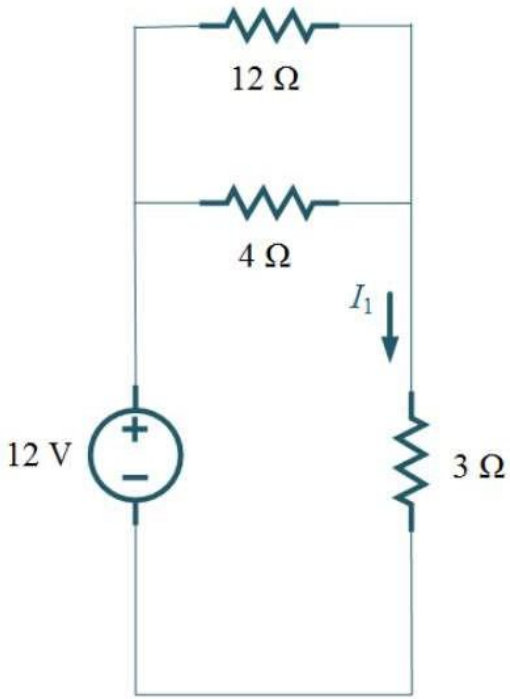
DC circuit problem with multiple sources

- Source transformation theorem
- Superposition principles
- Mesh current analysis
- Node voltage analysis
- Thevenin’s theorem
- Norton’s theorem

Superposition principle



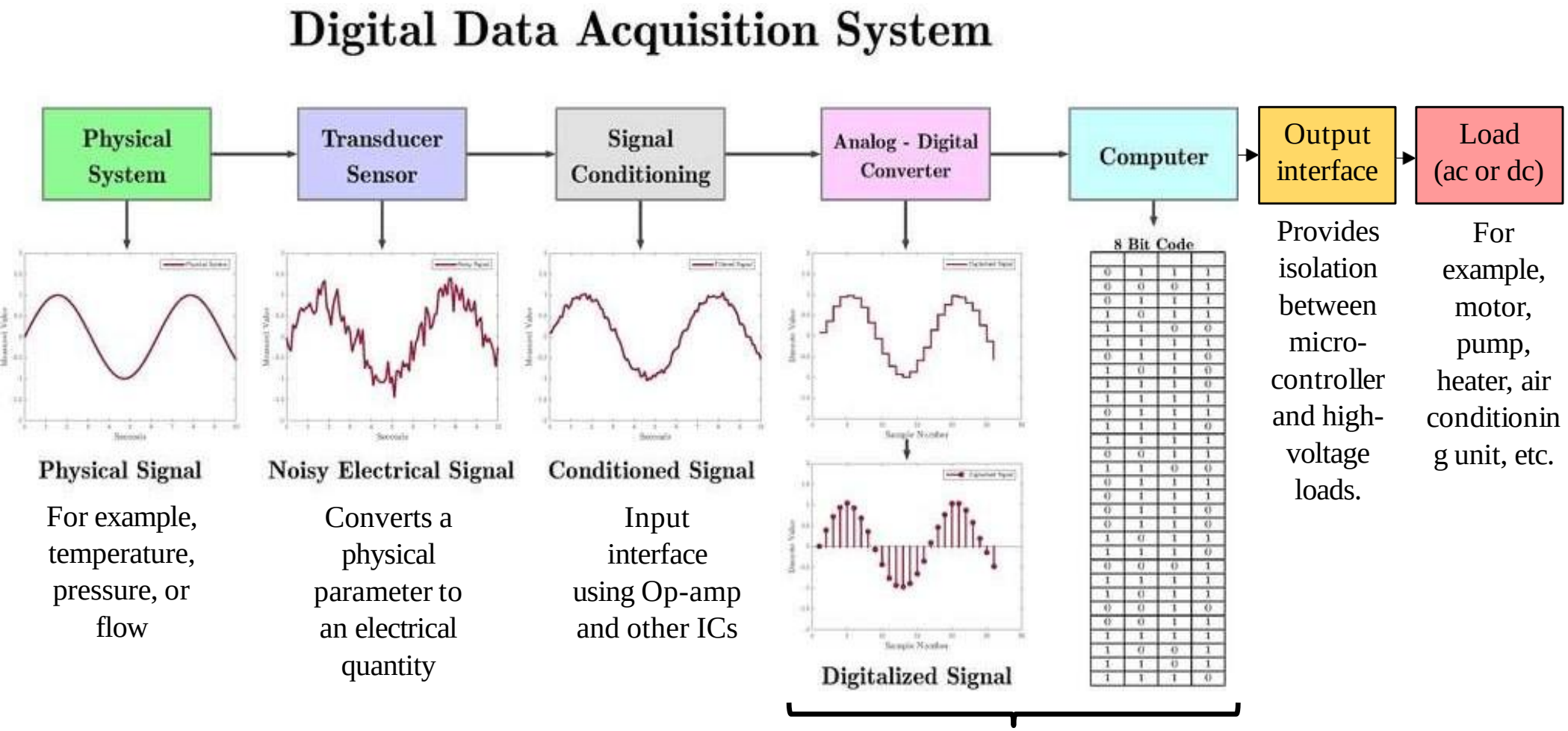




Problem Link: https://electricalworkbook.com/superposition-theorem-in-electric-circuits/

Source: Alexander and Sadiku

Why analog circuit in the digital world?

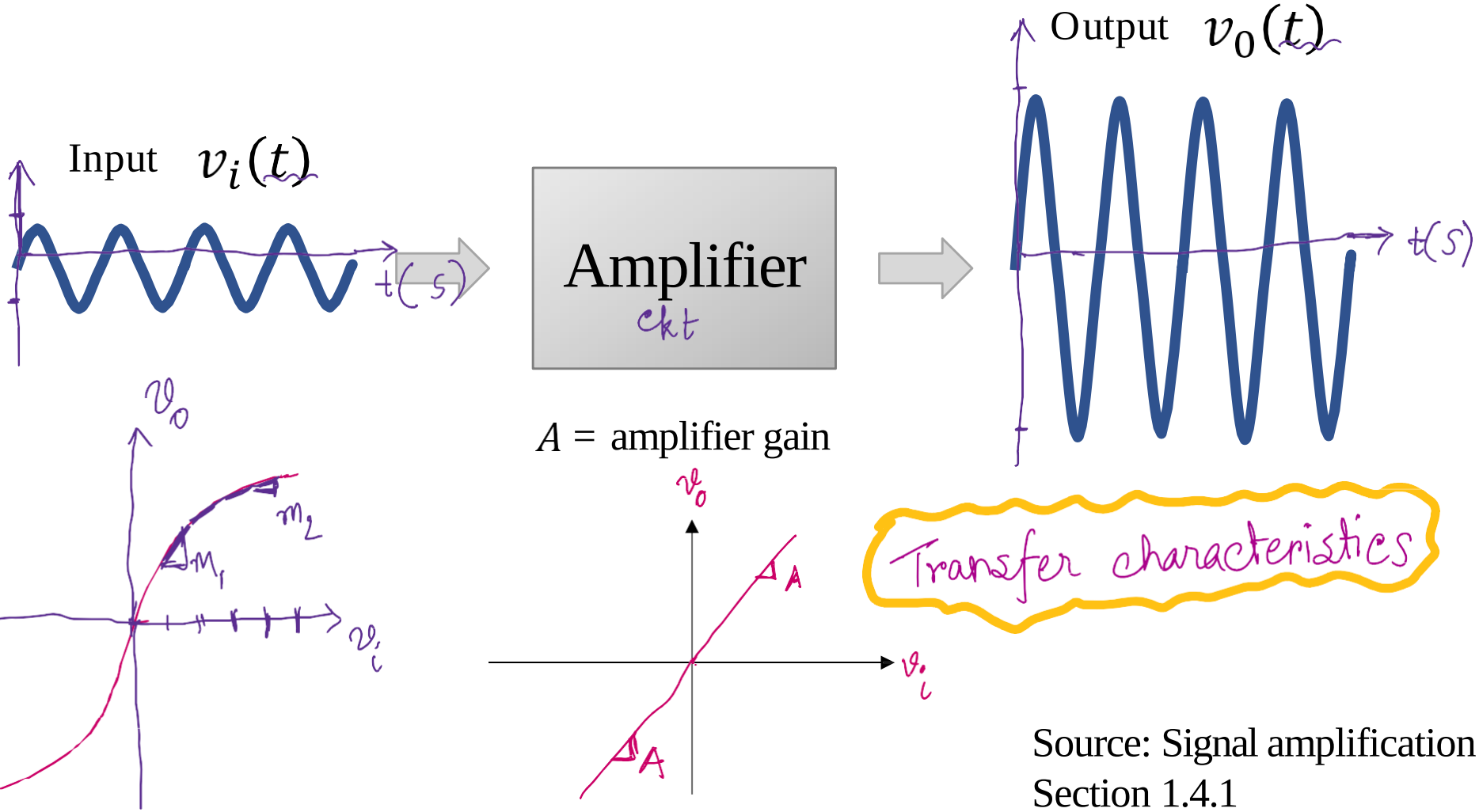


Source: Coughlin, section 1-1 and Wikipedia

Amplifier

For linear amplifier,

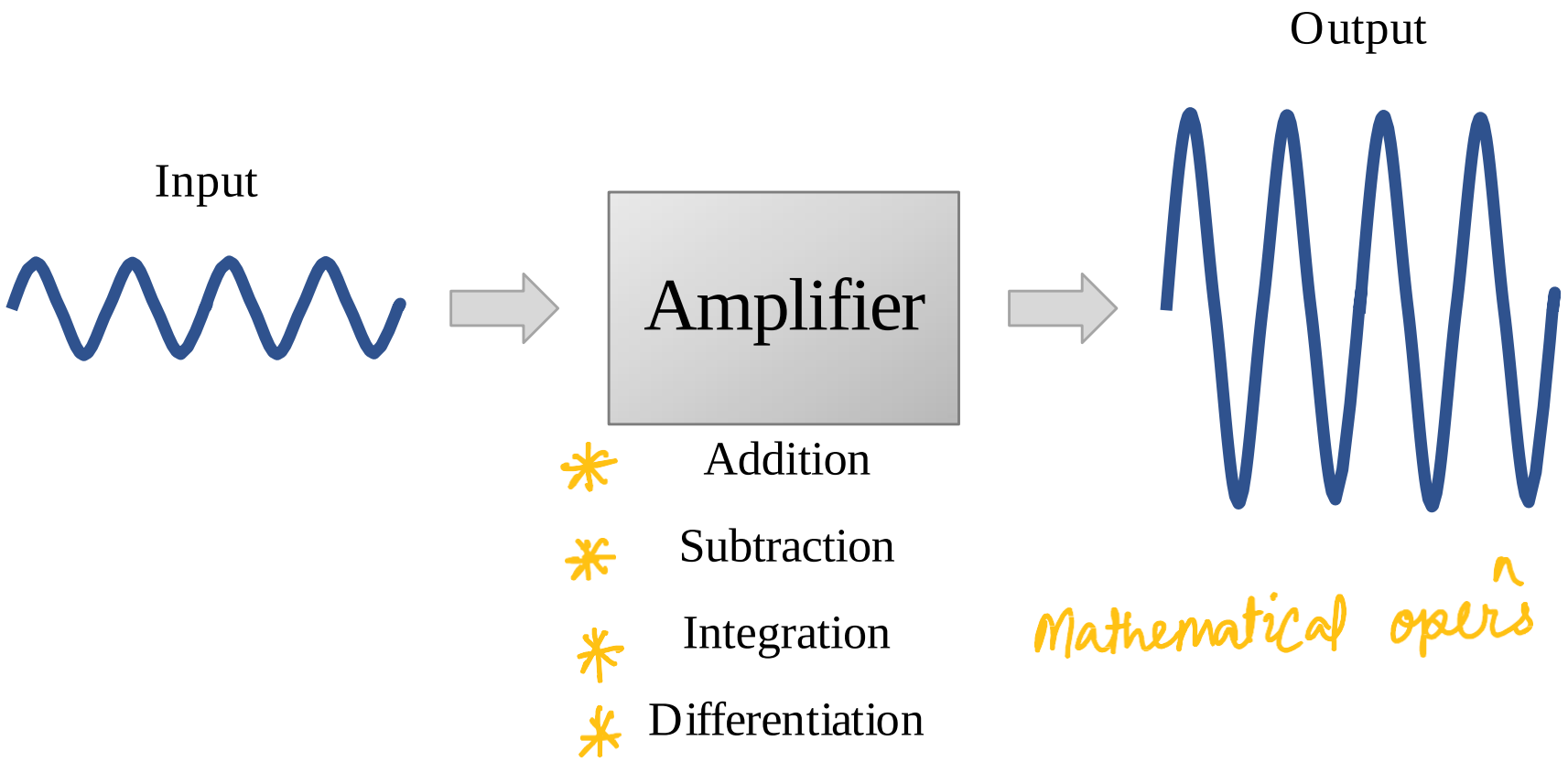
$$v_0(t) = Av_i(t)$$



Source: Signal amplification
Section 1.4.1
Sedra Smith

OP-AMP: an amplifier

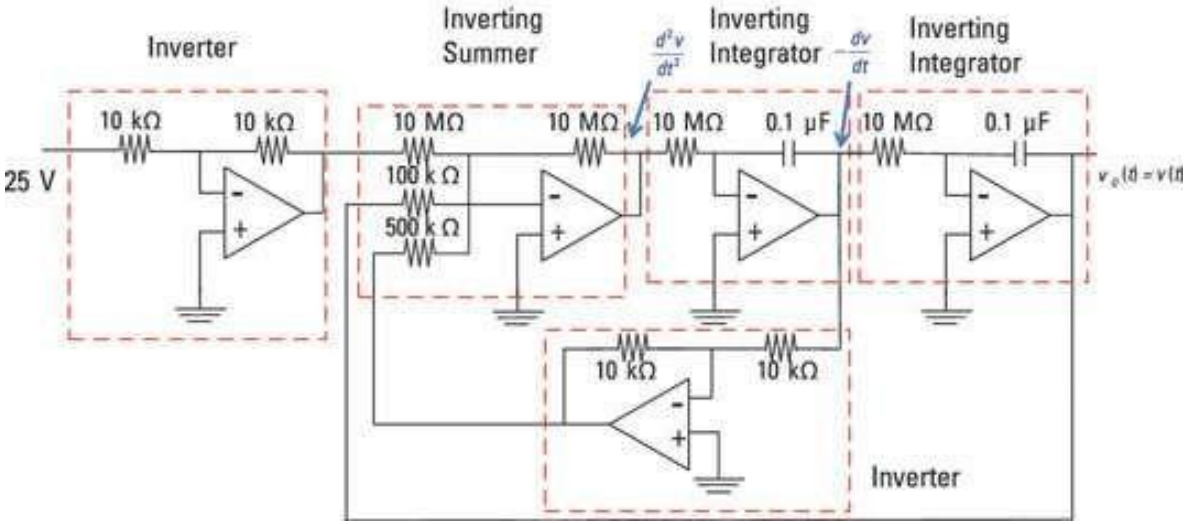
OP-AMP: Operational amplifier



Operational amplifier or Op-Amp

- One of the most versatile and widely used electronic devices in linear applications
- Low-cost and easy-to-use
- Many operations, such as addition, subtraction, multiplication, integration, differentiation, solving differential equations, and so on

$$\frac{d^2v}{dt^2} = -20\frac{dv}{dt} - 100\,v + 25\text{ V}$$



$v_o = v_1 + v_2$

$v_o = v_1 - v_2$

$v_o = 4 \cdot v_1 + 5 \cdot v_2$

$v_o = 2 \cdot v_1 - 8 v_2$

dummies

Source: Coughlin, section 1-0

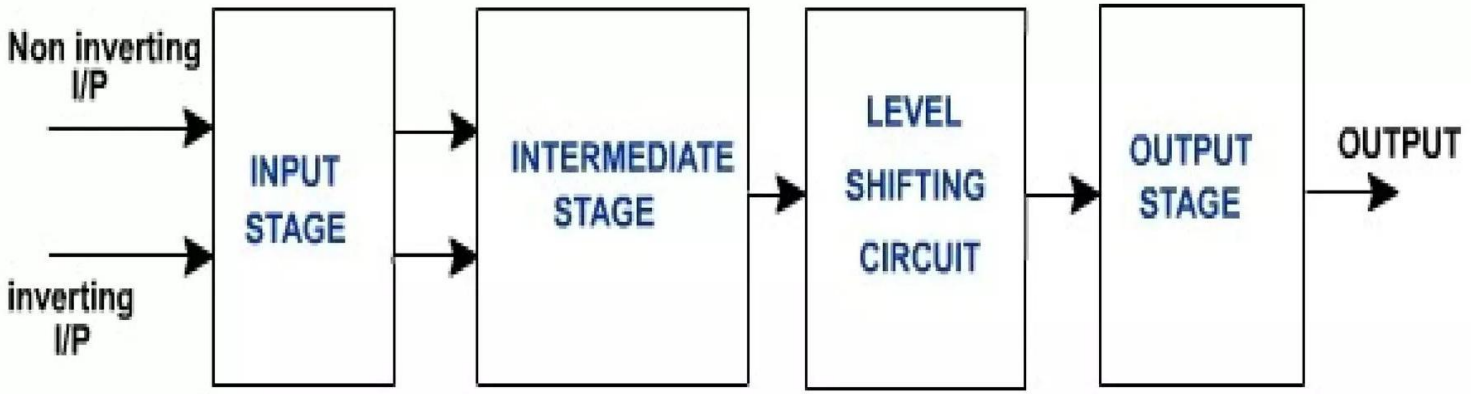
Operational amplifier or Op-Amp

✓An operational amplifier (op-amp) is a DC-coupled high- gain electronic voltage amplifier

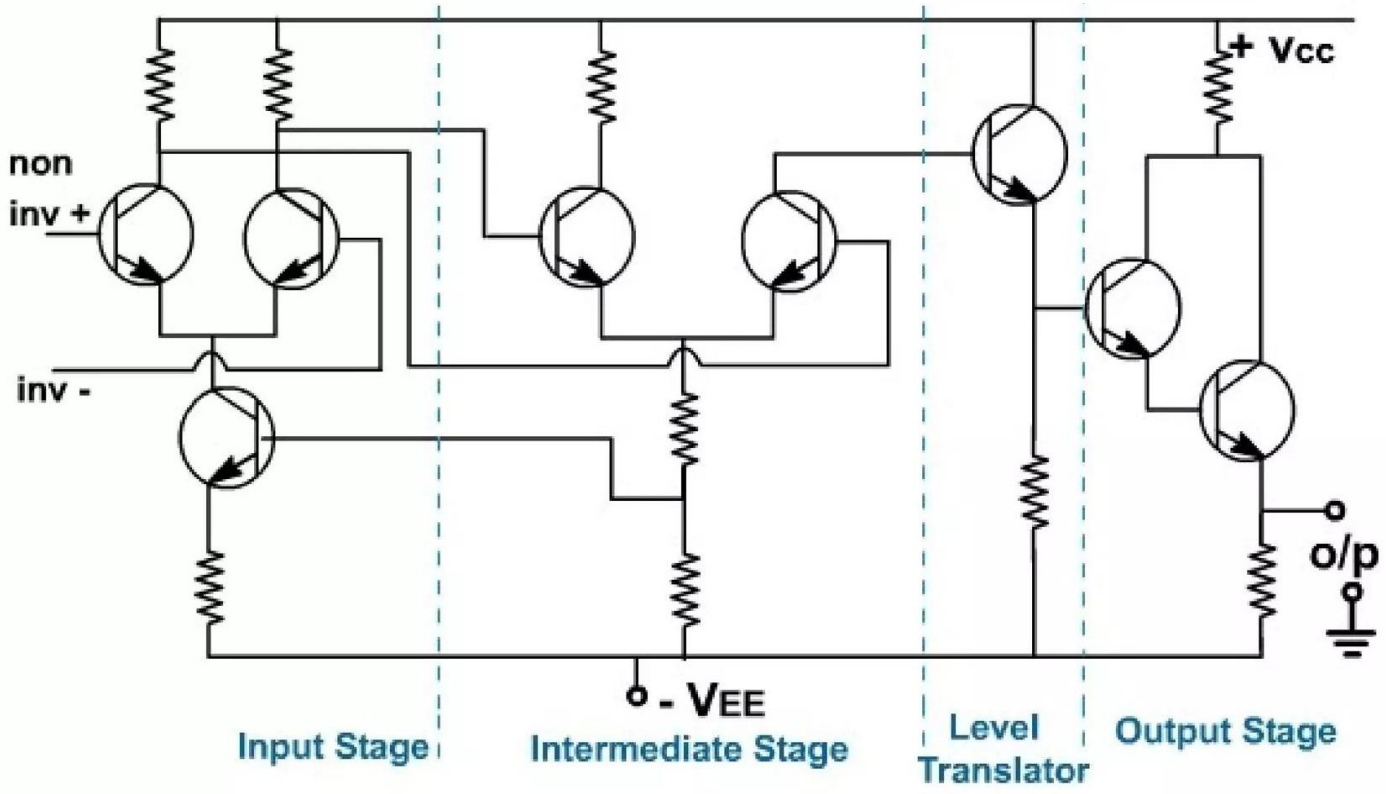
✓ Direct- coupled high gain amplifier usually consisting of one or more differential amplifiers

✓Designed for computing mathematical functions such as addition, subtraction ,multiplication, integration & differentiation

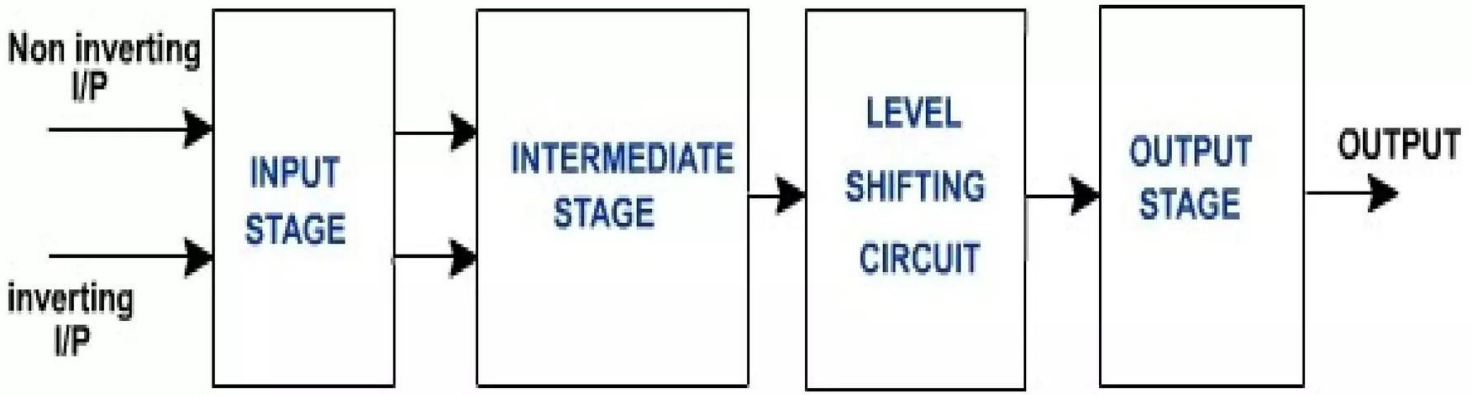
BLOCK DIAGRAM



INTERNAL CIRCUIT DIAGRAM



INTERNAL CIRCUIT DIAGRAM

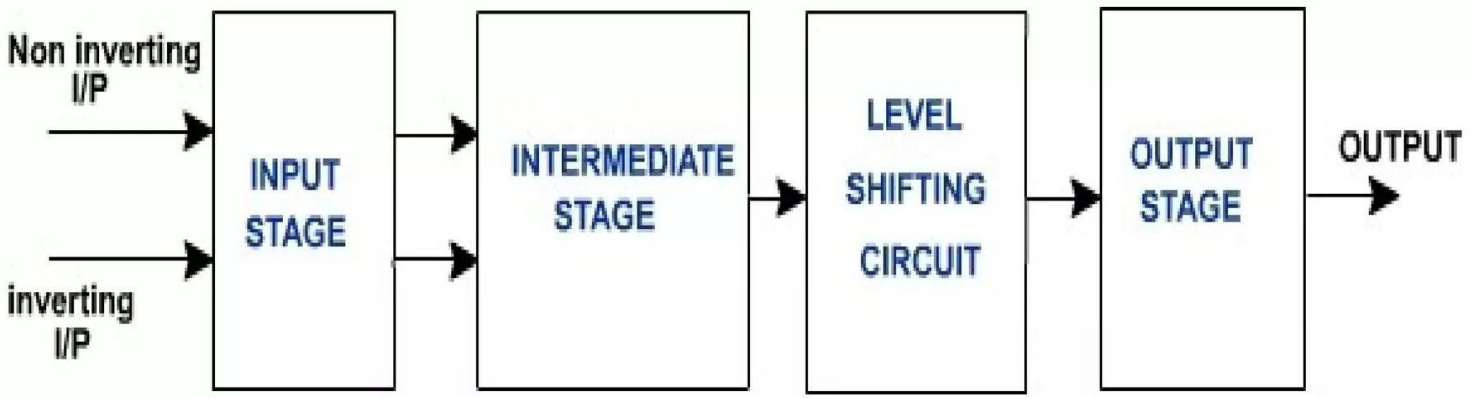


INPUT STAGE

The input stage is a dual input balanced output differential amplifier. This stage provides most of the voltage gain of the amplifier and also establishes the input resistance of the OPAMP.

INTERMEDIATE STAGE

The intermediate stage of OPAMP is another differential amplifier which is driven by the output of the first stage. This is usually dual input unbalanced output.



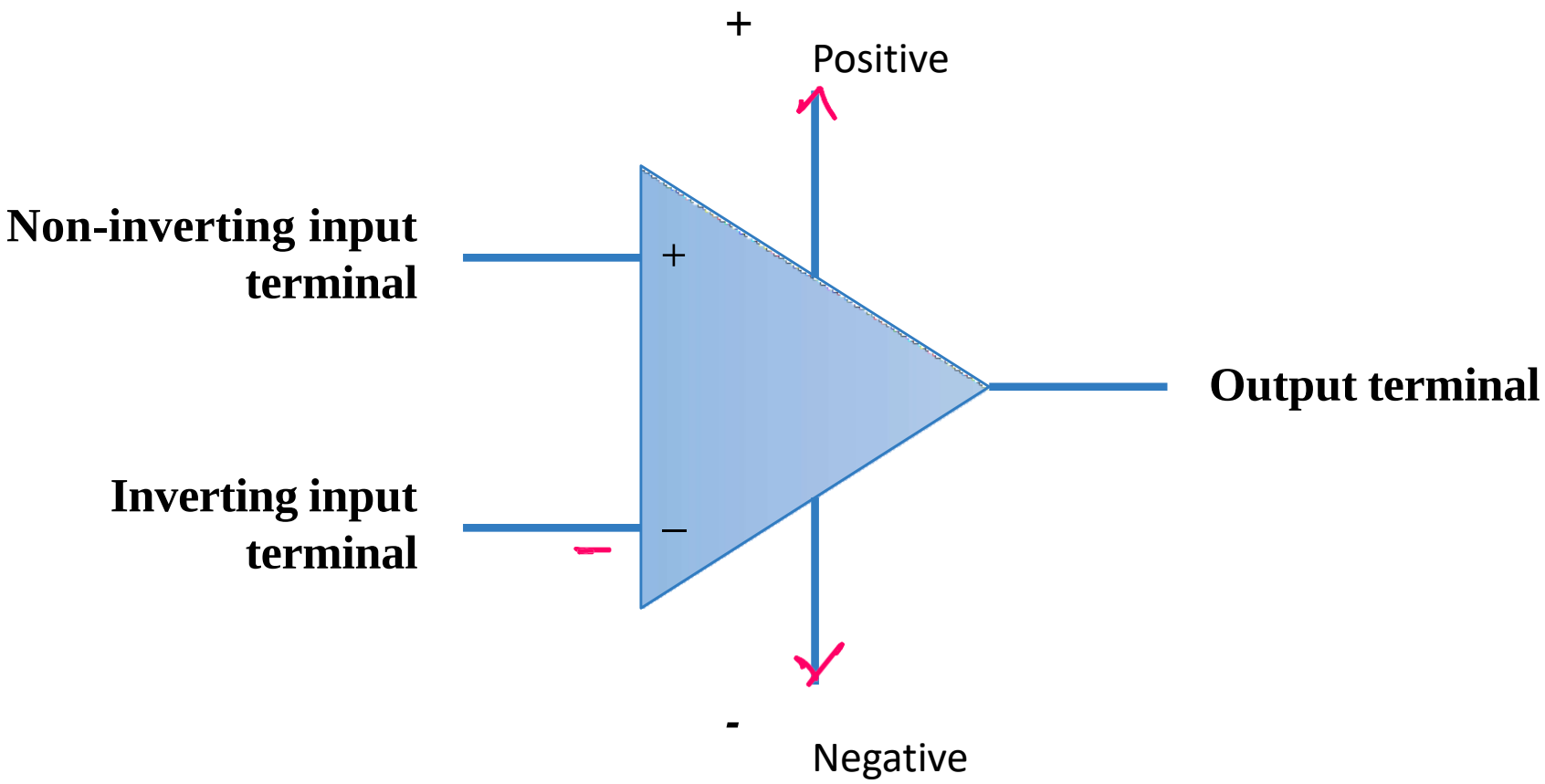
LEVEL SHIFTING STAGE

Because direct coupling is used, the dc voltage level at the output of intermediate stage is well above ground potential. Therefore level shifting circuit is used to shift the dc level at the output downward to zero with respect to ground.

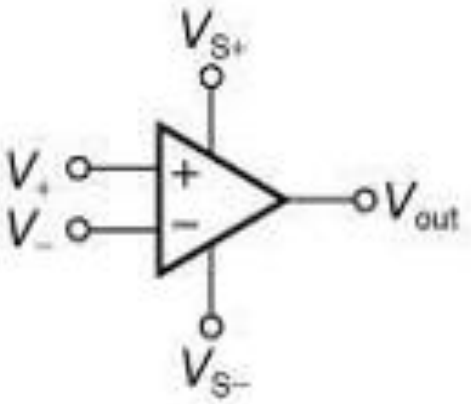
OUTPUT STAGE

The output stage is generally a push pull complementary amplifier. The output stage increases the output voltage swing and raises the current supplying capability of the OPAMP. It also provides low output resistance.

OP-AMP circuit symbol



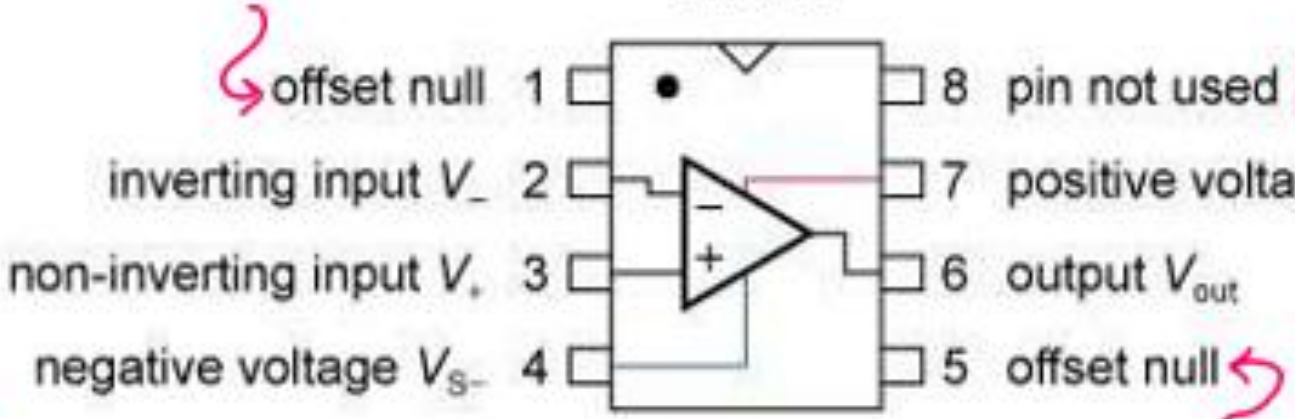
Circuit symbol



(a)

Dual-in-line package (DIP)

(b) Terminals are numbered counter-clockwise

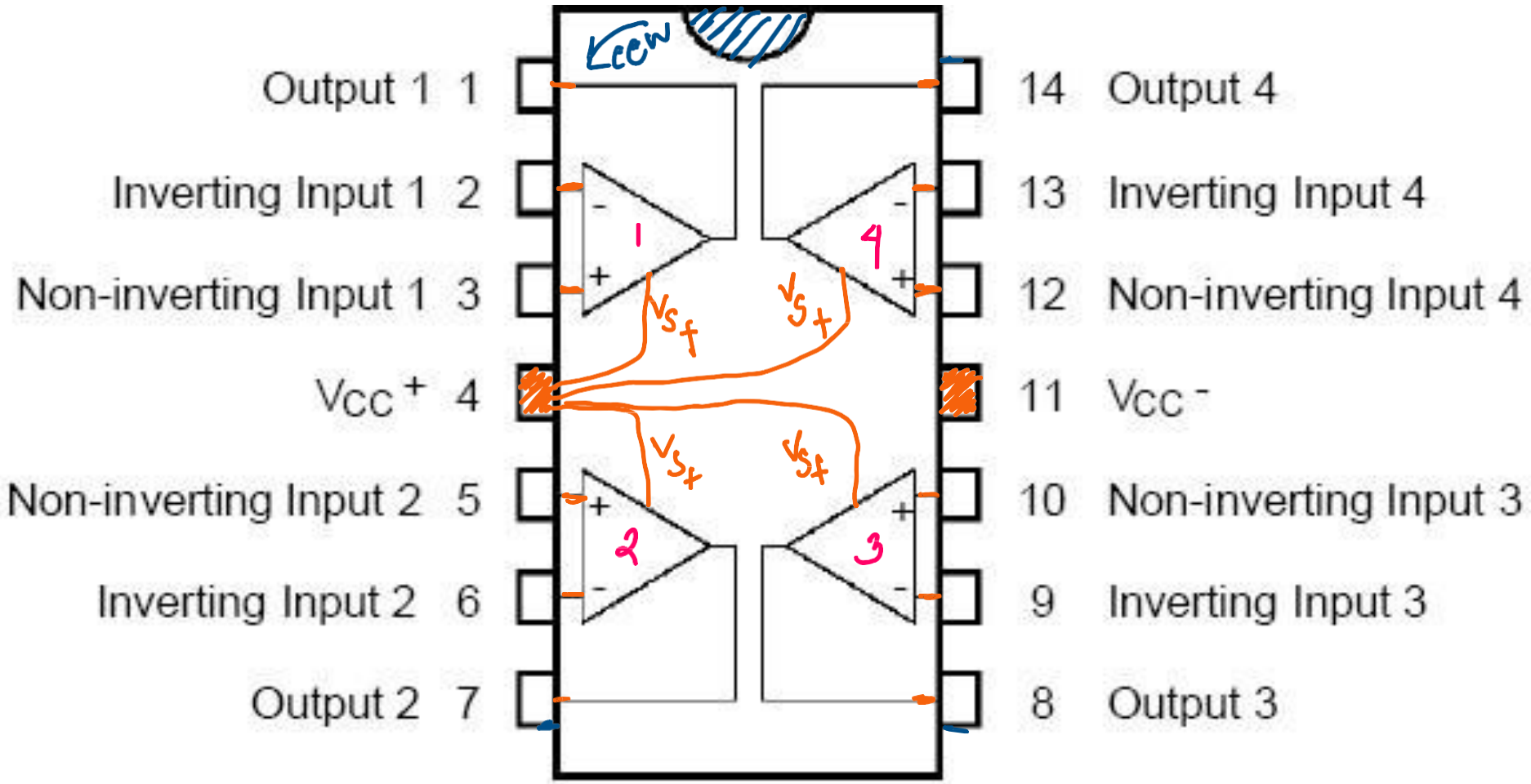


Connection diagram

(c)

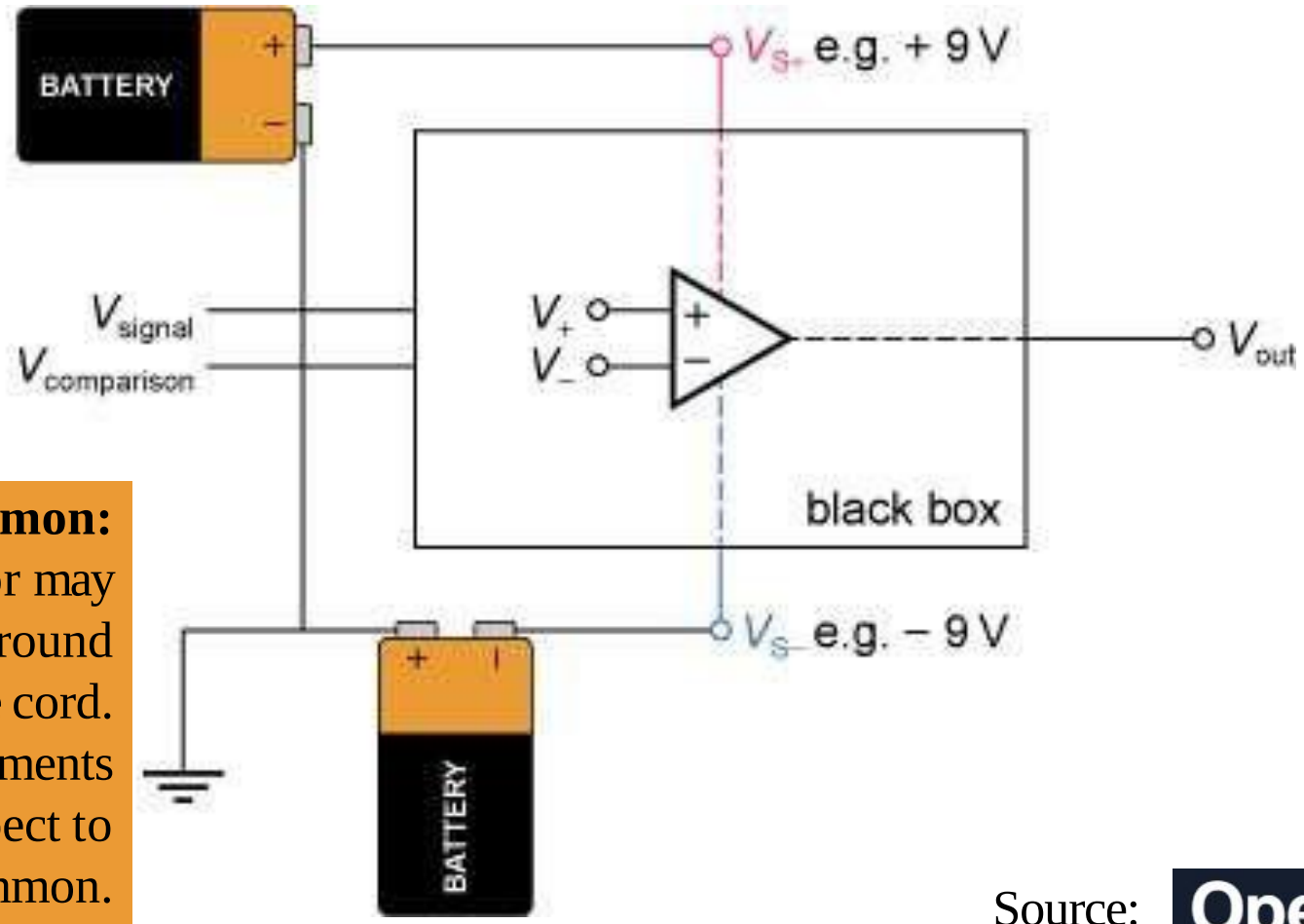
Source: OpenLearn

Quad op-amp



Source: Soundtronics.co.uk

Dual power supplies

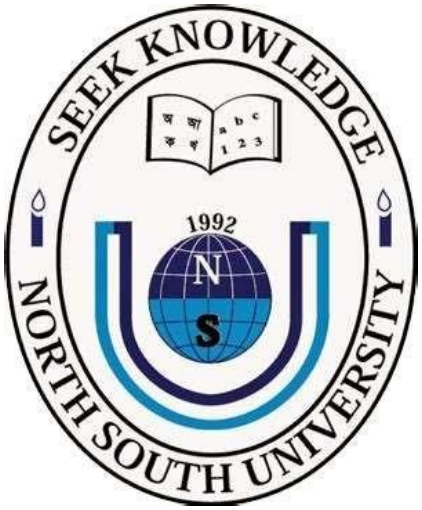


Power supply common: This terminal may or may not be wired to earth ground via the 3rd wire of line cord. All voltage measurements are made with respect to power supply common.

Source: OpenLearn

Coughlin 1-6.1, 2-1.1

Analog Electronics II



Course instructor:

Dr. Mohsin Sajjad

Assistant Professor

North South University

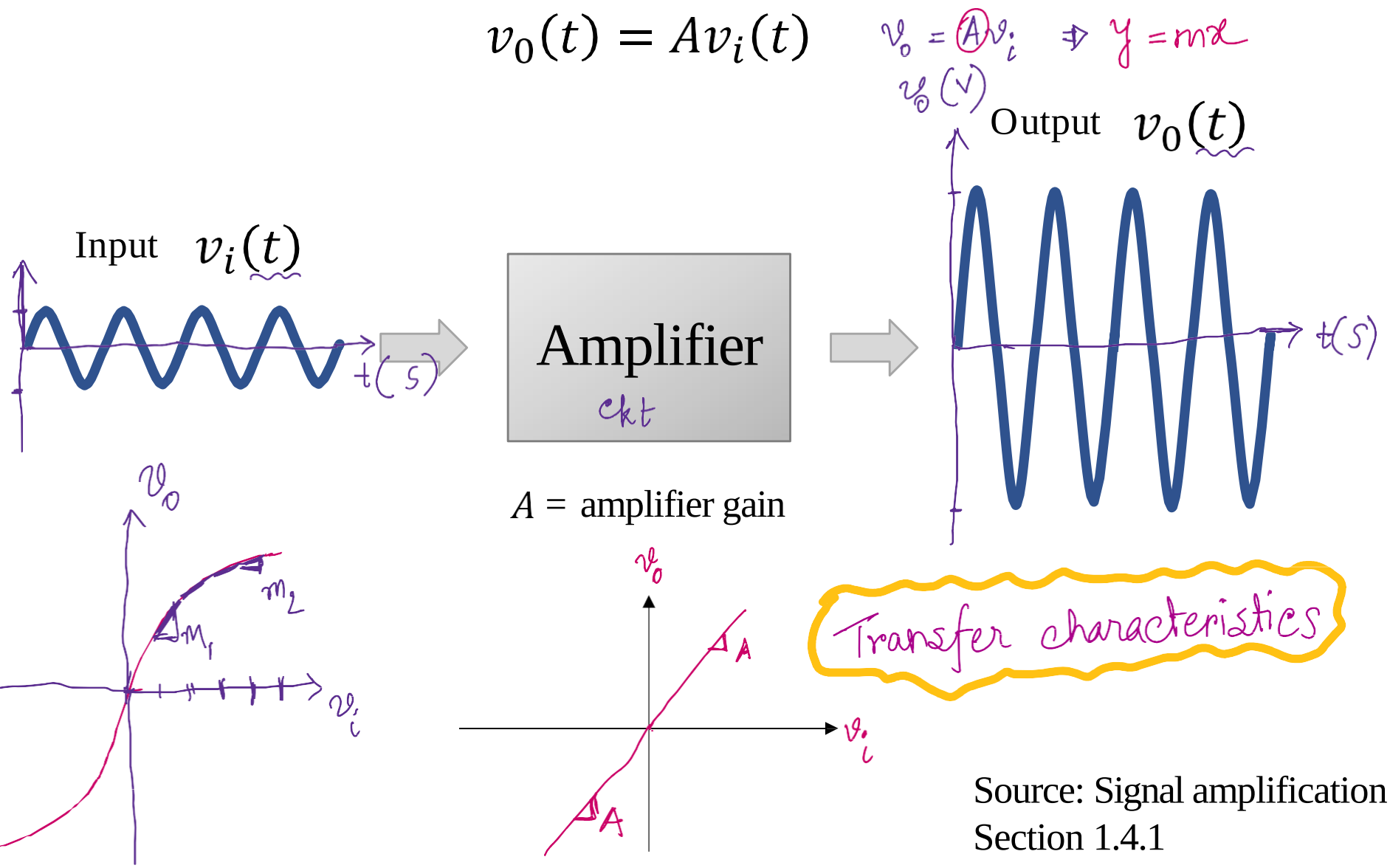
EEE 311

Lecture # 3

Amplifier

For linear amplifier,

$$v_0(t) = Av_i(t)$$

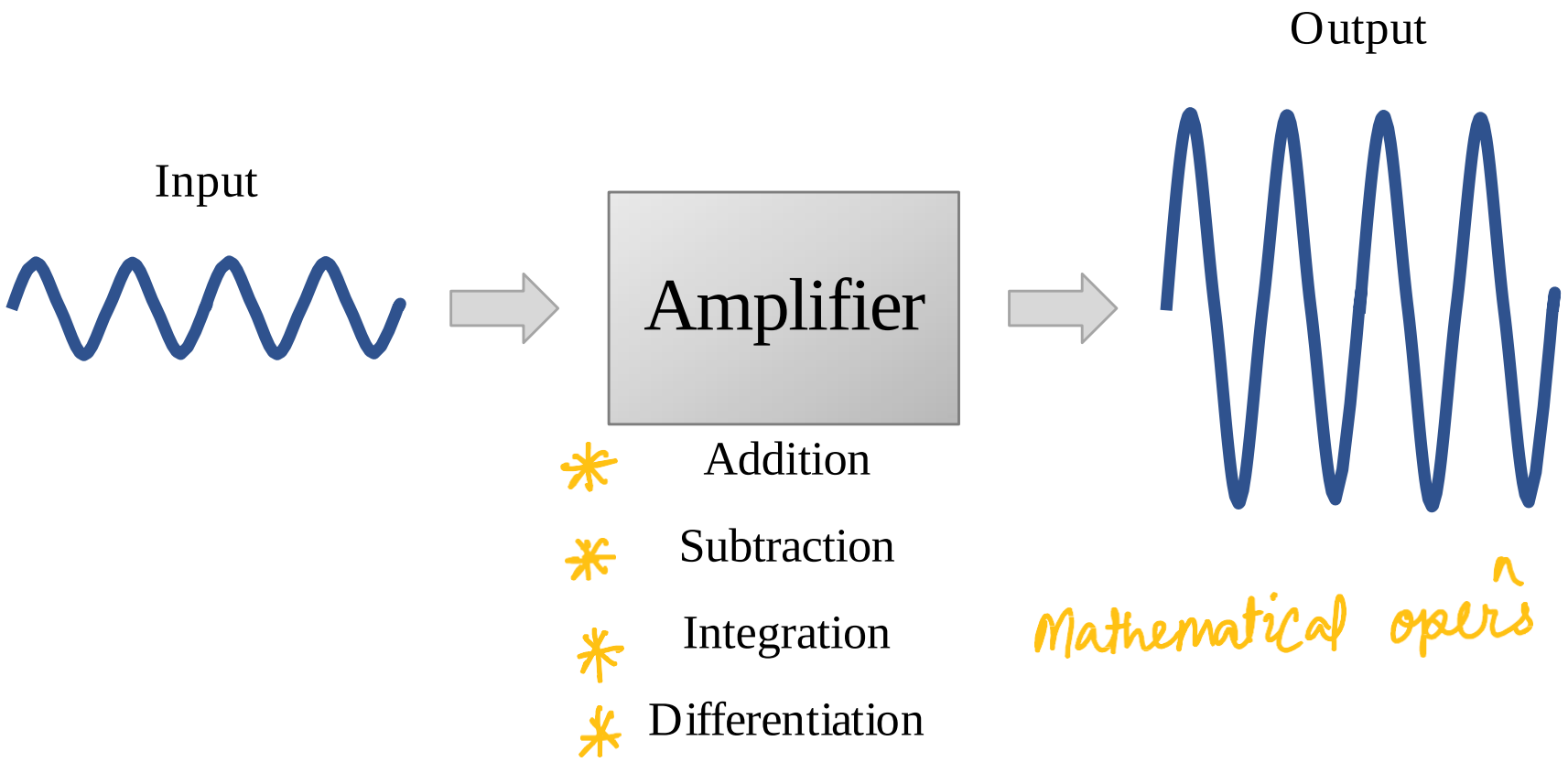


A = amplifier gain

Source: Signal amplification
Section 1.4.1
Sedra Smith

OP-AMP: an amplifier

OP-AMP: Operational amplifier



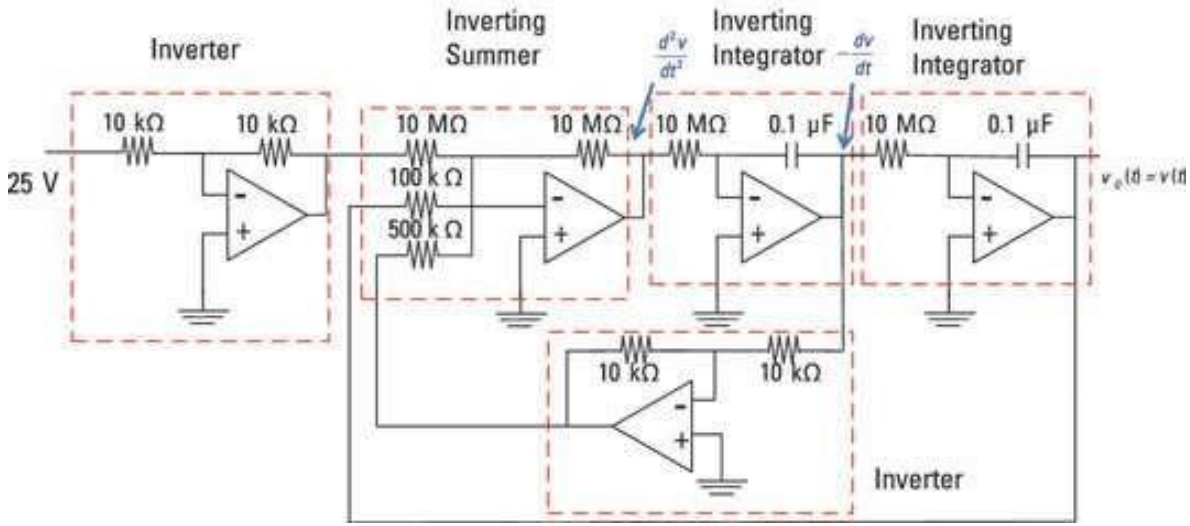
Operational amplifier or Op-Amp

- One of the most versatile and widely used electronic devices in linear applications $v_o \propto v_i$ (linear)
- Low-cost and easy-to-use
- Many operations, such as addition, subtraction, multiplication, integration, differentiation, solving differential equations, and so on

$v_o = 2 \cdot v_1 + 3 \cdot v_2 - 8 \cdot v_3$

$y'', y', y \rightarrow x$

$$\frac{d^2v}{dt^2} = -20\frac{dv}{dt} - 100\,v + 25\text{ V}$$



$v_o = v_1 + v_2$

$v_o = v_1 - v_2$

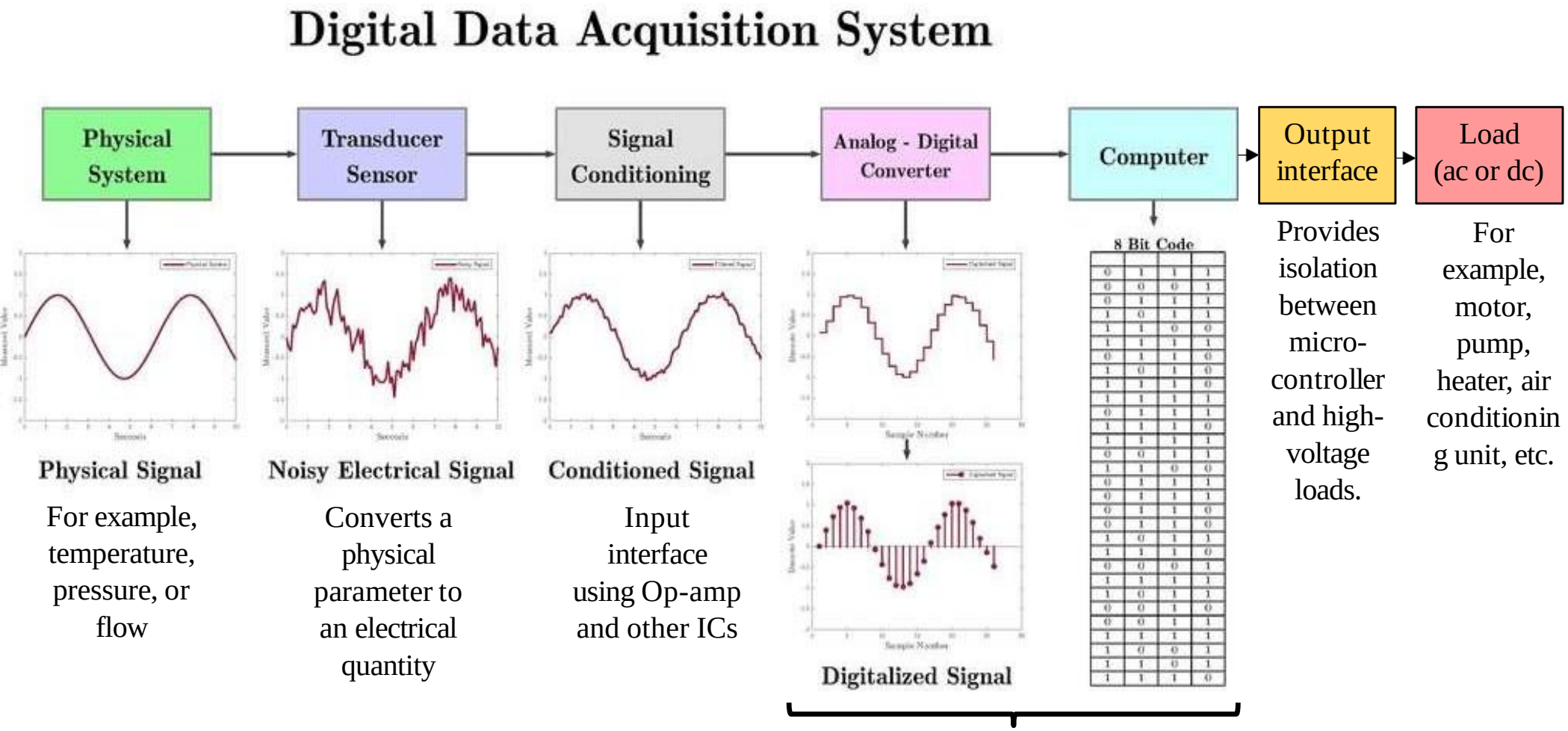
$v_o = 4 \cdot v_1 + 5 \cdot v_2$

$v_o = 2 \cdot v_1 - 8 v_2$

dummies

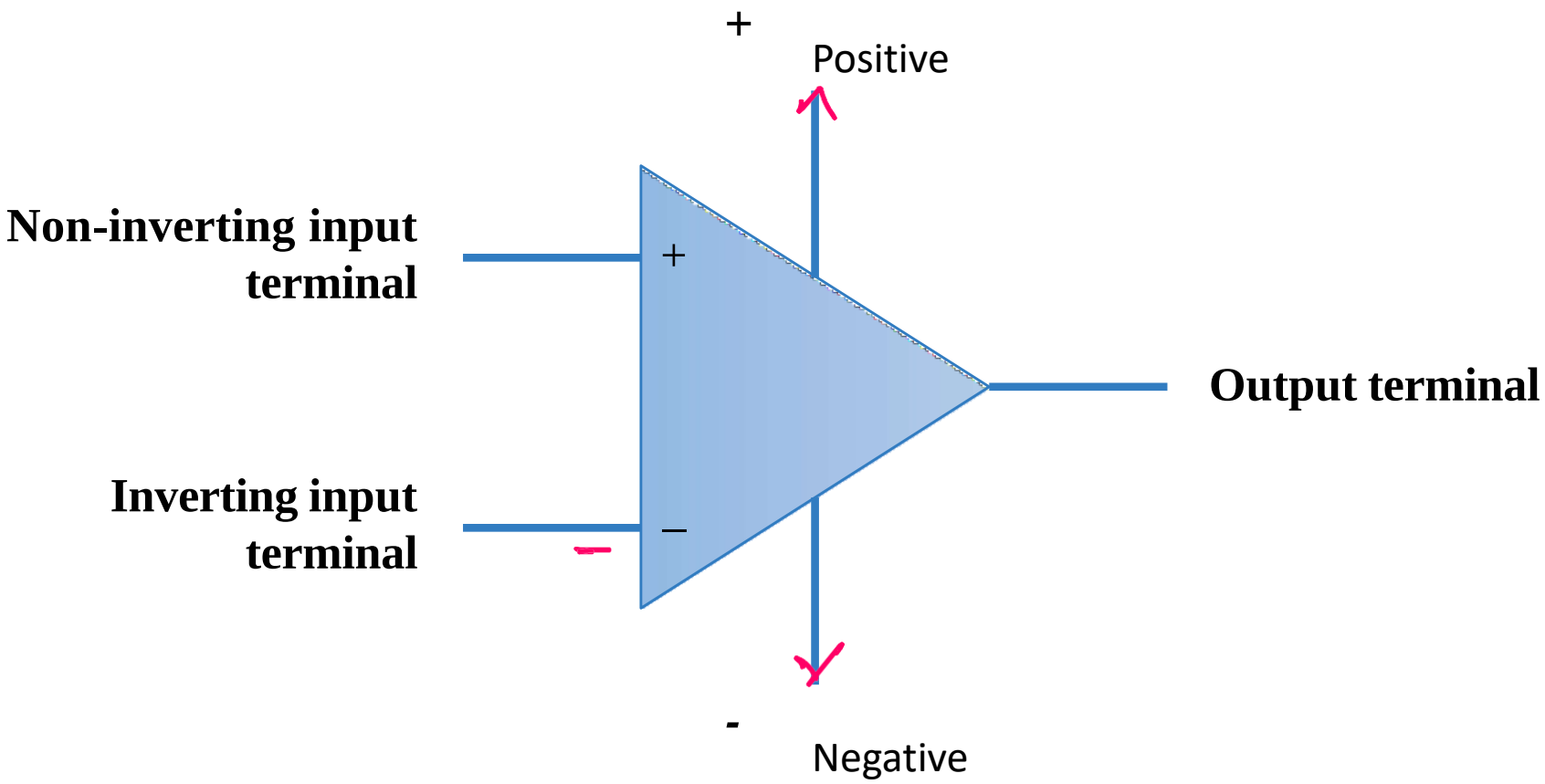
Source: Coughlin, section 1-0

Why analog circuit in the digital world?

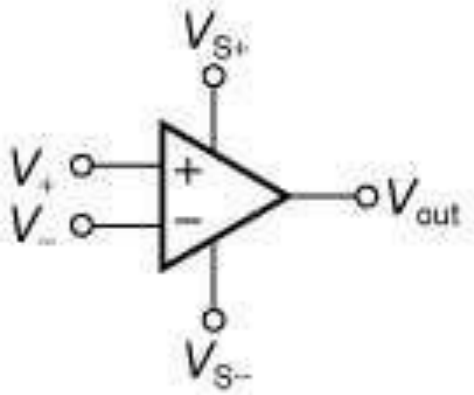


Source: Coughlin, section 1-1 and Wikipedia

OP-AMP circuit symbol



Circuit symbol



(a)

Dual-in-line package (DIP)

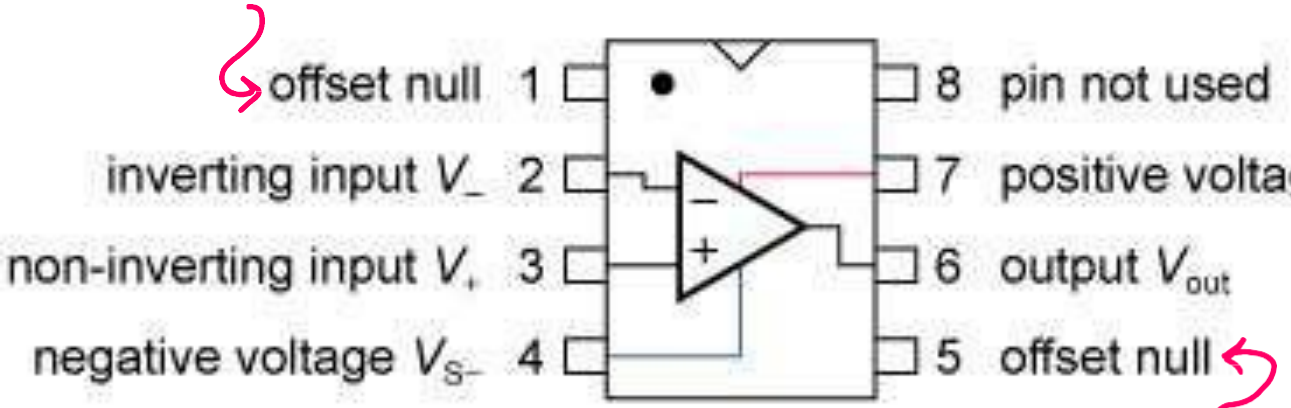
Notch
Or dot

Plastic / ceramic case

(b)

Terminals are numbered counter-clockwise

Notch
Or dot

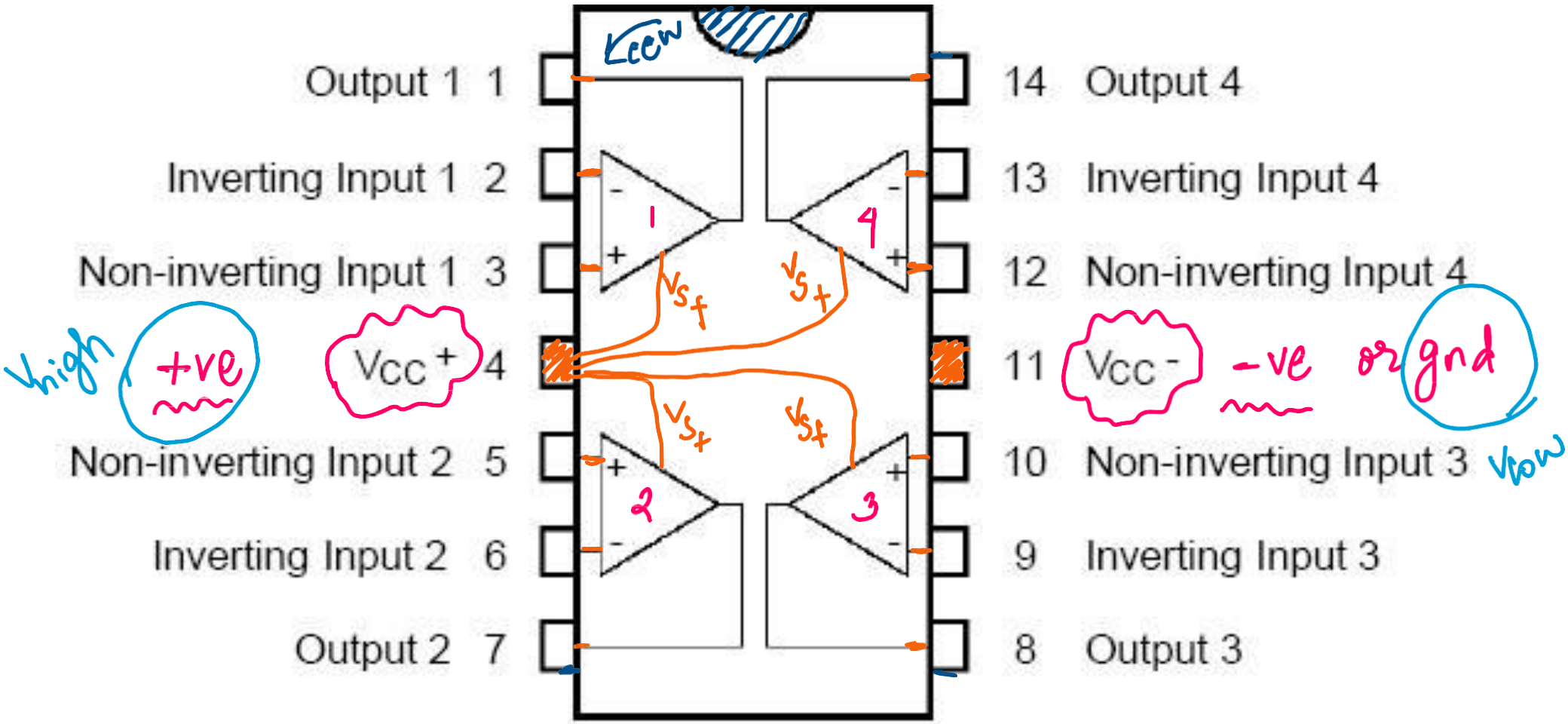


(c)

Connection diagram

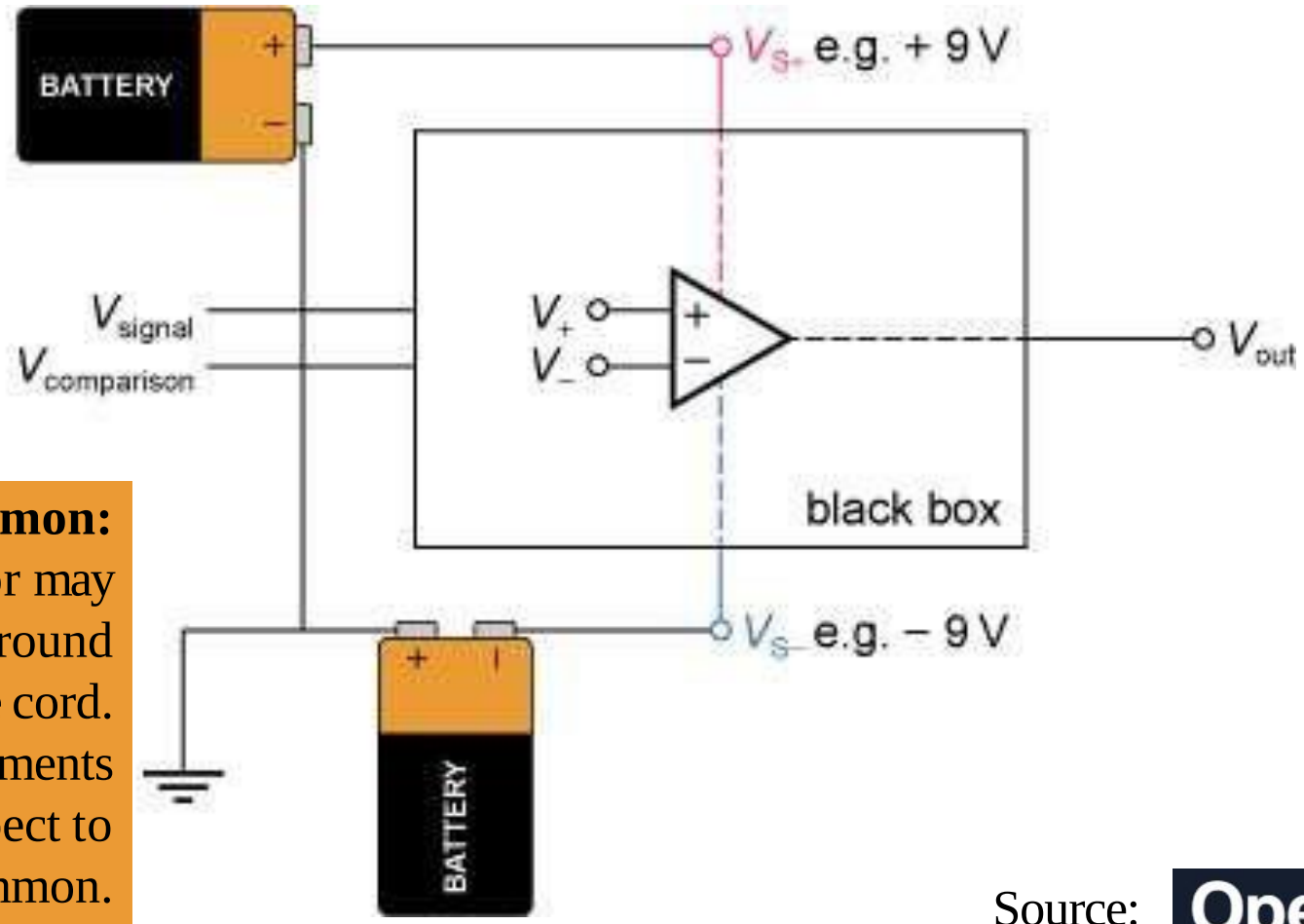
Source: OpenLearn

Quad op-amp



Source: Soundtronics.co.uk

Dual power supplies



Power supply common: This terminal may or may not be wired to earth ground via the 3rd wire of line cord. All voltage measurements are made with respect to power supply common.

Source: OpenLearn

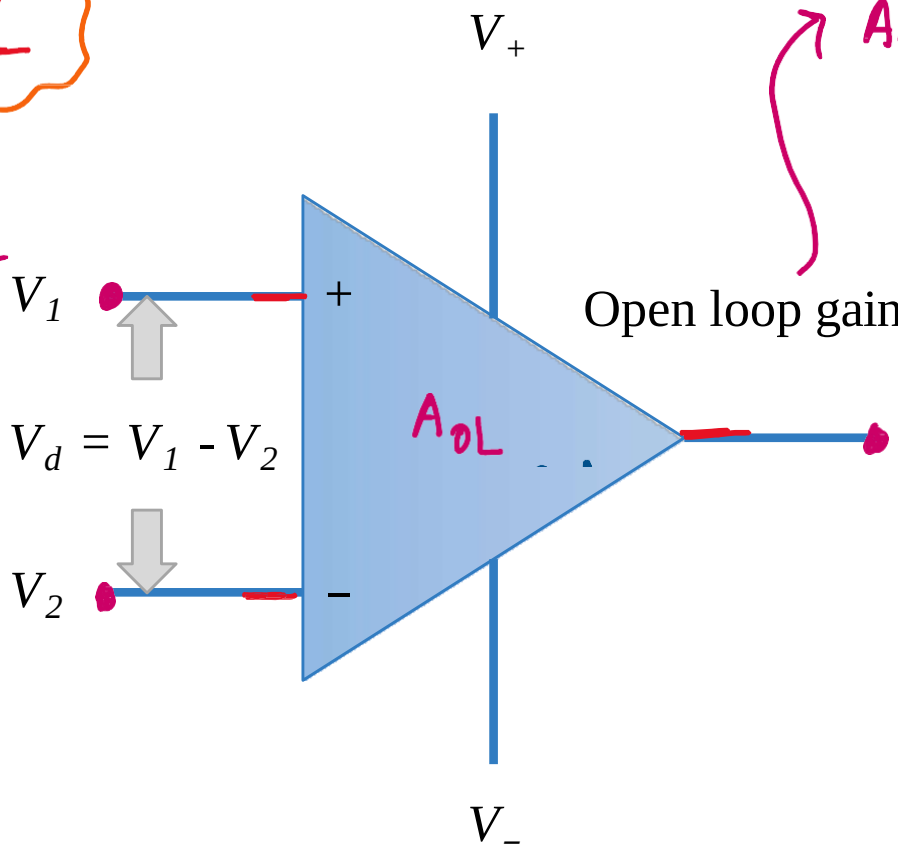
Coughlin 1-6.1, 2-1.1

OP-AMP: a differential amplifier

Op-amp senses the difference between the voltage signals applied at its two input terminals.

$V_d = V_+ - V_-$

$V_d = V_1 - V_2$



practically $A_{OL} = 10^{5-6}$

ideally $A_{OL} = \infty$

Open loop gain = A_{OL}

$V_{out} = A_{OL} \times (V_1 - V_2)$

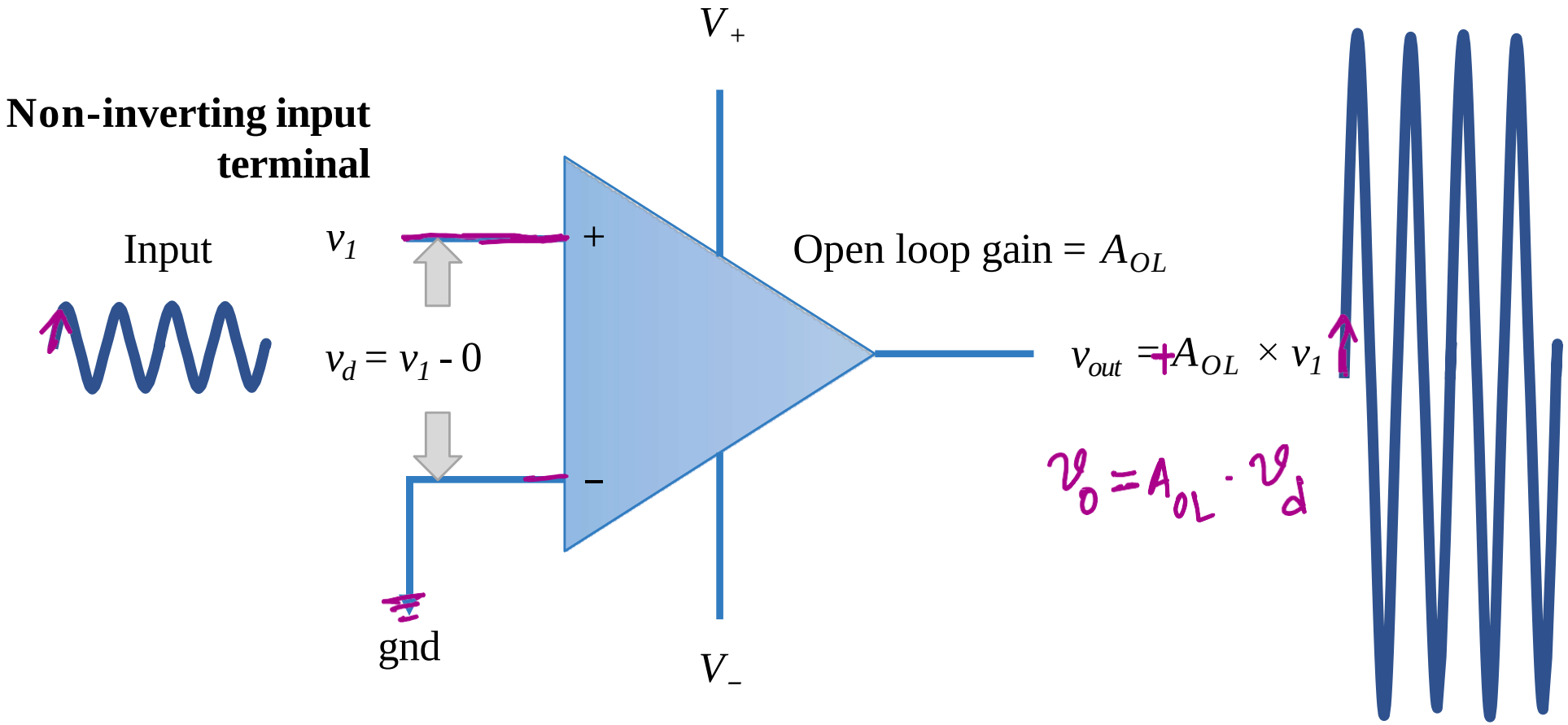
$= A_{OL} V_d$

open-loop configuration : No feedback

OP-AMP: a differential amplifier

$v_d = v_+ - v_- = v_1 - 0V = v_1$

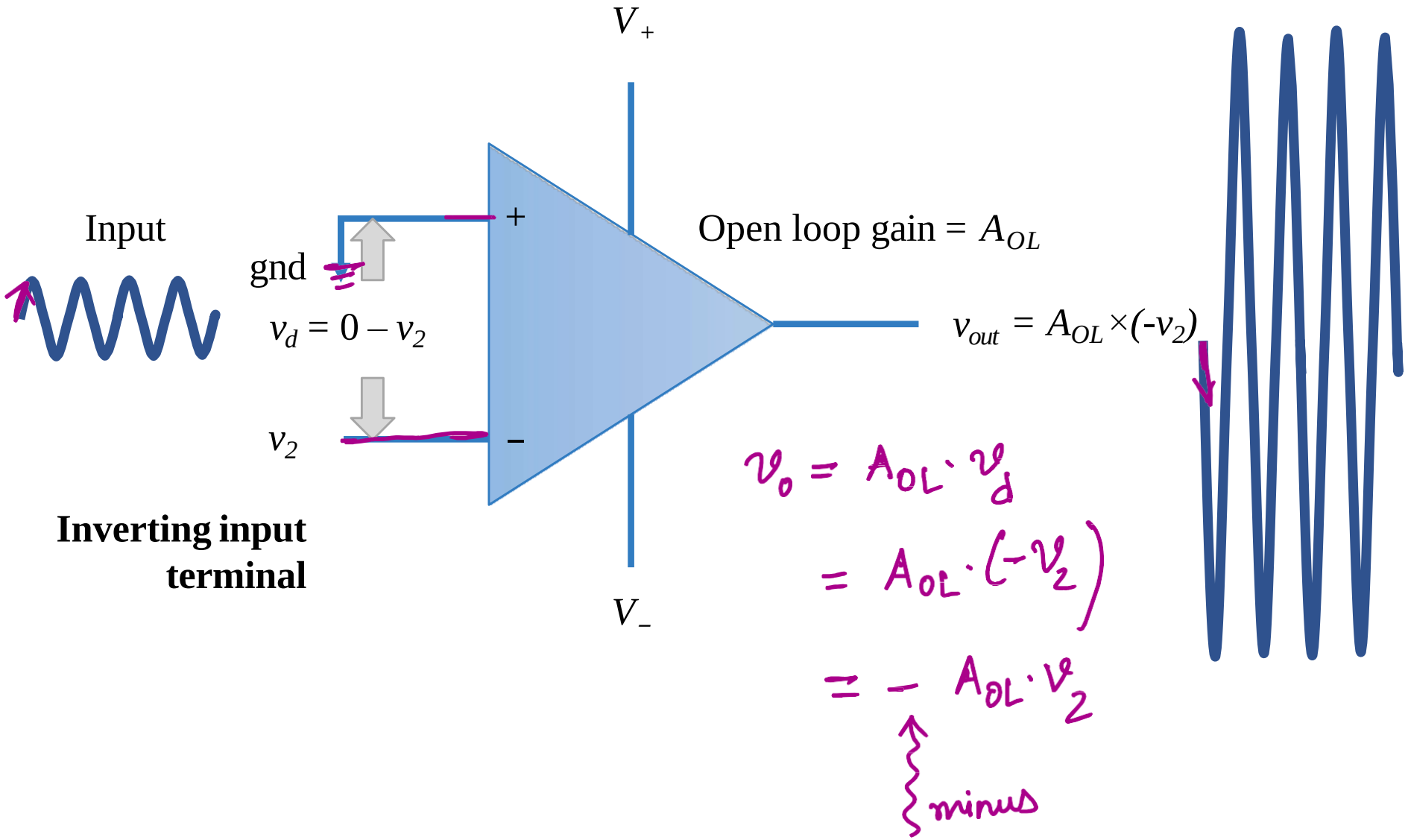
Output: Same phase



OP-AMP: a differential amplifier

$v_d \quad v_+ - v_- = 0V - v_2 = -v_2$

Output: 180° phase



$v_o = A_{OL} \cdot v_d$

$= A_{OL} \cdot (-v_2)$

$= - A_{OL} \cdot v_2$

minus

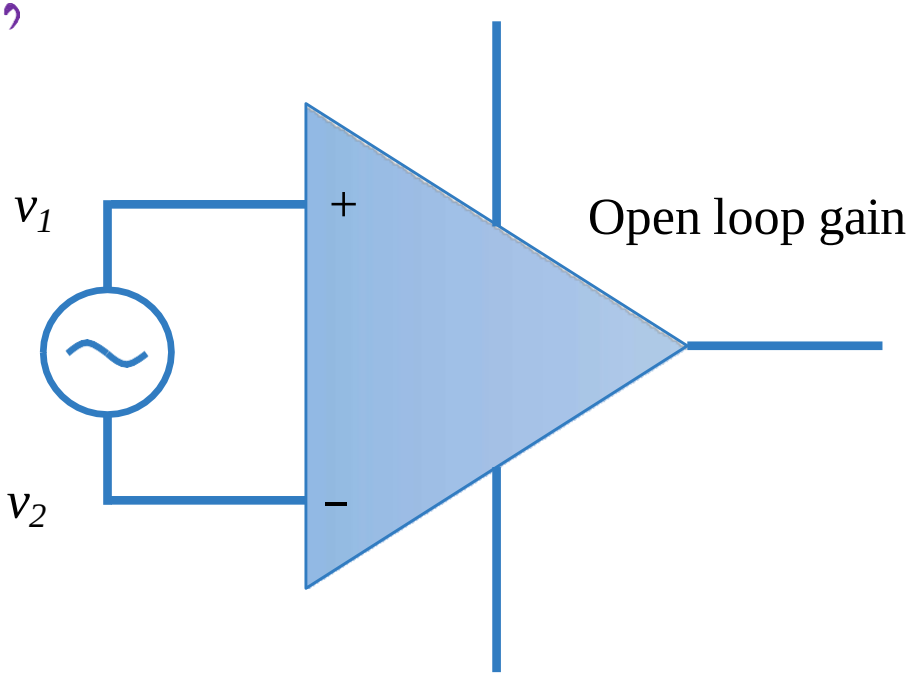
OP-AMP: a differential amplifier

Linear $v_d = 1\,\mu V = 10^{-6}\,V$

Linear $v_d = 10\,\mu V = 10^{-5}\,V$,

$v_d = 100\,\mu V = 10^{-4}\,V$,

Saturation $= 0.1\,mV$



Open loop gain = A_{OL} = 10^5 - 10^6

Saturation $v_d = v_1 - v_2$

= 1 mV

$v_{out} = A_{OL} \times (v_d)$

= 100 V to 1 kV
(theoretically)
(but not practically)
V_+

Saturation $v_d = 10\,mV$

Saturation $v_d = 100\,mV$

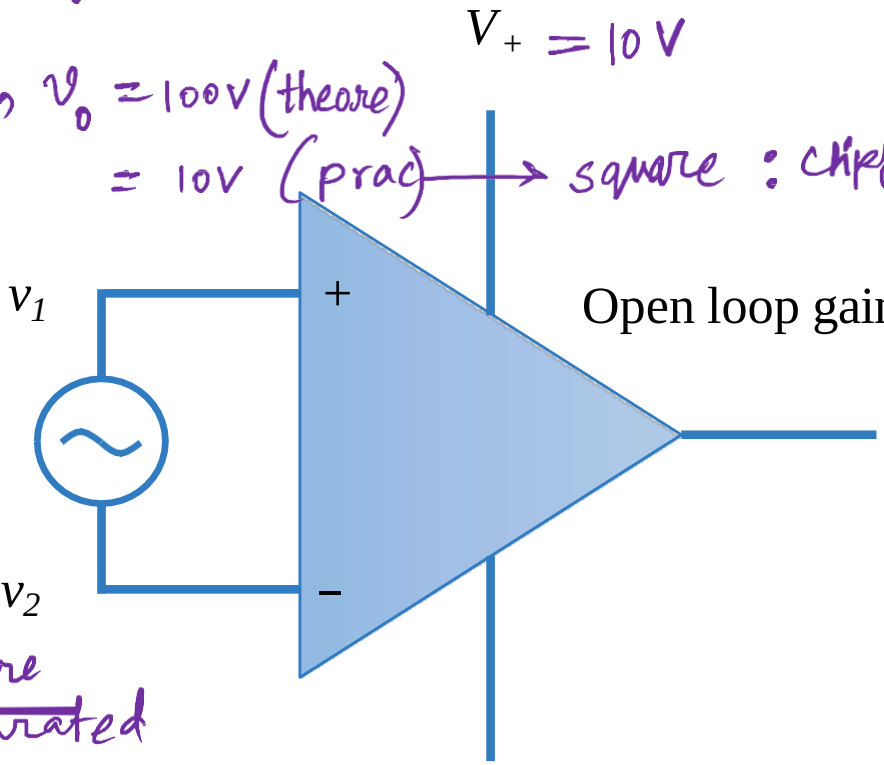
Saturation $v_d = 1\,V$

OP-AMP: a differential amplifier

Linear $v_d = 1\,\mu V = 10^{-6}\,V$, $v_o = 1\,V$

Linear $v_d = 10\,\mu V = 10^{-5}\,V$, $v_o = 10\,V$ } (both theore & prac) i/p & o/p waveforms will look similare

Saturation $v_d = 100\,\mu V = 10^{-4}\,V$, $v_o = 100\,V$ (theore)
$= 0.1\,mV$ $= 10\,V$ (prac) → square : clipped / saturated



Saturation $v_d = v_1 - v_2$

$= 1$ mV

v_o = Square saturated

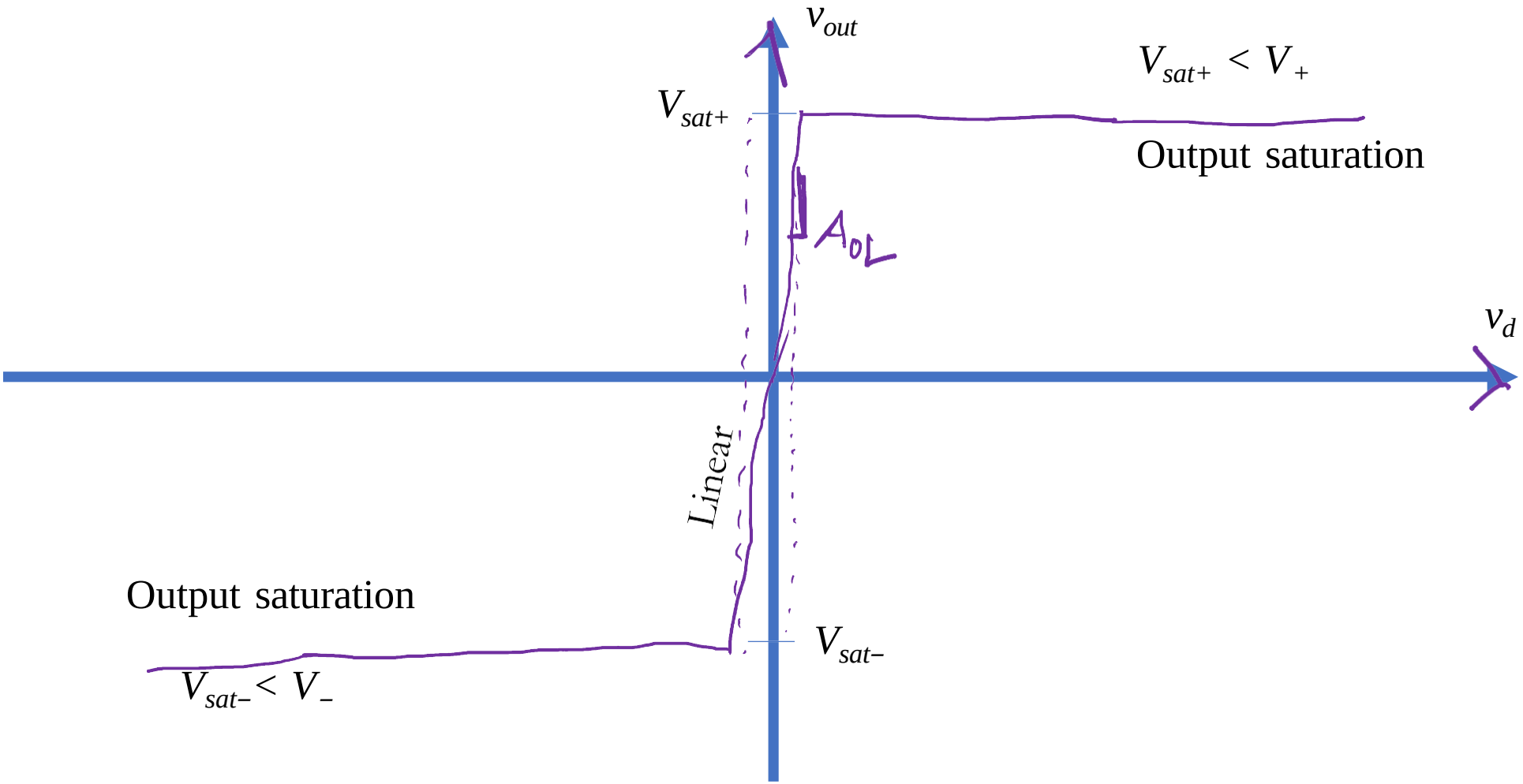
= 100 V to 1 kV
(theoretically)
(but not practically)
V_+

Saturation $v_d = 10\,mV \rightarrow v_o$ saturated

Saturation $v_d = 100\,mV$,,

Saturation $v_d = 1\,V$,,

OP-AMP: transfer characteristics

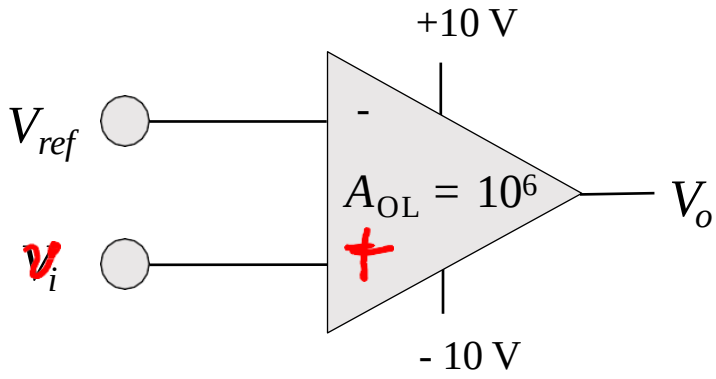


OP-AMP applications

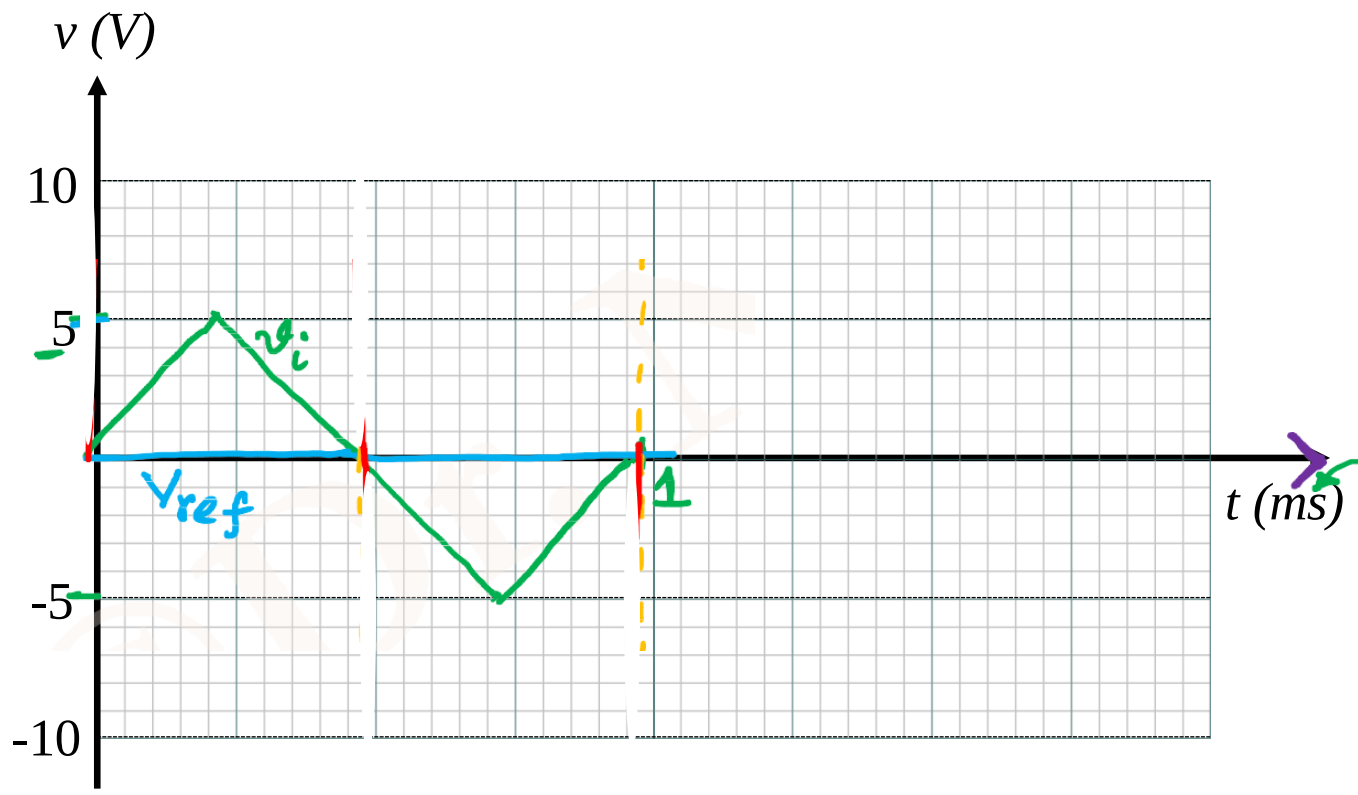
- Adder, subtractor, differentiator, integrator, etc.
- Comparator
- Active filter
- Oscillator
- Waveform convertor
- Analog-to-digital converter & Digital-to-analog convertor

Non-inverting zero crossing detector

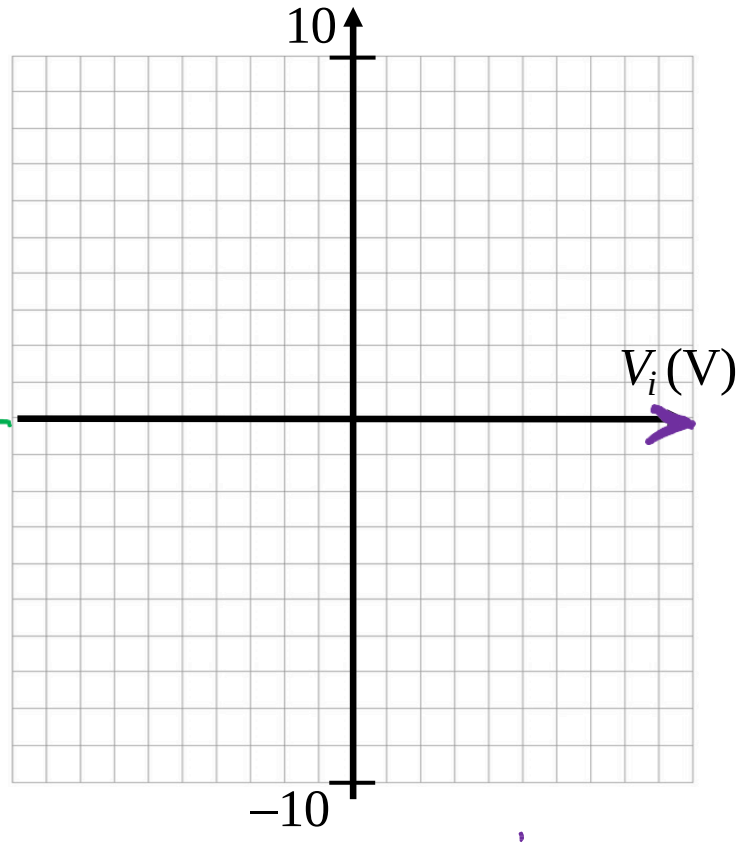
Problem 1. The input signal v_i is a triangular waveform of 1 kHz frequency and 5 V peak. The reference V_{ref} is 0 V. Draw (a) the input waveform v_i, (b) the output waveform v_o and (c) the transfer characteristics.



(a, b)

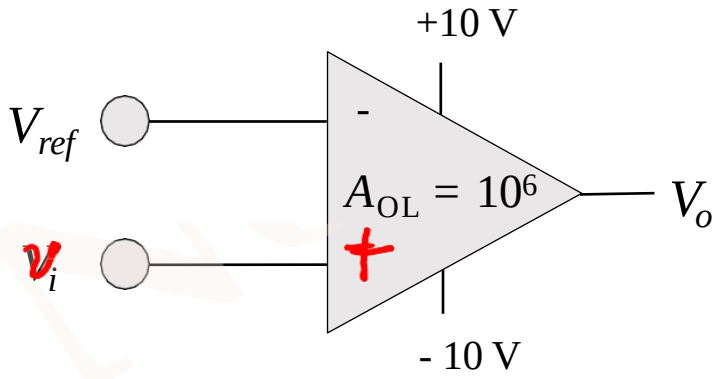


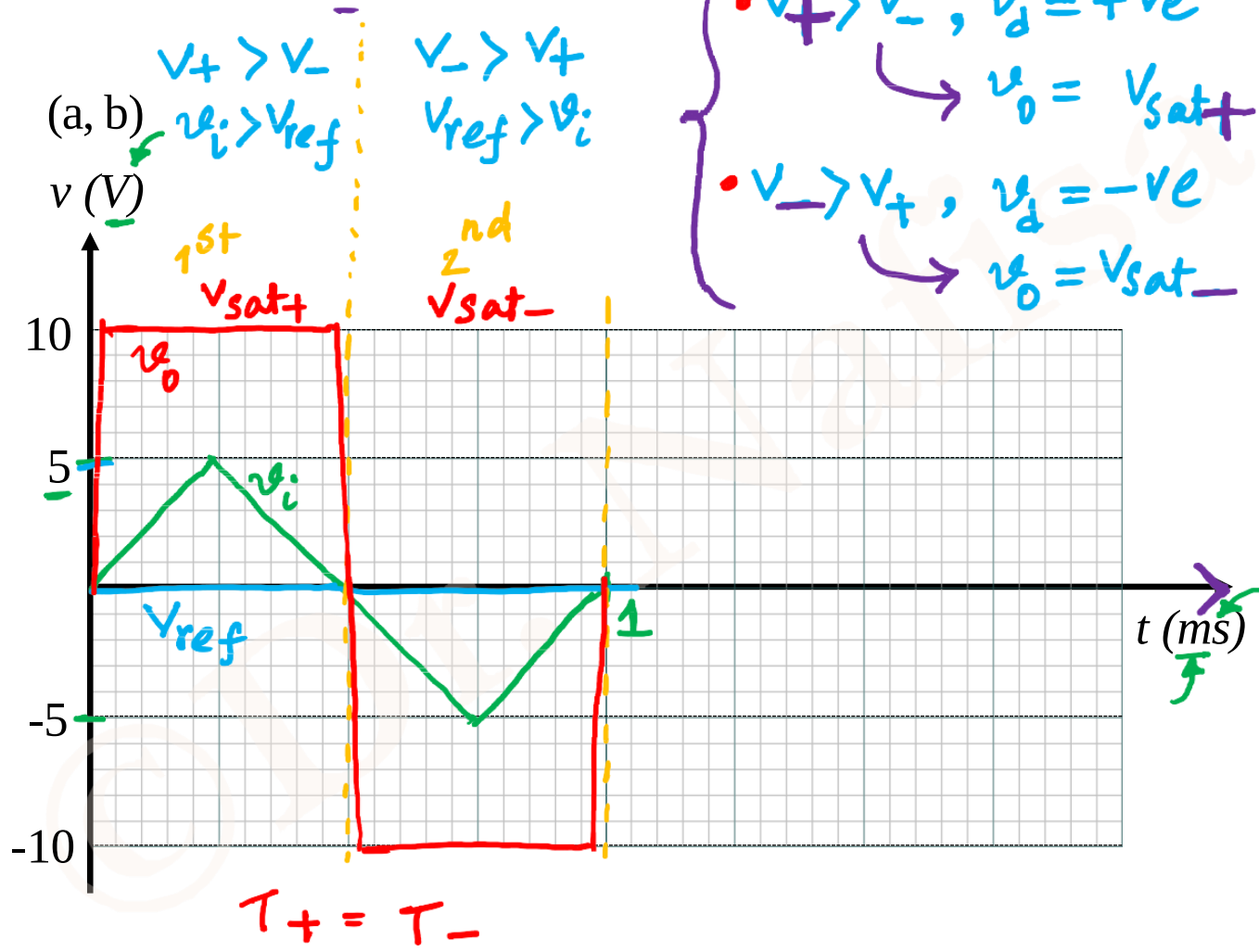
(c) V_O (V)

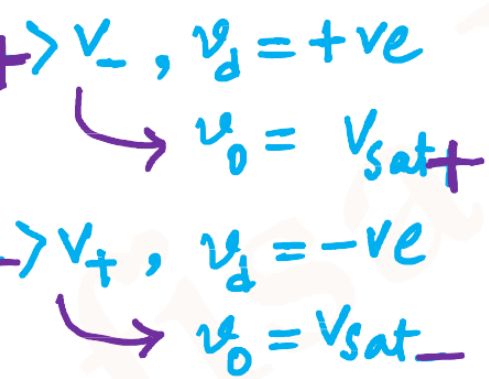


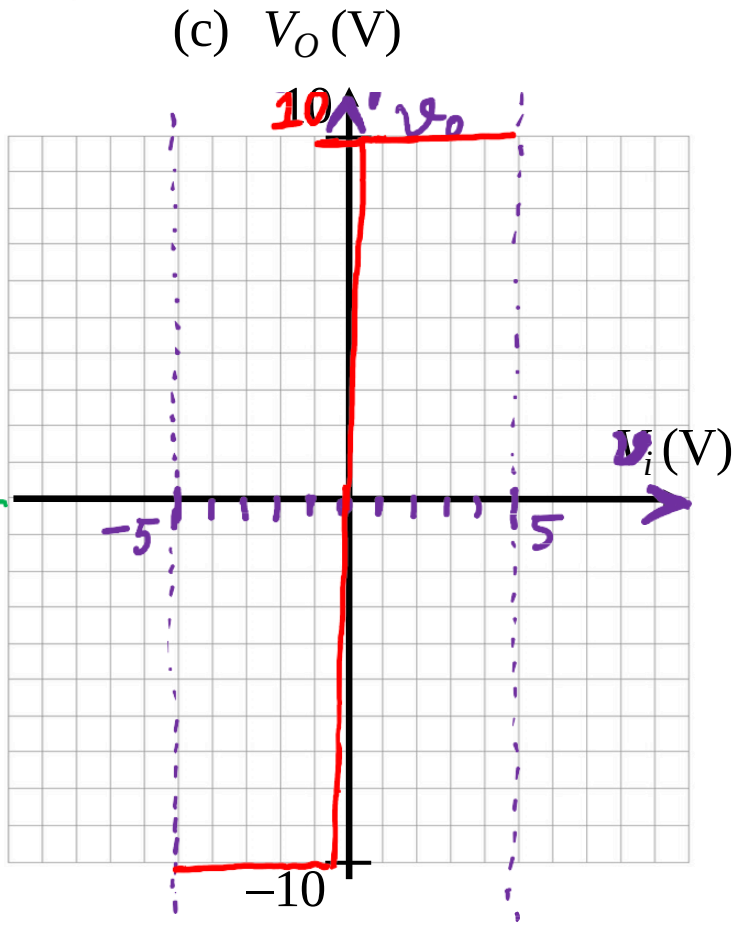
Non-inverting zero crossing detector

Problem 1. The input signal v_i is a triangular waveform of 1 kHz frequency and 5 V peak. The reference V_{ref} is 0 V. Draw (a) the input waveform v_i, (b) the output waveform v_o and (c) the transfer characteristics.



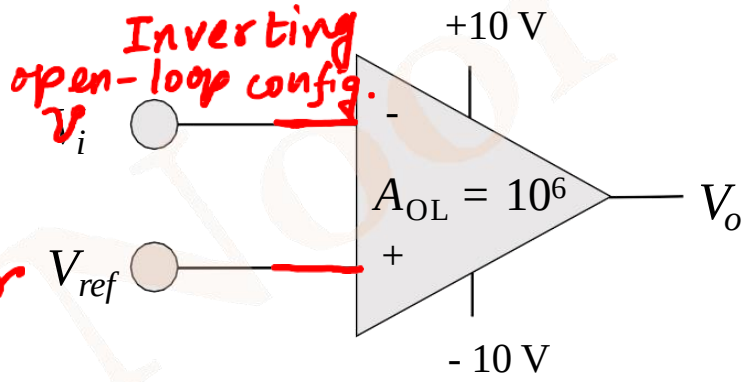






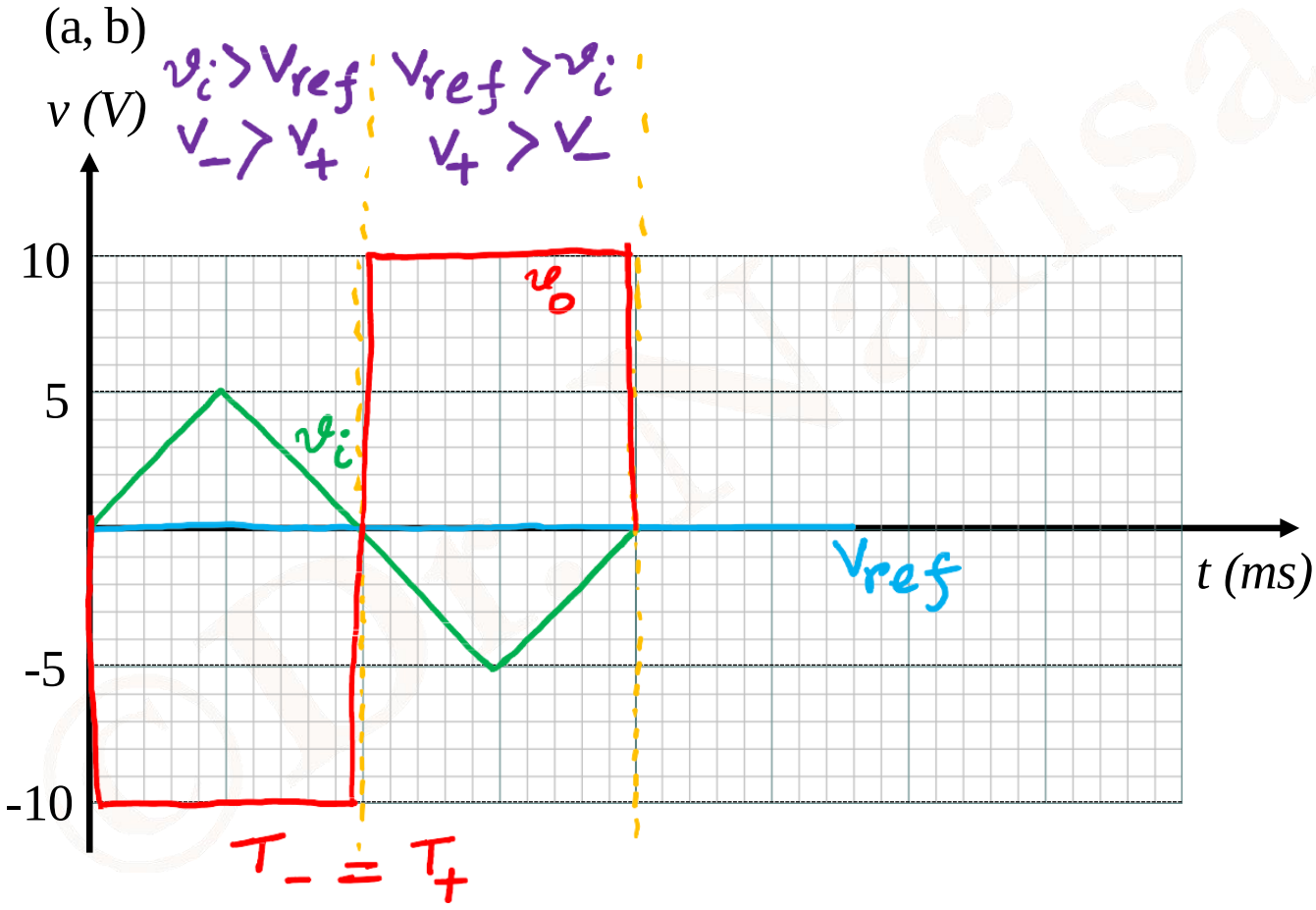
Inverting zero crossing detector

Problem 2. The input signal v_i is a triangular waveform of 1 kHz frequency and 5 V peak. The reference V_{ref} is 0 V. Draw (a) the input waveform v_i, (b) the output waveform v_o and (c) the transfer characteristics.

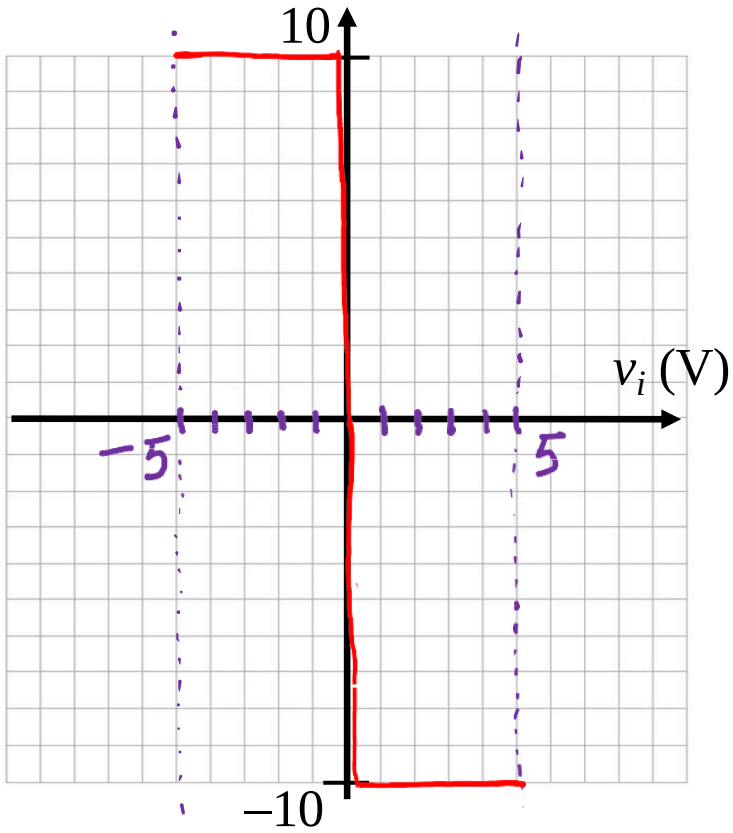


open-loop comparator

(a, b)

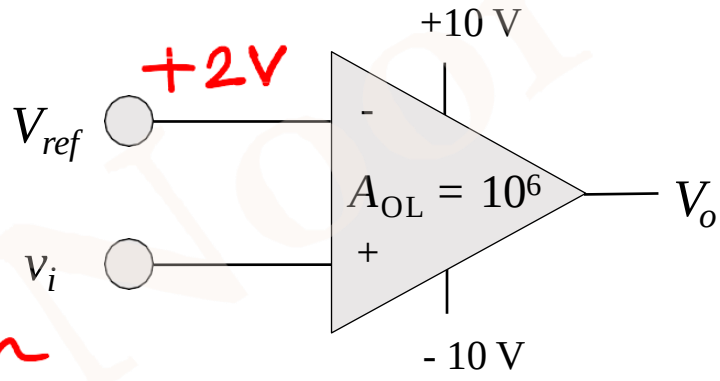


(c) v_O (V)



Non-inverting positive level detector

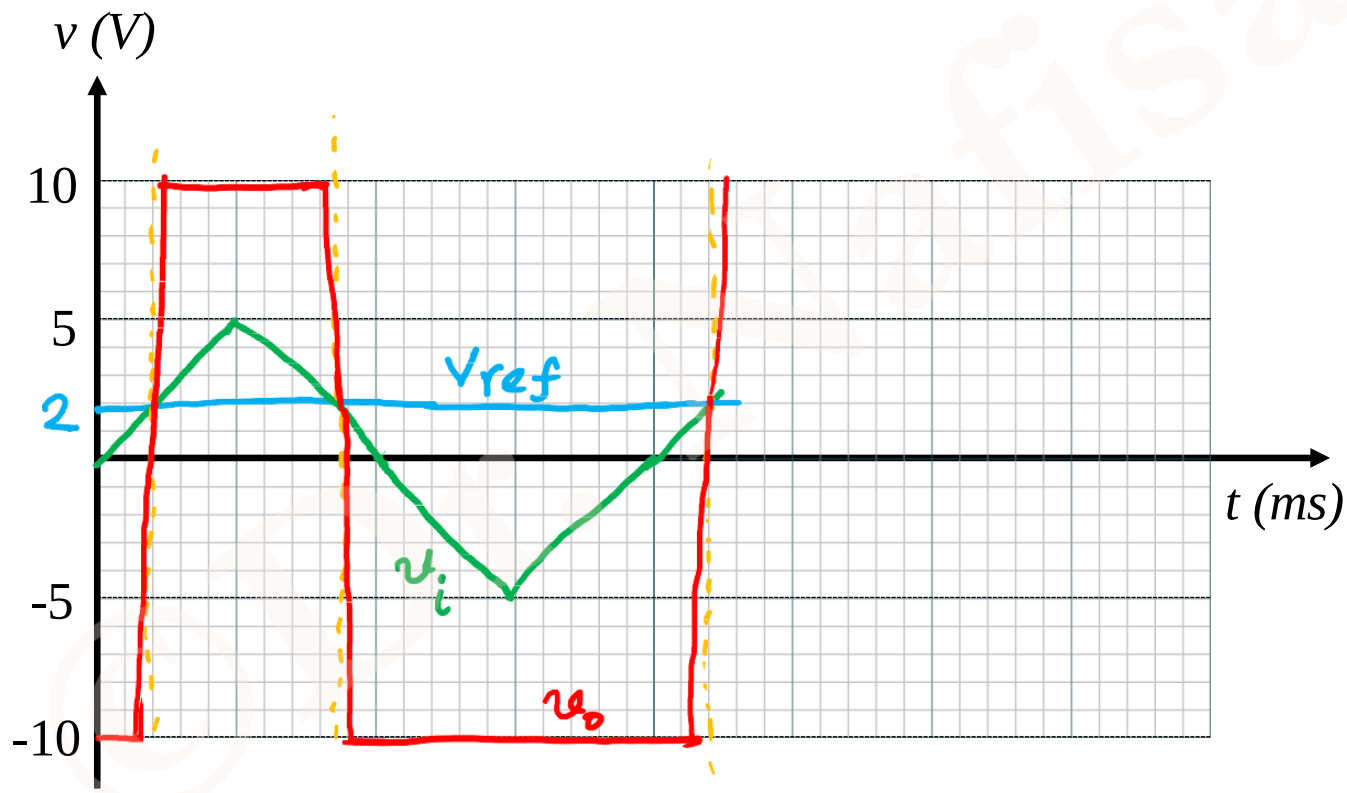
Problem 3. The input signal v_i is a triangular waveform of 1 kHz frequency and 5 V peak. The reference V_{ref} is +2 V. Draw (a) the input waveform v_i, (b) the output waveform v_o and (c) the transfer characteristics.



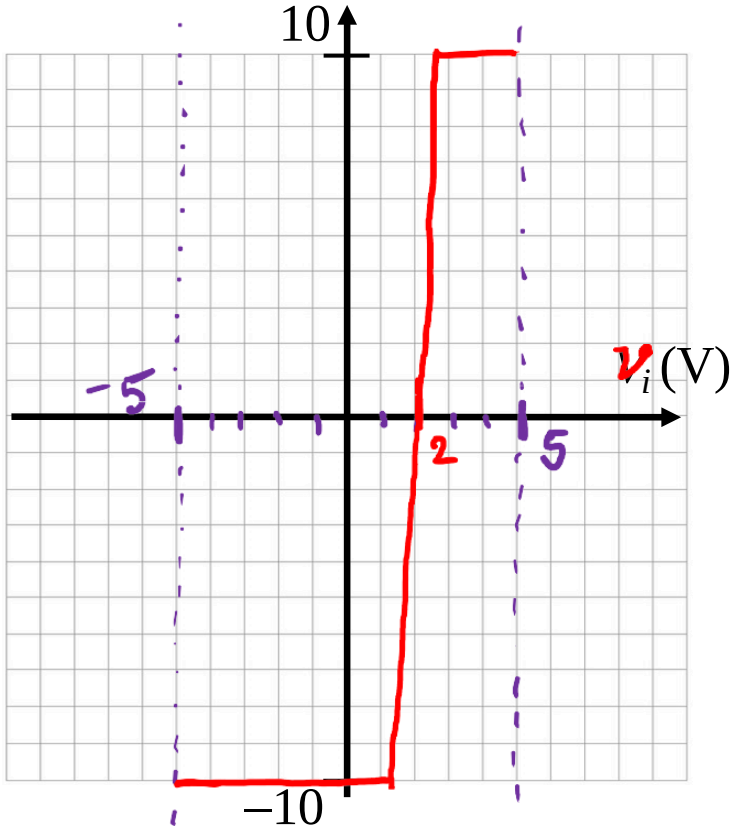
$T_+ < T_-$

open-loop comparator
Non-inv configuration

(a, b)

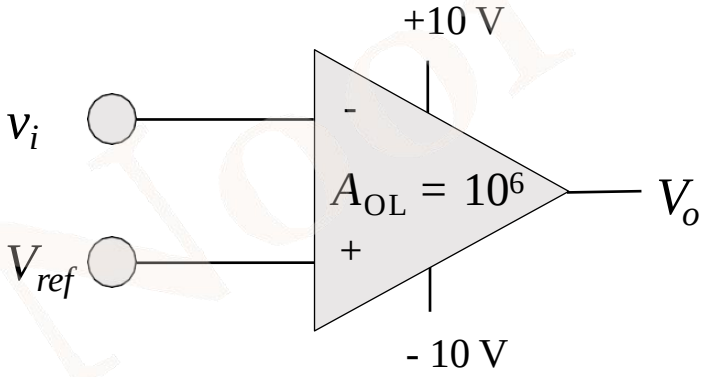


(c) v_O (V)

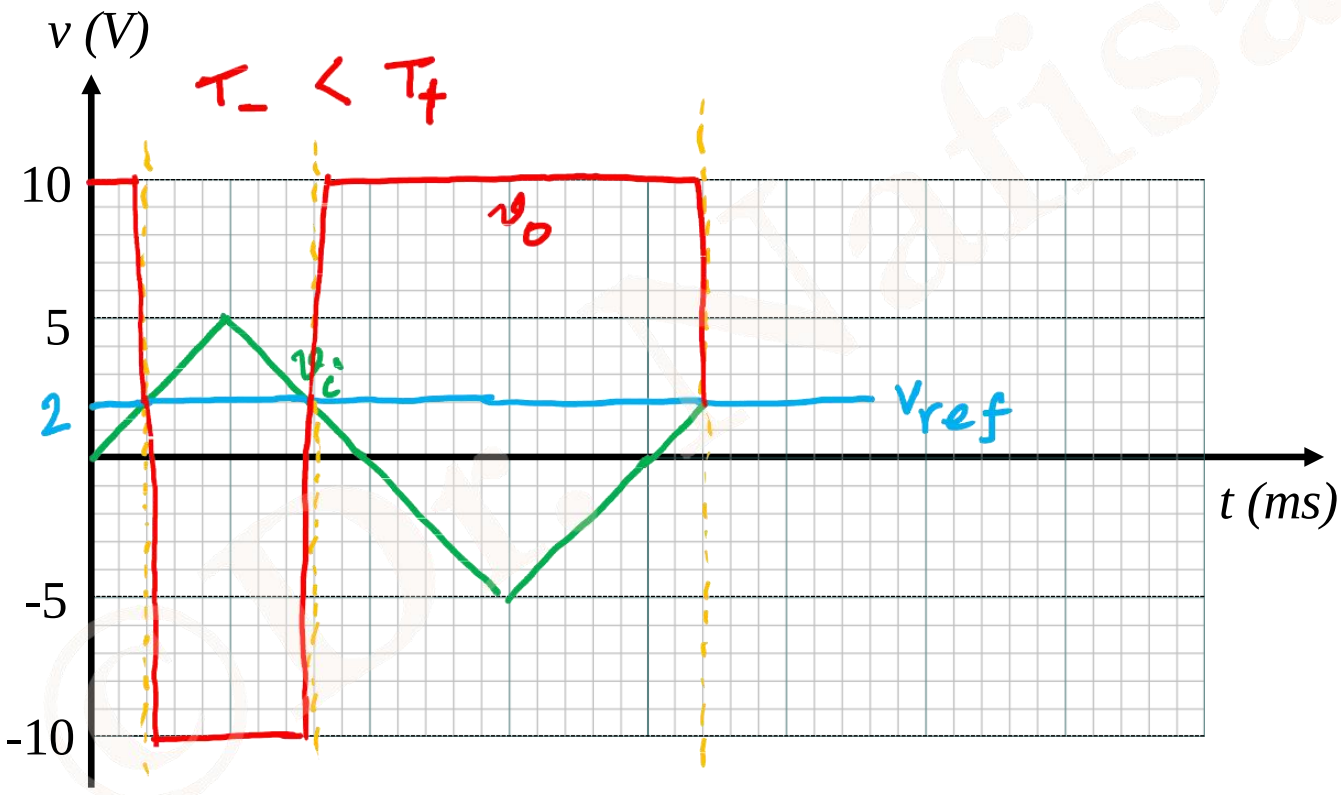


Inverting positive level detector

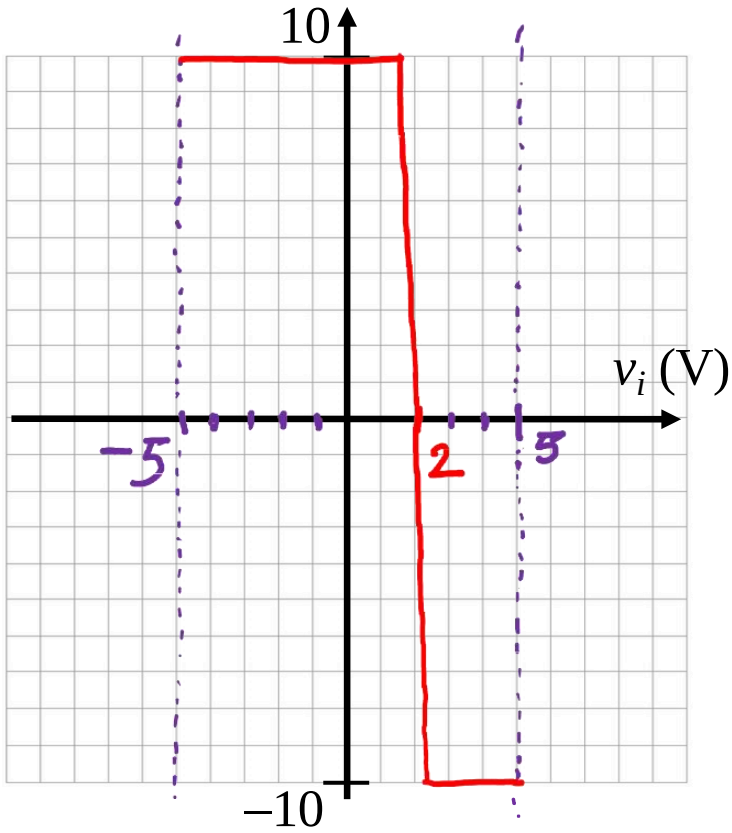
Problem 4. The input signal v_i is a triangular waveform of 1 kHz frequency and 5 V peak. The reference V_{ref} is +2 V. Draw (a) the input waveform v_i, (b) the output waveform v_o and (c) the transfer characteristics.



(a, b)

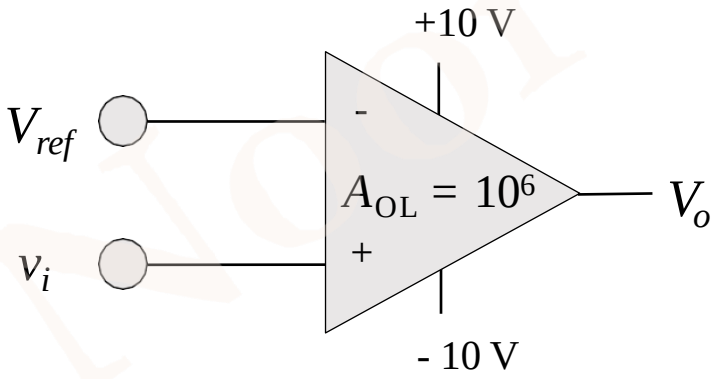


(c) v_O (V)

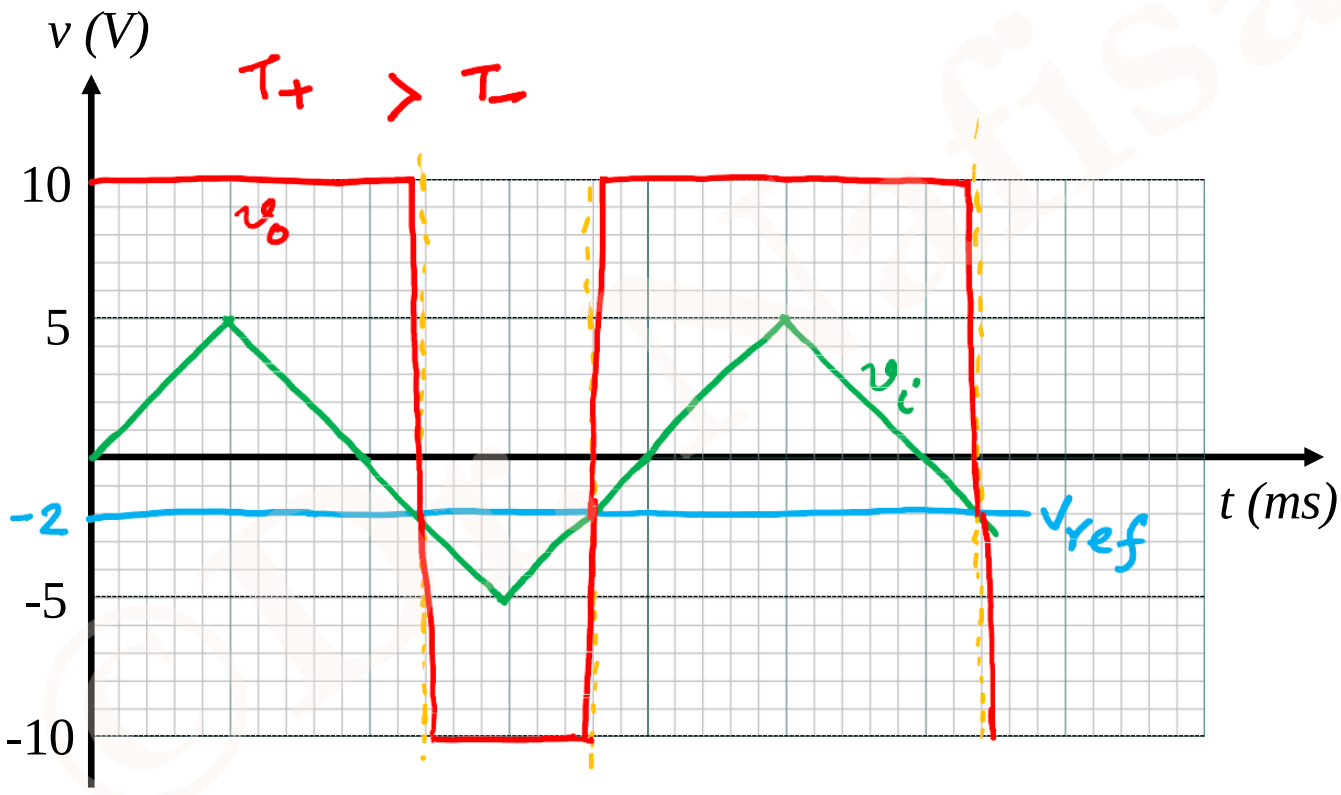


Non-inverting negative level detector

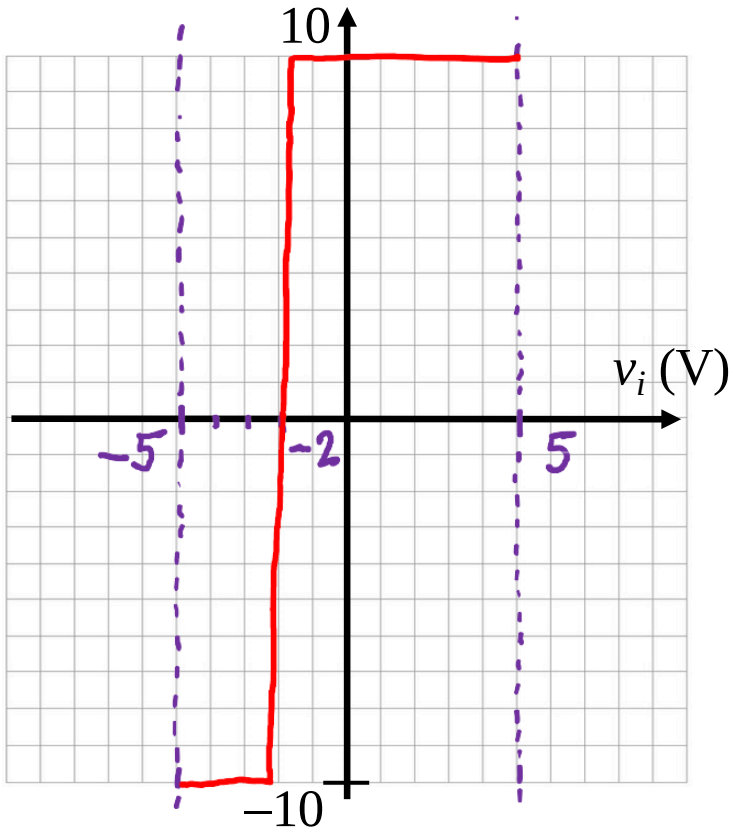
Problem 5. The input signal v_i is a triangular waveform of 1 kHz frequency and 5 V peak. The reference V_{ref} is -2 V. Draw (a) the input waveform v_i, (b) the output waveform v_o and (c) the transfer characteristics.



(a, b)

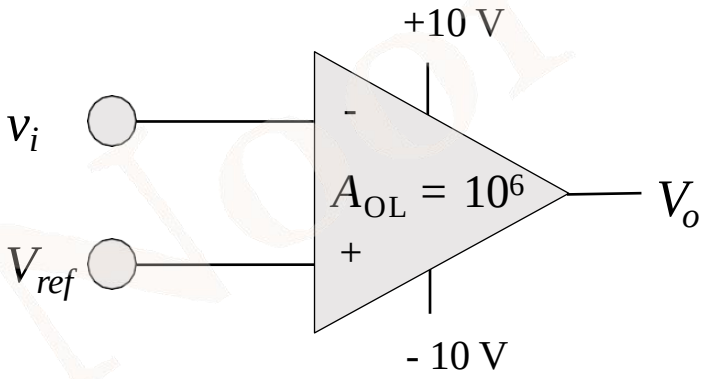


(c) v_O (V)

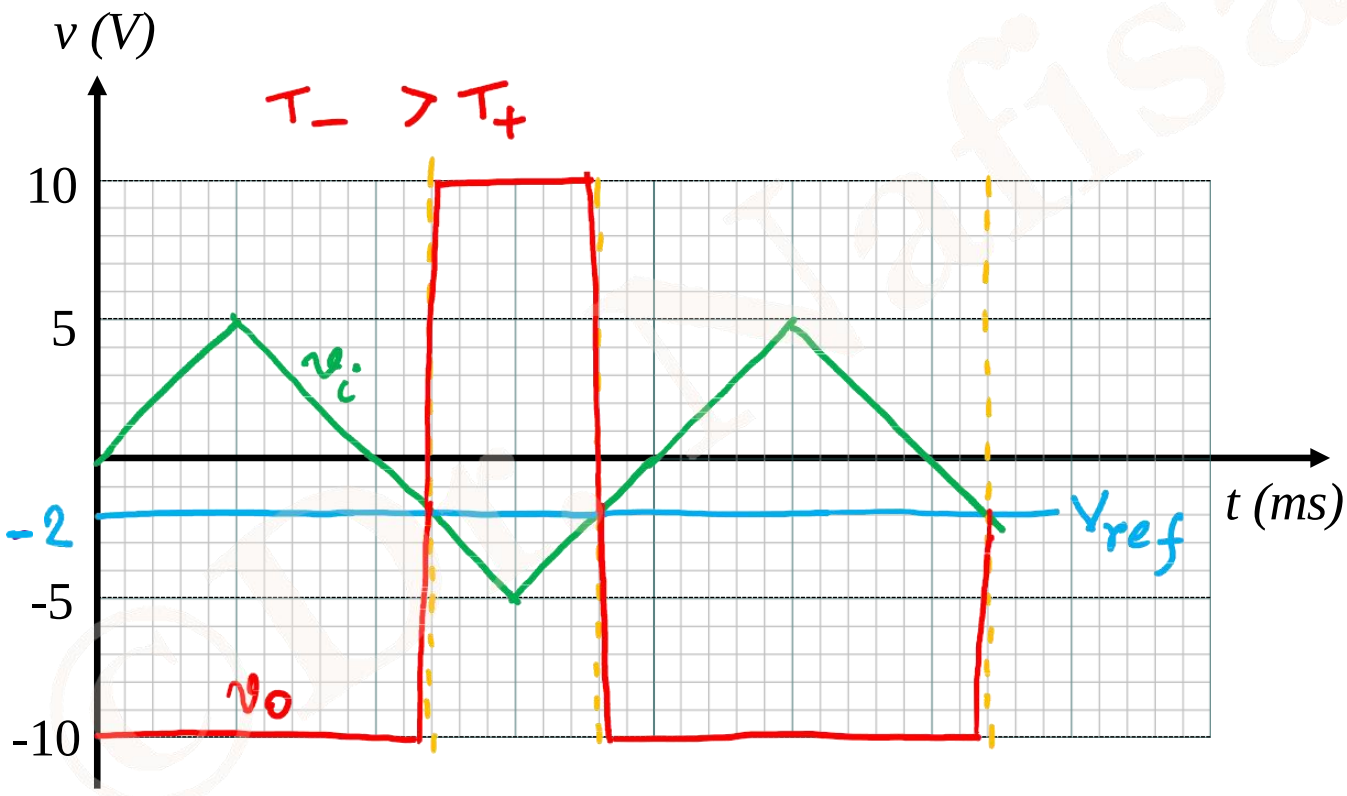


Inverting negative level detector

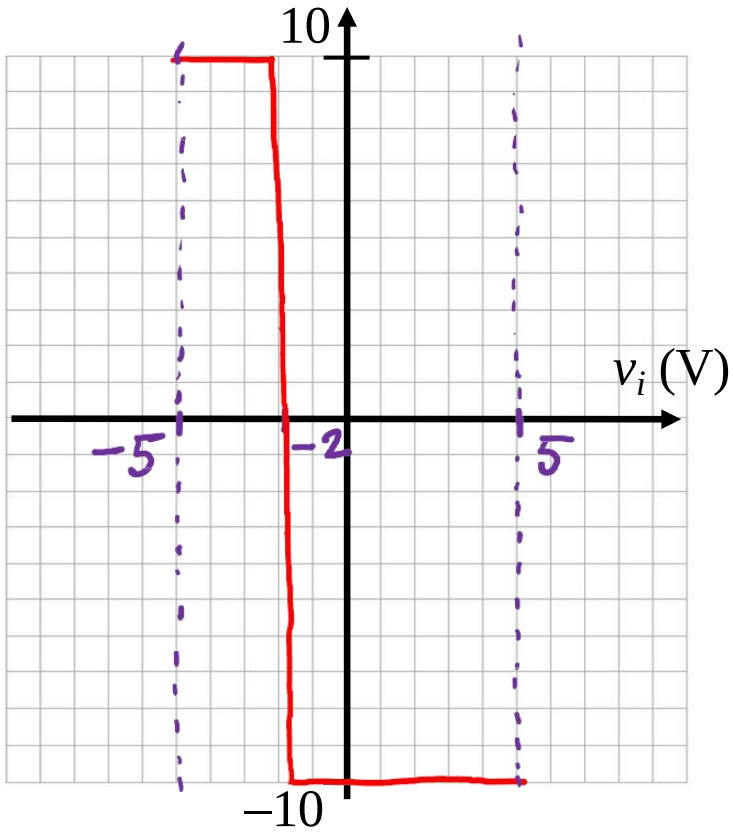
Problem 6. The input signal v_i is a triangular waveform of 1 kHz frequency and 5 V peak. The reference V_{ref} is -2 V. Draw (a) the input waveform v_i, (b) the output waveform v_o and (c) the transfer characteristics.



(a, b)

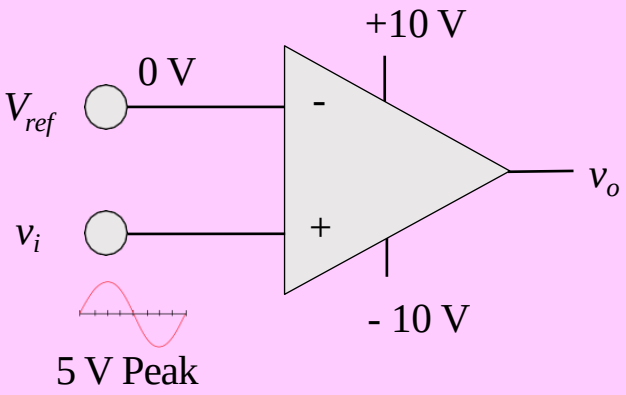


(c) v_O (V)

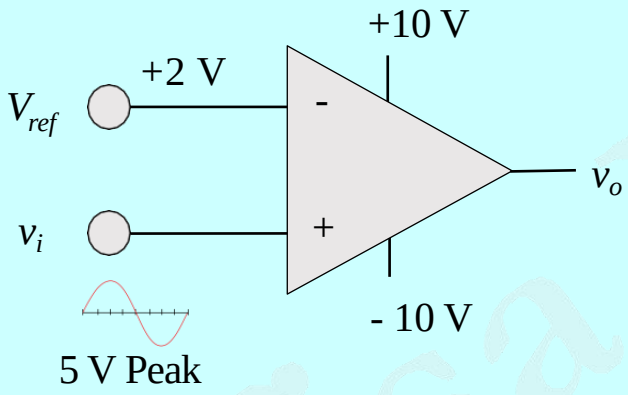


Review: Open-loop comparator circuits

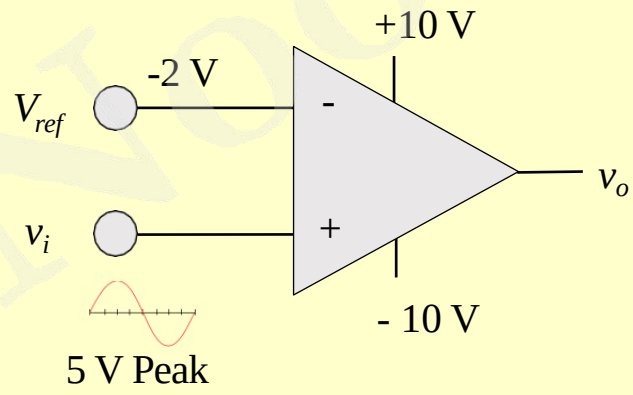
Non-inverting Zero Crossing Detector



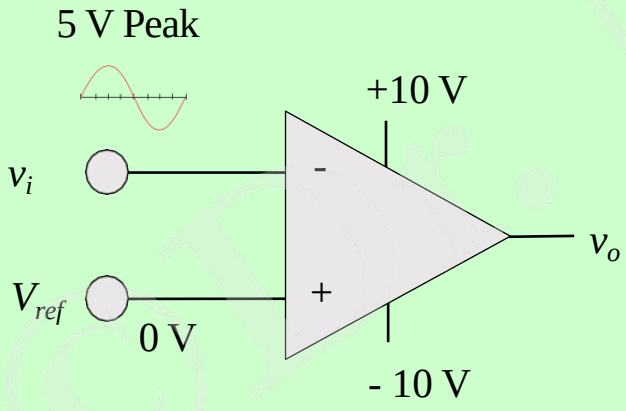
Non-inverting +ve level indicator



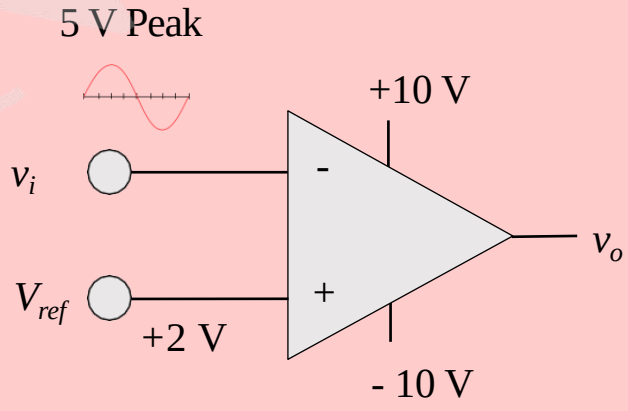
Non-inverting -ve level indicator



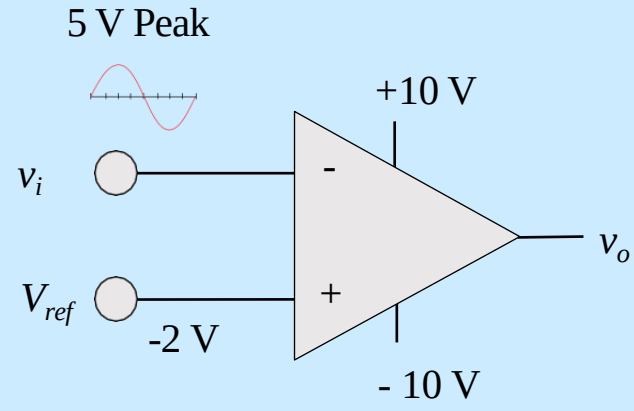
Inverting Zero Crossing Detector



Inverting +ve level indicator



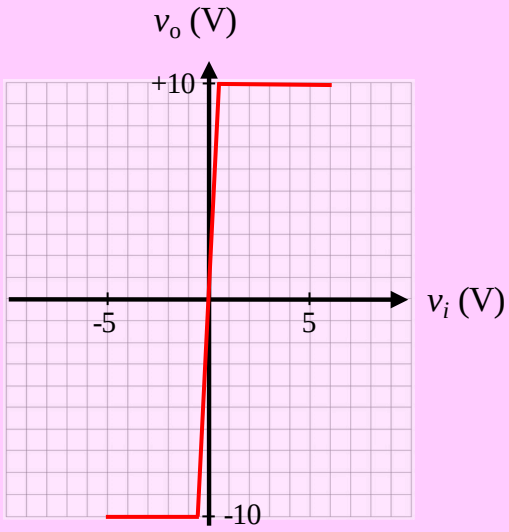
Inverting -ve level indicator



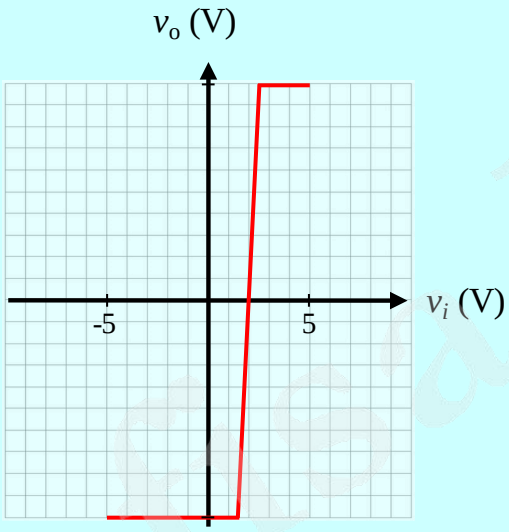
Review: Open-loop comparators

Transfer characteristics

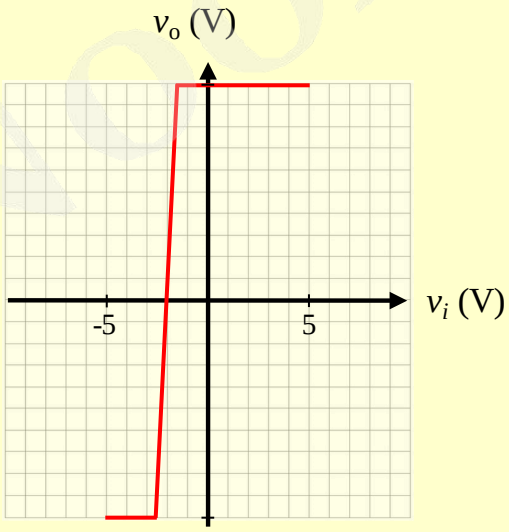
Non-inverting Zero Crossing Detector



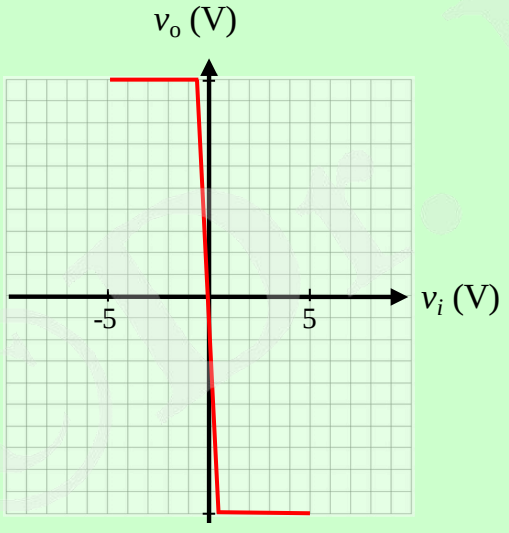
Non-inverting +ve level indicator



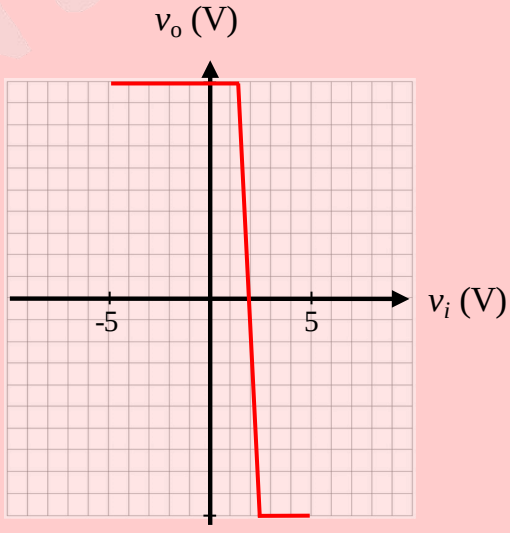
Non-inverting -ve level indicator



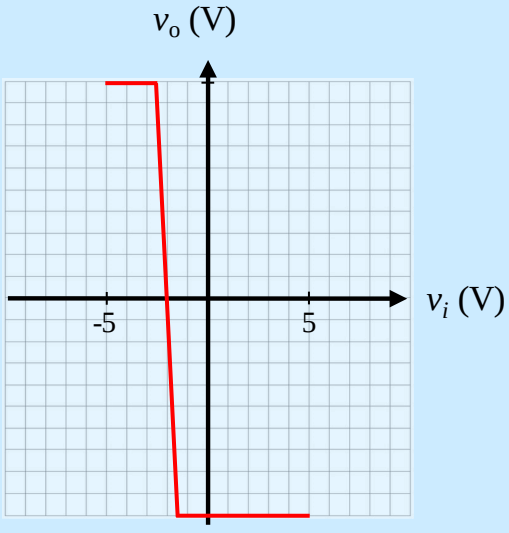
Inverting Zero Crossing Detector



Inverting +ve level indicator

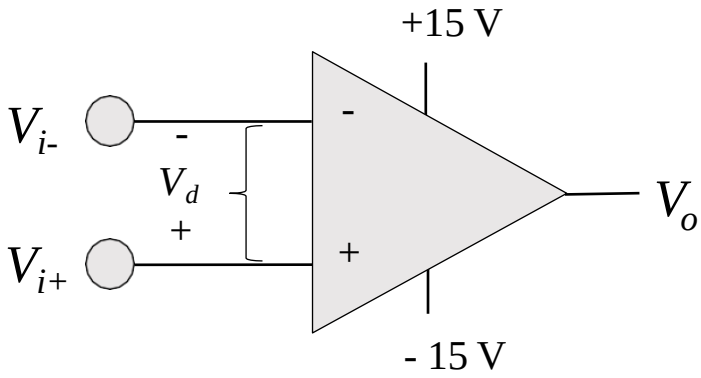


Inverting -ve level indicator



Open-loop comparator

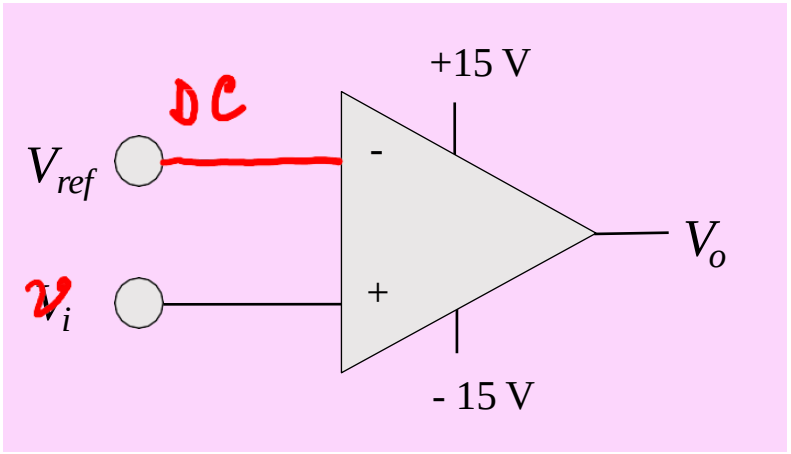
Problem 7. V_{i-} and V_{i+} are the dc inputs at the -ve and +ve input terminals, respectively. These voltage are given with respect to the ground. $A_{OL} = 2\times10^5$. $V_{sat+} = +15$ V, $V_{sat-} = -15$ V. Assume ideal conditions. Calculate the output voltage V_o for the following cases:

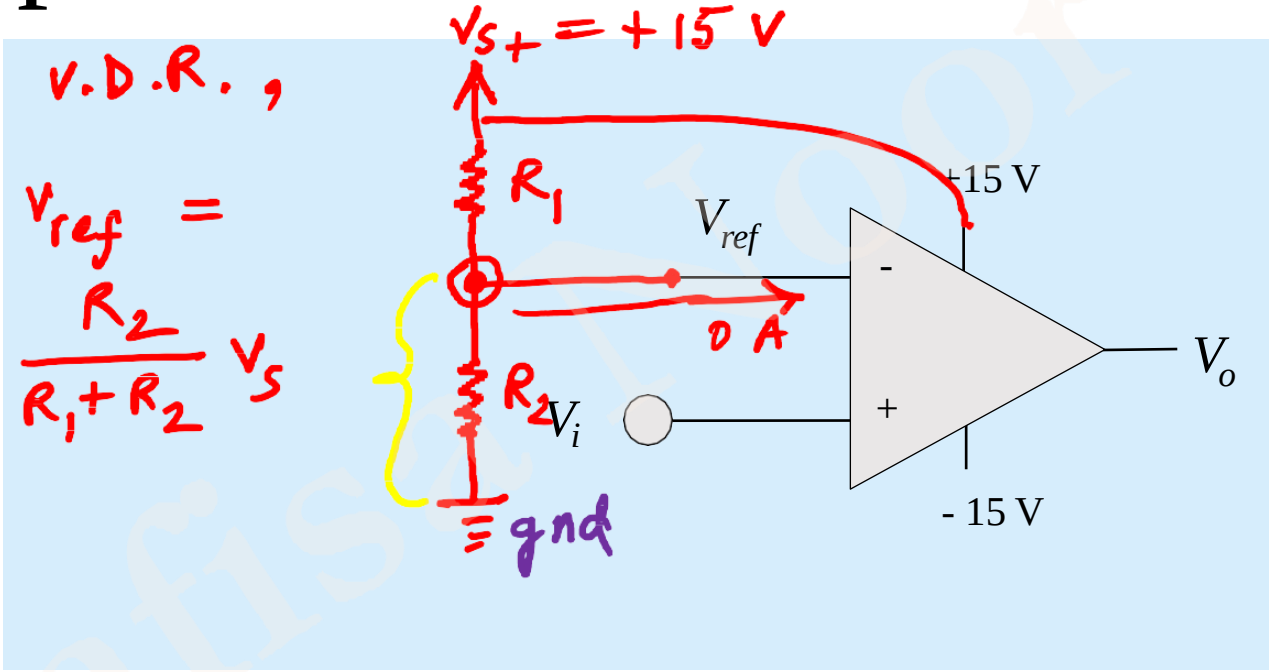


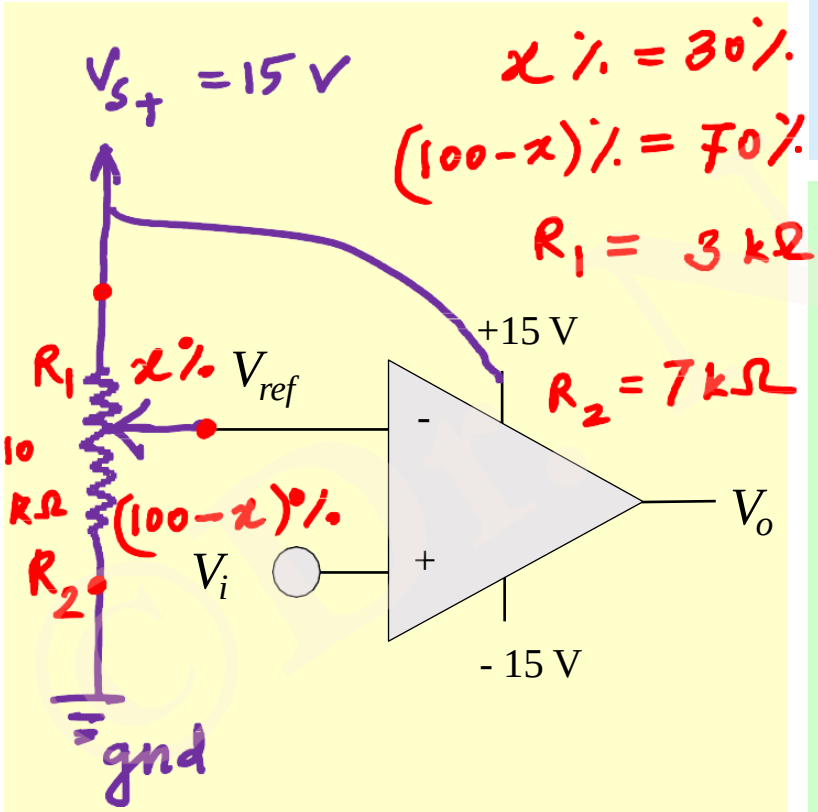
V_{i-}	V_{i+}	V_d	V_o
-10 µV	-15 µV		
-10 µV	+15 µV		
-10 µV	-5 µV		
+1.000001 V	+1.000000 V		
+5 mV	0		
0	+5 mV		

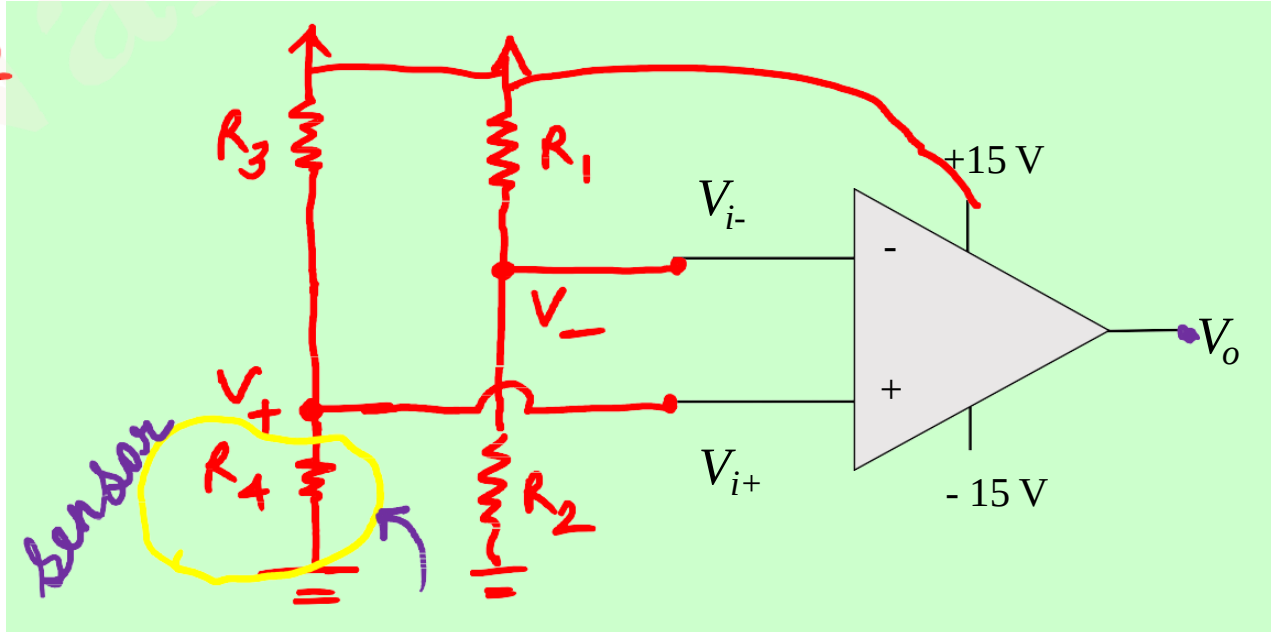
Source: Coughlin Example 2-1

Open-loop comparator - reference



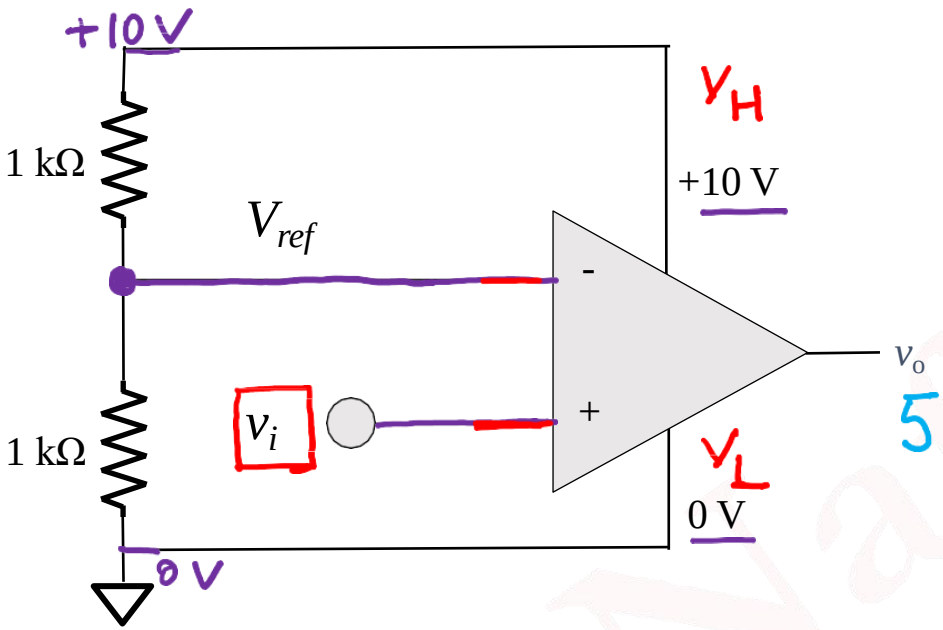






Open-loop comparator

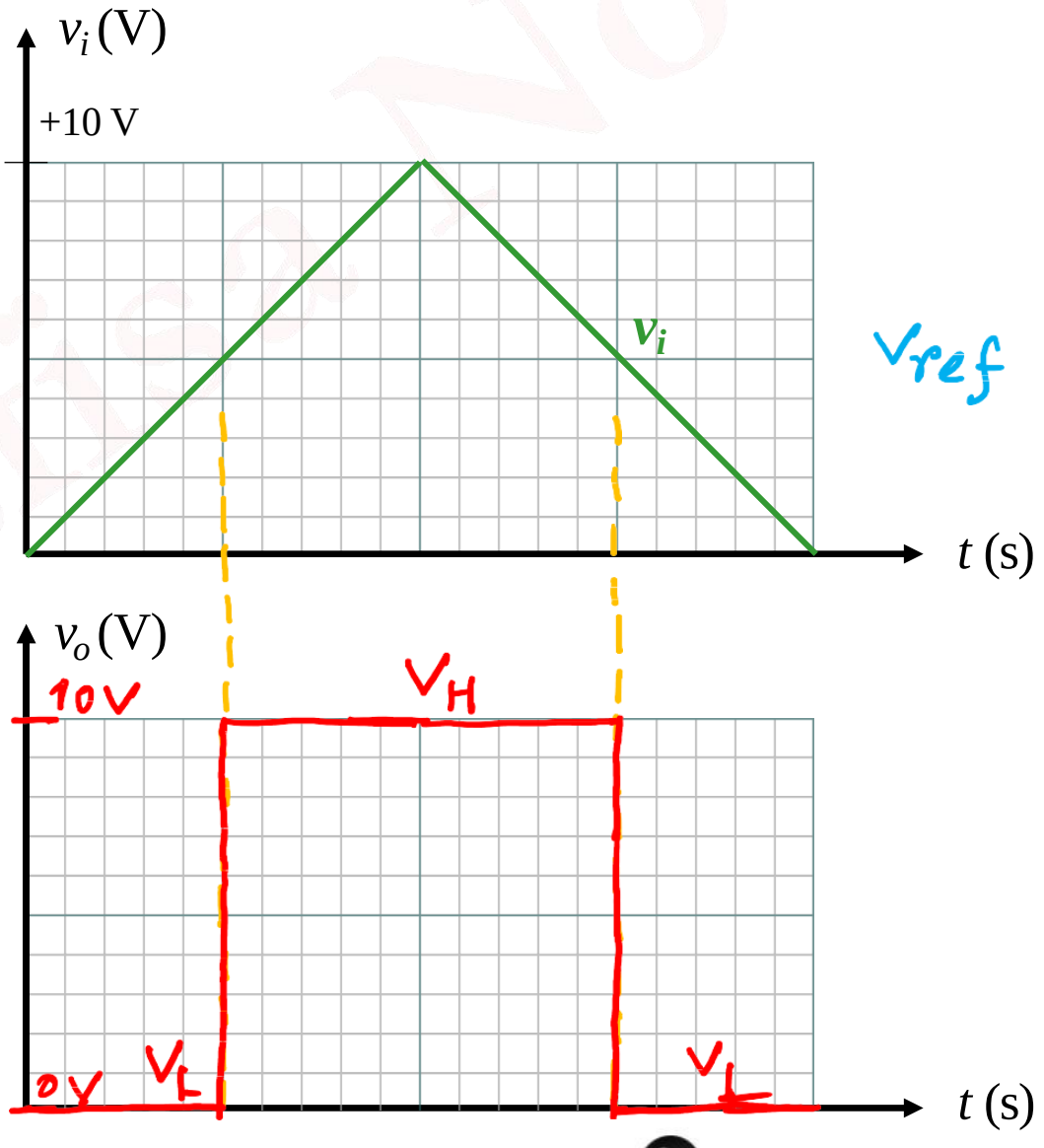
Problem 8. For the following circuit, $A_{OL} = 10^5$. Assume ideal conditions. Calculate the output voltage v_o.



$$V_{ref} = \frac{1}{1+1} \cdot 10\ V$$

$$V_{ref} = 5\ V$$

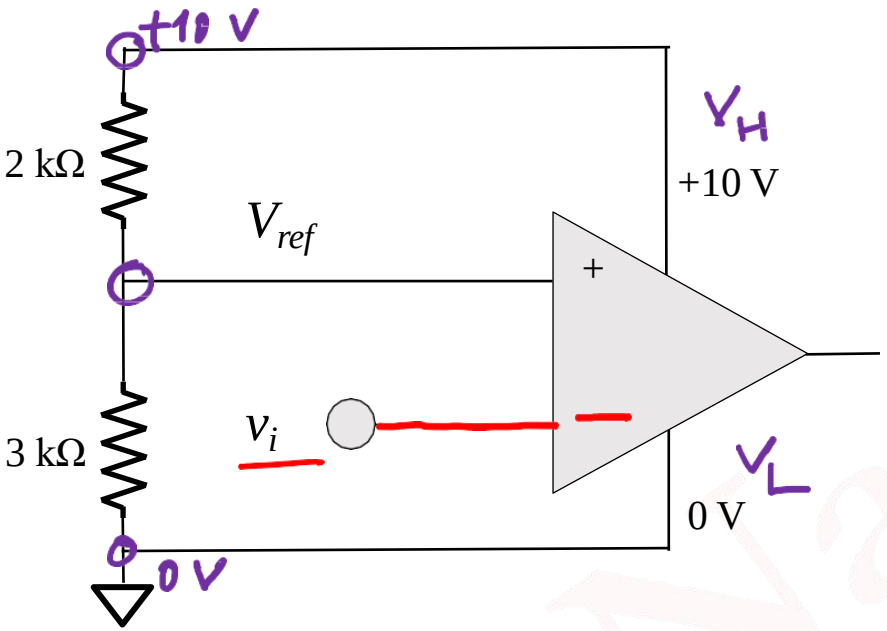
non-inv



Source: Electronics Tutorials

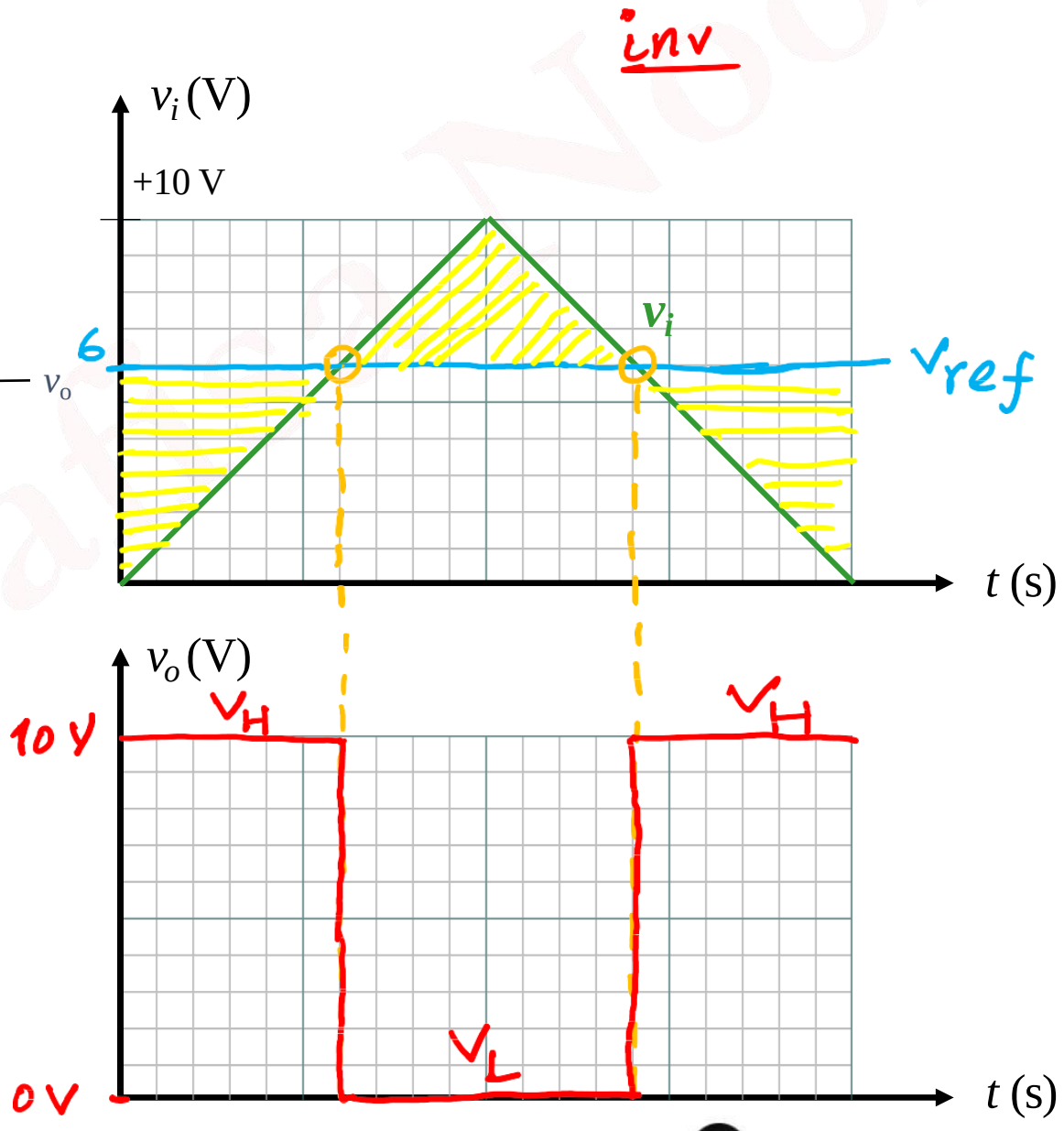
Open-loop comparator

Problem 9. For the following circuit, $A_{OL} = 10^5$. Assume ideal conditions. Calculate the output voltage v_o.



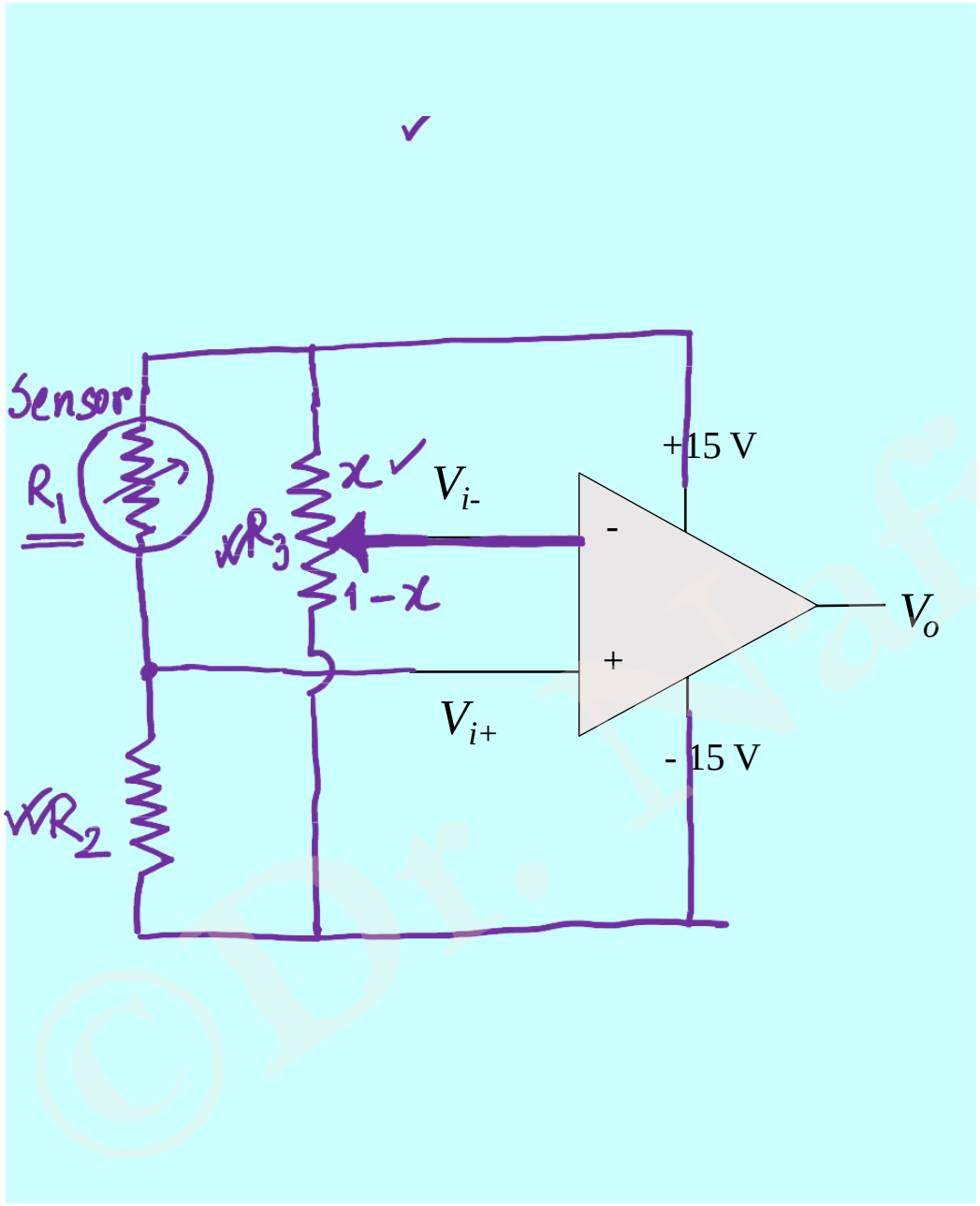
$$V_{ref} = \frac{3}{2+3} \times 10 \text{ V}$$

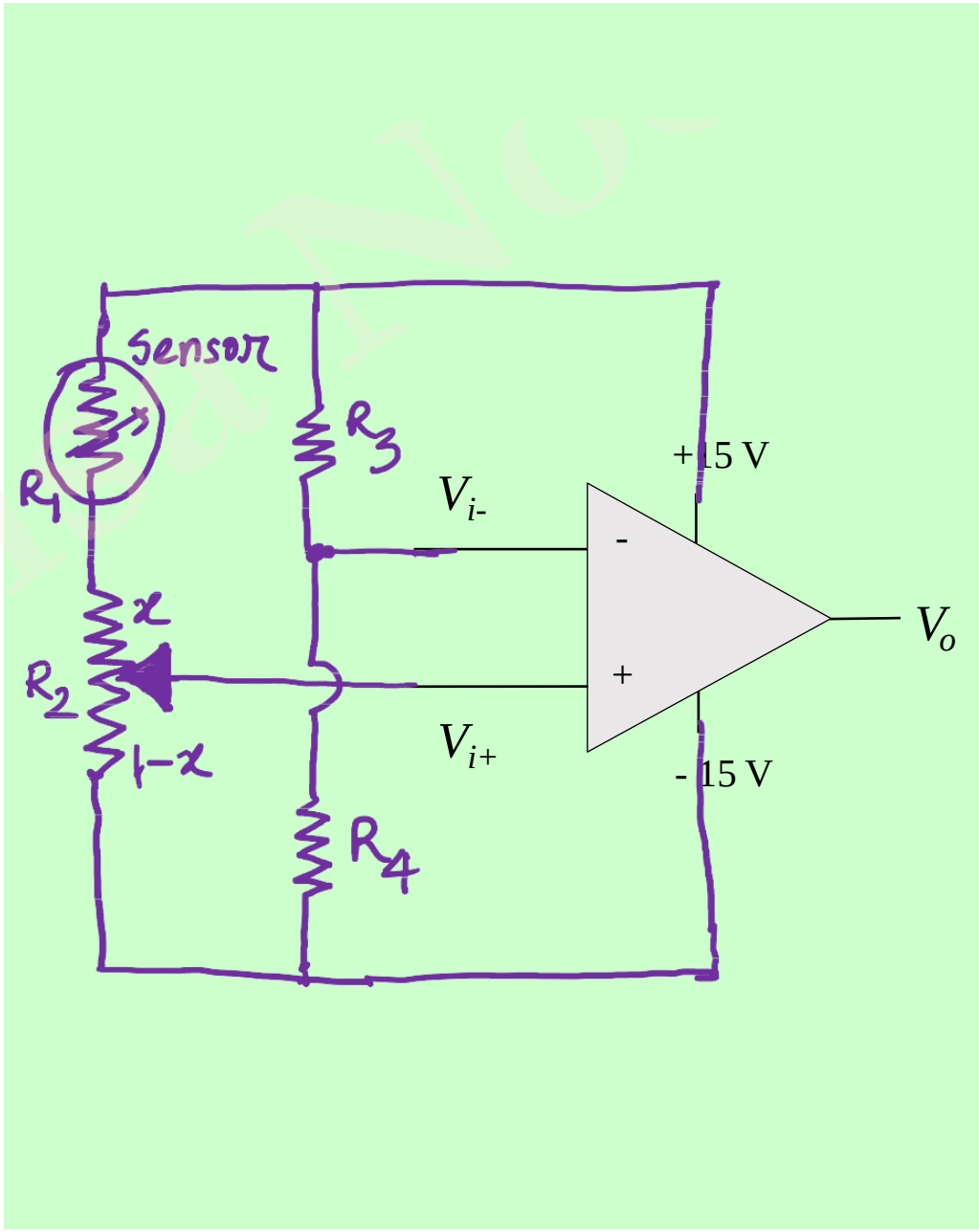
$$V_{ref} = 6 \text{ V}$$



Source: ElectronicsTutorials

Open-loop comparator - applications





Light Sensor Switch Circuit

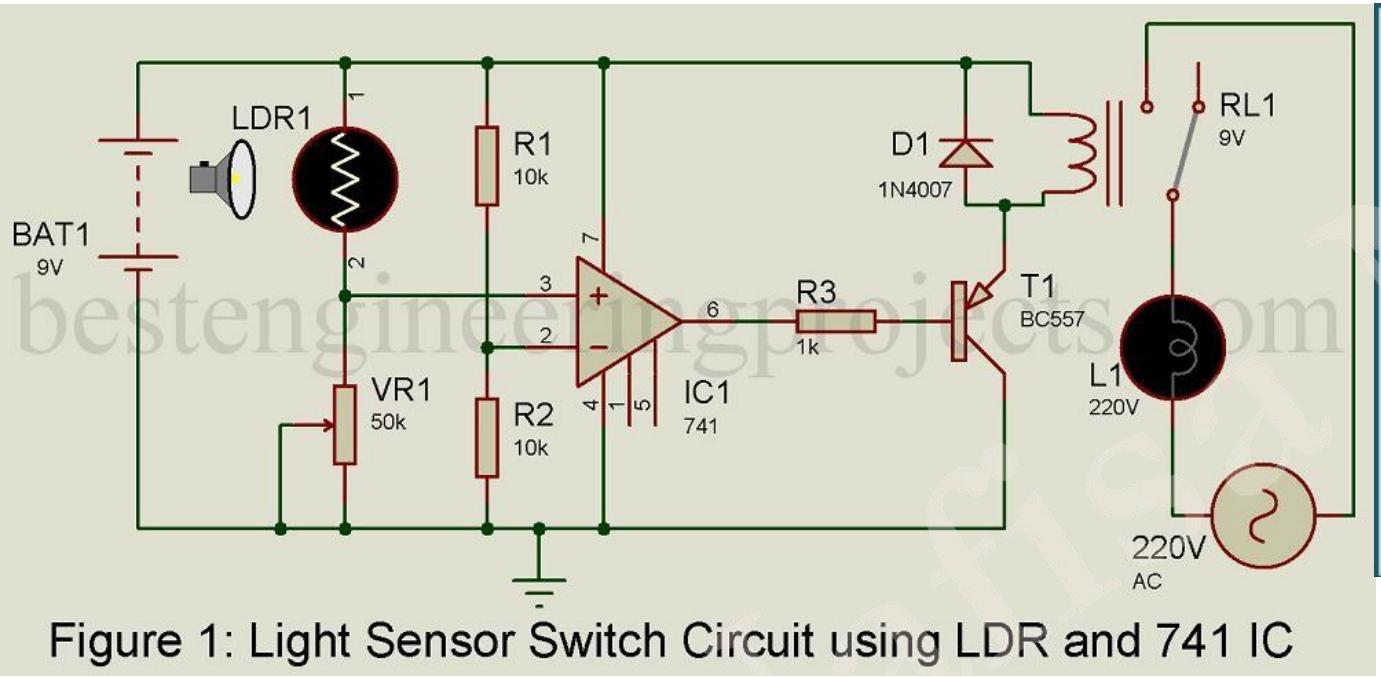
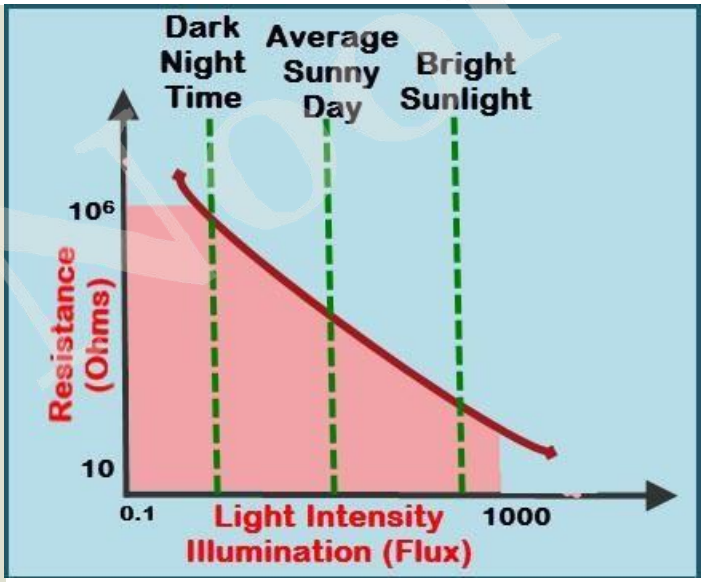


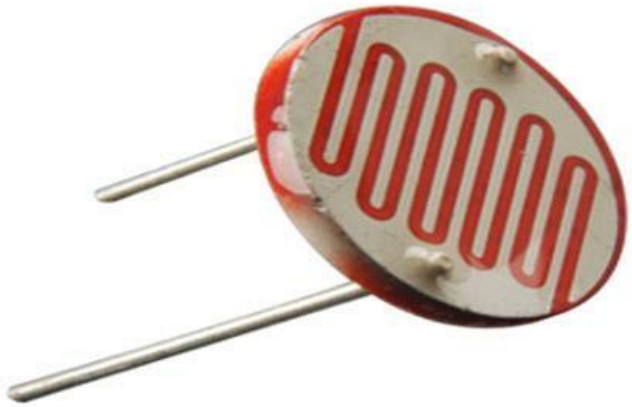
Figure 1: Light Sensor Switch Circuit using LDR and 741 IC

Source: Engineering Projects



WatElectronics.com

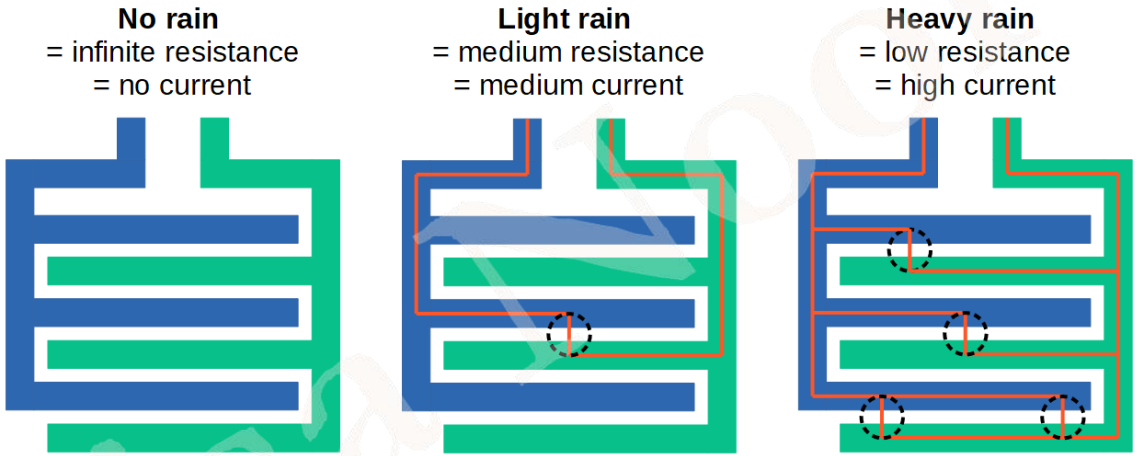
Light dependent resistor

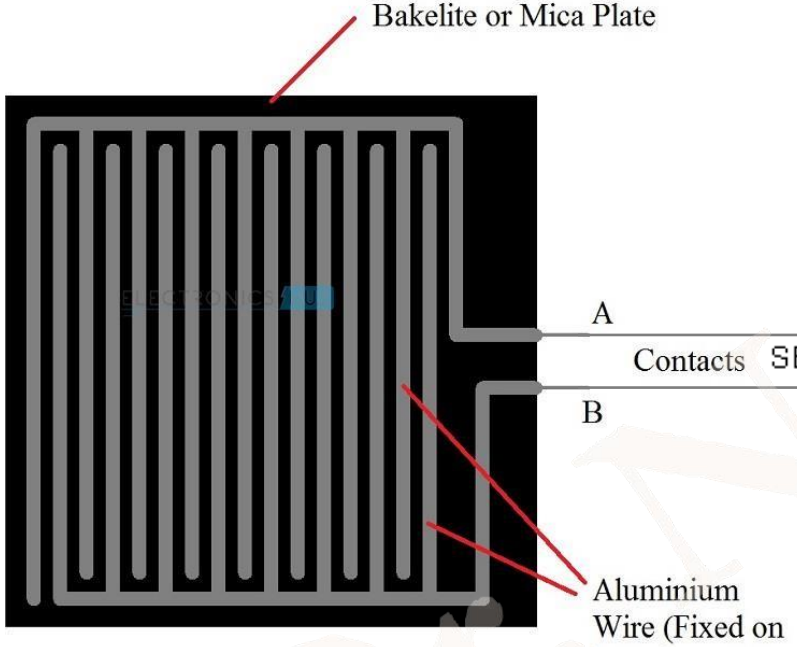


Source: EL-PRO-CUS ELECTRONICS | PROJECTS | FOCUS

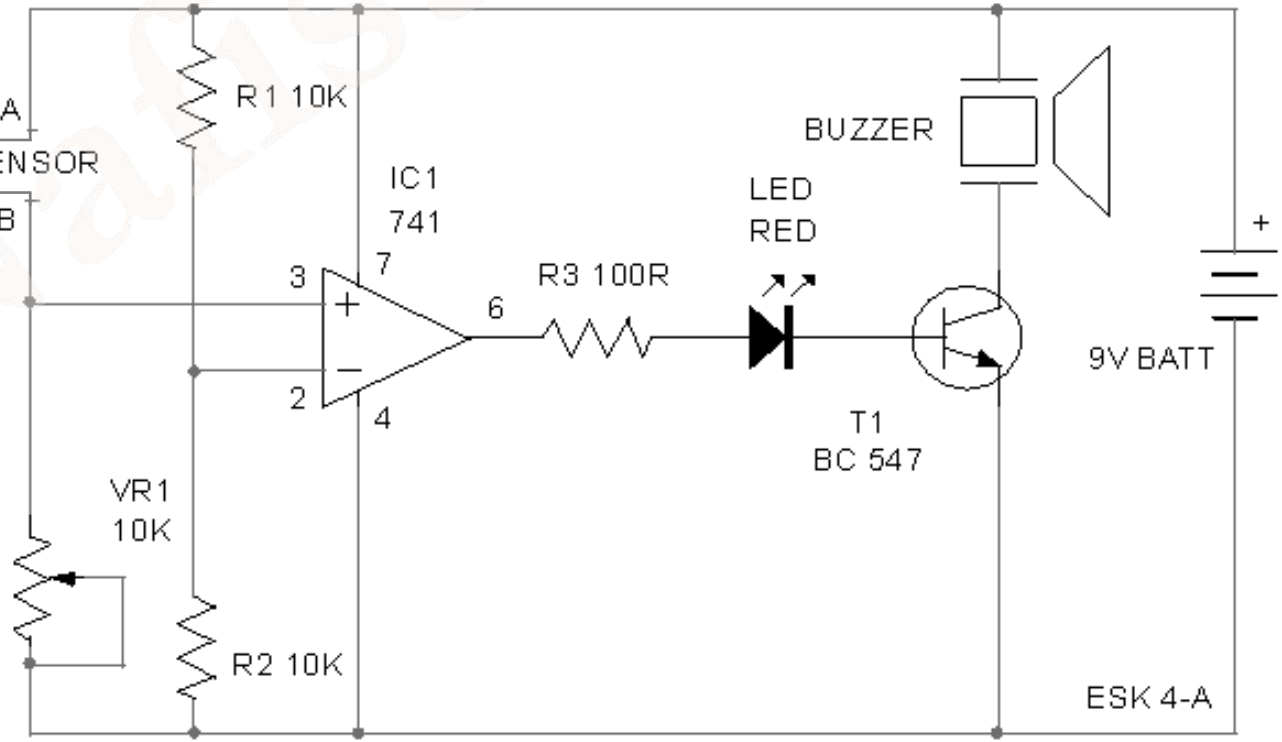
Rain detector circuit

Source: DiyIOt



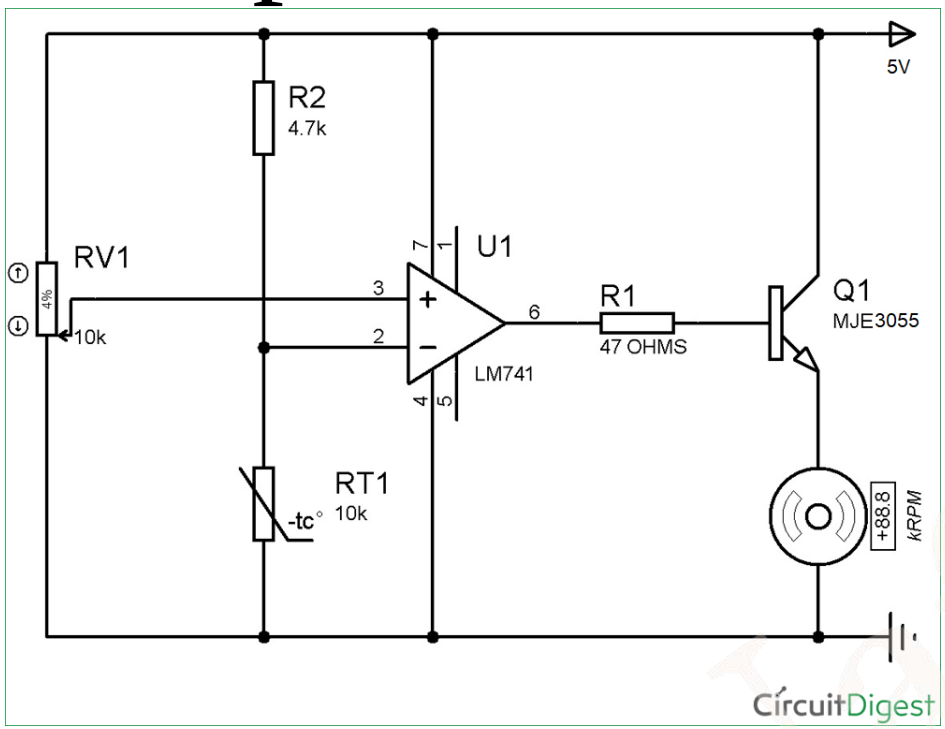


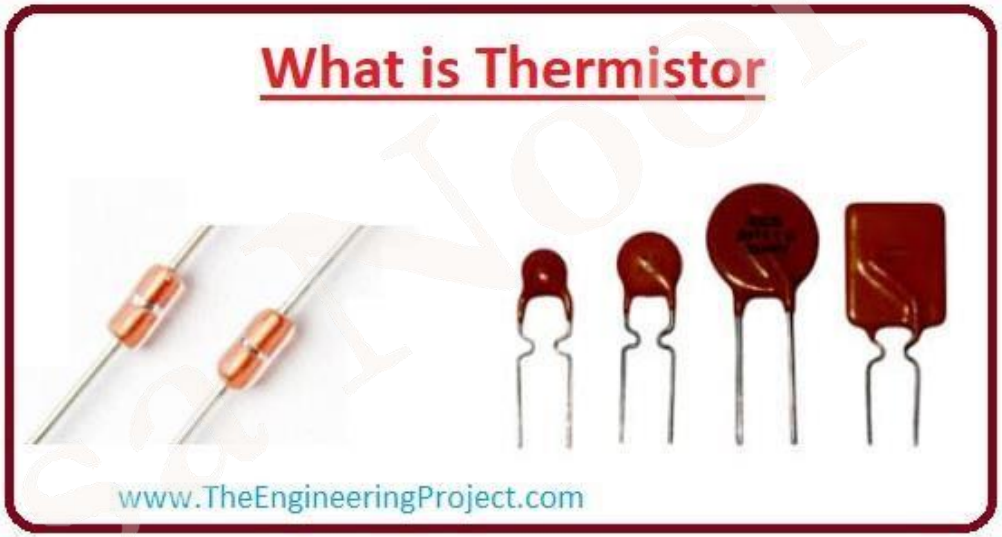
Source: ELECTRONICS HUB

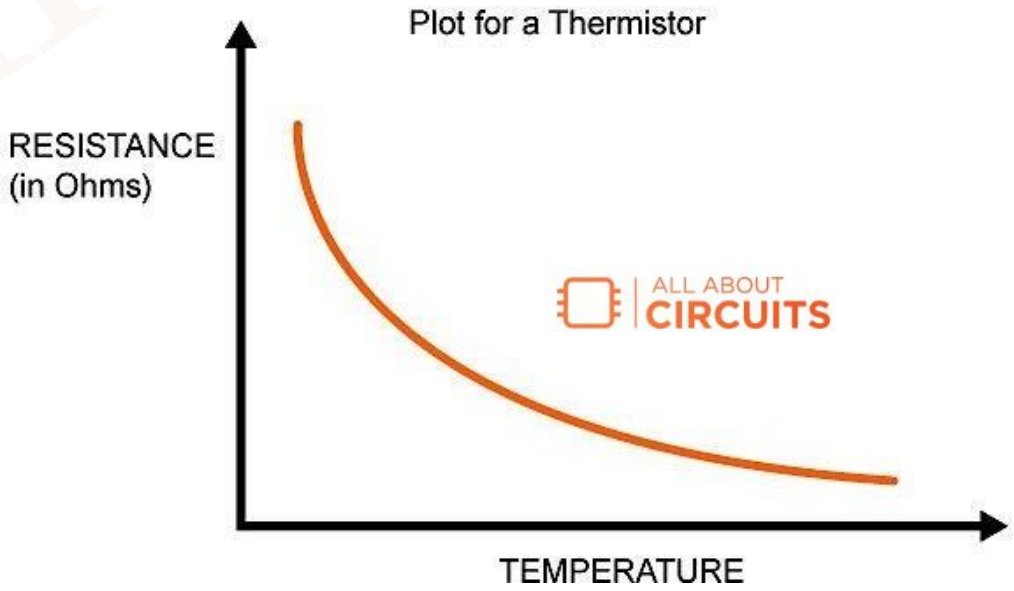


Source: Hobby Circuits and Projects

Temperature controlled DC fan

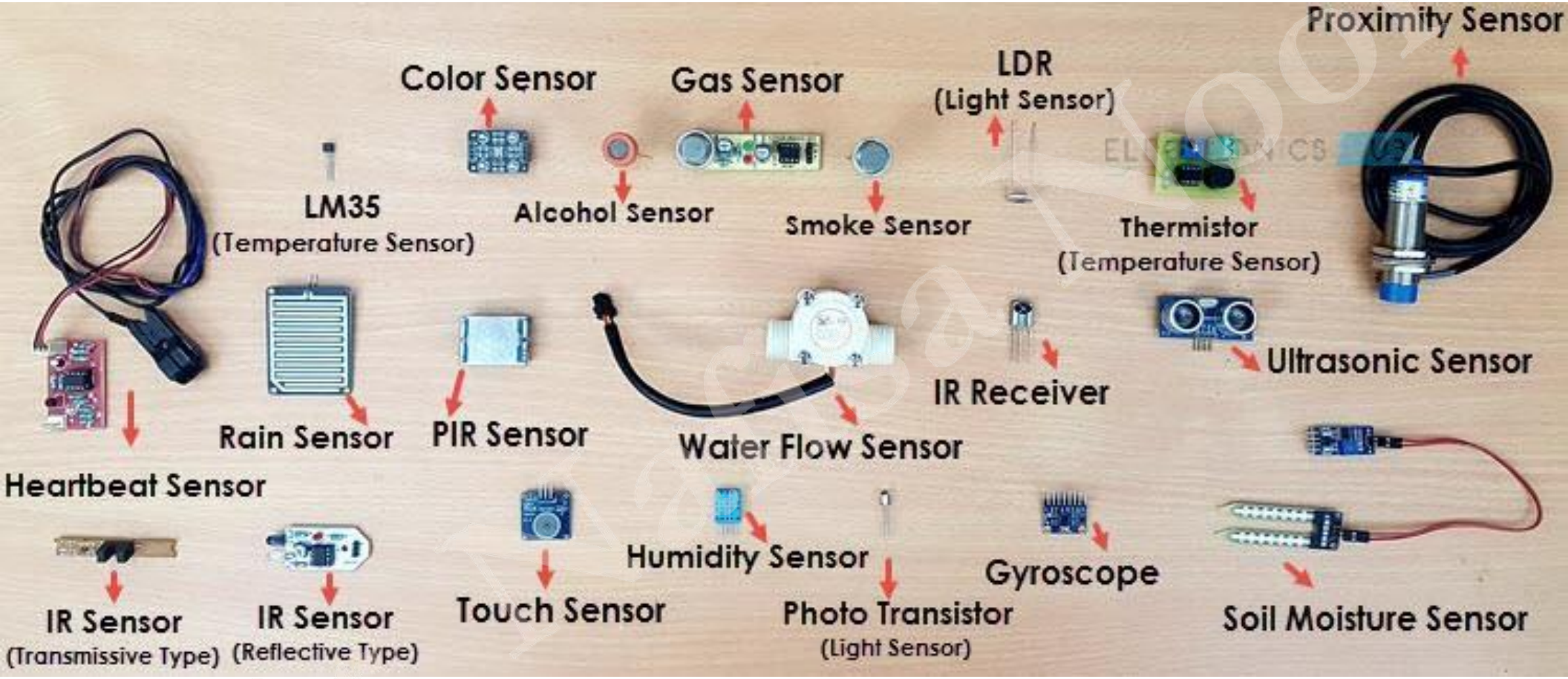






©Dr. Nafisa Noor

Various types of sensors



Source: ELECTRONICS HUB

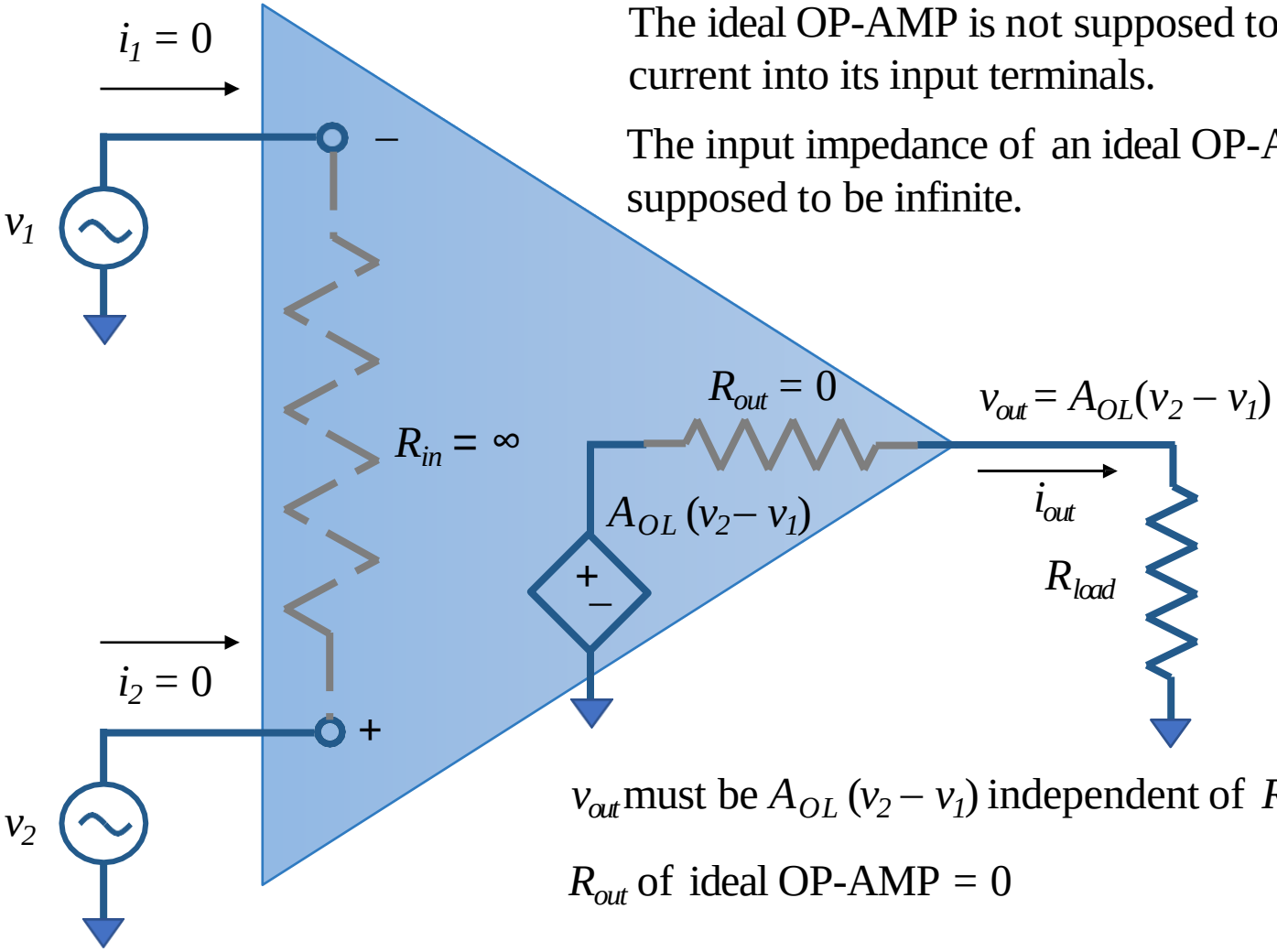
Analog Electronics II: Op-amp & 555 Timer

Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

Lecture # 4

Ideal OP-AMP equivalent circuit



The ideal OP-AMP is not supposed to draw any current into its input terminals.

The input impedance of an ideal OP-AMP is supposed to be infinite.

v_{out} must be $A_{OL}\,(v_2 - v_1)$ independent of R_{load} or i_{out}.

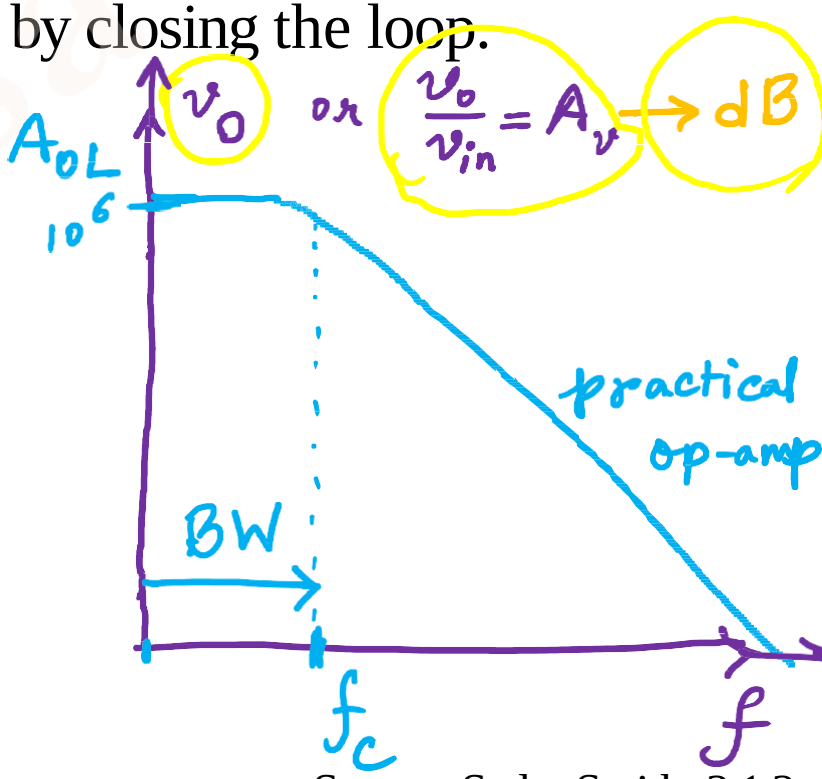
R_{out} of ideal OP-AMP = 0

Source: Sedra Smith, 2.1.2

Ideal OP-AMP: gain and bandwidth

- ✓ Infinite bandwidth: Gain remains constant down to zero frequency and up to infinite frequency.
- ✓ Infinite open-loop gain. But in most circuit designs, OP-AMP is not used in the open-loop configuration. OP-AMP is usually used with other circuit elements to apply feedback by closing the loop.

A_{OL}	10^{5-6} (practical)	∞ (ideal)
BW	Finite	∞



Source: Sedra Smith, 2.1.2

©Dr. Nafisa Noor

Characteristic	Ideal Op-Amp	Practical Op-Amp
Input Impedance (Z_{in})	∞ (Infinite)	High (e.g., MΩ to GΩ range)
Output Impedance (Z_{out})	0 (Zero)	Low (e.g., tens to hundreds of ohms)
Open-Loop Gain (A_{OL})	∞ (Infinite)	High but finite (e.g., 10^5 to 10^6)
Common-Mode Gain (A_{CM})	0 (Zero)	Small but nonzero
Common-Mode Rejection Ratio (CMRR)	∞ (Infinite)	High but finite (e.g., 80–120 dB)
Bandwidth	∞ (Infinite)	Limited (depends on Gain-Bandwidth Product)
Slew Rate	∞ (Infinite)	Limited (e.g., V/µs range)
Input Offset Voltage	0 V	Small (e.g., mV to µV range)
Input Bias Current	0 A	Small (nA to pA range)
Power Supply Requirement	None (assumed ideal)	Finite voltage range (e.g., ±15V, 0-5V)
Output Voltage Swing	Can rea↓pply rails	Limited (e.g., close but not equal to supply rails)

Ideal vs Practical OP-AMP: characteristics

Source: Sedra Smith, 2.1.2

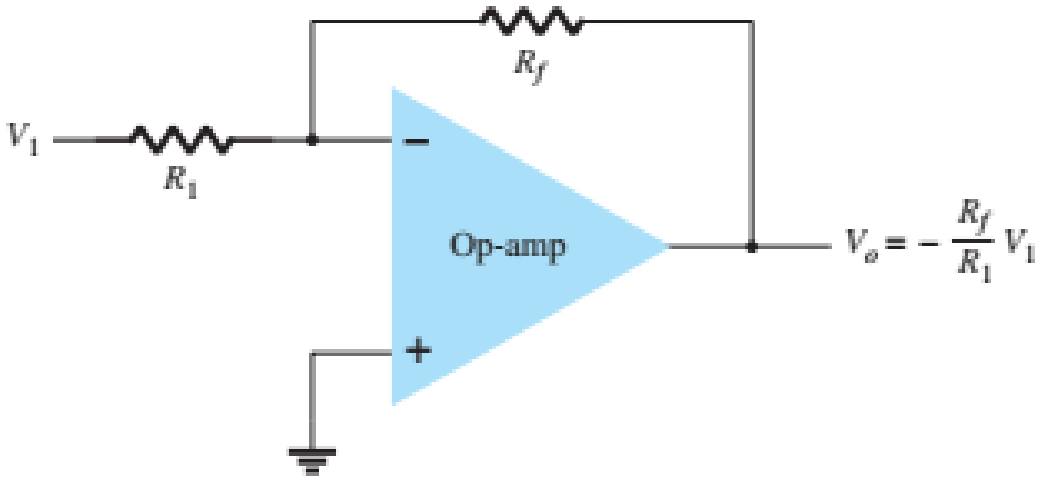
Inverting Amplifier

1. ***Negative feedback:*** between the output and the ***inverting*** input terminals. Due to the feedback, this is a closed-loop configuaration.
2. ***Inverting configuration:*** Signal is connected at the inverting input terminal. So, there will be phase inversion of signal at the output.

$$V_o = -\frac{R_f}{R_1} V_1$$

the voltage gain

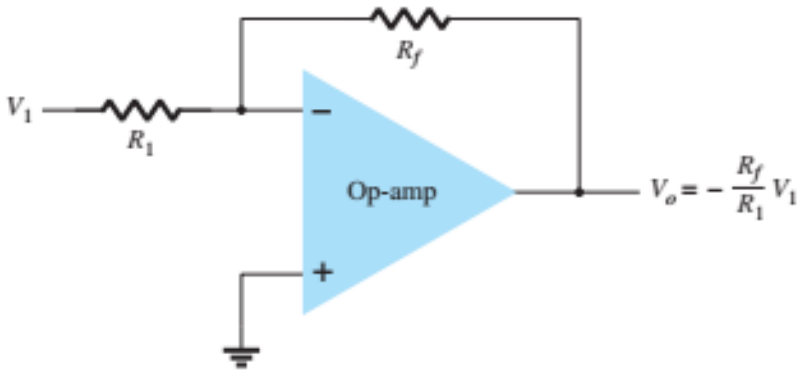
$$A_v = -\frac{R_f}{R_i}$$



Inverting Amplifier

The input impedance is

$$Z_{in} = R_1$$



Constant-Magnitude Gain

If R_f is some multiple of R_1, the overall amplifier gain is a constant. For example, if $R_f = 10R_1$, then

$$\text{Voltage gain} = -\frac{R_f}{R_1} = -10$$

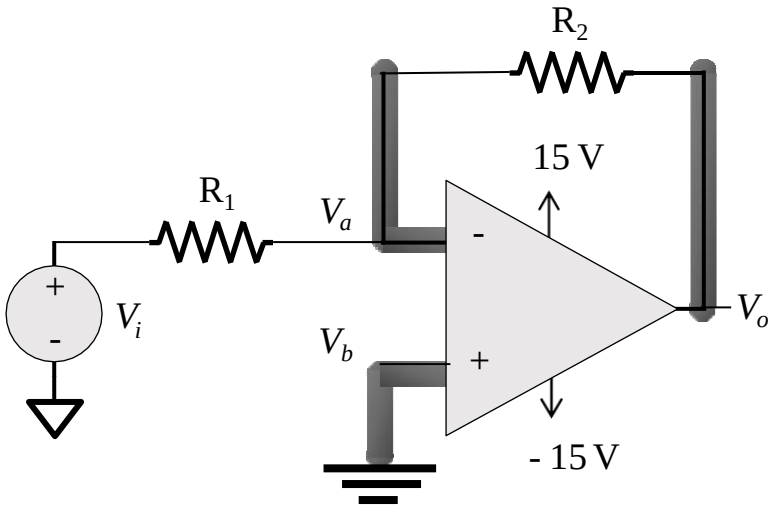
Unity Gain

If $R_f = R_1$, the gain is

$$\text{Voltage gain} = -\frac{R_f}{R_1} = -1$$

Virtual short circuit & virtual ground

1. ***Virtual short circuit*** (not physical): between the non-inverting and inverting input terminals
   - ✓ The two input terminals track each other in potential
2. ***Virtual ground*** (not physical): at the inverting input terminal

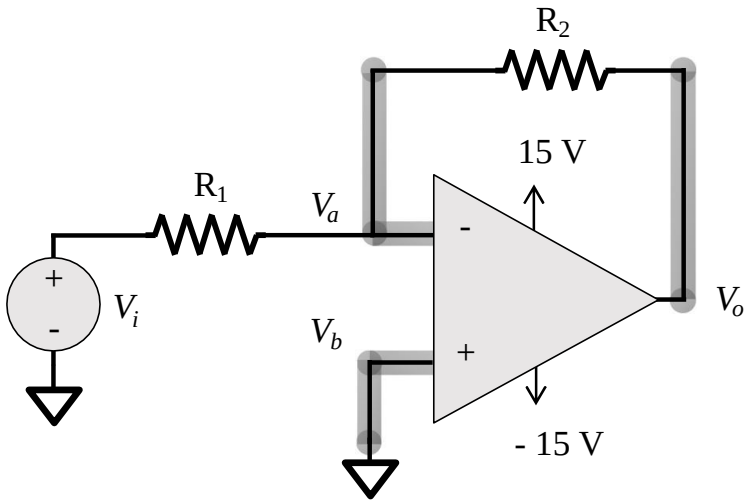


Source: Sedra Smith 2.2

Inverting amplifier: closed-loop gain

1. Open loop gain A_{OL} varies from one op-amp to the other $v_o \quad A_{OL} \cdot v_d$
2. Closed loop gain can be very accurate, stable, and predictable (trading gain for accuracy)

Closed-loop gain, $A_{CL} = \frac{v_0}{v_I}$ $v_0 = A_{CL} \cdot v_i$



Inverting amplifier: closed-loop gain

KCL @ node a, $I_1 = I_2 + 0A \Rightarrow I_1 = I_2$

(R_1 & R_2 are in series)

$$\frac{v_o}{v_I} = -\frac{R_2}{R_1}$$

inversion

$$\Rightarrow \frac{V_i - 0V}{R_1} \qquad \frac{0V - V_o}{R_2}$$

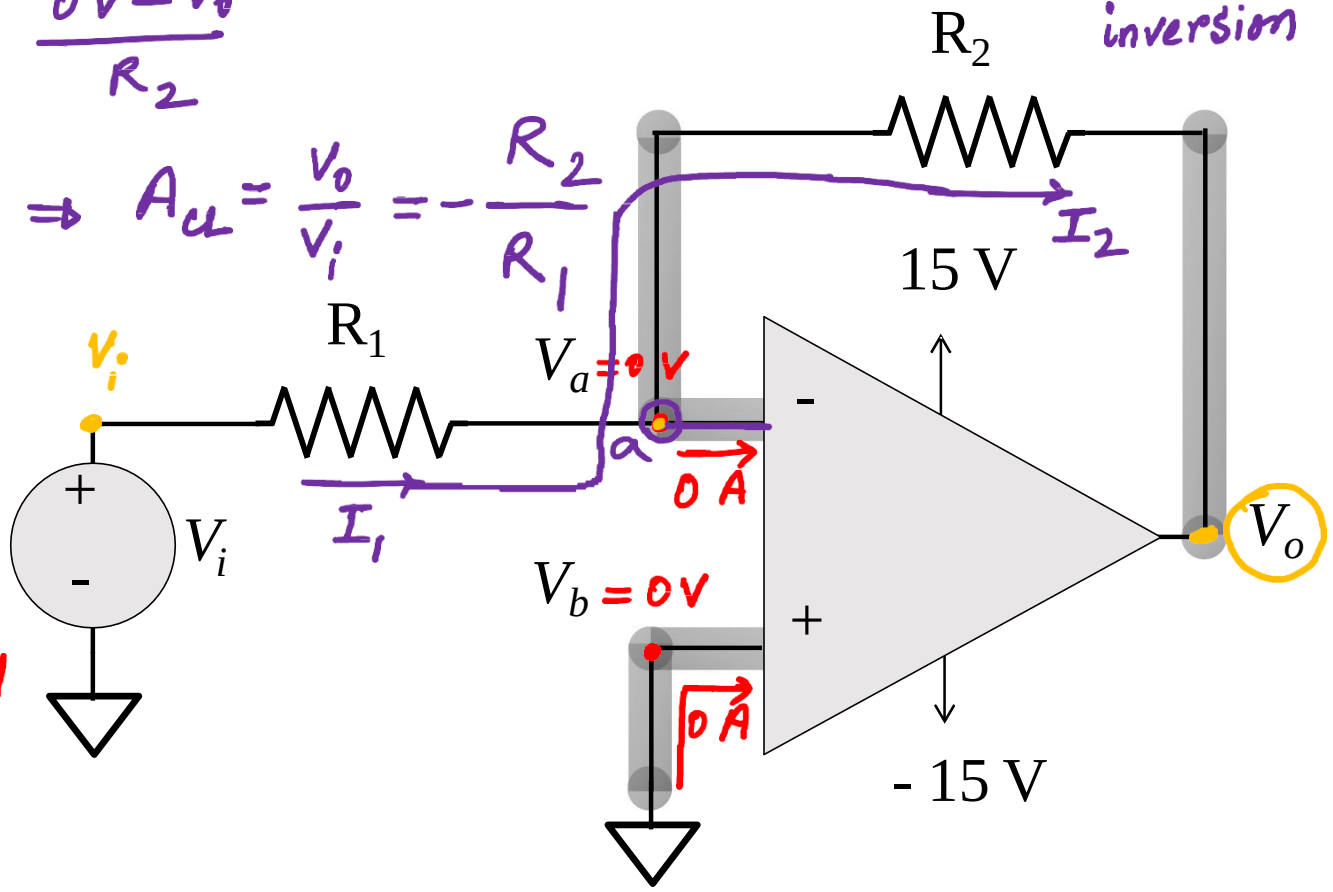
$$\Rightarrow \frac{V_i}{R_1} = -\frac{V_o}{R_2} \Rightarrow A_{CL} = \frac{V_o}{V_i} = -\frac{R_2}{R_1}$$

CL virtual S.C.

$V_a = V_b$

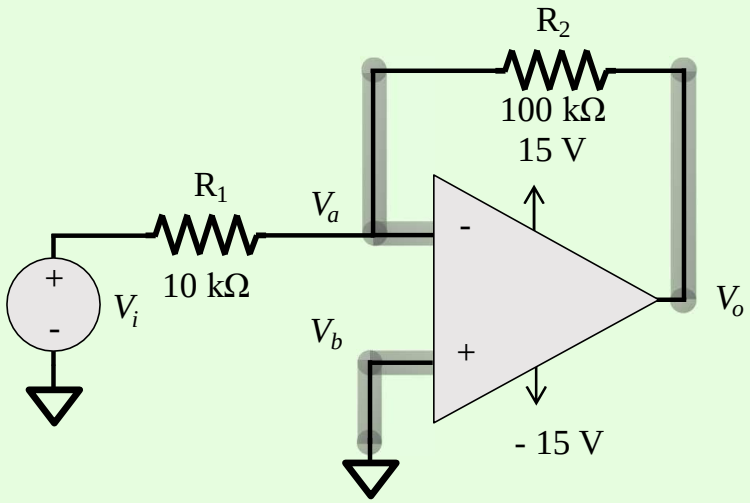
inv. amp.: virtual gnd

$V_a = 0V = V_b$



Inverting amplifier gain $A_{CL} = -\frac{R_2}{R_1}$

(a)



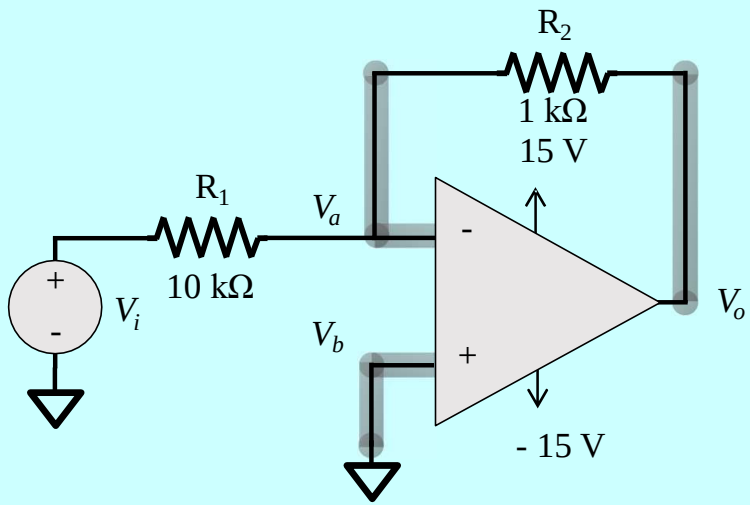
Amplification when $R_2 > R_1$

$$A_{CL} = -\frac{R_2}{R_1} = -\frac{100\ k\Omega}{10\ k\Omega} = -10\ \frac{V}{V}$$

$$\Rightarrow |A_{CL}| = 10\ \frac{V}{V} > 1$$

$$\Rightarrow \left|\frac{v_o}{v_i}\right| > 1 \Rightarrow |v_o| > |v_i|$$

(b)



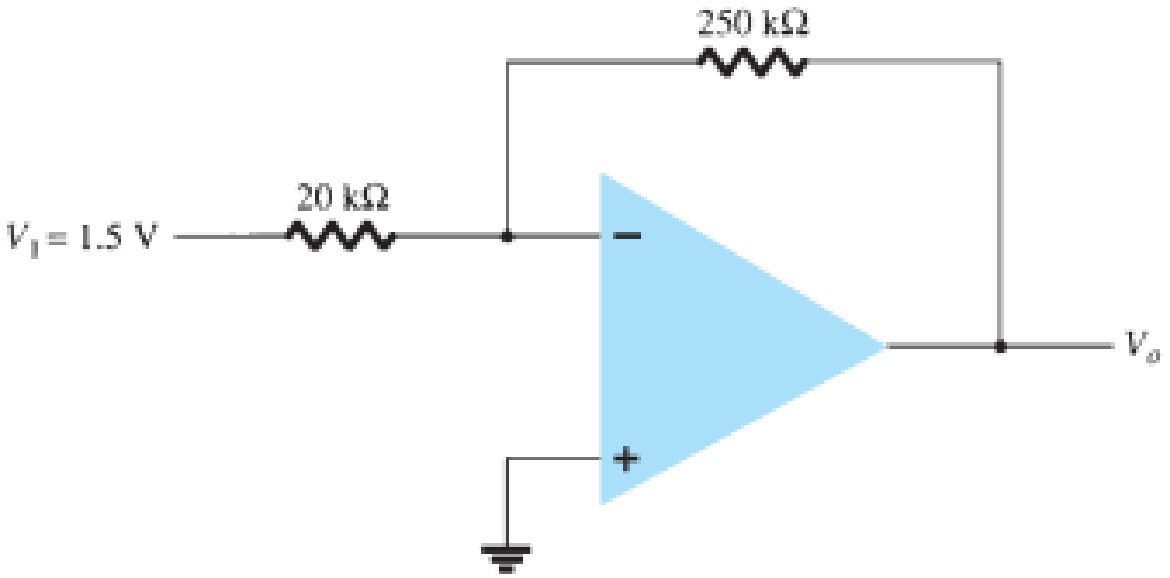
Attenuation when $R_2 < R_1$

$$A_{CL} = -0.1\ \frac{V}{V} \Rightarrow |A_{CL}| = 0.1\ \frac{V}{V} < 1$$

$$\Rightarrow \left|\frac{v_o}{v_i}\right| < 1 \Rightarrow |v_o| < |v_i|$$

Practice Problem

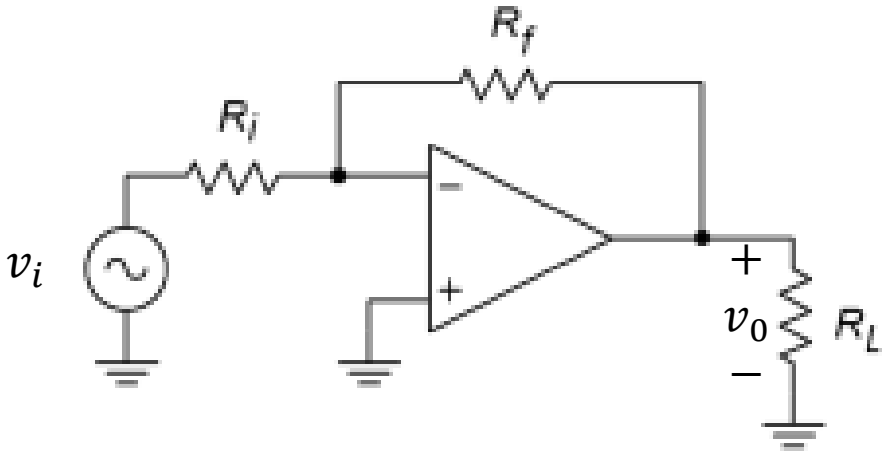
What is the output voltage in the circuit?



$$V_o = -\frac{R_f}{R_1} V_1$$

Practice Problem

Design an inverting amplifier with a gain of 10 and an input impedance of 15 kΩ.



The input impedance tells us what R_i must be

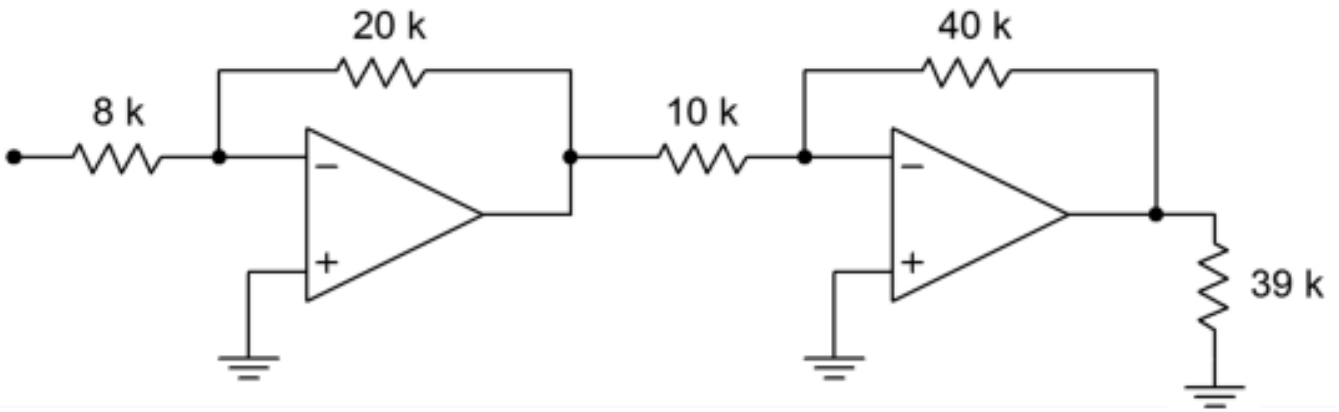
$$Z_{in} = R_i$$
$$R_i = 15\,\text{k}$$

Knowing R_i, solve for R_f:

$$A_v = -\frac{R_f}{R_i}$$
$$R_f = R_i(-A_v)$$
$$R_f = 15\,\text{k} \times (-(-10))$$
$$R_f = 150\,\text{k}$$

Practice Problem

What is the system input impedance ? What is the system gain?



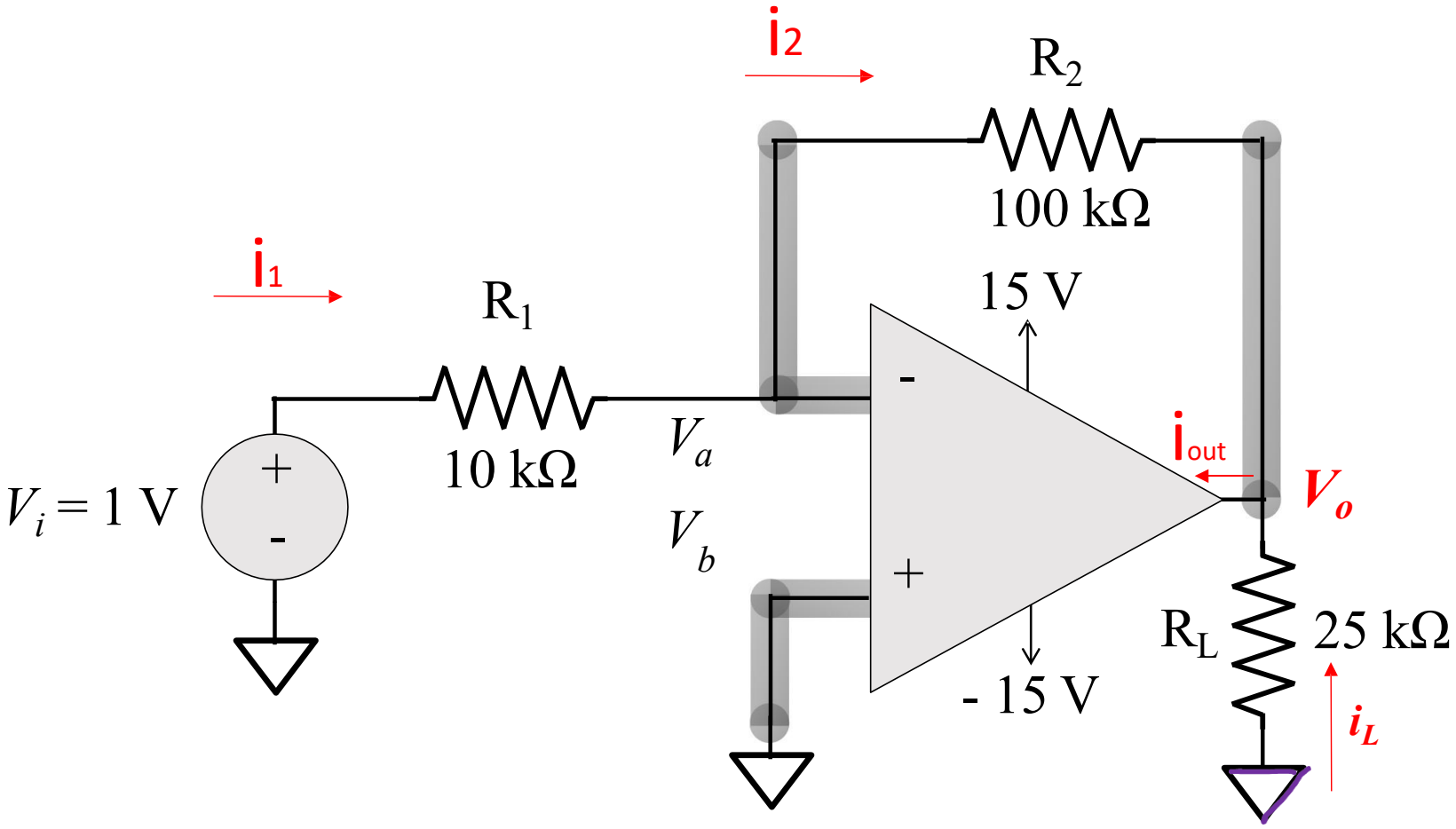
Analog Electronics II: Op-amp & 555 Timer

Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

Lecture # 5

Problem 10. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions. Calculate the A_{CL}, i_1, i_2, V_o, i_L, and i_{out}.



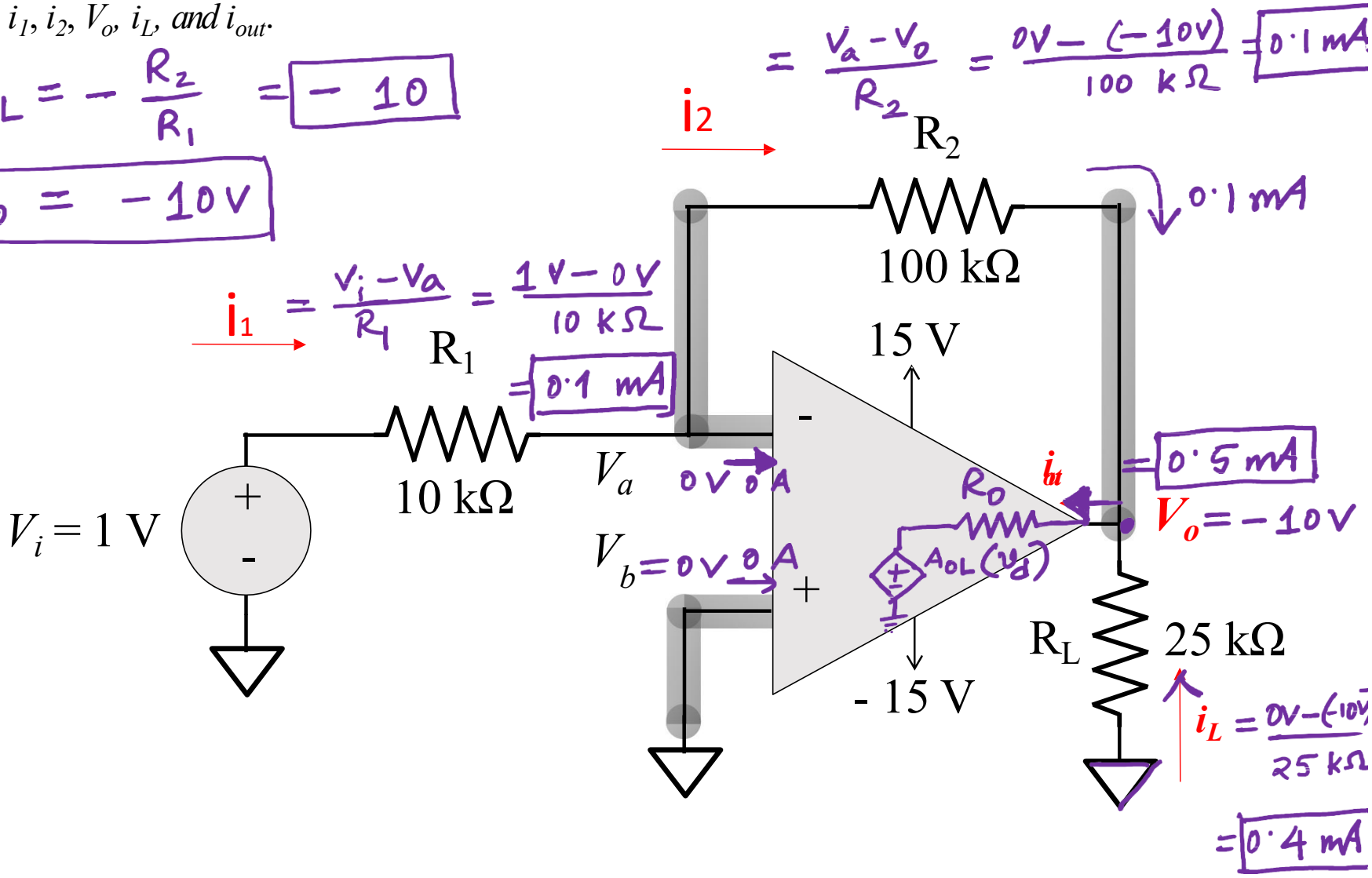
Problem 10. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions. Calculate the A_{CL}, i_1, i_2, V_o, i_L, and i_{out}.

$A_{CL} = -\frac{R_2}{R_1} = -10$

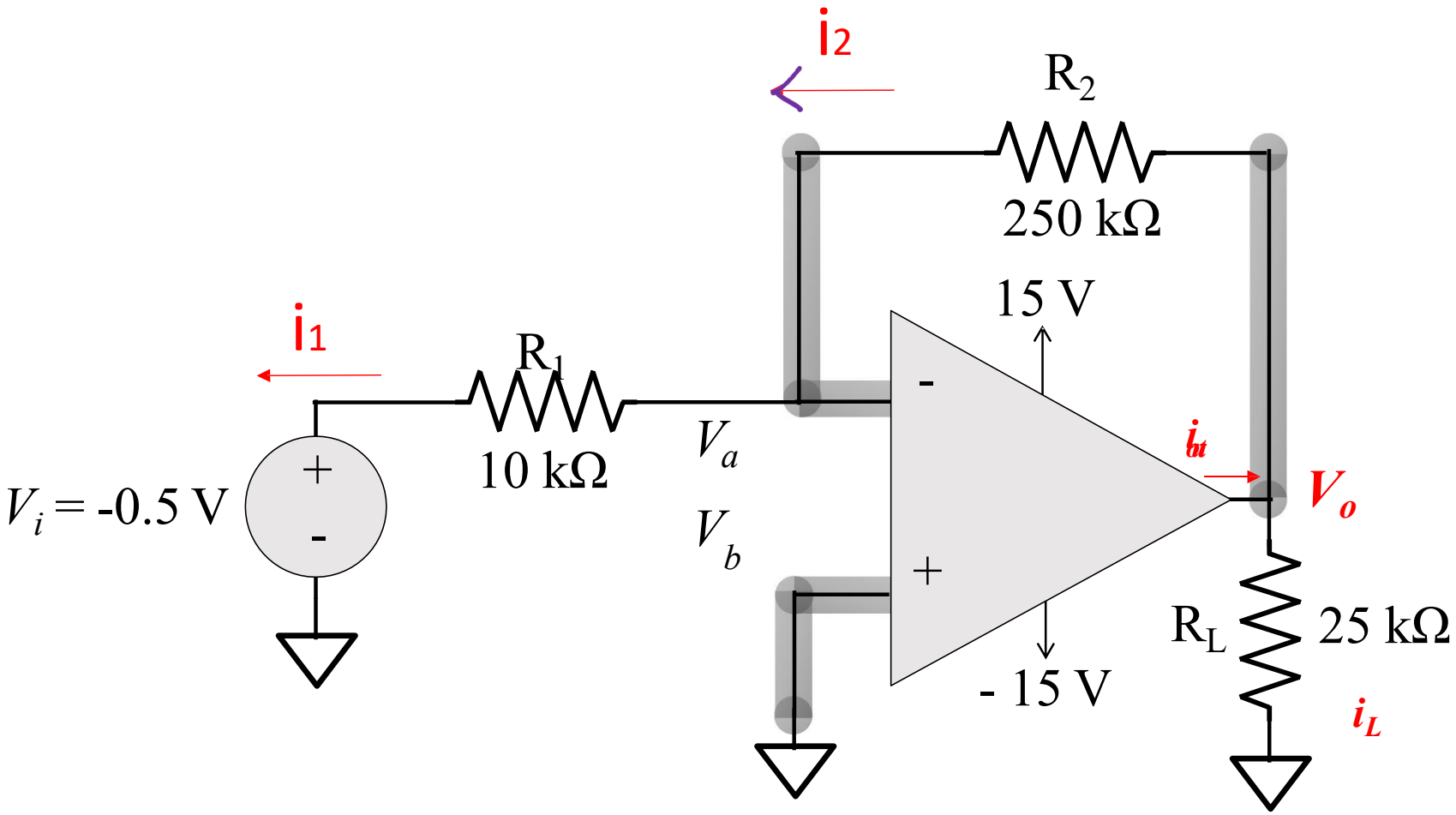
$V_o = -10$ V

$i_2 = \frac{V_a - V_o}{R_2} = \frac{0\text{V} - (-10\text{V})}{100\ \text{k}\Omega} = 0.1$ mA

$i_1 = \frac{V_i - V_a}{R_1} = \frac{1\text{V} - 0\text{V}}{10\ \text{k}\Omega} = 0.1$ mA



Problem 11. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions. Calculate the A_{CL}, i_1, i_2, V_o, i_L, and i_{out}.



Problem 11. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions. Calculate the A_{CL}, i_1, i_2, V_o, i_L, and i_{out}.

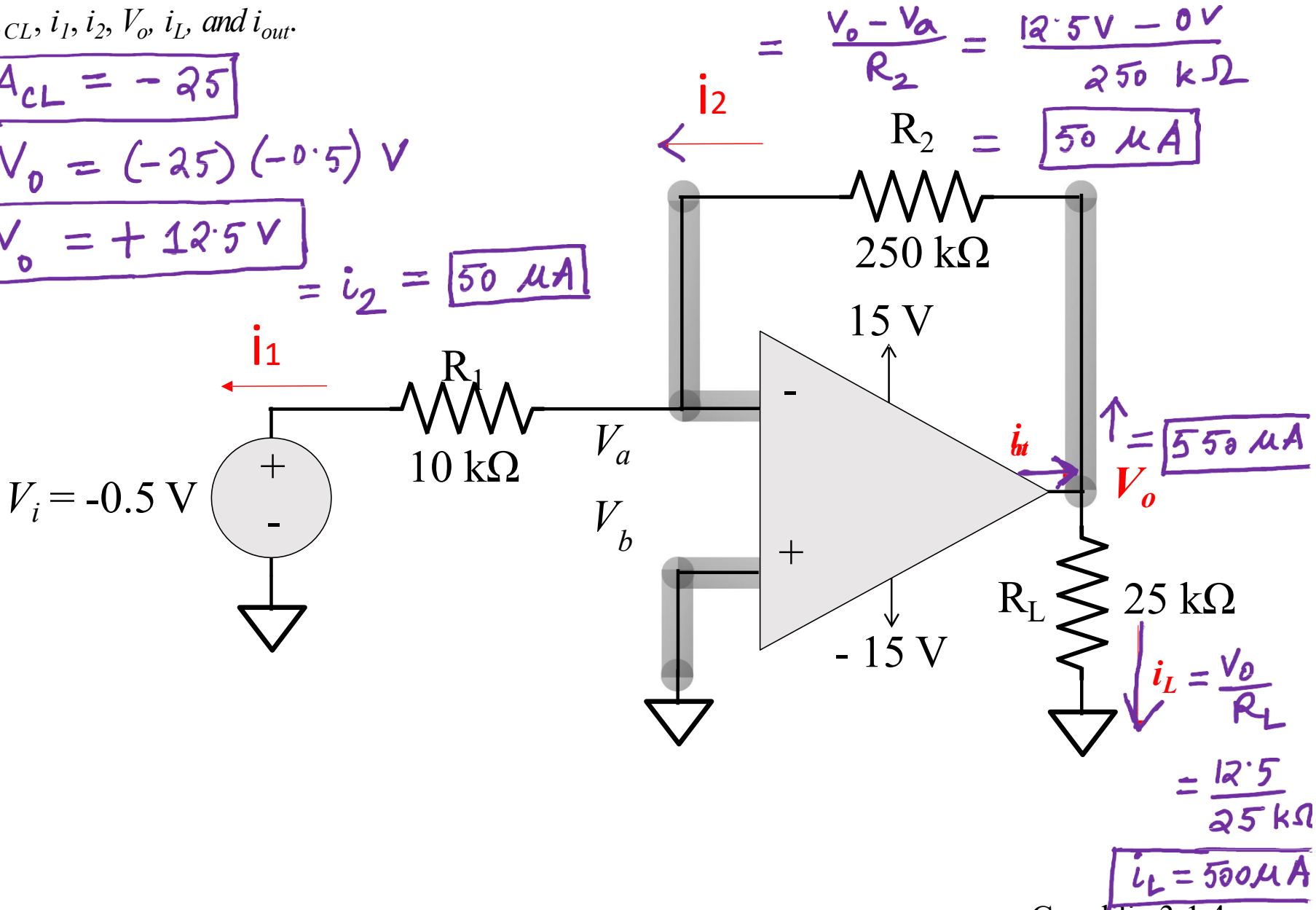
$A_{CL} = -25$

$V_o = (-25)(-0.5)$ V

$V_o = +12.5$ V

$= \frac{V_o - V_a}{R_2} = \frac{12.5\text{ V} - 0\text{ V}}{250\text{ k}\Omega}$

$= i_2 = 50\ \mu\text{A}$



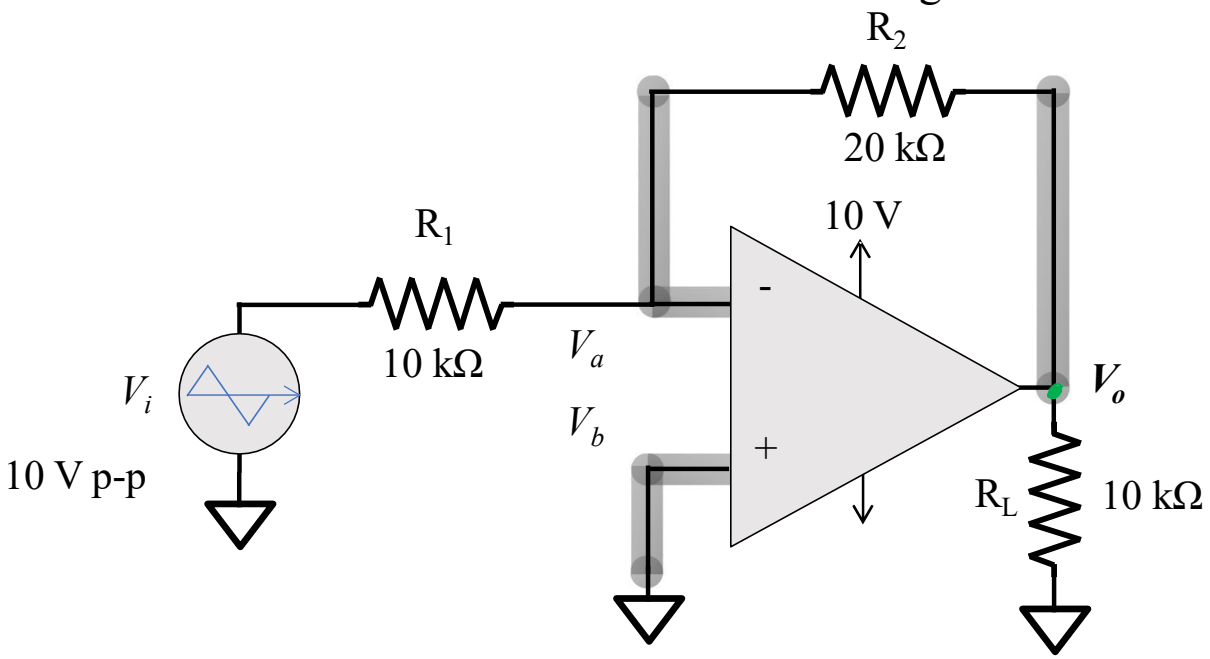
$= \frac{12.5}{25\text{ k}\Omega}$

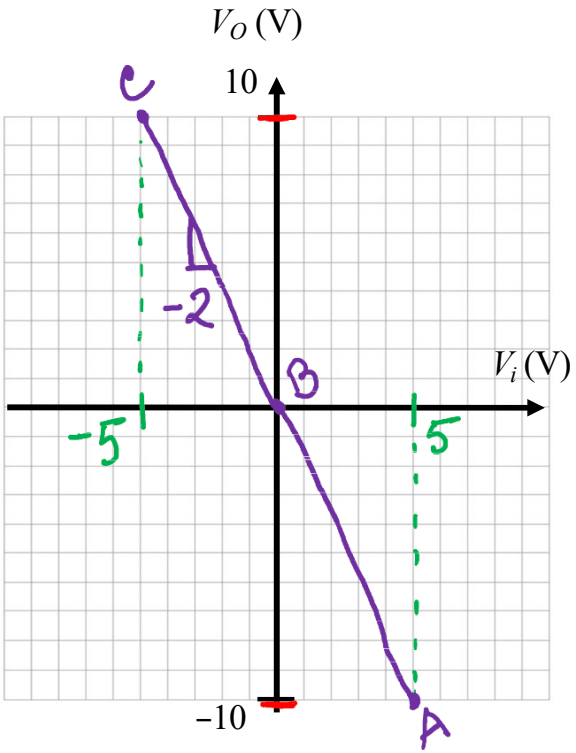
$i_L = 500\ \mu\text{A}$

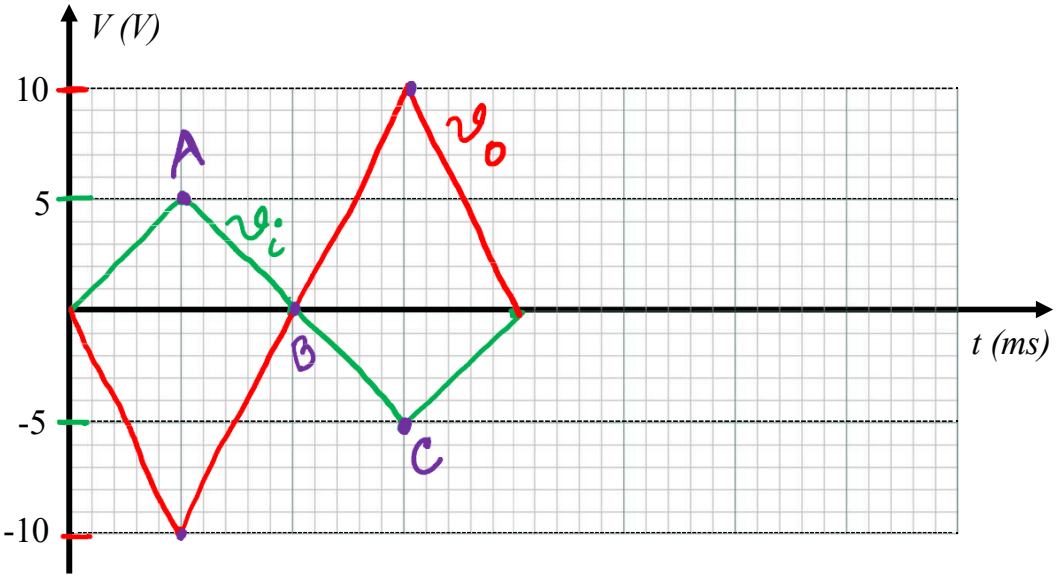
Problem 12. $A_{OL} = 2\times10^5$. $V_{sat+} = +10$ V, $V_{sat-} = -10$ V. Assume ideal conditions. Calculate the A_{CL}. Draw the V_i and V_o. Draw the transfer characteristics.

Coughlin 3-1.5

$A_{CL} = -2$

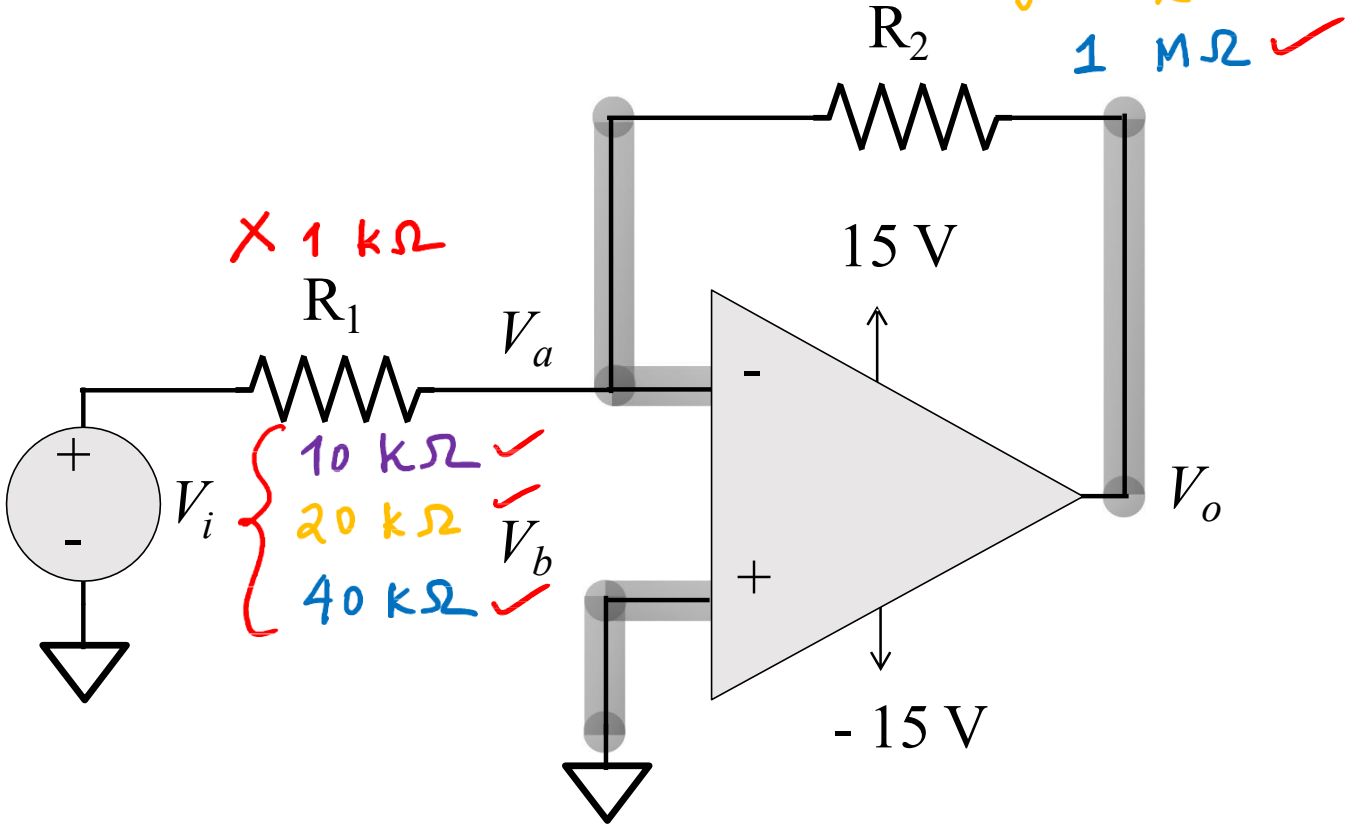




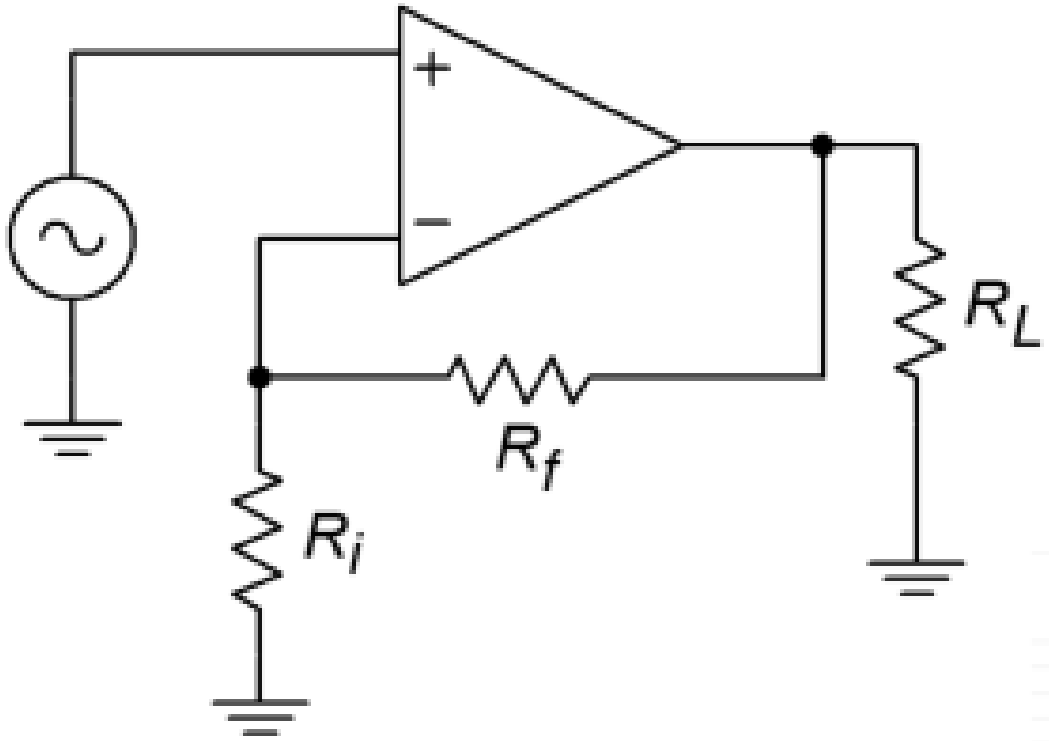


Problem 13. Design an amplifier with a gain of -25. The input resistance should be at least 10 kΩ. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions.

25 kΩ
250 kΩ ✓
500 kΩ ✓
1 MΩ ✓



The Noninverting Voltage Amplifier



$$A_v = \frac{V_{out}}{V_{in}}$$

$$V_{in} = V_{Ri}$$

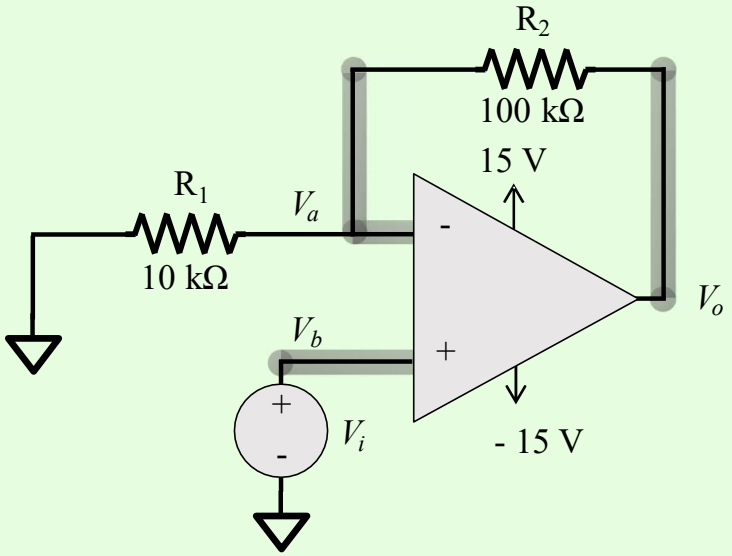
$$V_{out} = V_{Ri} + V_{Rf}$$

$$A_v = \frac{V_{Ri} + V_{Rf}}{V_{Ri}}$$

$$A_v = \frac{R_i I_{Ri} + R_f I_{Rf}}{R_i I_{Ri}} = \frac{R_i + R_f}{R_i} = 1 + \frac{R_f}{R_i}$$

Non-inverting amplifier gain $A_{CL} = 1 + \frac{R_2}{R_1}$

(a)

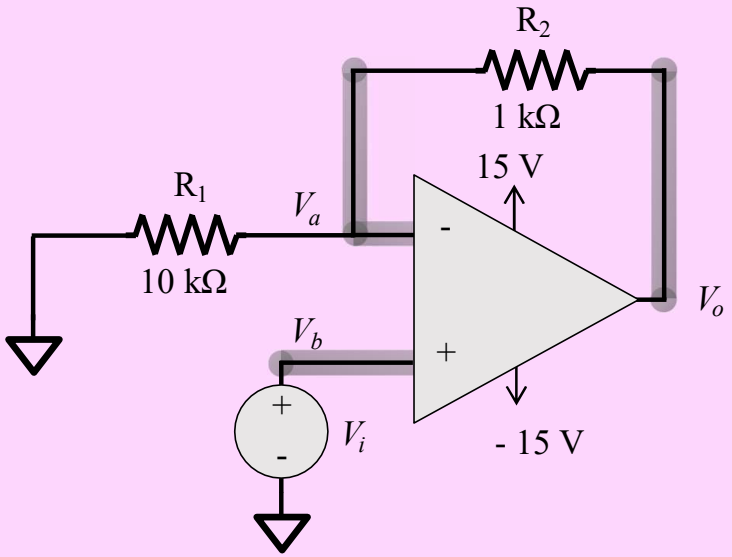


$A_{CL} = 1 + 10$

$= 11 \frac{V}{V}$

$A_{CL} > 1$: amplific^n

(b)



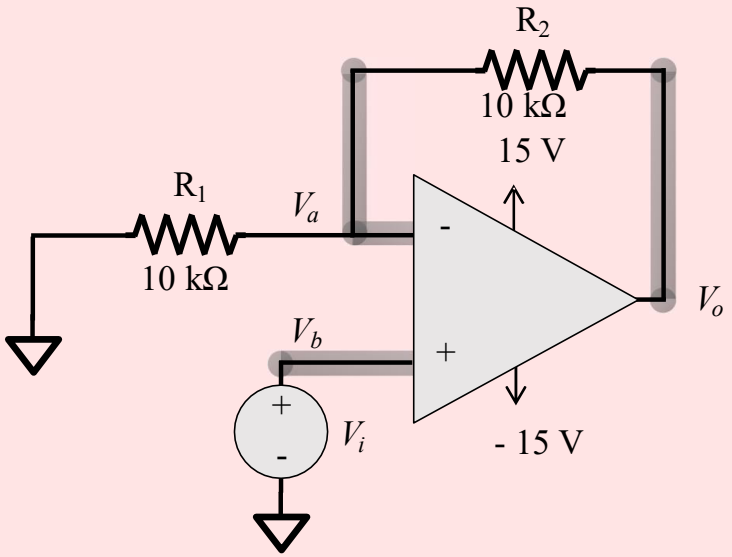
$A_{CL} = 1 + \frac{1}{10}$

$= 1.1 \frac{V}{V}$

$A_{CL} > 1$: amplific^n

Non-inverting amplifier gain A_{CL}

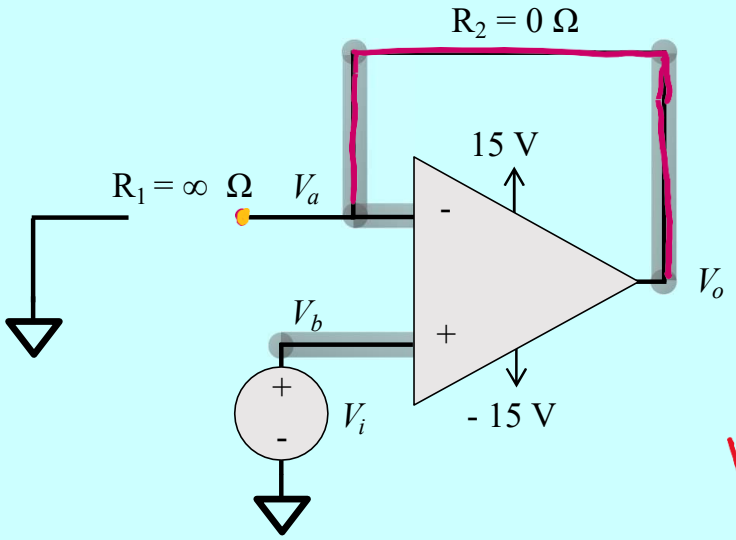
(c)



$A_{CL} = 1 + 1 = 2$

$A_{CL} > 1$: amplific^{n}

(d)

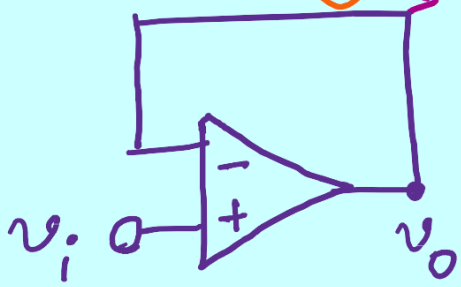


$A_{CL} = 1 + \frac{R_2}{R_1}$

$R_2 = 0\,\Omega$ (S.C.)

$R_1 = \infty\,\Omega$ (O.C.)

Both



Voltage follower / Buffer

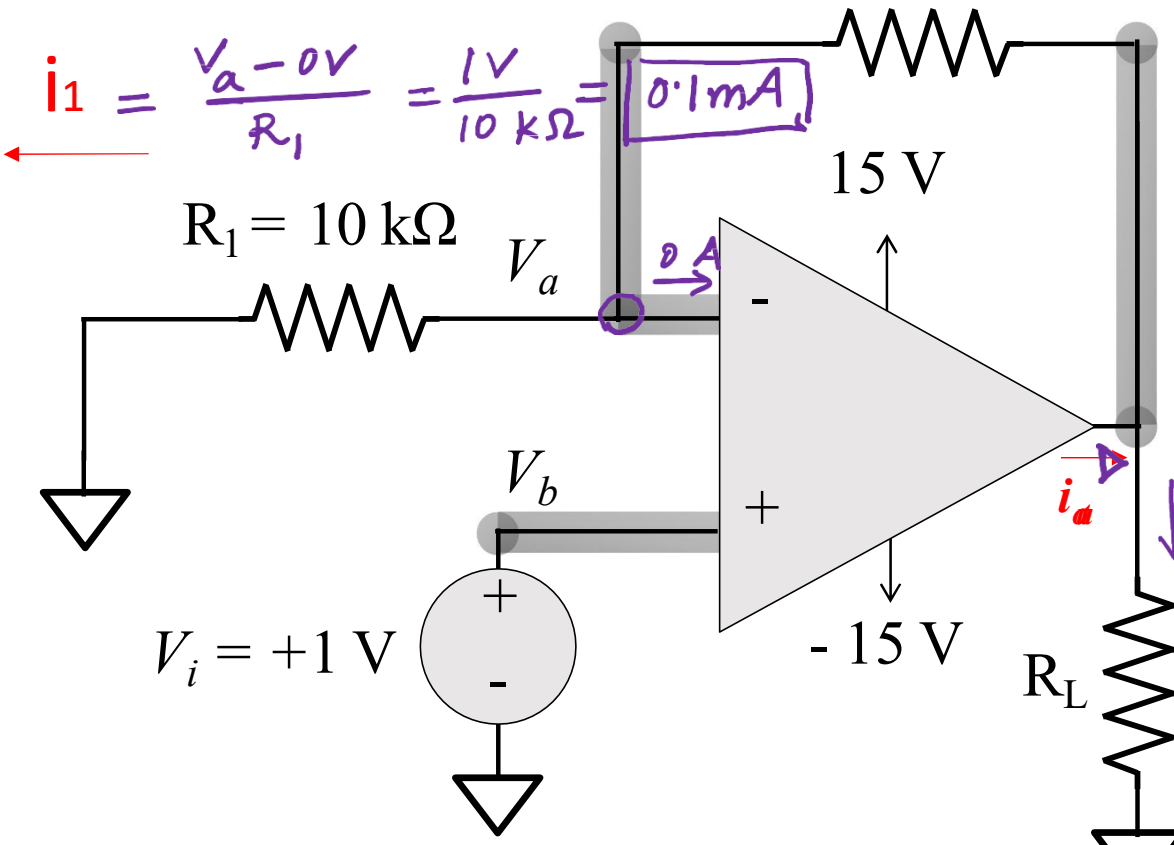
Problem 14. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions. Calculate the A_{CL}, i_1, i_2, V_o, i_L, and i_{out}.

$A_{CL} = 1 + \frac{R_2}{R_1} = 5$

$i_2 = i_1 = 0.1\text{ mA} = \frac{V_o - V_a}{R_2}$

$R_2 = 40\text{ k}\Omega = \frac{5\text{V} - 1\text{V}}{40\text{ k}\Omega} = \frac{4\text{V}}{40\text{ k}\Omega} = 0.1\text{ mA}$

$i_1 = \frac{V_a - 0\text{V}}{R_1} = \frac{1\text{V}}{10\text{ k}\Omega} = 0.1\text{mA}$



$V_o = +5\text{ V}$

$= i_2 + i_L$

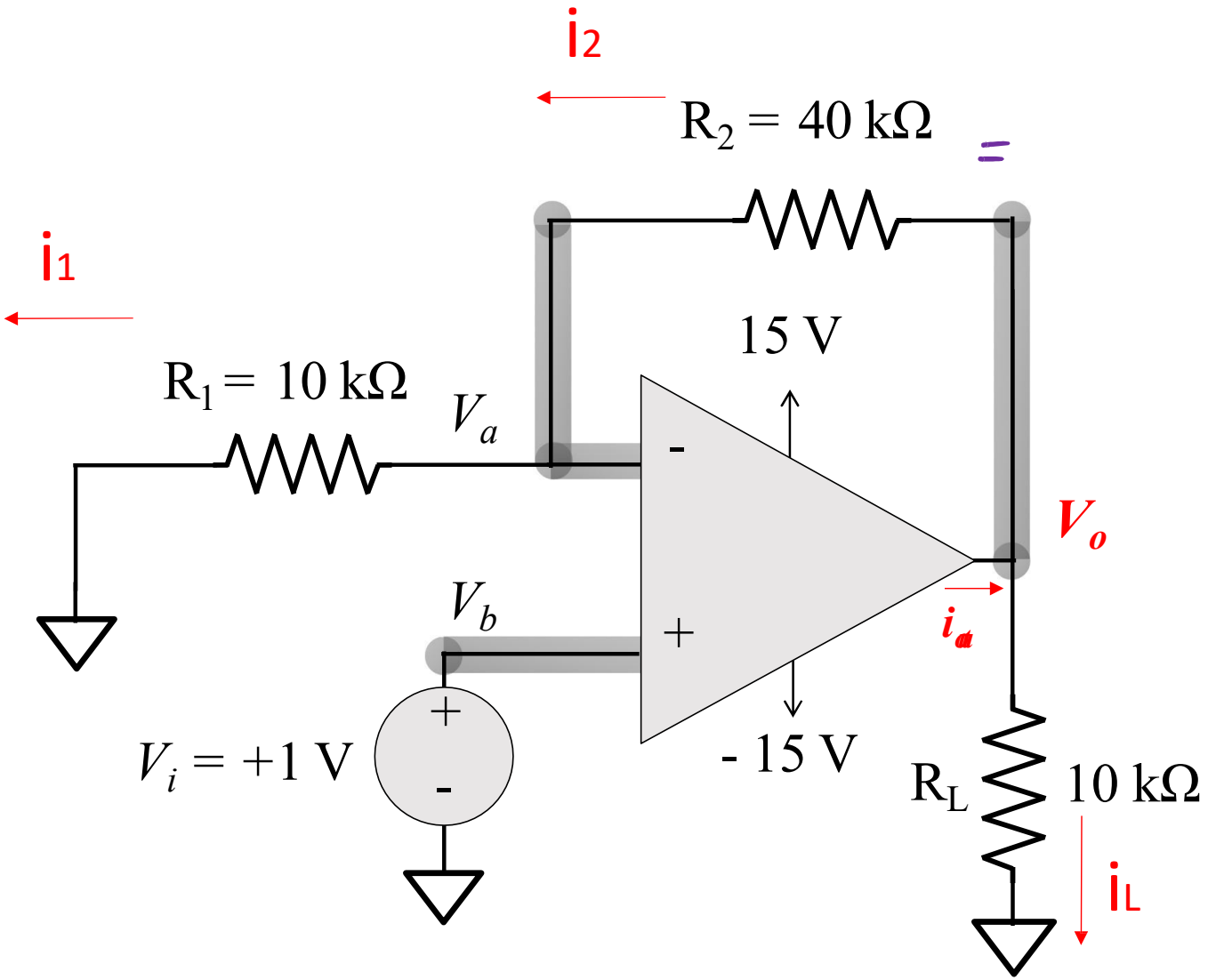
$= 0.6\text{ mA}$

R_L 10 kΩ

$i_L = \frac{V_o}{R_L} = \frac{5\text{V}}{10\text{ k}\Omega}$

$= 0.5\text{ mA}$

Problem 14. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions. Calculate the A_{CL}, i_1, i_2, V_o, i_L, and i_{out}.

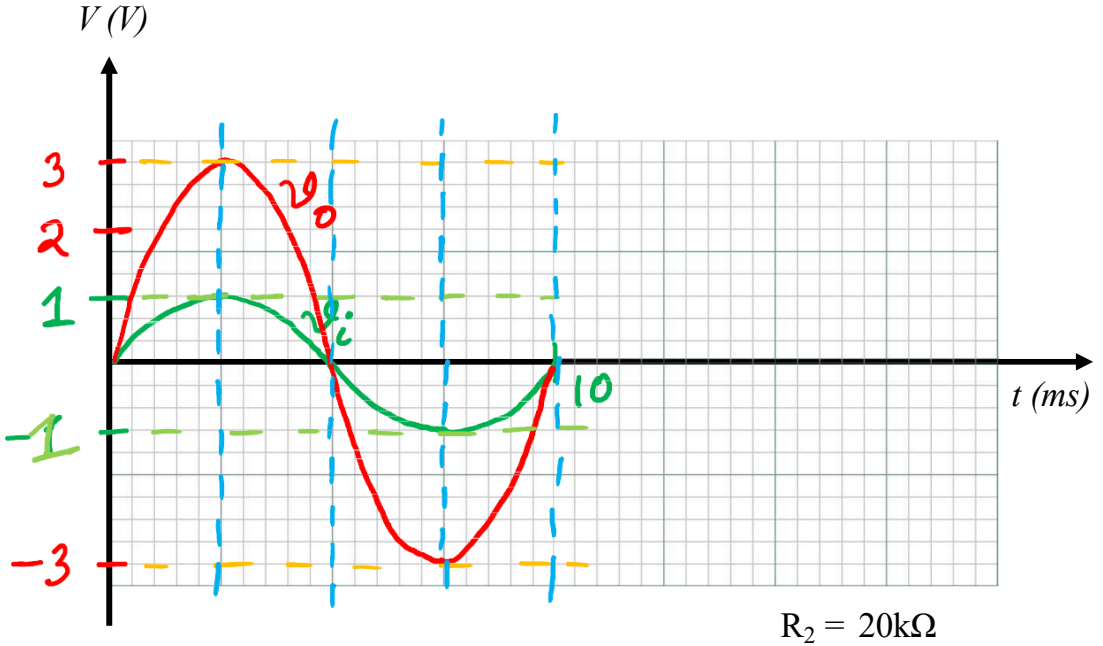


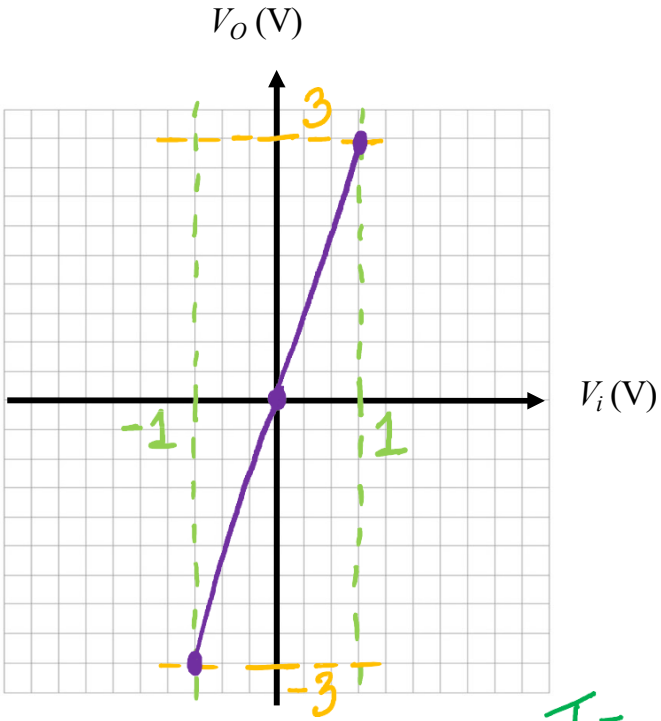
Problem 15. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions. Calculate the A_{CL}. Draw the V_i and V_o. Draw the transfer characteristics.

$A_{CL} = 1 + \frac{R_2}{R_1} = 1 + 2 = 3$

Non-inv

+ve slope



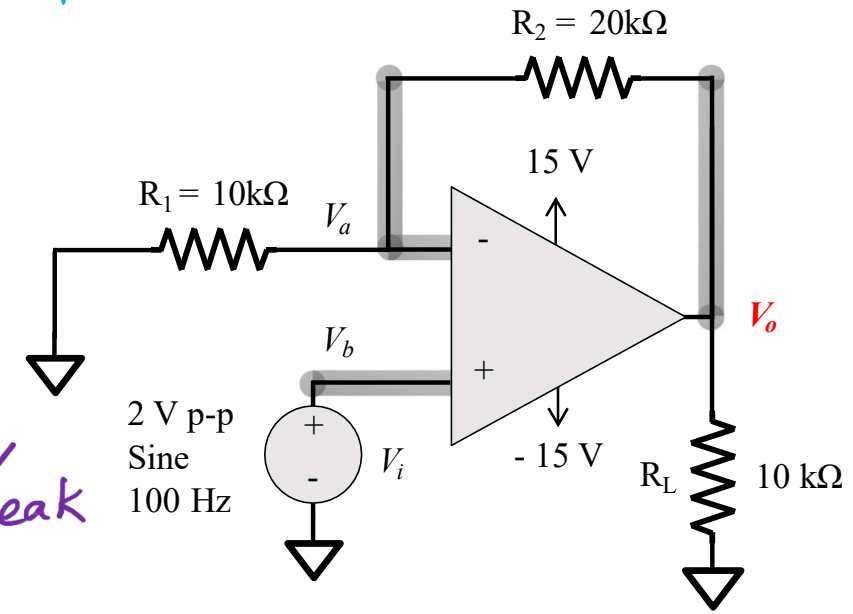


$T = \frac{1}{10}$ s

$= 0.01$ s

10 ms

1V peak



Problem 16. Design an amplifier with a gain of +25. $A_{OL} = 2\times10^5$. $V_{sat+} = +13$ V, $V_{sat-} = -13$ V. Assume ideal conditions.

Coughlin Example 3-14

Practice Problem: Design a three-stage amplifier with gain +10, -18 and -27. Use a 270kΩ feedback resistor for all three stages. What will be the output voltage if input voltage is 150μV?

For the gain of +10,

$$A_1 = 1 + \frac{R_f}{R_1} = +10$$

$$\frac{R_f}{R_1} = 10 - 1 = 9$$

$$R_1 = \frac{R_f}{9} = \frac{270\,\text{k}\Omega}{9} = 30\,\text{k}\Omega$$

For the gain of −18,

$$A_2 = -\frac{R_f}{R_2} = -18$$

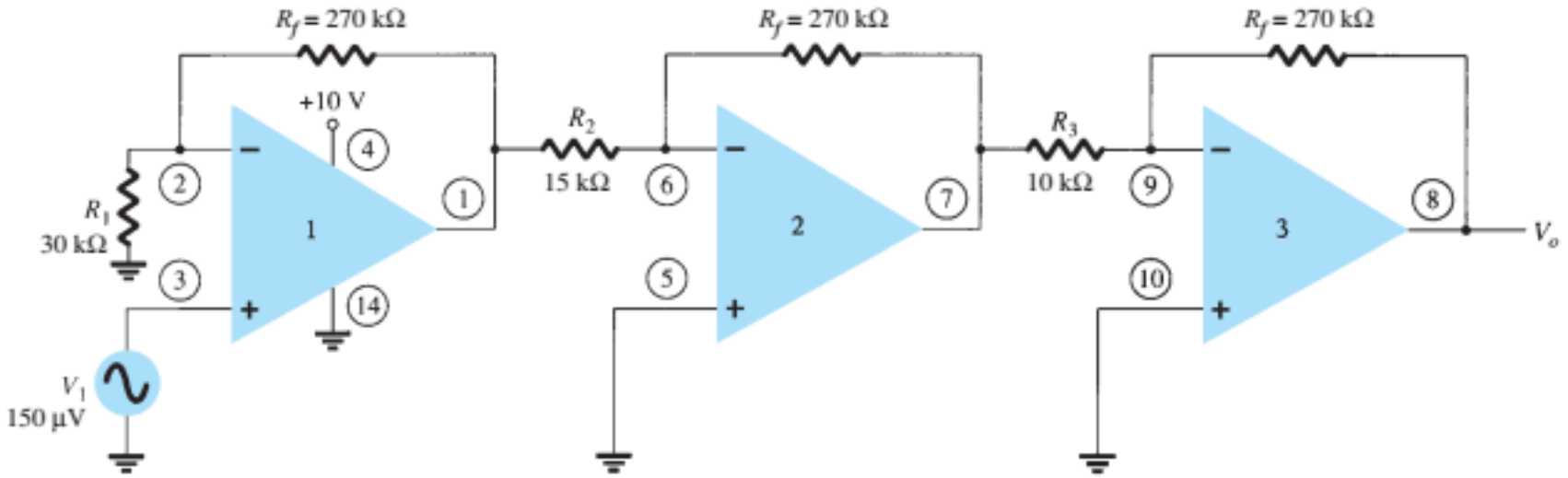
$$R_2 = \frac{R_f}{18} = \frac{270\,\text{k}\Omega}{18} = 15\,\text{k}\Omega$$

For the gain of −27,

$$A_3 = -\frac{R_f}{R_3} = -27$$

$$R_3 = \frac{R_f}{27} = \frac{270\,\text{k}\Omega}{27} = 10\,\text{k}\Omega$$

$$V_o = A_1A_2A_3V_1 = (10)(-18)(-27)(150\ \mu\text{V}) = 4860(150\ \mu\text{V})$$
$$= \mathbf{0.729\ V}$$



Analog Electronics II: Op-amp & 555 Timer

Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

Lecture # 6

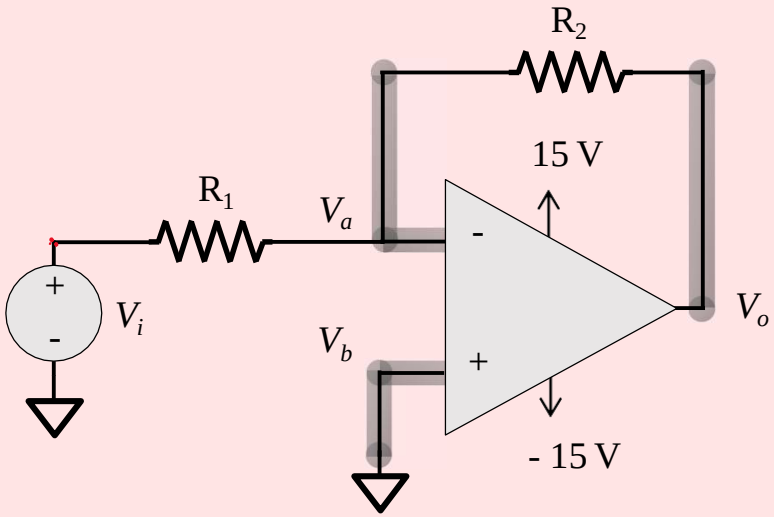
Inverting & non-inverting configurations

Inverting configuration

$$A_{CL} = -\frac{R_2}{R_1}$$

Input impedance = R_1

Output impedance = 0

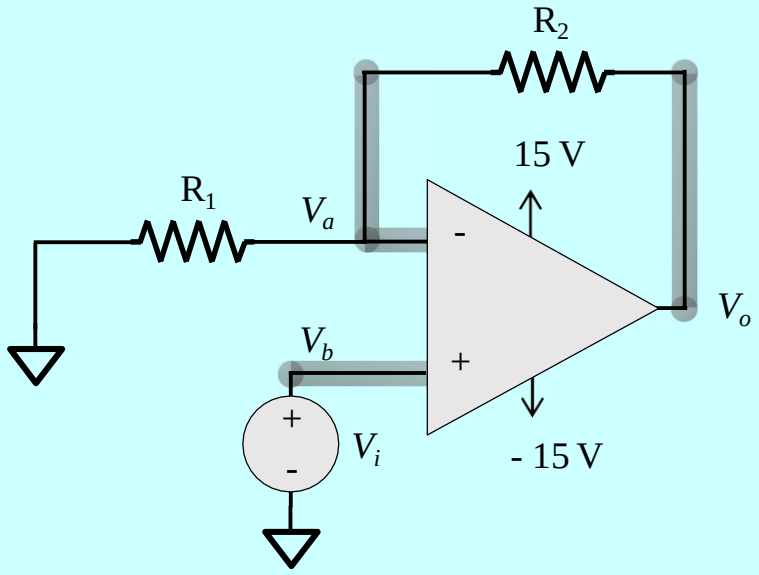


Non-inverting configuration

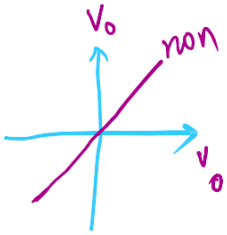
$$A_{CL} = 1 + \frac{R_2}{R_1}$$

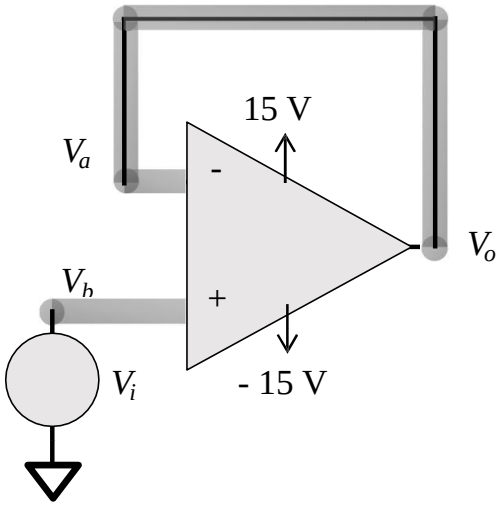
Input impedance = ∞

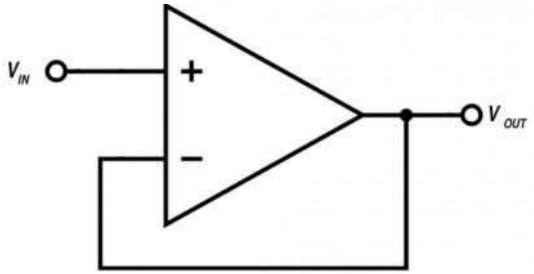
Output impedance = 0



Voltage follower – circuit diagrams



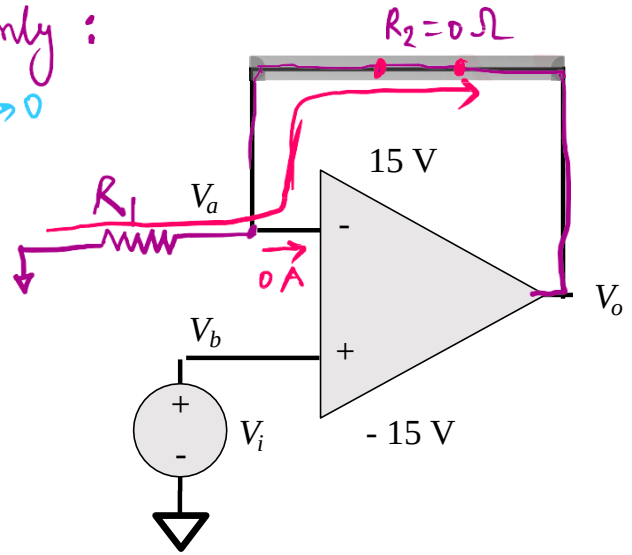




• $R_2 = 0\,\Omega$ only :

$A_{CL} = 1 + \frac{R_2}{R_1} \to 0$

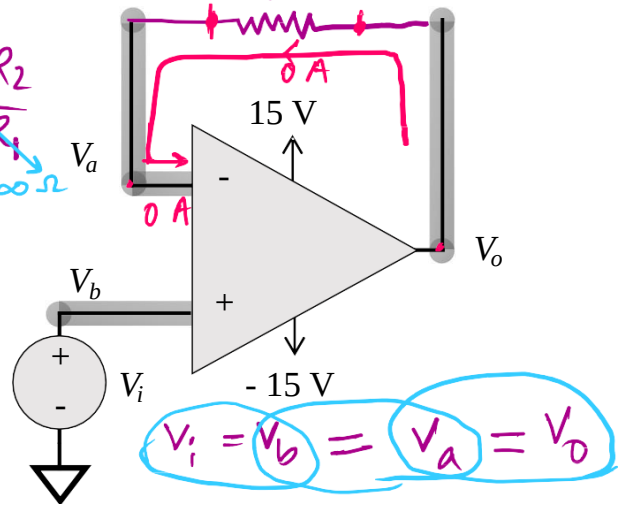
$= 1$



• $R_1 = \infty\,\Omega$ only : + 0 V −

$A_{CL} = 1 + \frac{R_2}{R_1} \to \infty\,\Omega$

$= 1$



ALL ABOUT CIRCUITS

Voltage Follower

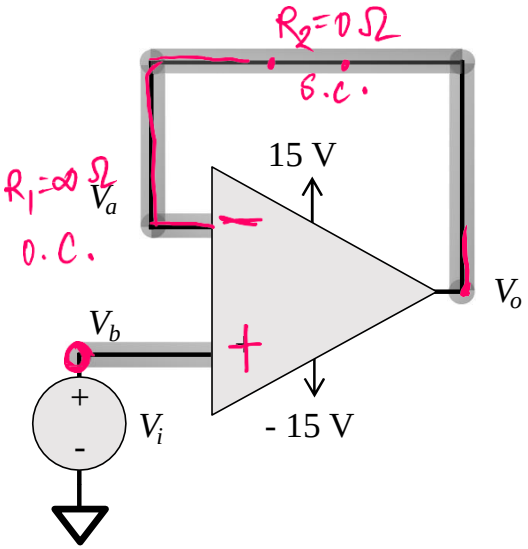
- -ve feedback
- Non-inverting configuration
- Unity gain

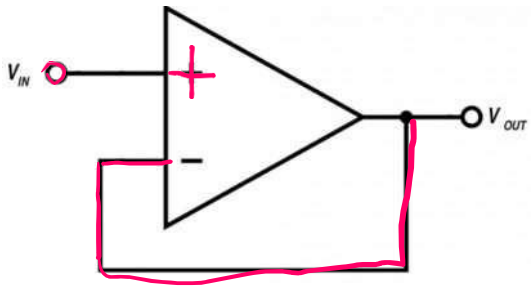
- Also called: **Unity-gain buffer** or **noninverting buffer.**
- The only necessary component is the op-amp itself.
- A voltage follower produces an output signal that is equal in amplitude to the input signal.
- Since the input signal is applied to the noninverting input terminal, no inversion takes place.

$A_{CL} = 1$ (keep)

$\frac{R_2}{R_1}$ • 0 Ω : S.C. only • Both (most common) • ∞ Ω : O.C. only

• both $R_2 = 0\,\Omega$ & $R_1 = \infty\,\Omega$



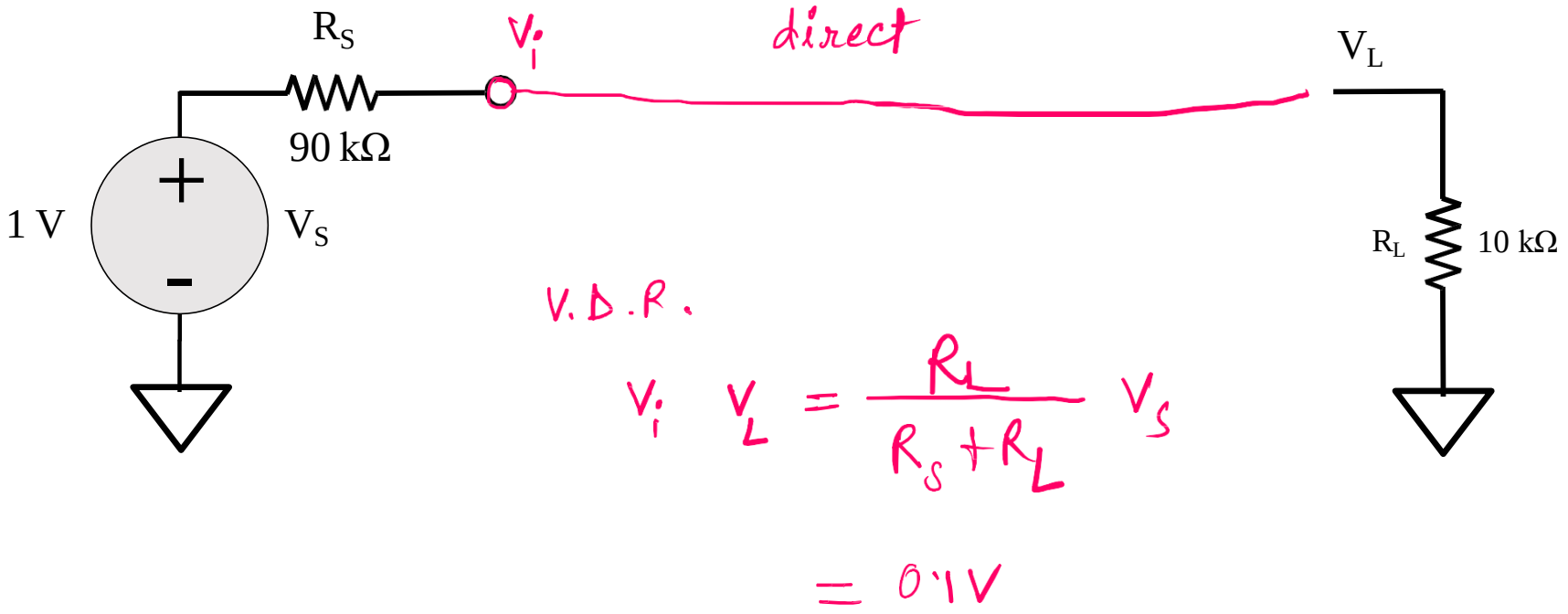


Source: ALL ABOUT CIRCUITS

Voltage follower -applications

- Effective voltage transfer requires a source circuit with low output impedance and a load circuit with high input impedance.

Model of a whole bunch ckt element



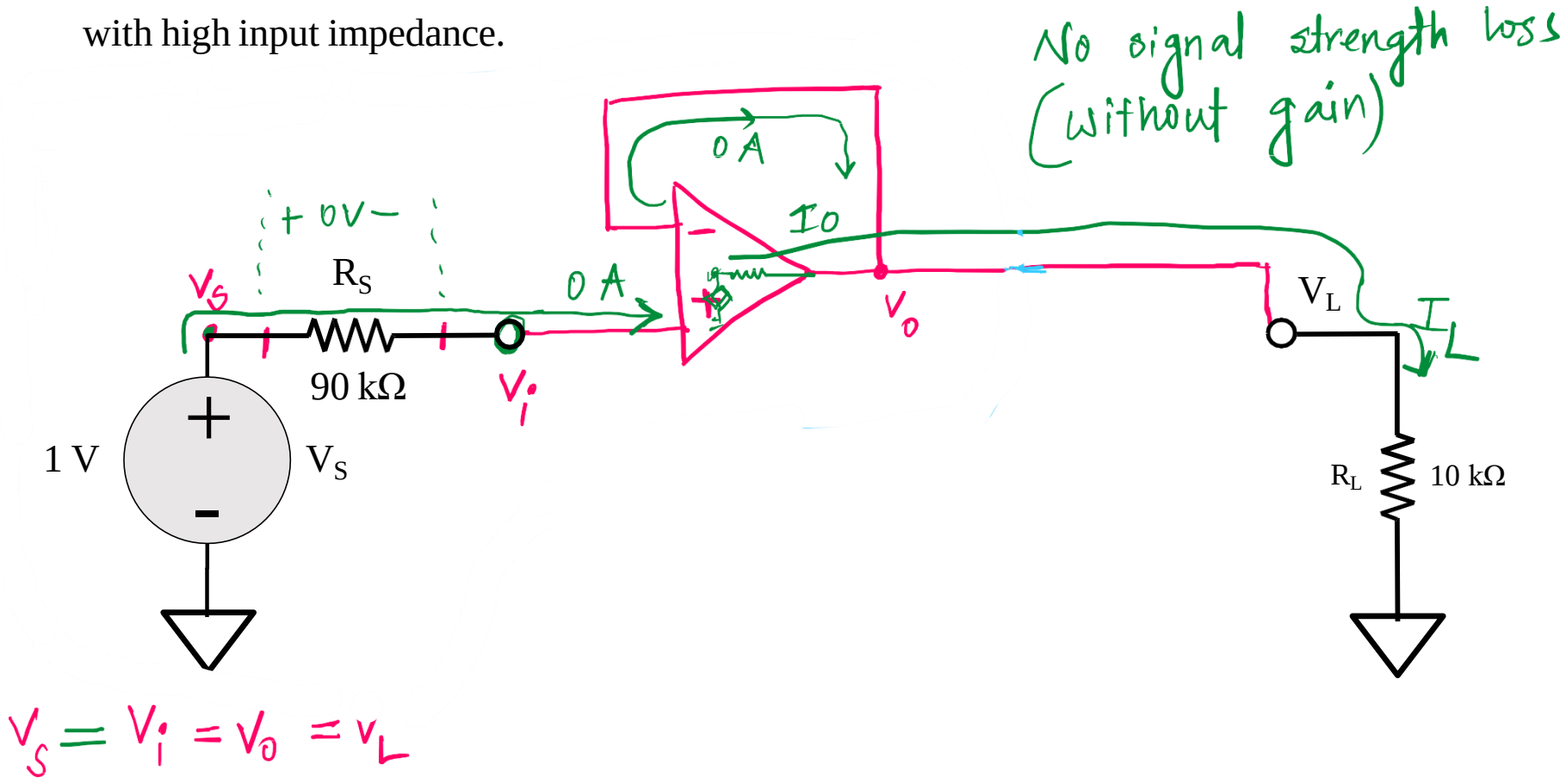
V.D.R.

$$V_i \quad V_L = \frac{R_L}{R_S + R_L} V_S$$

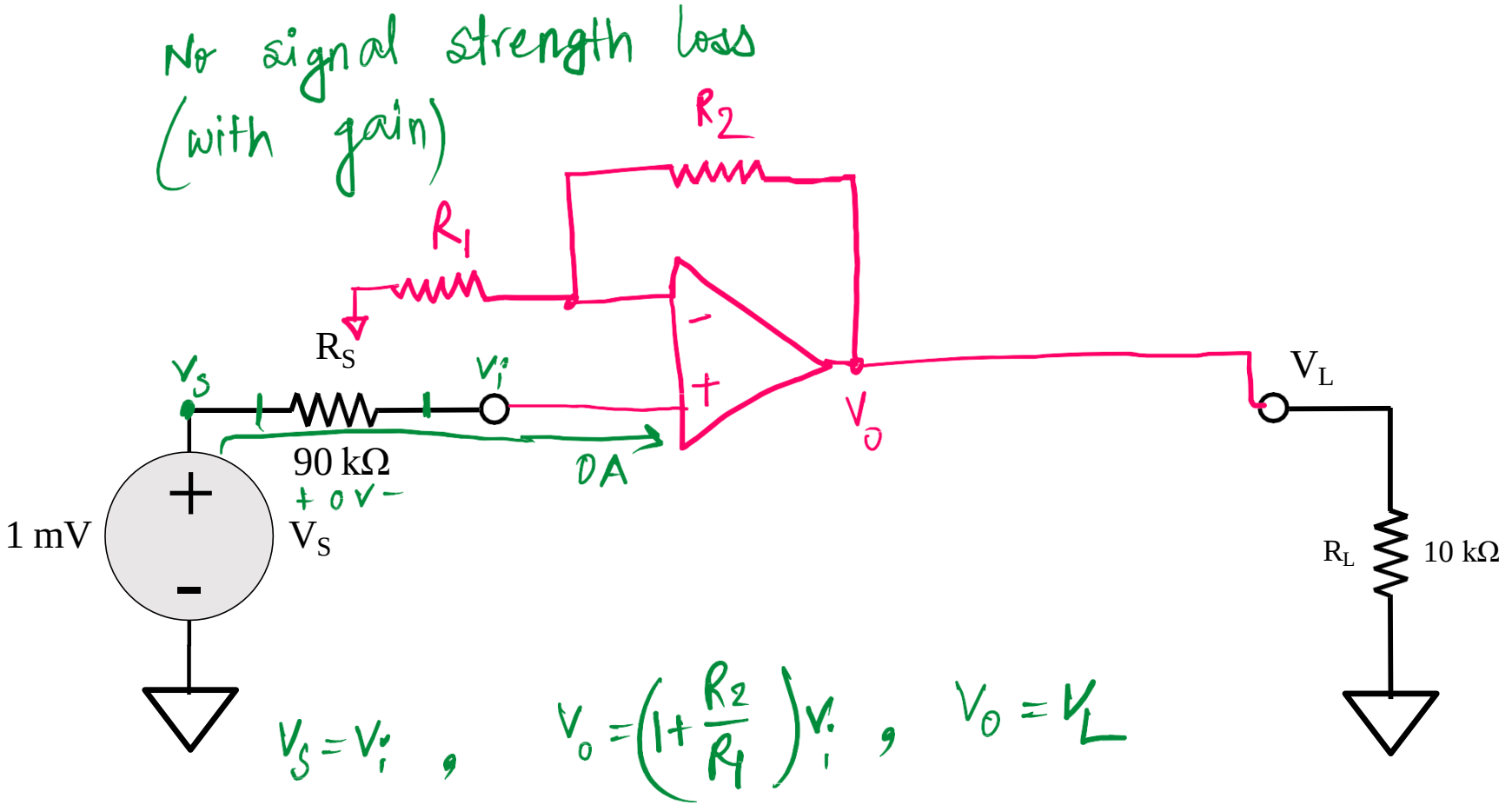
$$= 0.1V$$

Voltage follower -applications

- Effective voltage transfer requires a source circuit with low output impedance and a load circuit with high input impedance.

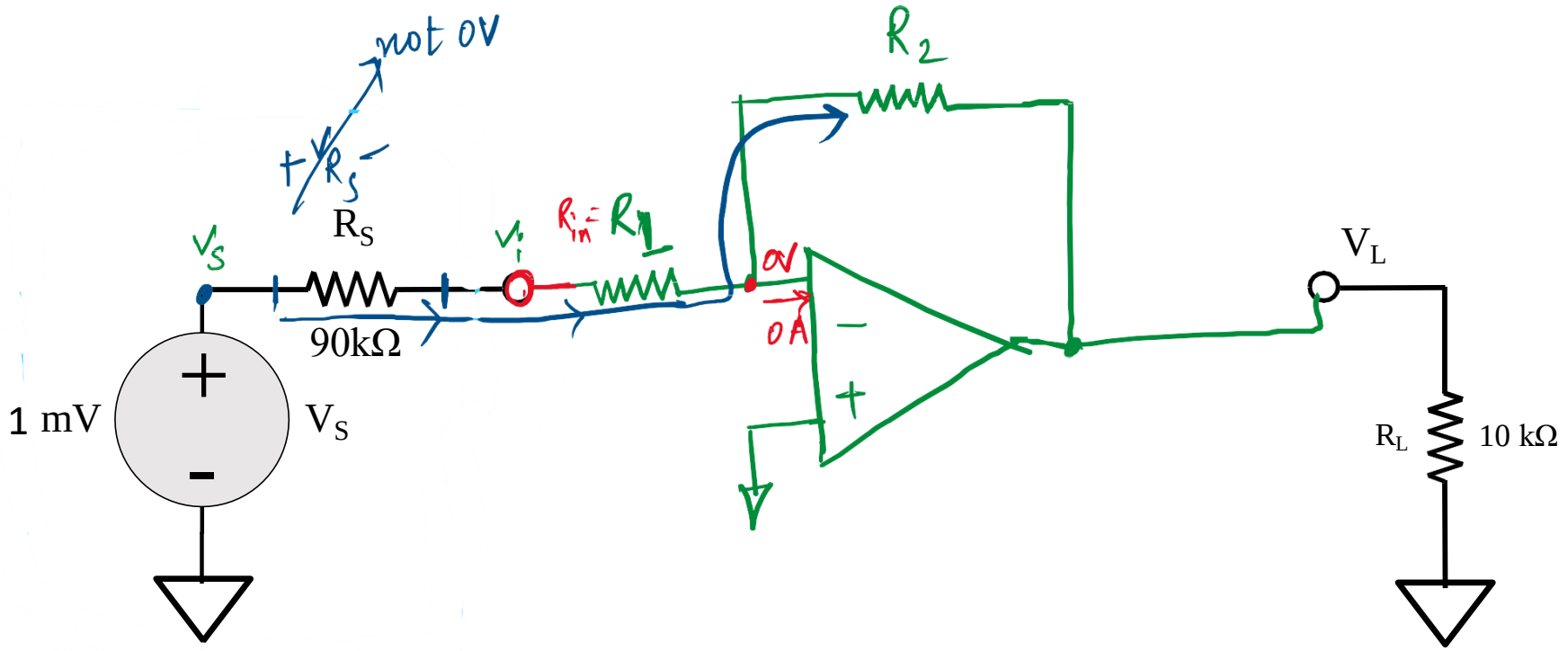


Non-inverting configuration - advantage



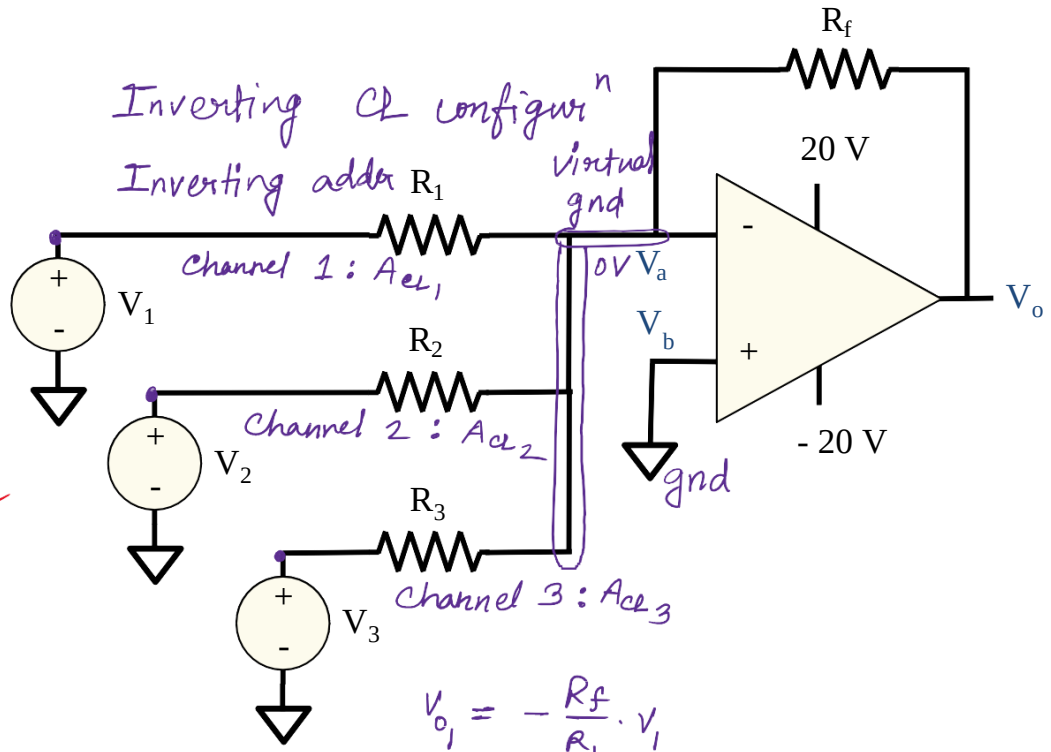
Inverting configuration - disadvantage

'inv. amp. ⇒ signal strength loss



Weighted Adder

Also called Multi-channel amplifier.



$$v_o = A_{CL_1} v_1 + A_{CL_2} \cdot v_2 + A_{CL_3} \cdot v_3$$

$$v_o = \left(-\frac{R_f}{R_1}\right) v_1 + \left(-\frac{R_f}{R_2}\right) v_2 + \left(-\frac{R_f}{R_3}\right) v_3$$

Memorize

$$A_{CL_1} = -\frac{R_f}{R_1}$$

$$A_{CL_2} = -\frac{R_f}{R_2}$$

$$A_{CL_3} = -\frac{R_f}{R_3}$$

$$V_{o_1} = -\frac{R_f}{R_1} \cdot V_1$$

$$V_{o_2} = -\frac{R_f}{R_2} \cdot V_2$$

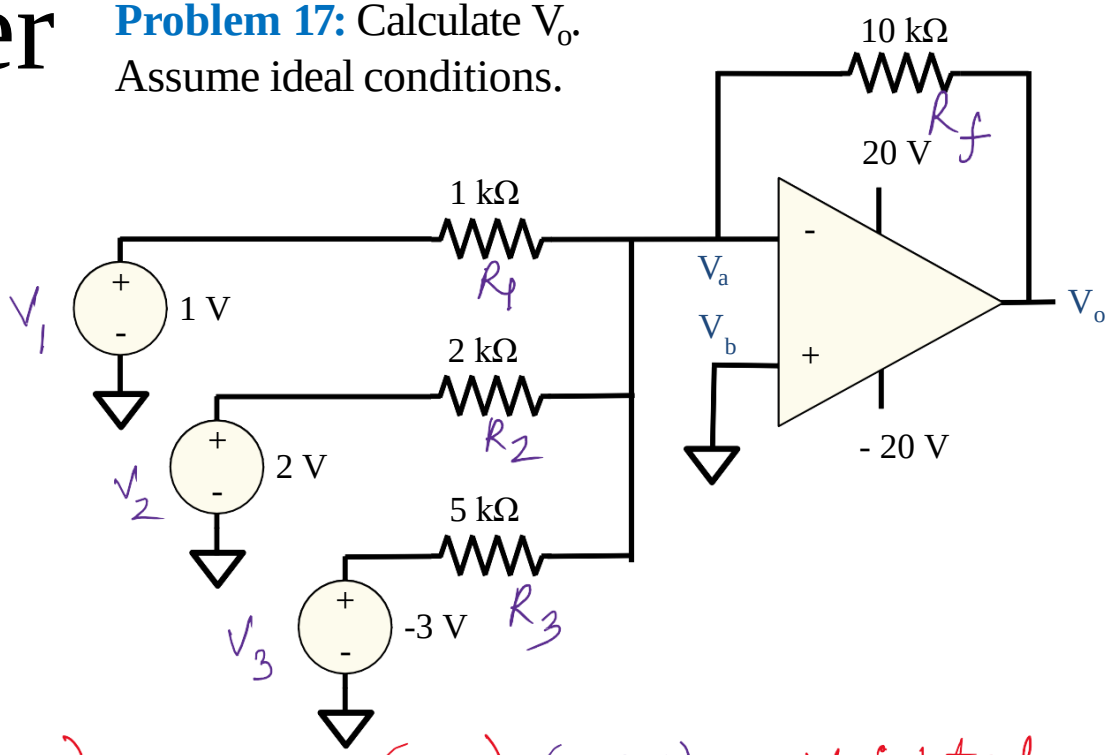
$$V_{o_3} = -\frac{R_f}{R_3} \cdot V_3$$

4 unknown variables, 3 eqn

assume 1 variable → most likely R_f

Weighted Adder

Problem 17: Calculate V_o. Assume ideal conditions.



$$v_o = \left(-\frac{10\,k\Omega}{1\,k\Omega}\right) \times 1\,V +$$

$$\left(-\frac{10\,k\Omega}{2\,k\Omega}\right) \times 2V +$$

$$\left(-\frac{10\,k\Omega}{5\,k\Omega}\right) \times (-3V)$$

$$v_o = (-10) \cdot 1V + (-5) \cdot 2V + (-2)(-3V)$$

Weighted Addition

$$= -14\,V$$

Source: Coughlin 3-2.1

Weighted Adder

Problem 18a: Design an amplifier so that the output v_o is related with the inputs v_1, v_2, v_3 in the following way:

$v_o = -(10v_1 + 5v_2 + 2v_3)$

Assume ideal conditions.

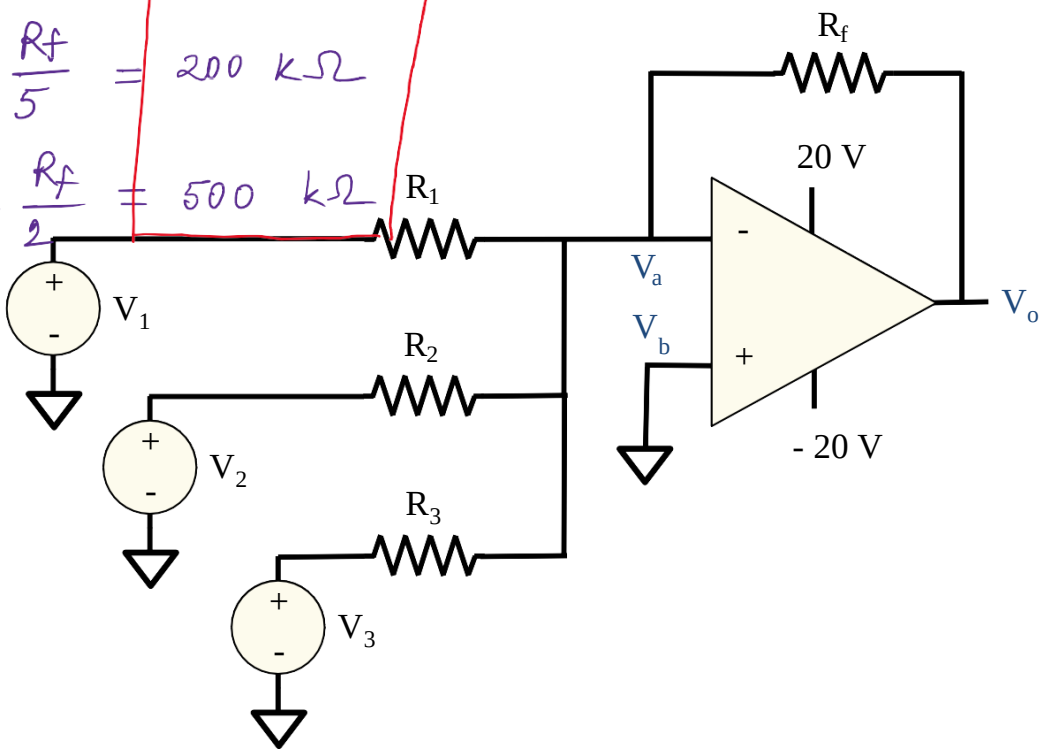
$R_f = 1\,M\Omega$ (assume)

$v_o = (-10)v_1 + (-5)v_2 + (-2)v_3$

$A_{CL_1} = -10 = -\frac{R_f}{R_1} \Rightarrow R_1 = \frac{R_f}{10} = 100\,k\Omega$

$A_{CL_2} = -5 = -\frac{R_f}{R_2} \Rightarrow R_2 = \frac{R_f}{5} = 200\,k\Omega$

$A_{CL_3} = -2 = -\frac{R_f}{R_3} \Rightarrow R_3 = \frac{R_f}{2} = 500\,k\Omega$



Design Steps

1. Identify the type of ckt from the given transfer eq^n.

$v_0 = -10v_1 - 7v_2$ — inverting adder

$v_0 = +10v_1 + 5v_2$ — non-inverting adder

$v_0 = -10v_{in}$ — inverting CL amp

$v_0 = +10v_{in}$ — noninv CL amp

$v_0 = +15V$ or $-15V$ — rectangular

saturating ckt : open-loop configur^n

2. Draw the ckt.
3. Label all resistors, voltages, currents that you are using in the calculation.
4. Write the relavant eq^n. Assume, one or more variables. Calculate rest of the variable.

Weighted Adder

Problem 18b: Design an amplifier so that the output v_o is related with the inputs v_1, v_2, v_3 in the following way:
$v_o = -(7v_1 + 5v_2 + 3v_3)$
Assume ideal conditions.

Use the method of the previous solution.

For the assumption of R_f in these problems,
you can take the largest multiple of LCM $\leq 10^6$

LCM: $7 \times 5 \times 3 = 105$

You can assume $R_f = 105 \times 9\ k\Omega = 945\ k\Omega$
to stay within range

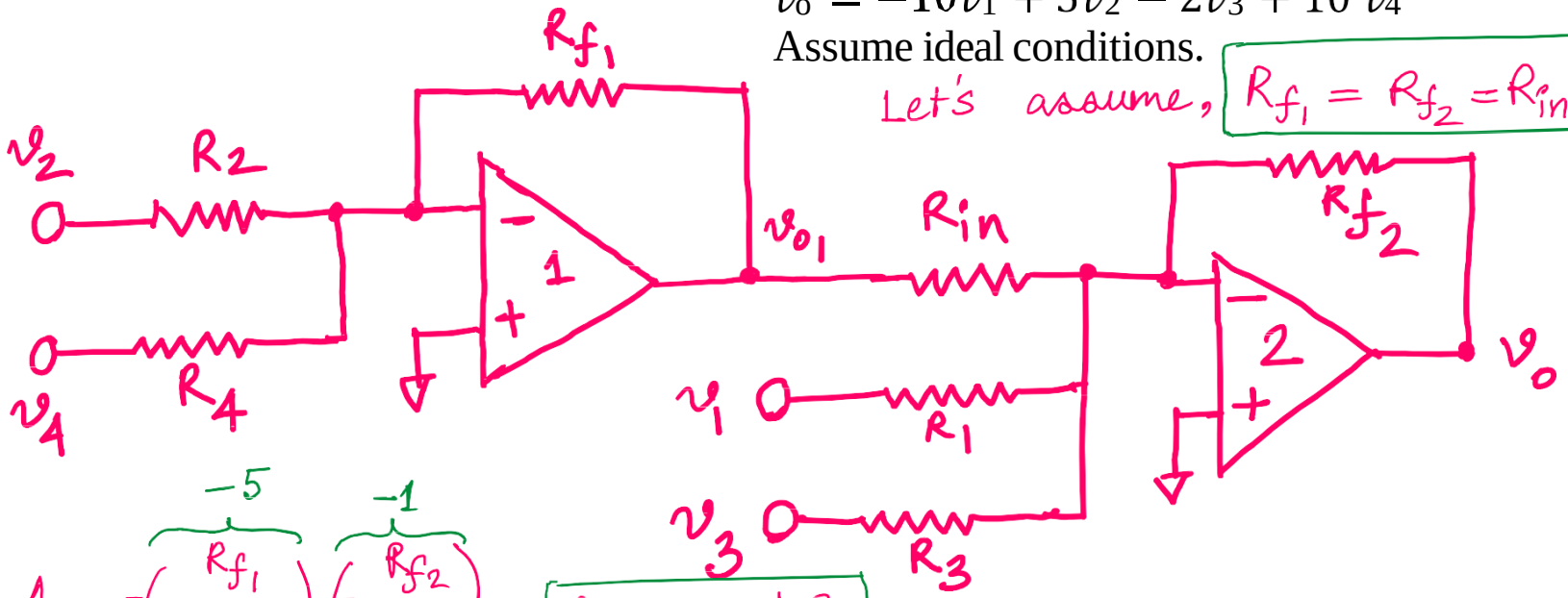
Weighted Adder

Problem 18c: Design an amplifier so that the output v_o is related with the inputs v_1, v_2, v_3 in the following way:

$v_o = -10v_1 + 5v_2 - 2v_3 + 10\ v_4$

Assume ideal conditions.

Let's assume, $R_{f_1} = R_{f_2} = R_{in} = 1\ M\Omega$



$A_{CL_2} = \left(-\frac{R_{f_1}}{R_2}\right)\left(-\frac{R_{f_2}}{R_{in}}\right)$ (-5, -1) $\Rightarrow$ $R_2 = 200\ k\Omega$

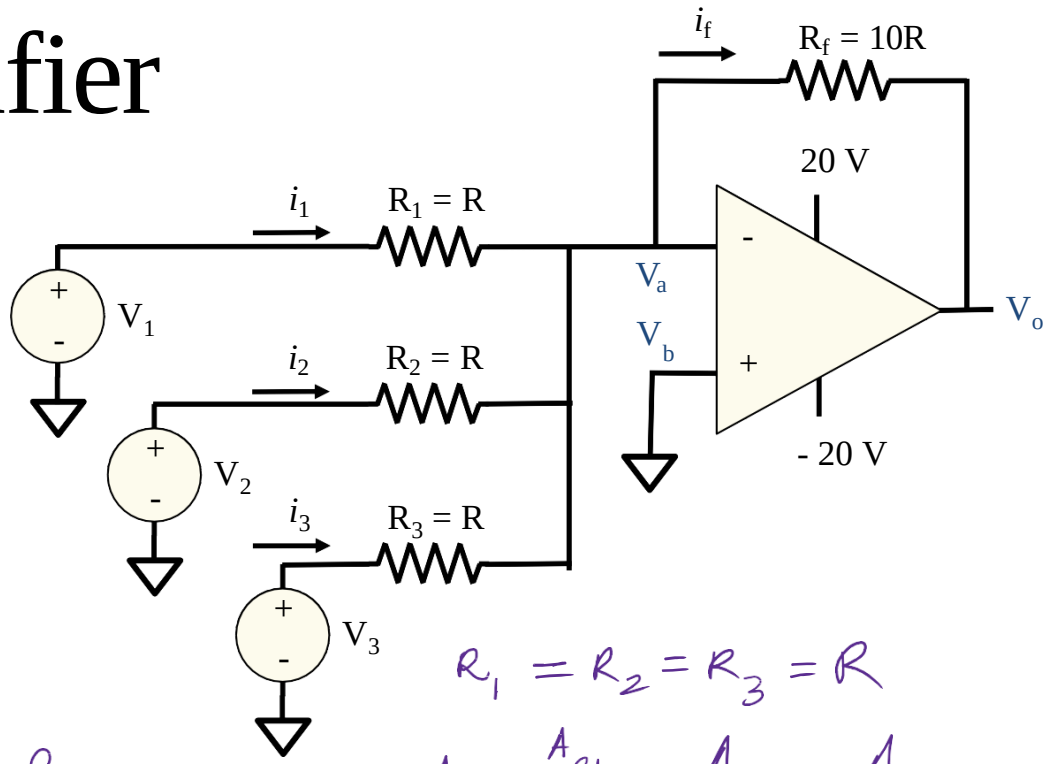
$A_{CL_4} = \left(-\frac{R_{f_1}}{R_4}\right)\left(-\frac{R_{f_2}}{R_{in}}\right)$ (-10, -1) $\Rightarrow$ $R_4 = 100\ k\Omega$

$A_{CL_1} = -\frac{R_{f_2}}{R_1} = -10 \Rightarrow R_1 = 100\ k\Omega$

$A_{CL_3} = -\frac{R_{f_2}}{R_3} = -2 \Rightarrow R_3 = 500\ k\Omega$

Summing Amplifier

Problem 19: Calculate V_o as a function of V_1, V_2, and V_3. Assume ideal conditions.



$$v_o = -\frac{R_f}{R_1}V_1 - \frac{R_f}{R_2}V_2 - \frac{R_f}{R_3}V_3$$

$$R_1 = R_2 = R_3 = R$$

$$A_{CL} = A_{CL_1} = A_{CL_2} = A_{CL_3}$$

$$v_o = -\frac{R_f}{R}V_1 - \frac{R_f}{R}V_2 - \frac{R_f}{R}V_3$$

$$= -\frac{R_f}{R}(V_1 + V_2 + V_3)$$

$$= \underbrace{A_{CL}}_{\text{amp}} \underbrace{(V_1 + V_2 + V_3)}_{\text{add/sum}} = -10(V_1 + V_2 + V_3)$$

Inverting Adder (direct addition)

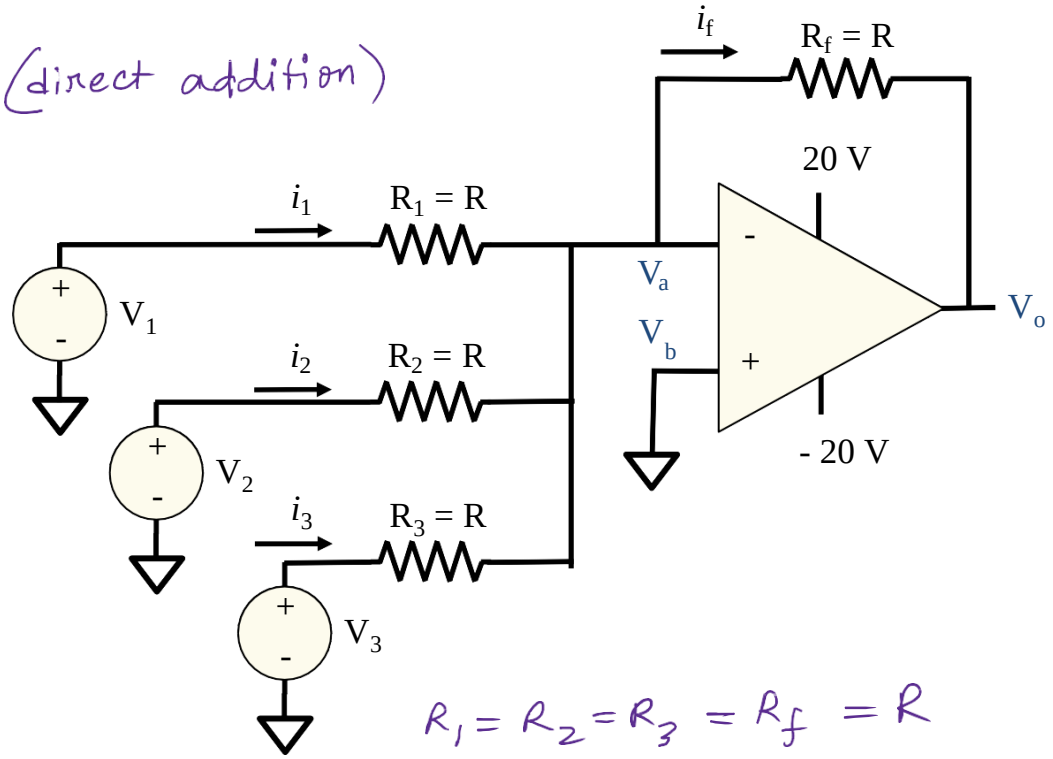
$$v_o = -\frac{R_f}{R_1} v_1 - \frac{R_f}{R_2} v_2 - \frac{R_f}{R_3} v_3$$

$$= -\frac{R}{R} v_1 - \frac{R}{R} v_2 - \frac{R}{R} v_3$$

$$= -v_1 - v_2 - v_3$$

$$= -(v_1 + v_2 + v_3)$$

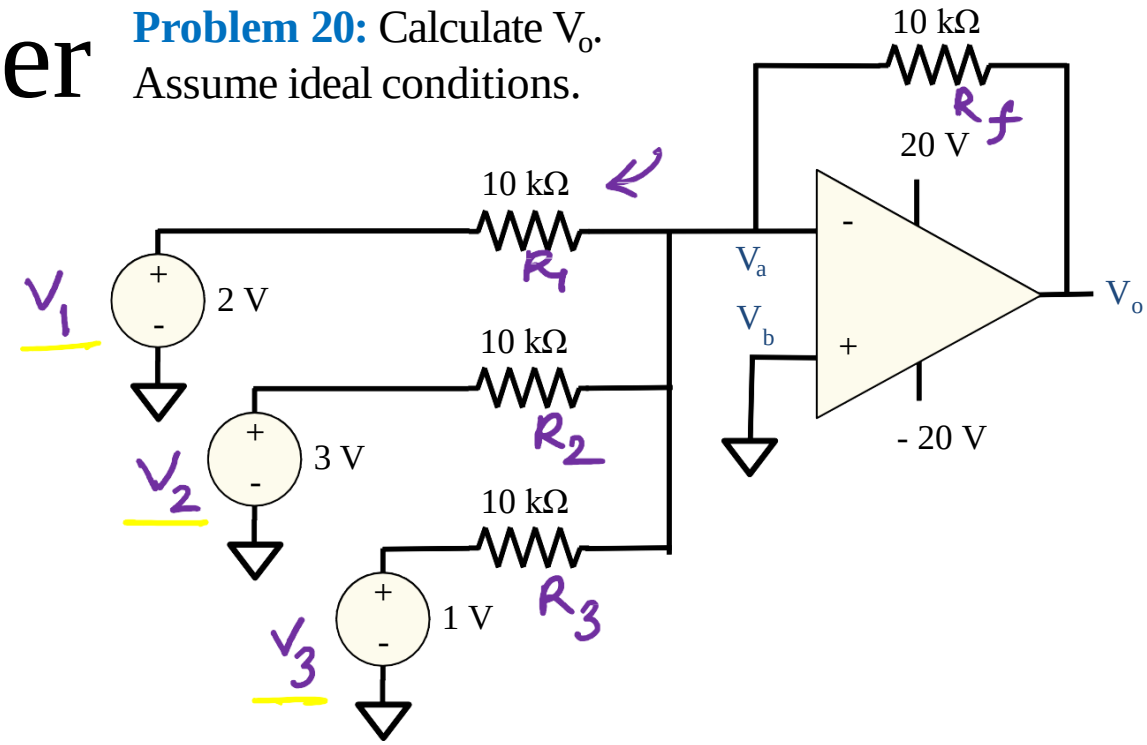
direct addition



$R_1 = R_2 = R_3 = R_f = R$

Inverting Adder

Problem 20: Calculate V_o. Assume ideal conditions.



$$V_o = -(V_1 + V_2 + V_3)$$

$$= -(2+3+1)\ V$$

$$= -6\ V$$

Analog Electronics II: Op-amp & 555 Timer

Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

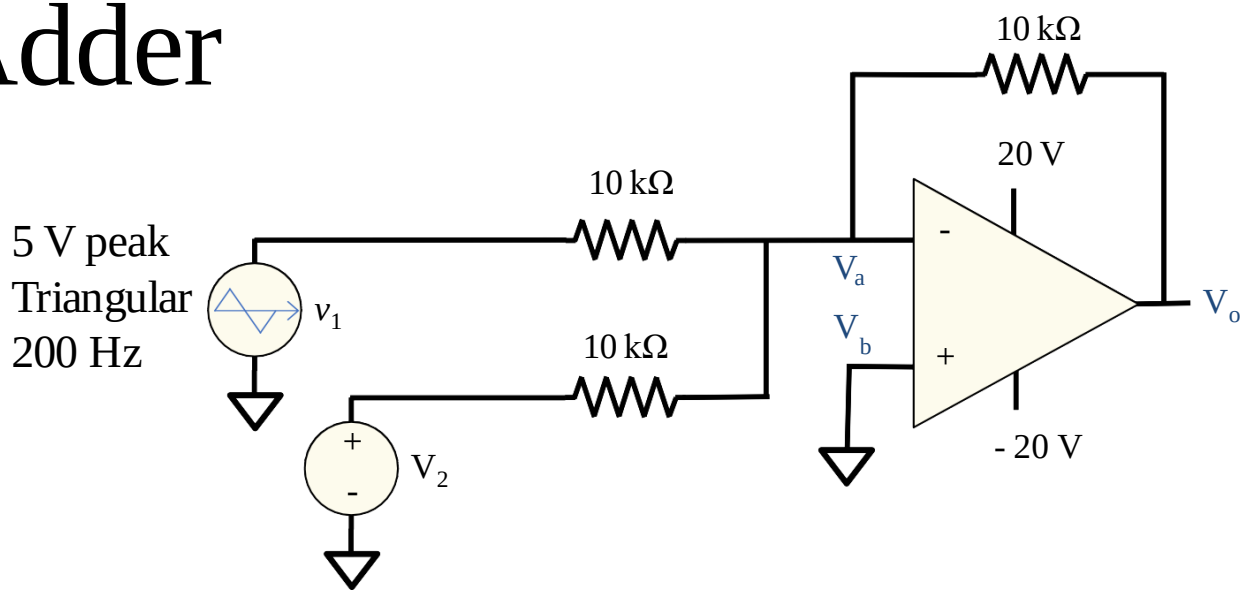
Lecture # 7

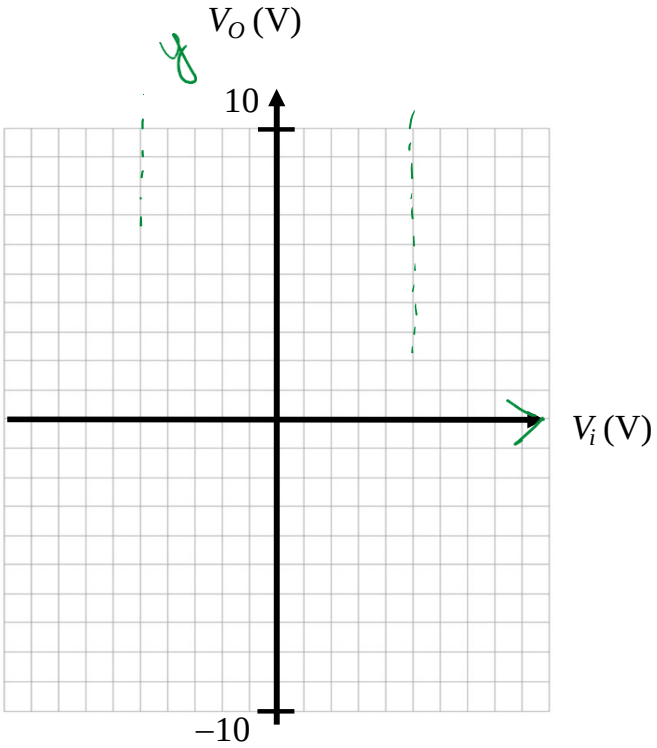
DC Offset Adder

Problem 21. Draw the output waveform and transfer characteristics for

i. $V_2 = +5$ V and

ii. $V_2 = -5$ V





Coughlin 3-2.3

Subtractor

$v_o = 5v_1 - 2v_2$ ⇒ we learn this with inv amp. & inv adder

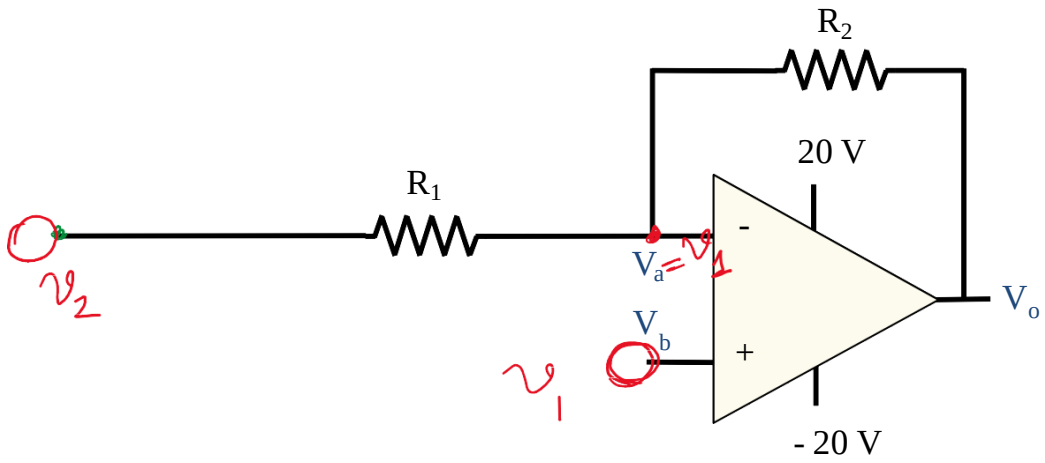
i) weighted subtrac^n:

ii) Difference amplifier:

$v_o = 5(v_1 - v_2)$

iii) Direct subtrac^n: (with unity gain)

$v_o = v_1 - v_2$



Mingle the inv amp. & non-inv amp. ?

Superpos^n:

v_1 active alone:

non-inv. amp.

$v_{o_1} = \left(1 + \frac{R_2}{R_1}\right) v_1$

* v_2 active alone

inv amp.

$v_{o_2} = -\frac{R_2}{R_1} \cdot v_2$

$v_o = v_{o_1} + v_{o_2}$

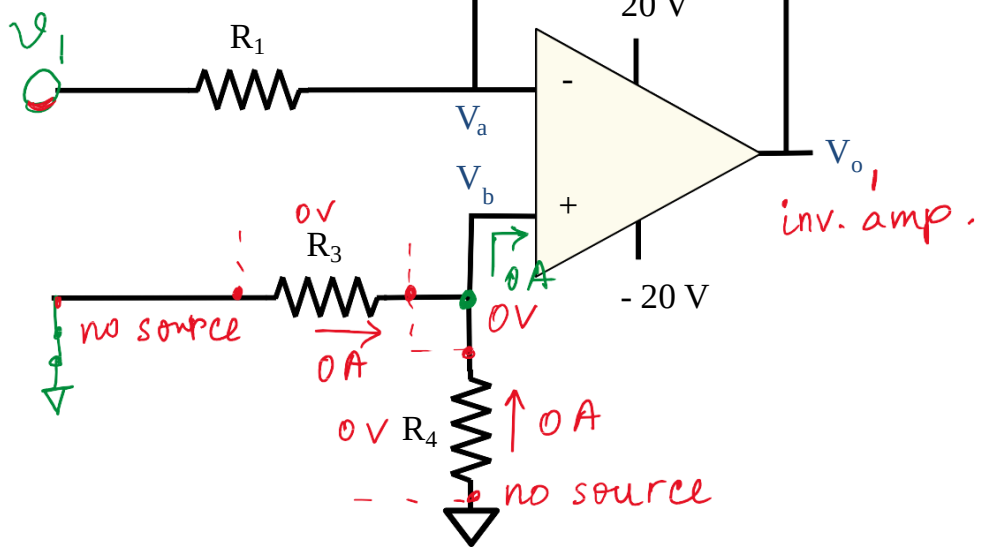
$= \left(1 + \frac{R_2}{R_1}\right) v_1 + \left(-\frac{R_2}{R_1}\right) v_2$

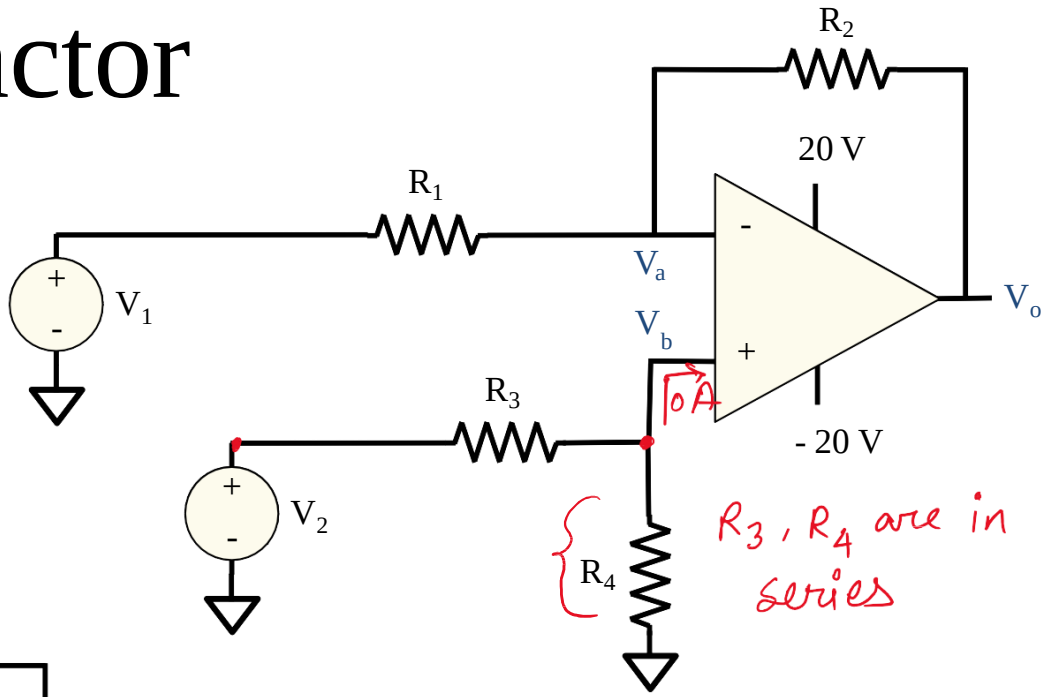
Source: Coughlin 3-2.3

Weighted Subtractor

Superposition:

* v_1 active alone:

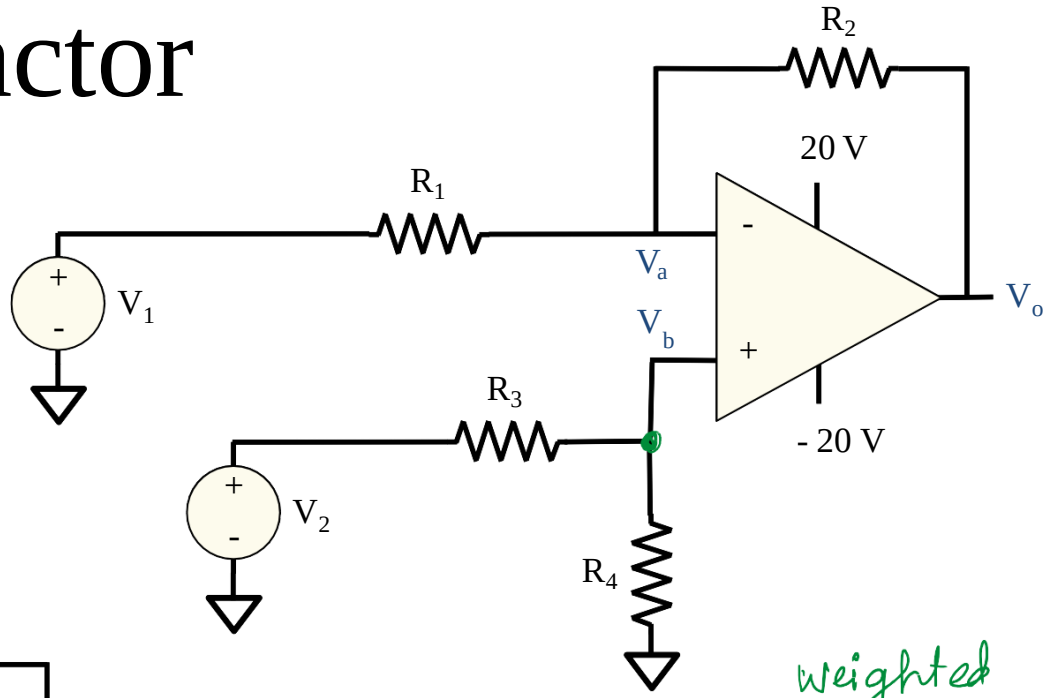




$$v_{o_1} = -\frac{R_2}{R_1} \cdot v_1$$

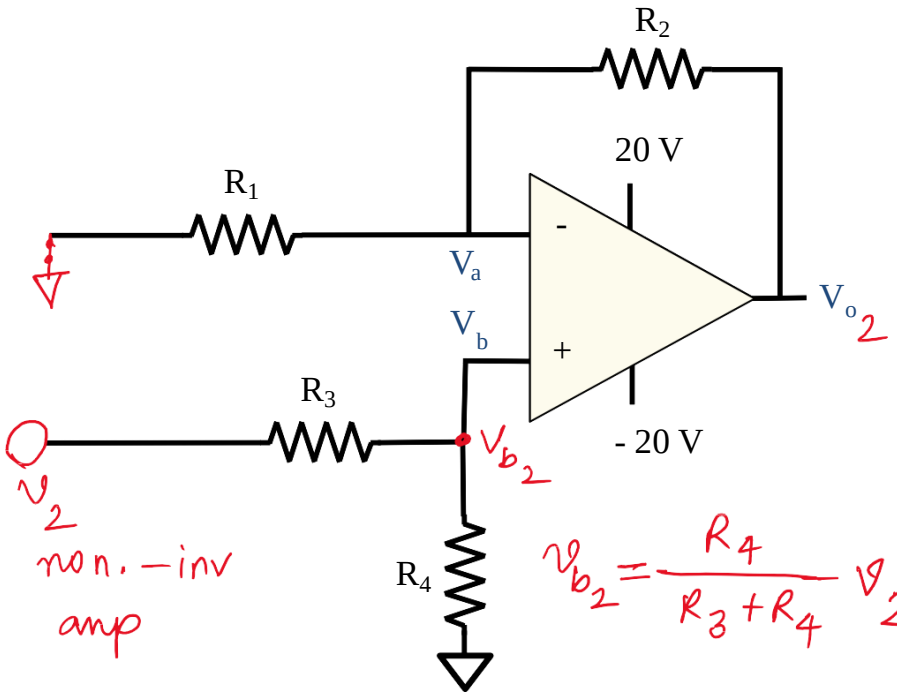
Source: Coughlin 3-2.3

Weighted Subtractor



$v_{o_2} = \left(1 + \frac{R_2}{R_1}\right) v_{b_2}$

$v_{o_2} = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{R_4}{R_3 + R_4}\right) v_2$



$v_{b_2} = \frac{R_4}{R_3 + R_4} v_2$

weighted subtractor

$v_o = v_{o_1} + v_{o_2}$

$= \underbrace{-\frac{R_2}{R_1}}_{A_{CL_1}} \cdot v_1 + \underbrace{\left(1 + \frac{R_2}{R_1}\right) \cdot \left(\frac{R_4}{R_3 + R_4}\right)}_{A_{CL_2}} v_2$

Source: Coughlin 3-2.3

Weighted Subtractor

design condⁿ

$R_4 = R_2$

$R_3 = R_1$

} diff. amp.

weighted subtractor ✓

$v_o = v_{o_1} + v_{o_2}$

$= \underbrace{-\frac{R_2}{R_1}}_{A_{cl_1}} \cdot v_1 + \underbrace{\left(1+\frac{R_2}{R_1}\right)\cdot\left(\frac{R_4}{R_3+R_4}\right)}_{A_{cl_2}} v_2$

$\Rightarrow \left(\frac{R_1+R_2}{R_1}\right)\left(\frac{R_4}{R_3+R_4}\right)\cdot v_2$ (R_2 → R_4, R_3+R_4)

$\Rightarrow \frac{R_1+R_2}{R_1}\cdot\frac{R_2}{R_1+R_2}\, v_2$

$v_o = -\frac{R_2}{R_1}\cdot v_1 + \frac{R_2}{R_1}\cdot v_2 = \frac{R_2}{R_1}(v_2 - v_1)$: Difference amp.

$v_o = v_2 - v_1$: Subtractor (unity gain) →

design condⁿ

$R_4 = R_2 = R_3 = R_1$

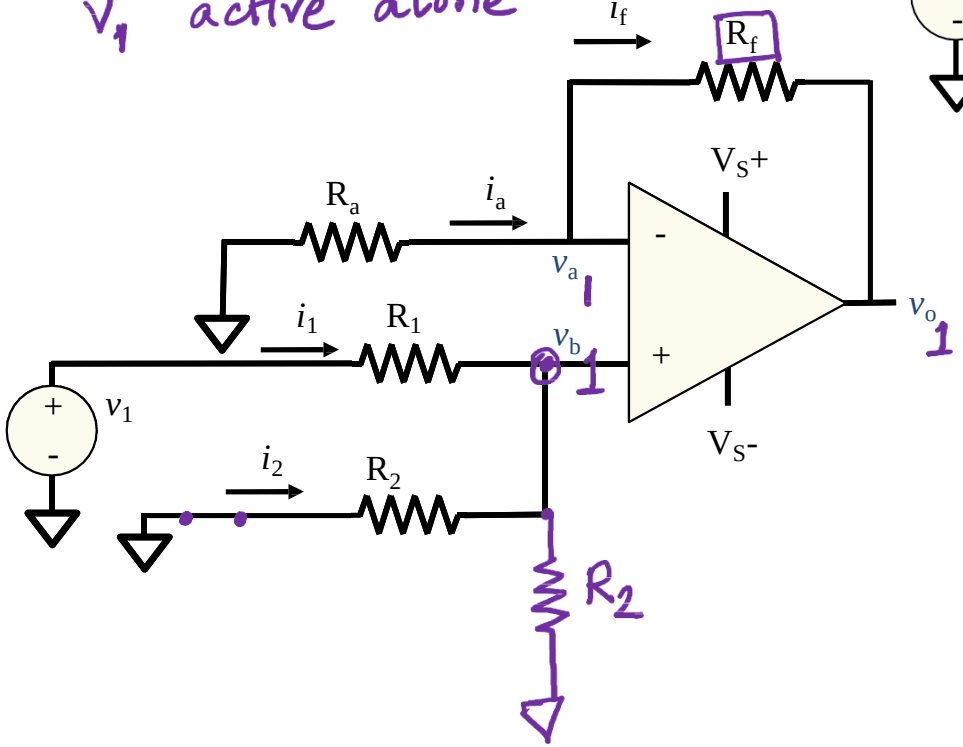
Source: Coughlin 3-2.3

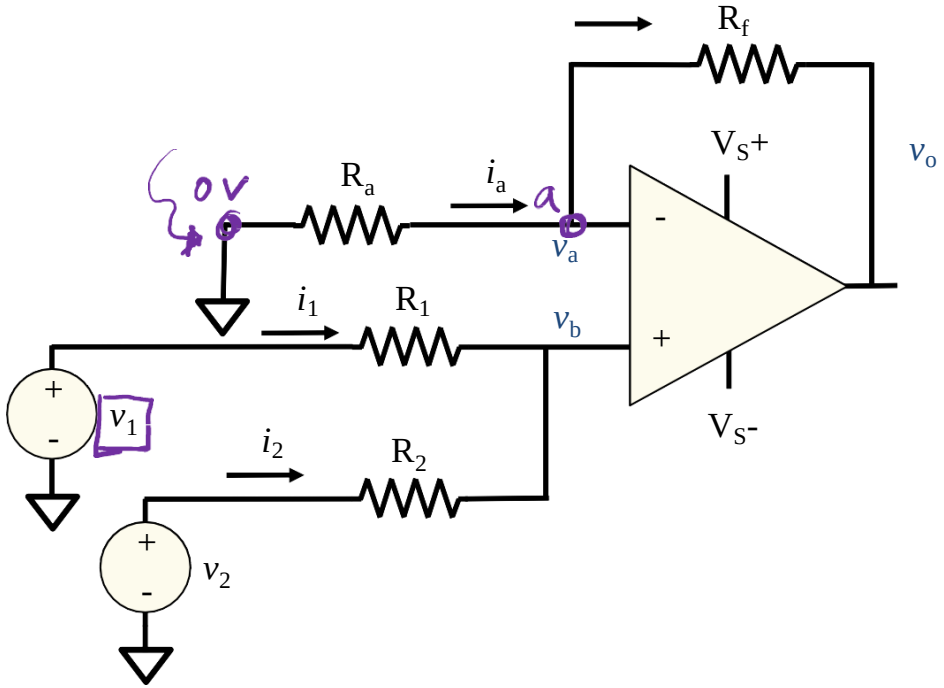
Non-inverting Adder

$$v_o = \left(1 + \frac{R_f}{R_a}\right) v_b \quad \cdots\cdots (I)$$

Superposition

V_1 active alone





$$v_{b_1} = \frac{R_2}{R_1 + R_2} \cdot v_1 \quad \cdots\cdots (II)$$

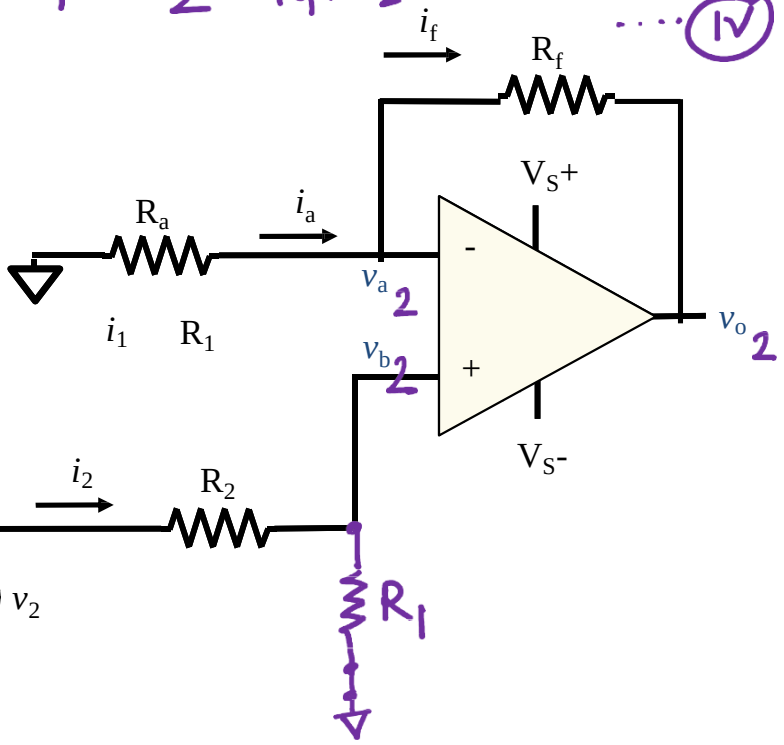
Non-inverting Adder

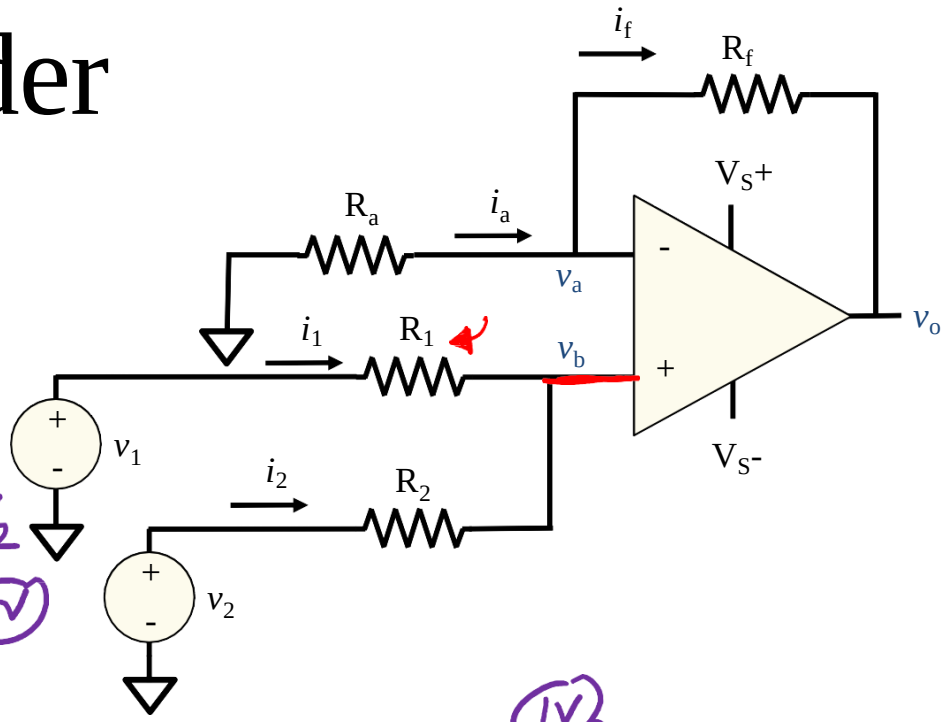
V_2 active alone

$$v_{b_2} = \frac{R_1}{R_1+R_2} \cdot v_2 \quad \cdots (III)$$

(II), (III) $\Rightarrow$

$$v_b = v_{b_1} + v_{b_2} = \frac{R_2}{R_1+R_2} \cdot v_1 + \frac{R_1}{R_1+R_2} \cdot v_2 \quad \cdots (IV)$$





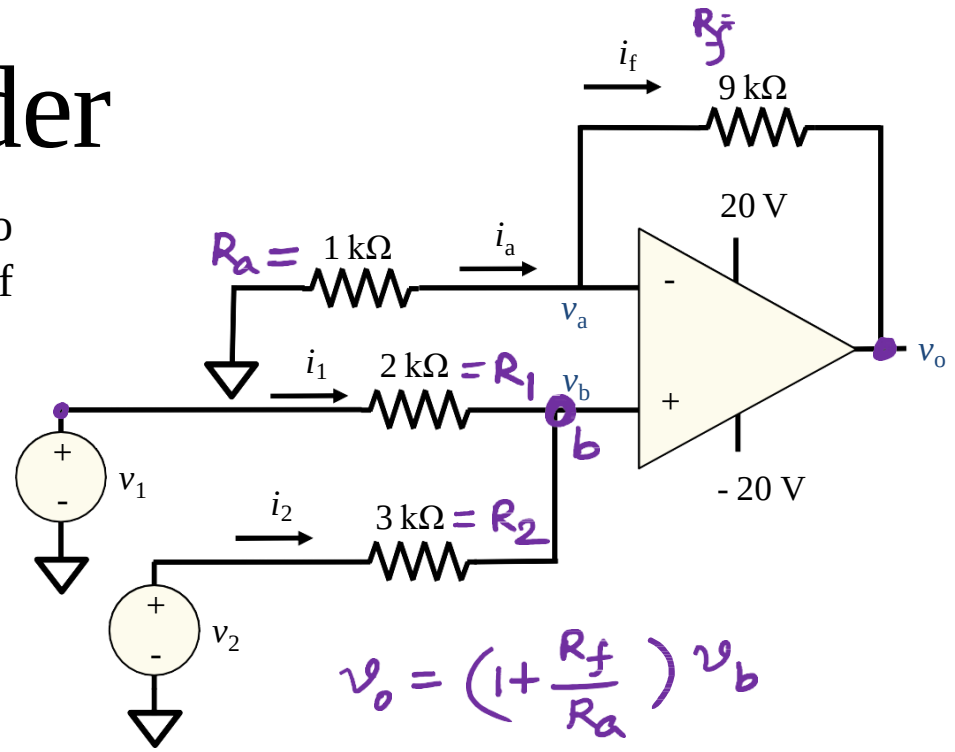
(IV) in (I) $\Rightarrow$

$$V_o = \left(1 + \frac{R_f}{R_a}\right)\left\{\frac{R_2}{R_1+R_2}V_1 + \frac{R_1}{R_1+R_2} \cdot V_2\right\}$$

$$= \left\{\left(1 + \frac{R_f}{R_a}\right)\left(\frac{R_2}{R_1+R_2}\right)V_1\right\} + \left\{\left(1 + \frac{R_f}{R_a}\right)\left(\frac{R_1}{R_1+R_2}\right)V_2\right\}$$

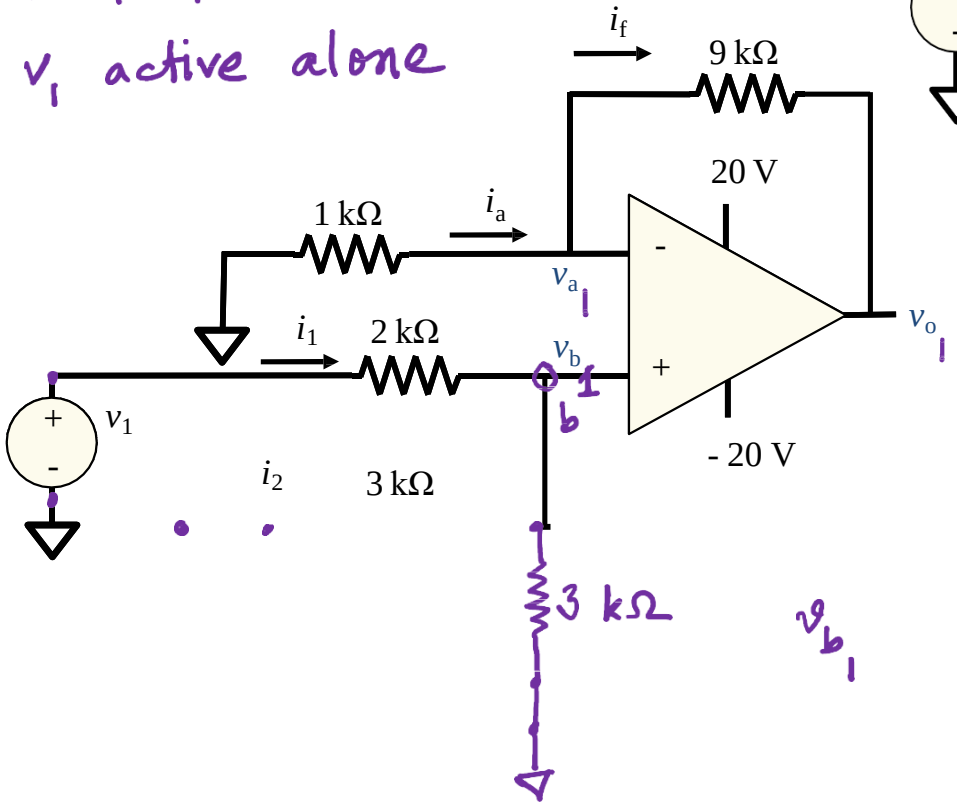
Non-inverting Adder

Problem 22. Use the superposition principle to find the output voltage of the circuit in terms of v_1 and v_2. Assume ideal conditions.



Superposition

v_1 active alone



v_{b_1}

$$\frac{3\,k\Omega}{(2+3)\,k\Omega} V_1 = \frac{3}{5} V_1 \quad \cdots\cdots (II)$$

$$v_o = \left(1 + \frac{R_f}{R_a}\right) v_b$$

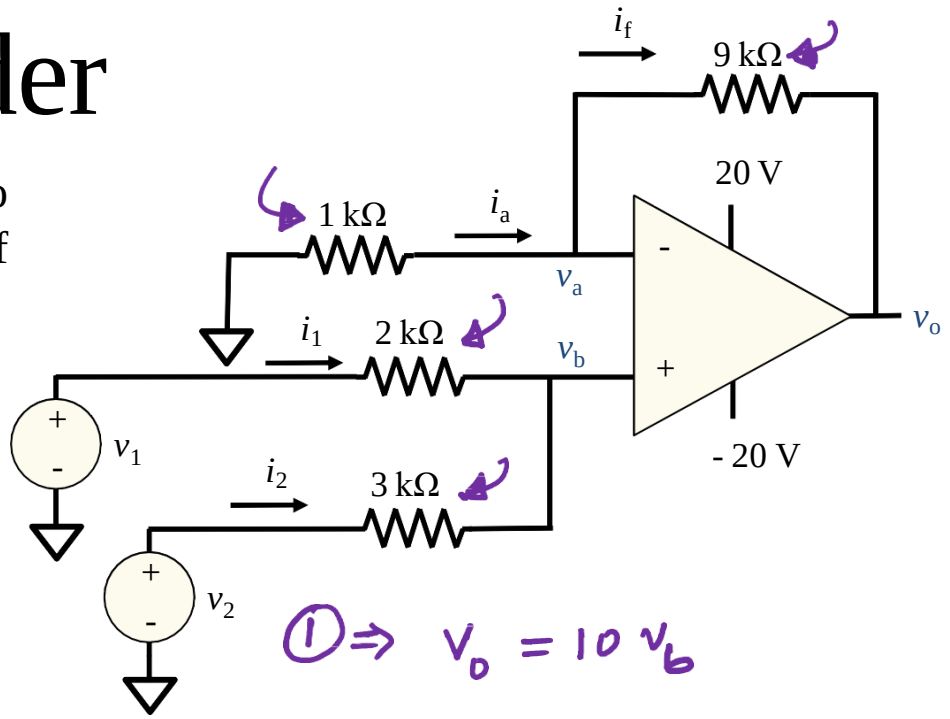
$$= \left(1 + \frac{9\,k\Omega}{1\,k\Omega}\right) v_b$$

$$v_o = 10\, v_b \quad \cdots\cdots (I)$$

Ans. $v_o = 6v_1 + 4v_2$

Non-inverting Adder

Problem 22. Use the superposition principle to find the output voltage of the circuit in terms of v_1 and v_2. Assume ideal conditions.

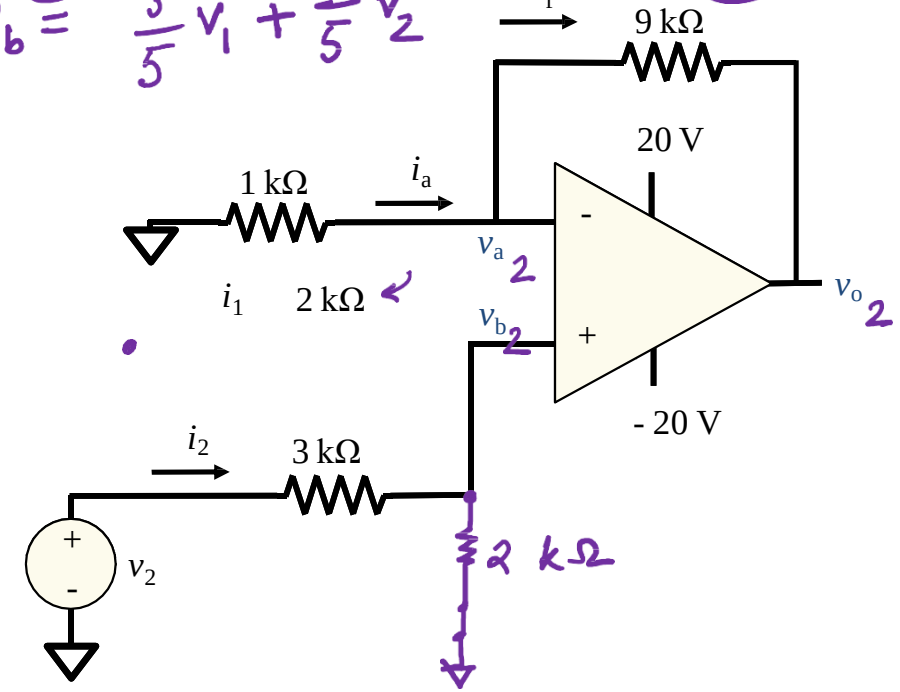


v_2 active alone

$$v_{b_2} = \frac{2\text{ k}\Omega}{(2+3)\text{k}\Omega} v_2 = \frac{2}{5} v_2 \quad \cdots (III)$$

(II), (III) $\Rightarrow$

$$v_b = \frac{3}{5} v_1 + \frac{2}{5} v_2 \quad \cdots (IV)$$



(I) $\Rightarrow$ $V_o = 10\, v_b$

$$= 10\left(\frac{3}{5} v_1 + \frac{2}{5} v_2\right)$$

$$\Rightarrow V_o = 6 v_1 + 4 v_2$$

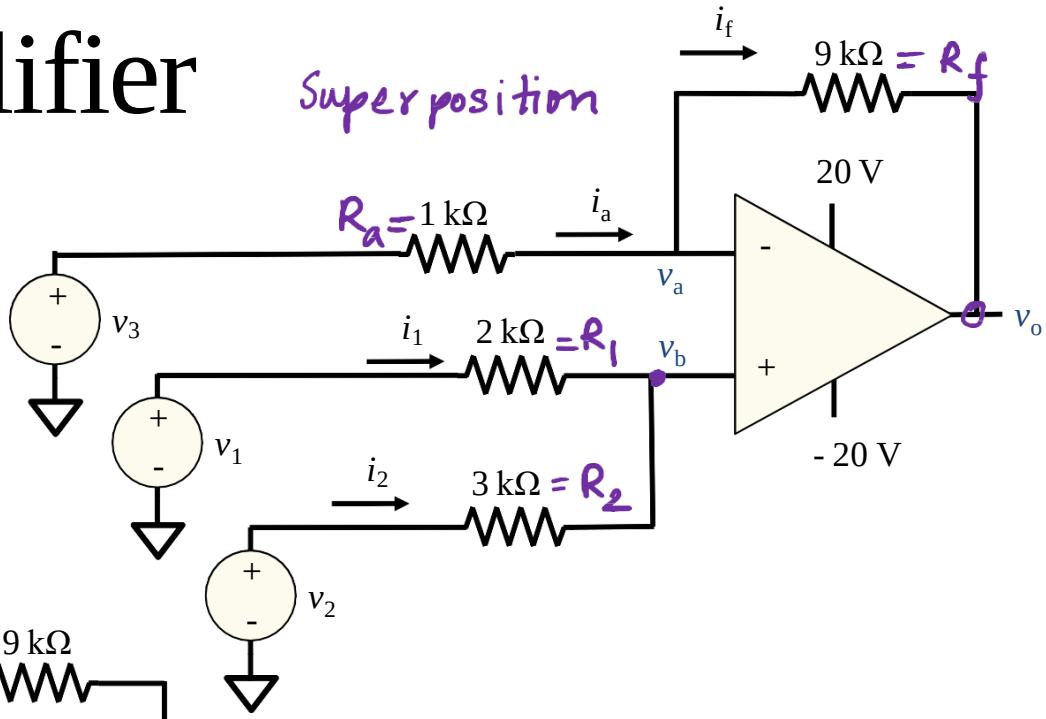
weighted adder

Ans. $v_o = 6v_1 + 4v_2$

Difference Amplifier

Superposition

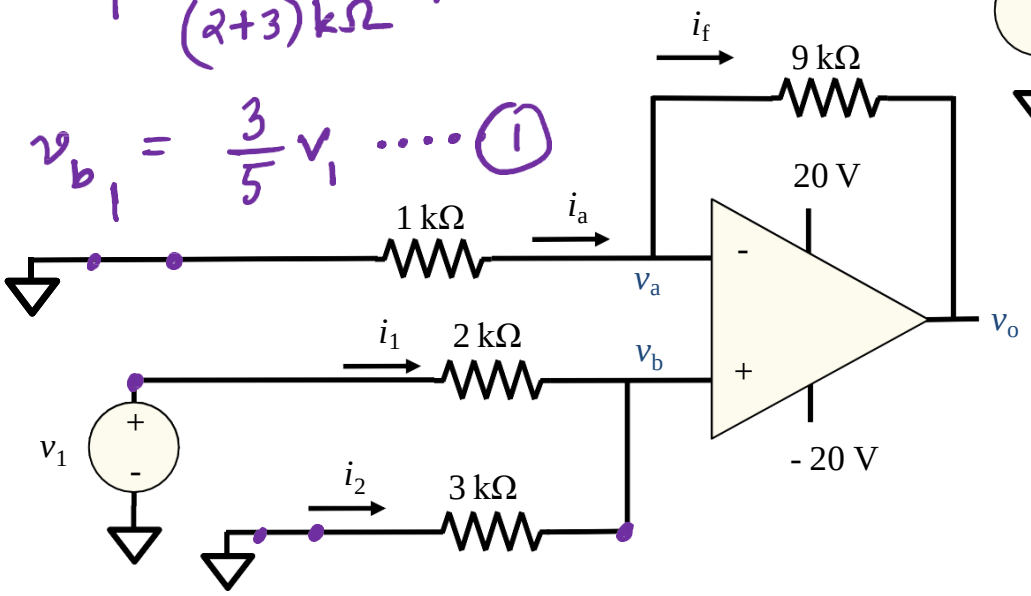
Problem 23. Use the superposition principle to find the output voltage of the circuit in terms of v_1, v_2, and v_3. Assume ideal conditions.



v_1 active alone

$$v_{b_1} = \frac{3\,k\Omega}{(2+3)\,k\Omega} \cdot v_1$$

$$v_{b_1} = \frac{3}{5} v_1 \quad \cdots (I)$$



$$v_{o_1} = \left(1 + \frac{9\,k\Omega}{1\,k\Omega}\right) v_{b_1}$$

$$= 10 \cdot \frac{3}{5} v_1$$

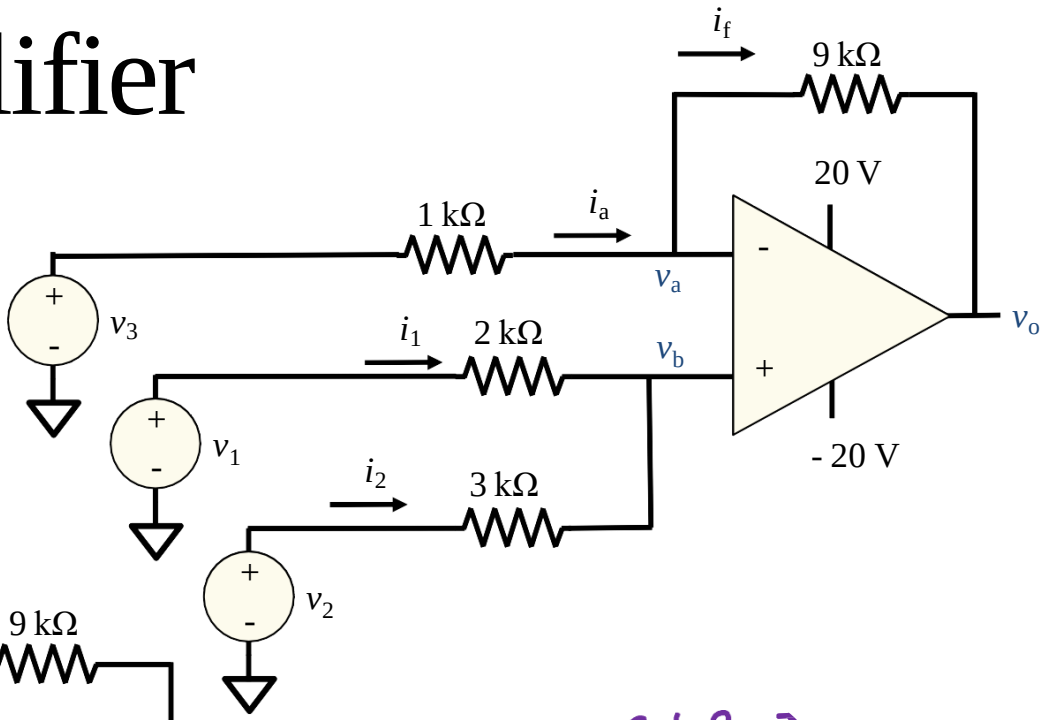
$$v_{o_1} = v_1 \quad \cdots (II)$$

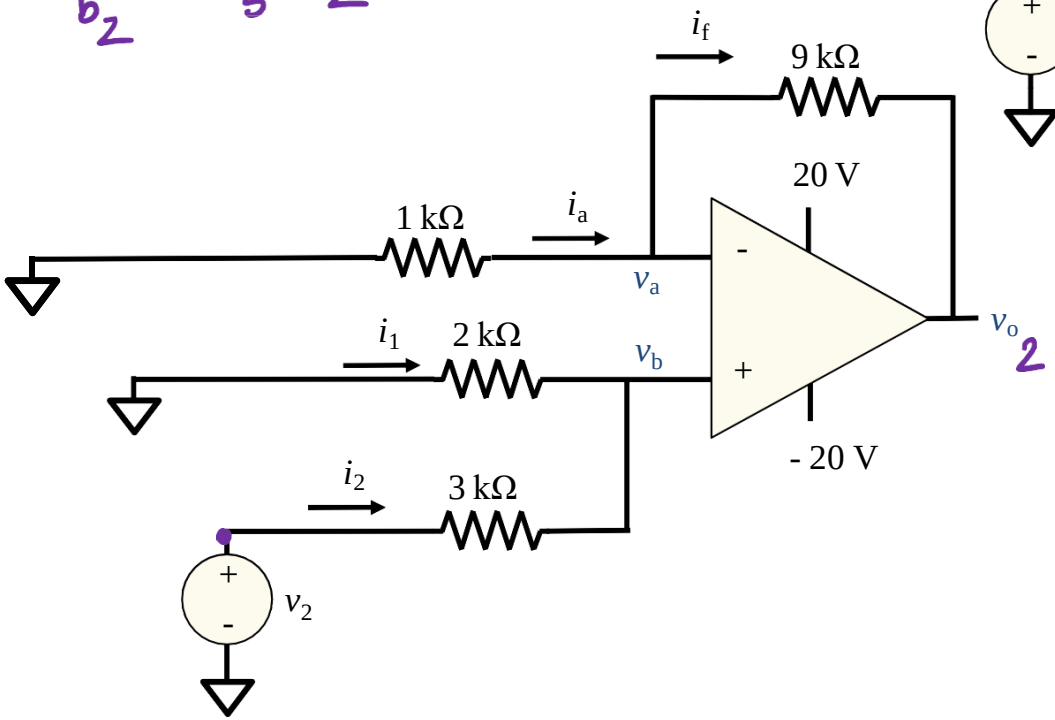
Difference Amplifier

V_2 active alone

$$v_{b_2} = \frac{2\,k\Omega}{(2+3)\,k\Omega} \cdot V_2$$

$$v_{b_2} = \frac{2}{5} V_2 \quad \cdots \text{(III)}$$





$$v_{o_2} = \left(1 + \frac{9\,k\Omega}{1\,k\Omega}\right) v_{b_2}$$

$$= 10 \cdot \frac{2}{5} V_2$$

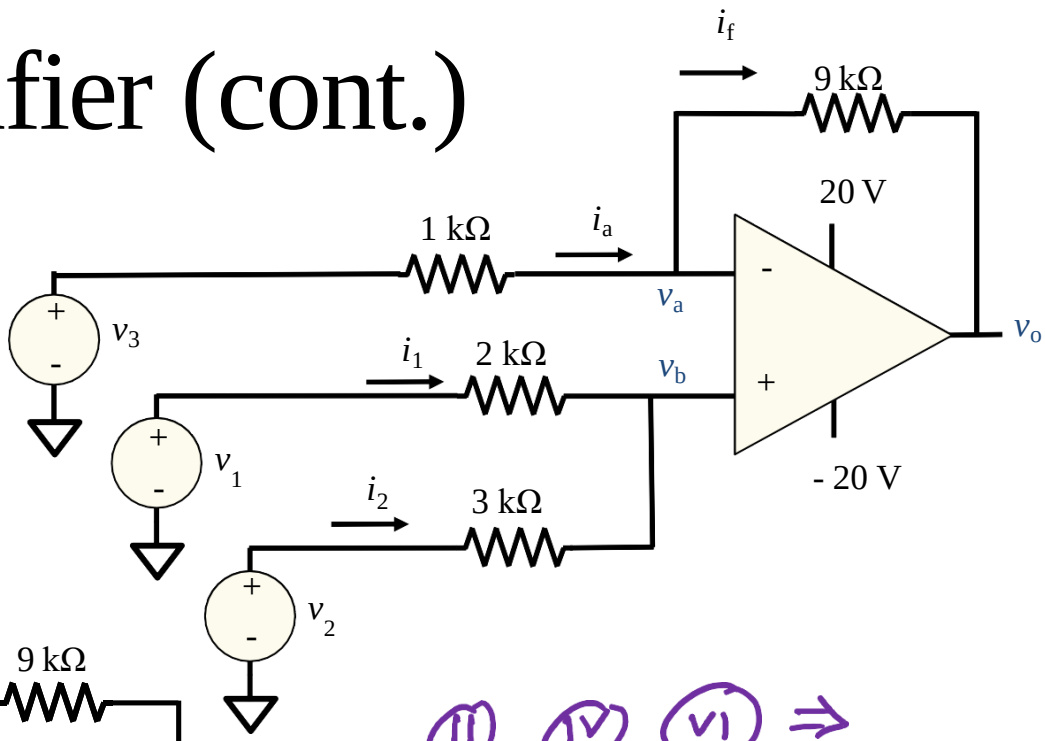
$$v_{o_2} = 4 V_2 \quad \cdots \text{(IV)}$$

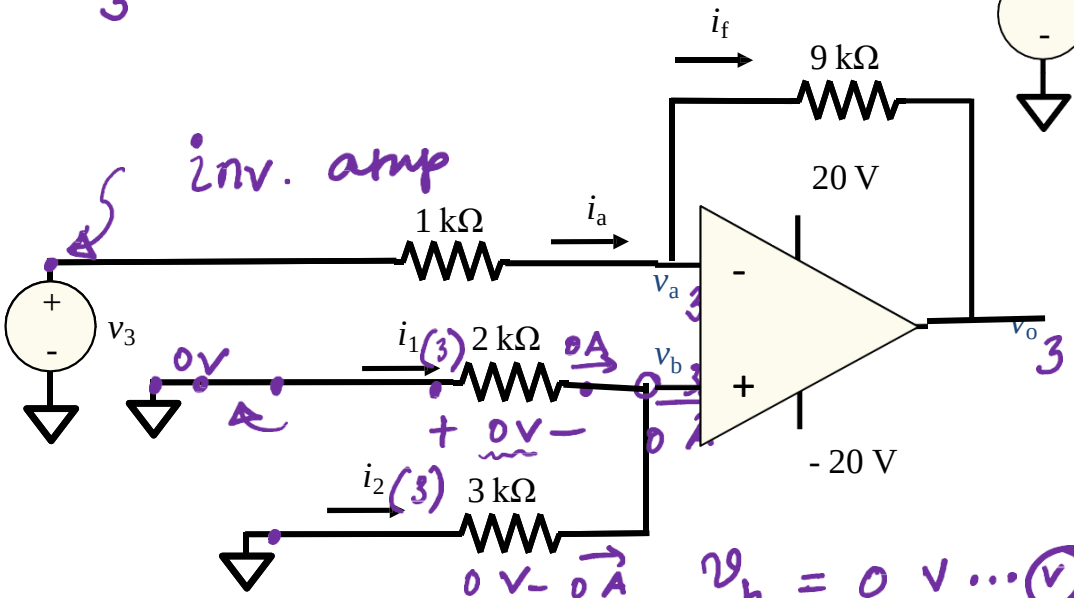
Difference Amplifier (cont.)

V_3 active alone

$$v_{o_3} = -\frac{9\,k\Omega}{1\,k\Omega} \cdot v_3$$

$$v_{o_3} = -9 v_3 \quad \cdots (VI)$$





$v_{b_3} = 0$ V ⋯ (V)

(II), (IV), (VI) ⇒

$$v_o = 6v_1 + 4v_2 - 9v_3$$

Ans. $v_o = 6v_1 + 4v_2 - 9v_3$

Analog Electronics II: Op-amp & 555 Timer

Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

Lecture # 7b

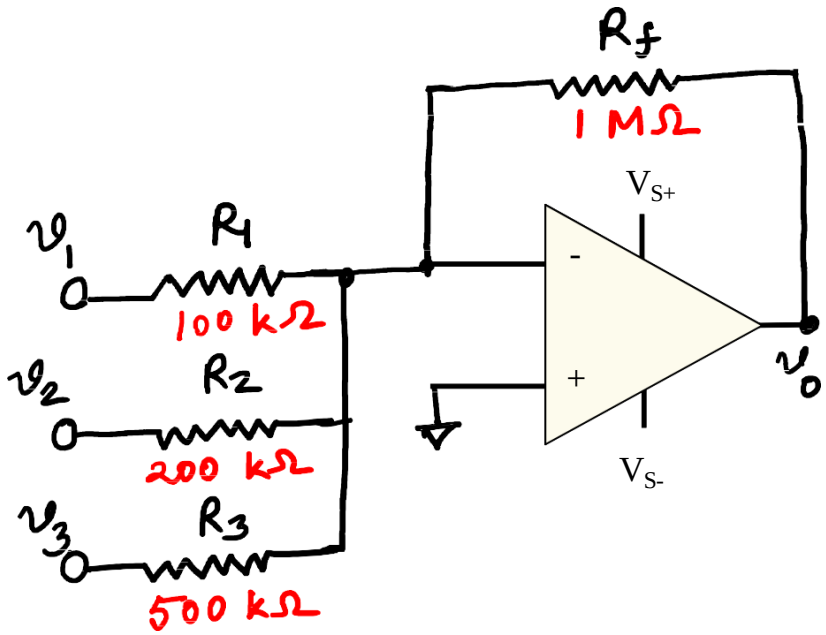
Practice Problems

Inverting Weighted Adder

R_f should be a common multiple of 10, 5, 2.

Design an op-amp-based circuit so that the output voltage v_o is related to the input voltages v_1, v_2, v_3 as follows:

$$v_o = -10v_1 - 5v_2 - 2v_3 = A_{cl_1} \cdot v_1 + A_{cl_2} \cdot v_2 + A_{cl_3} \cdot v_3$$



$$\Rightarrow v_o = \left(-\frac{R_f}{R_1}\right)v_1 + \left(-\frac{R_f}{R_2}\right)v_2 + \left(-\frac{R_f}{R_3}\right)v_3$$

$$A_{cl_1} = -\frac{R_f}{R_1} = -10 \Rightarrow R_1 = \frac{R_f}{10} \quad \text{(i)}$$

$$A_{cl_2} = -\frac{R_f}{R_2} = -5 \Rightarrow R_2 = \frac{R_f}{5} \quad \text{(ii)}$$

$$A_{cl_3} = -\frac{R_f}{R_3} = -2 \Rightarrow R_3 = \frac{R_f}{2} \quad \text{(iii)}$$

4 unknown variables: R_f, R_1, R_2, R_3

3 eq^{n}s

From eq^n (i) – (iii),

Assume, $R_f = 1\ M\Omega$

$R_1 = 100\ k\Omega$

$R_2 = 200\ k\Omega$

$R_3 = 500\ k\Omega$

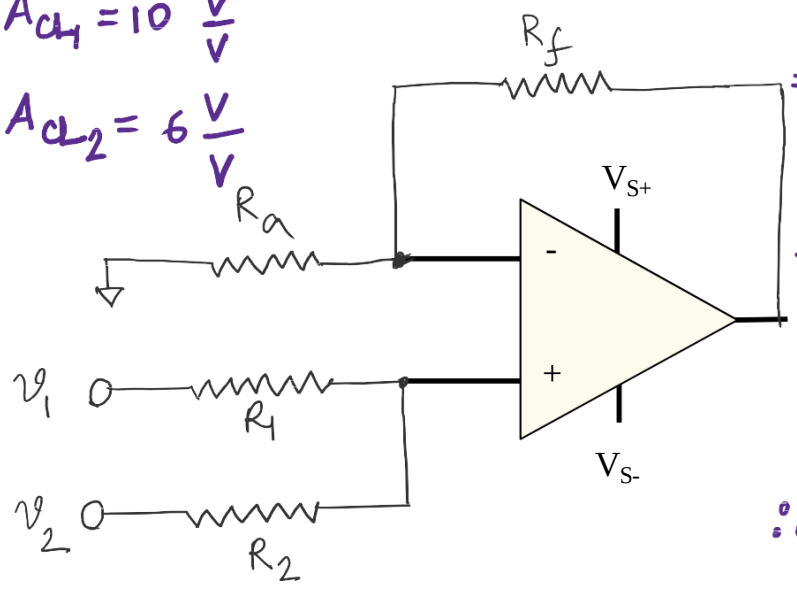
Non-inverting Weighted Adder

Design an op-amp-based weighted non-inverting adder circuit so that the output voltage as follows:

$v_o = 10v_1 + 6v_2 = A_{CL_1} \cdot v_1 + A_{CL_2} \cdot v_2 = \left\{\left(1+\frac{R_f}{R_a}\right)\cdot\frac{R_2}{R_1+R_2}\right\} v_1 + \left\{\left(1+\frac{R_f}{R_a}\right)\cdot\frac{R_1}{R_1+R_2}\right\} v_2$

$A_{CL_1} = 10 \frac{V}{V}$

$A_{CL_2} = 6 \frac{V}{V}$



$= \left(1+\frac{R_f}{R_a}\right)\cdot\frac{1}{R_1+R_2}\cdot\left\{R_2\cdot v_1 + R_1\cdot v_2\right\}$

$= A_x \cdot \left\{R_2\cdot v_1 + R_1\cdot v_2\right\}$

$\left[\text{Assuming, } A_x = \left(1+\frac{R_f}{R_a}\right)\cdot\frac{1}{R_1+R_2}\right]$

$\therefore A_{CL_1} = A_x \cdot R_2 = 10\frac{V}{V} \Rightarrow R_2 \propto 10$

$A_{CL_2} = A_x \cdot R_1 = 6\frac{V}{V} \Rightarrow R_1 \propto 6$

Hence, we should start the design with assuming R_1 and R_2. $\therefore R_2 \propto A_{CL_1}$ and $R_1 \propto A_{CL_2}$

If we assume R_f and R_a, R_1 and R_2 will not be proportional to A_{CL_2} and A_{CL_1}, respectively.

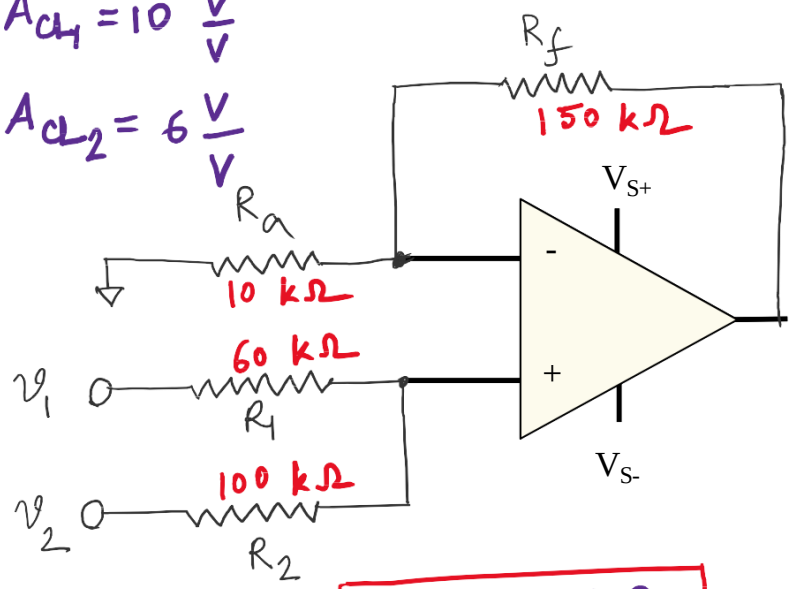
Non-inverting Weighted Adder

Design an op-amp-based weighted non-inverting adder circuit so that the output voltage as follows:

$v_o = 10v_1 + 6v_2 = A_{CL_1} \cdot v_1 + A_{CL_2} \cdot v_2 = \left\{\left(1+\frac{R_f}{R_a}\right)\cdot\frac{R_2}{R_1+R_2}\right\}v_1 + \left\{\left(1+\frac{R_f}{R_a}\right)\cdot\frac{R_1}{R_1+R_2}\right\}v_2$

$A_{CL_1} = 10\ \frac{V}{V}$

$A_{CL_2} = 6\ \frac{V}{V}$



Let's assume,

$R_2 = 100\ k\Omega$

$R_1 = 60\ k\Omega$

Assume,

$R_a = 10\ k\Omega$

$= \left\{\left(1+\frac{R_f}{R_a}\right)\cdot\frac{100}{60+100}\right\}\cdot v_1 + \left\{\left(1+\frac{R_f}{R_a}\right)\cdot\frac{60}{60+100}\right\}v_2$

$= \left\{\left(1+\frac{R_f}{R_a}\right)\times 0.625\right\}\cdot v_1 + \left\{\left(1+\frac{R_f}{R_a}\right)\times 0.375\right\}\cdot v_2$

Now, $\left(1+\frac{R_f}{R_a}\right)\times 0.625 = A_{CL_1} = 10\ \frac{V}{V}$

$\Rightarrow 1+\frac{R_f}{R_a} = \frac{10}{0.625} = 16$

$\Rightarrow \frac{R_f}{R_a} = 16-1 = 15 \Rightarrow R_f = 15R_a \cdots\cdots (1)$

Then, $\left(1+\frac{R_f}{R_a}\right)\times 0.375 = 6 \Rightarrow 1+\frac{R_f}{R_a} = 16$

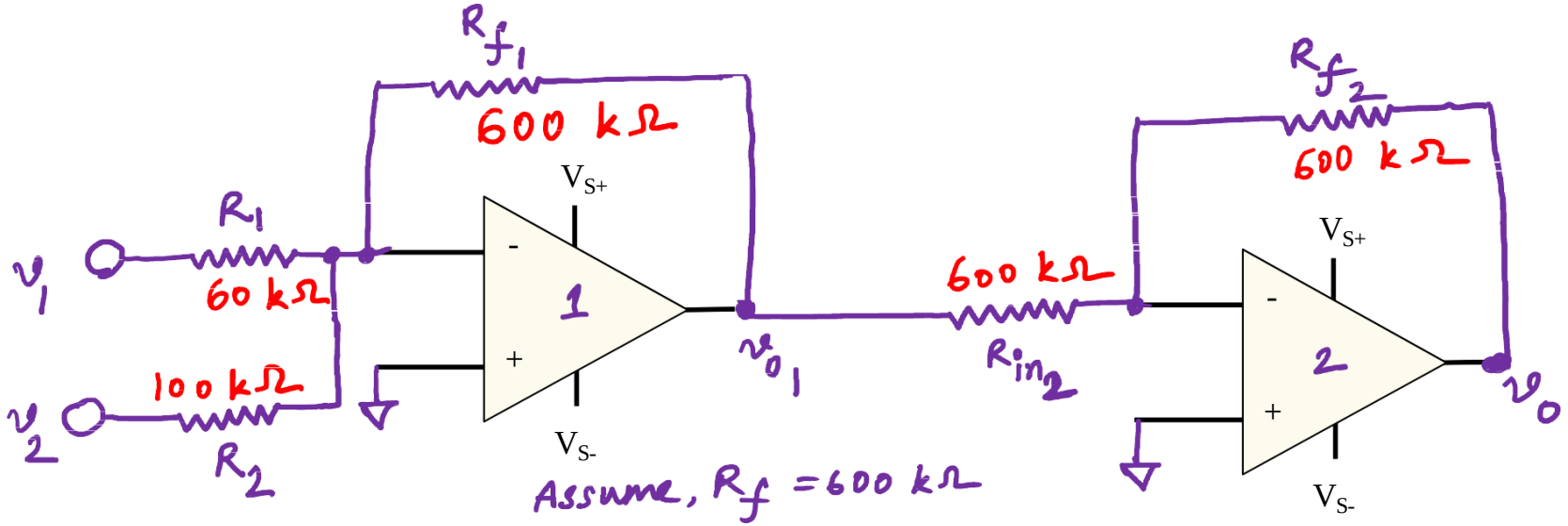
$\Rightarrow R_f = 15R_a \cdots\cdots (11)$

$\therefore R_f = 15\times 10\ k\Omega \Rightarrow \boxed{R_f = 150\ k\Omega}$

Cascaded Inverting Adders & Amplifiers

Design an op-amp-based circuit so that the output voltage v_o is related to the input voltages v_1, v_2 as follows:

$v_o = 10v_1 + 6v_2$ = $(-10v_1 - 6v_2)\cdot(-1)$ (-ve × -ve = +ve)



Assume, $R_f = 600\ k\Omega$

$A_{cl_1} = +10$

$-\frac{R_f}{R_1} = -10 \Rightarrow R_1 = \frac{R_f}{10}$

$-\frac{R_{f_2}}{R_{in_2}} = -1 \Rightarrow R_{f_2} = R_{in_2}$

$A_{cl_2} = +6$

$-\frac{R_f}{R_2} = -6 \Rightarrow R_2 = \frac{R_f}{6}$

Assume,

$R_{f_2} = 600\ k\Omega = R_{in_2}$

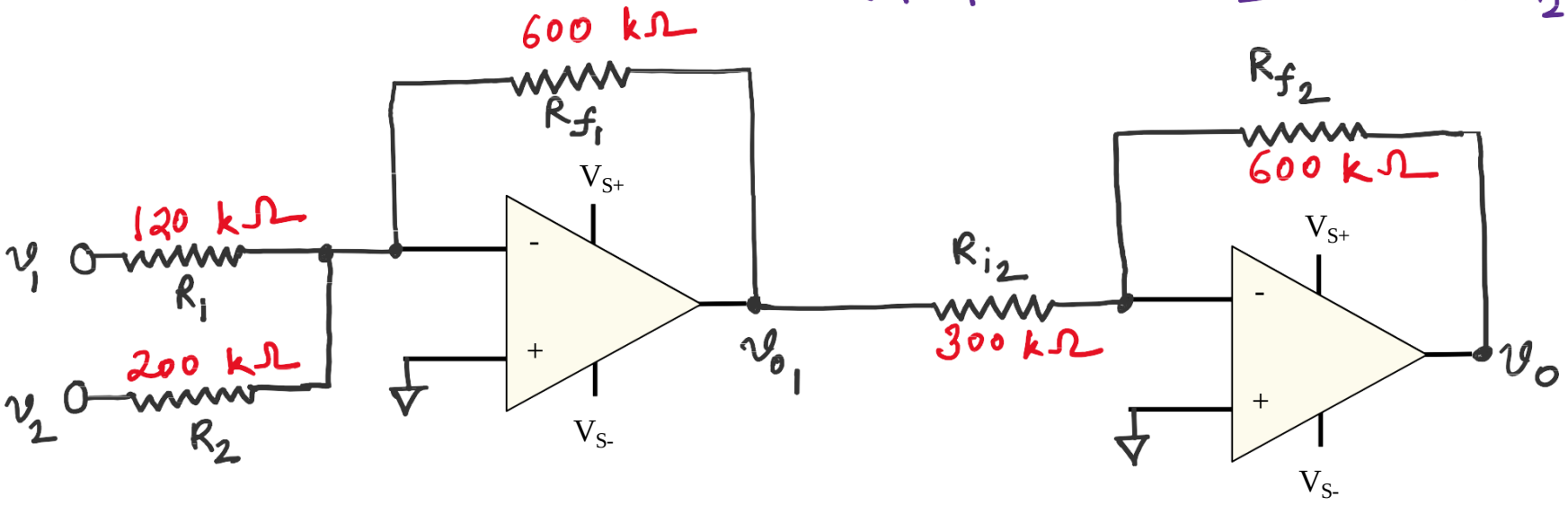
©Dipl
Assistant Professor, North South University

Cascaded Inverting Adders & Amplifiers

Design an op-amp-based circuit so that the output voltage v_o is related to the input voltages v_1, v_2 as follows:

Problem 29. Alternative solution (2)

$$v_o = 10v_1 + 6v_2 = \underbrace{(-5v_1 - 3v_2)}_{\text{inverting}}\underbrace{(-2)}_{\text{inv. amp.}} = \left\{\left(-\frac{R_{f1}}{R_1}\right)v_1 + \left(-\frac{R_{f1}}{R_2}\right)\cdot v_2\right\}\cdot\left(-\frac{R_{f2}}{R_{i2}}\right)$$



$A_{CL_1} = +10 \ \frac{V}{V}$

Let's assume, $R_{f1} = 600\ k\Omega$ and $R_{f2} = 600\ k\Omega$

$+\frac{R_{f1}}{R_1} = +5 \Rightarrow R_1 = \frac{R_{f1}}{5} \Rightarrow R_1 = 120\ k\Omega$

$+\frac{R_{f1}}{R_2} = +3 \Rightarrow R_2 = \frac{R_{f1}}{3} \Rightarrow R_2 = 200\ k\Omega$

$-\frac{R_{f2}}{R_{i2}} = -2 \Rightarrow R_{i2} = \frac{R_{f2}}{2}$

$\therefore R_{i2} = 300\ k\Omega$

©DPN **Dr. Assistant Professor, North South University**

Cascaded Differential Amplifier

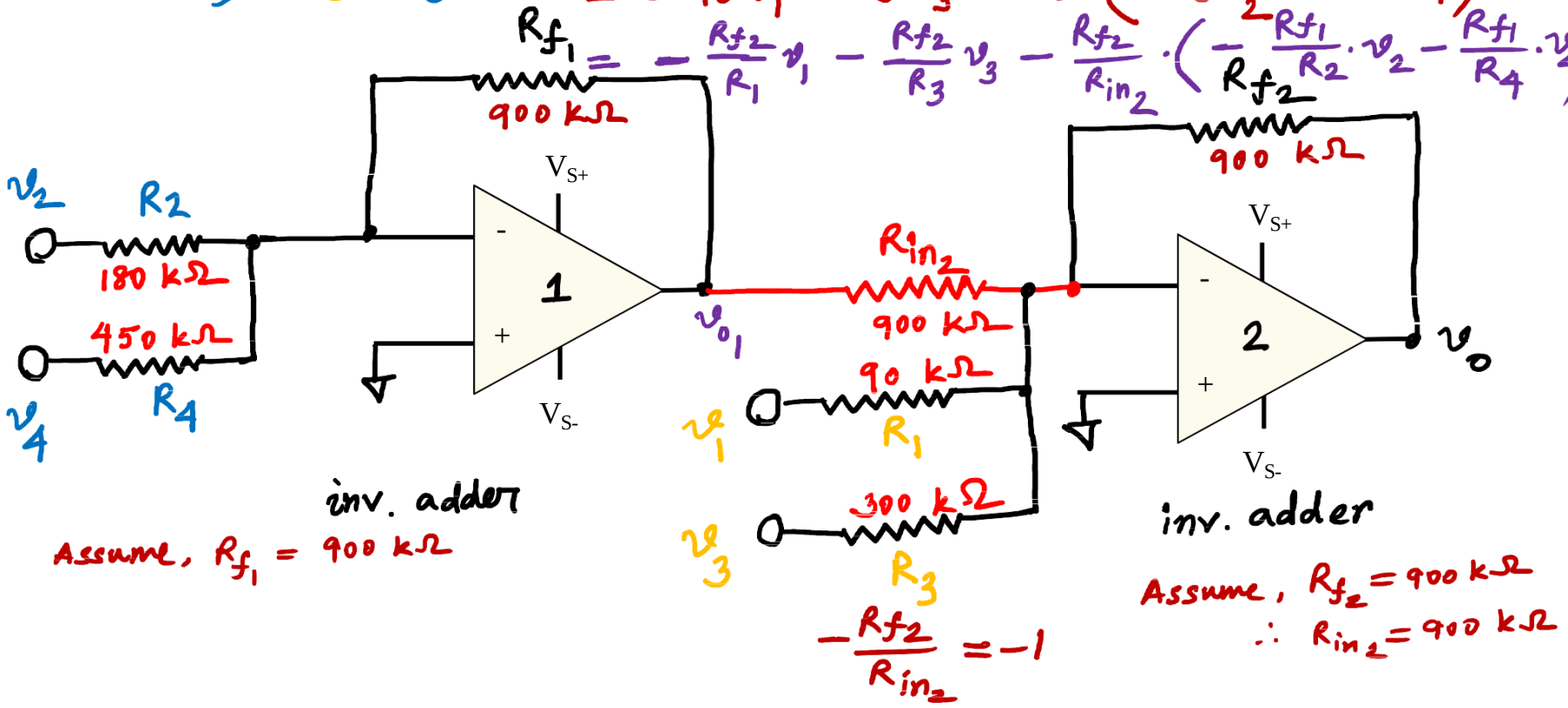
Design an op-amp-based circuit so that the output voltage v_o v_1, v_2, v_3, v_4 as follows:

$$v_o = -10v_1 + 5v_2 - 3v_3 + 2v_4$$

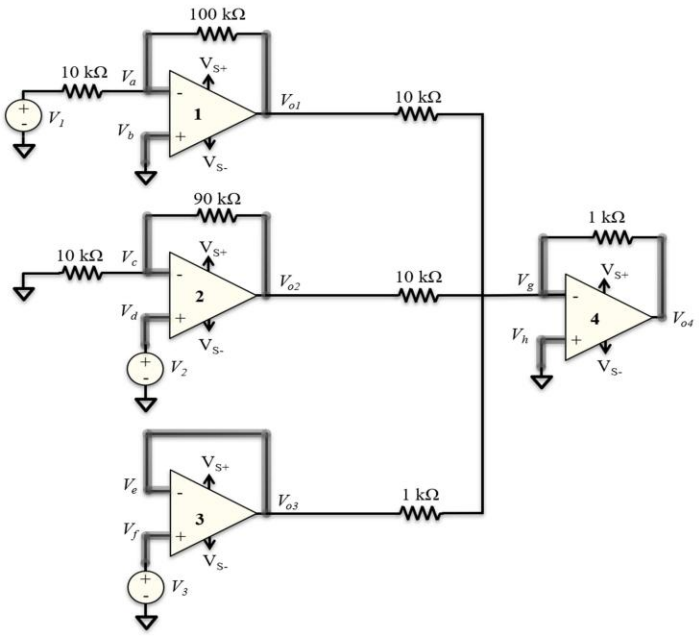
Do all calculation as shown in the above solutions.

$$= -10v_1 - 3v_3 - 1\left(-5v_2 - 2v_4\right)$$

$$= -\frac{R_{f2}}{R_1}v_1 - \frac{R_{f2}}{R_3}v_3 - \frac{R_{f2}}{R_{in_2}}\cdot\left(-\frac{R_{f1}}{R_2}\cdot v_2 - \frac{R_{f1}}{R_4}\cdot v_4\right)$$



Question. In the following circuit, calculate V_{o1}, V_{o2}, V_{o3}, and V_{o4}.



Question. Design an adder / subtractor circuit to achieve the following output voltage expression:

$$V_0 = V_1 + 2V_2 + \frac{1}{2} V_3 - V_4$$

Assume ideal op-amp with bias voltages of ±15 V. All input voltages are DC. You may use multiple op-amps.

Make sure that there is no chance of signal strength loss at any point of the circuit.

Question. Design an adder / subtractor circuit to achieve the following output voltage expression:

$$V_0 = V_1 + V_2 - 2V_3 - \frac{1}{2}V_4$$

Assume ideal op-amp with bias voltages of ±15 V. All input voltages are DC. You may use multiple op-amps.

Make sure that there is no chance of signal strength loss at any point of the circuit.

Question. Analyze the circuit using KVL, KCL, and Ohm's law. Then, calculate all branch currents and node voltages for the following circuit

$I_a =$

$I_f =$

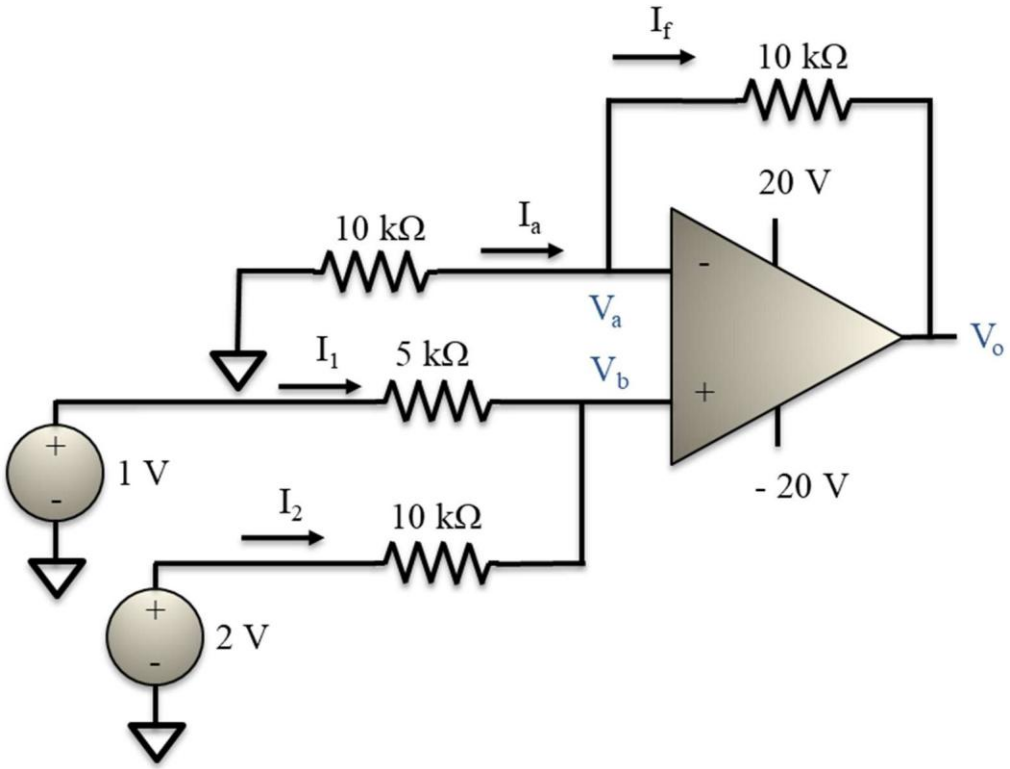
$I_1 =$

$I_2 =$

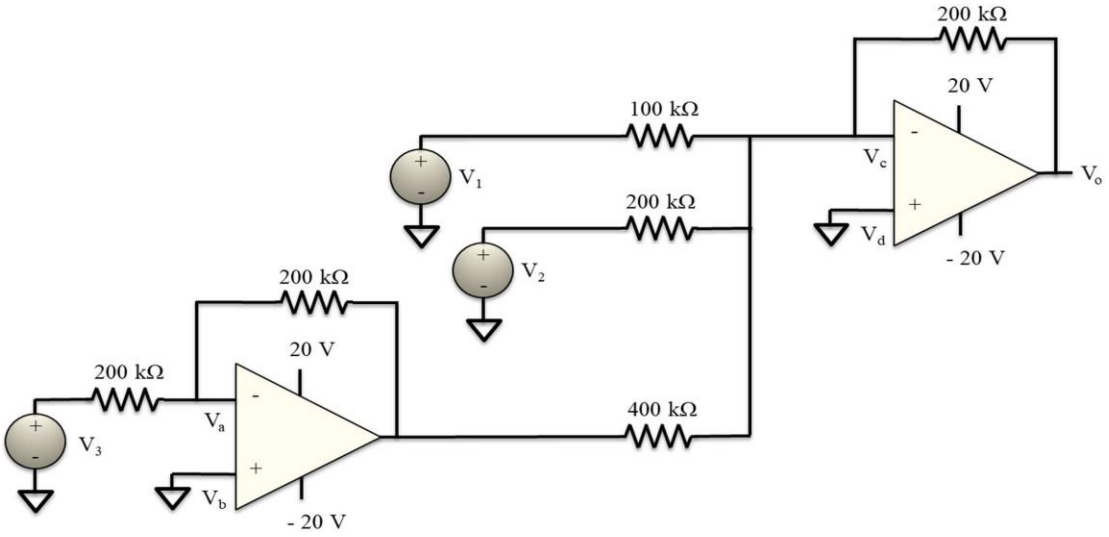
$V_a =$

$V_b =$

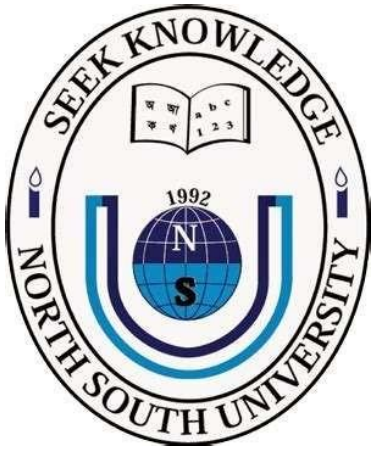
$V_0 =$



Question. Express V_o as a function of V_1, V_2, and V_3 for the following circuit.



Analog Electronics II: Op-amp & 555 Timer



Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

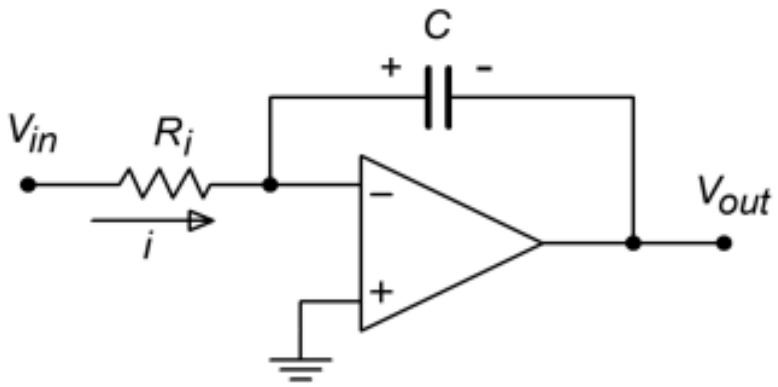
Lecture # 8

Textbook for Integrator and Differentiator

1. Fundamentals of Electric Circuits, Charles Alexander and Matthew Sadiku, 5th Edition: Chapter 6.6.1, 6.6.2 and relevant exercise problems.
2. Operational Amplifiers & Linear Integrated Circuits: Theory and Application, James M. Fiore 3rd Edition Chapter 10.1, 10.2, 10.3, and relevant exercise problems.

Integrator

An **integrator** is an op amp circuit whose output is proportional to the integral of the input signal.

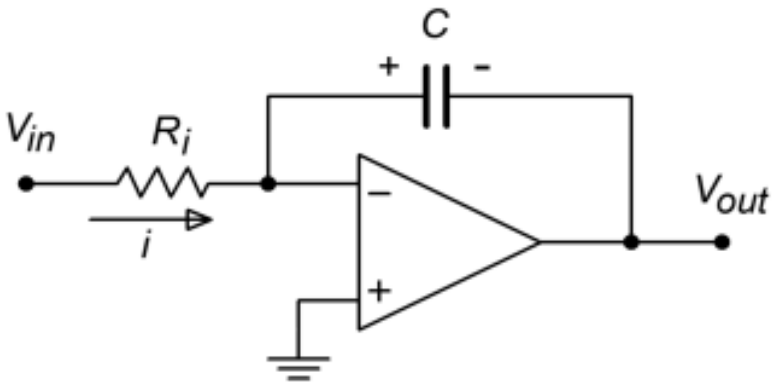


$$V_{out}(t) = -V_c(t)$$

$$i_R = \frac{v_i}{R}, \qquad i_C = -C\frac{dv_o}{dt}$$

$$i_R = i_C \Rightarrow \frac{v_i}{R} = -C\frac{dv_o}{dt}$$

$$dv_o = -\frac{1}{RC}v_i\,dt$$



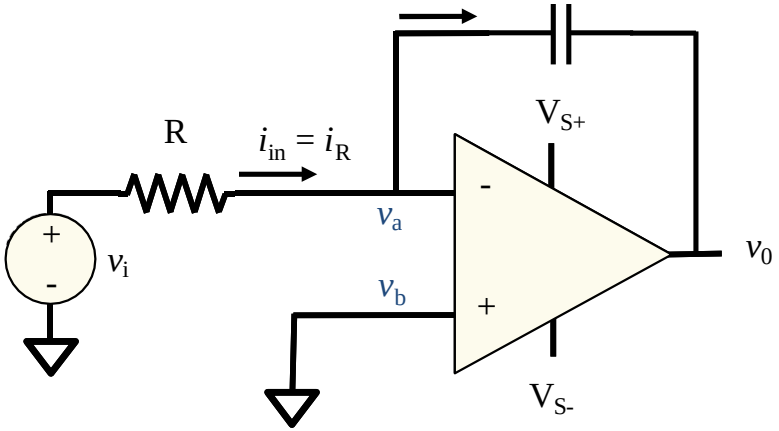
$$v_o(t) - v_o(0) = -\frac{1}{RC}\int_0^t v_i(\tau)\,d\tau$$

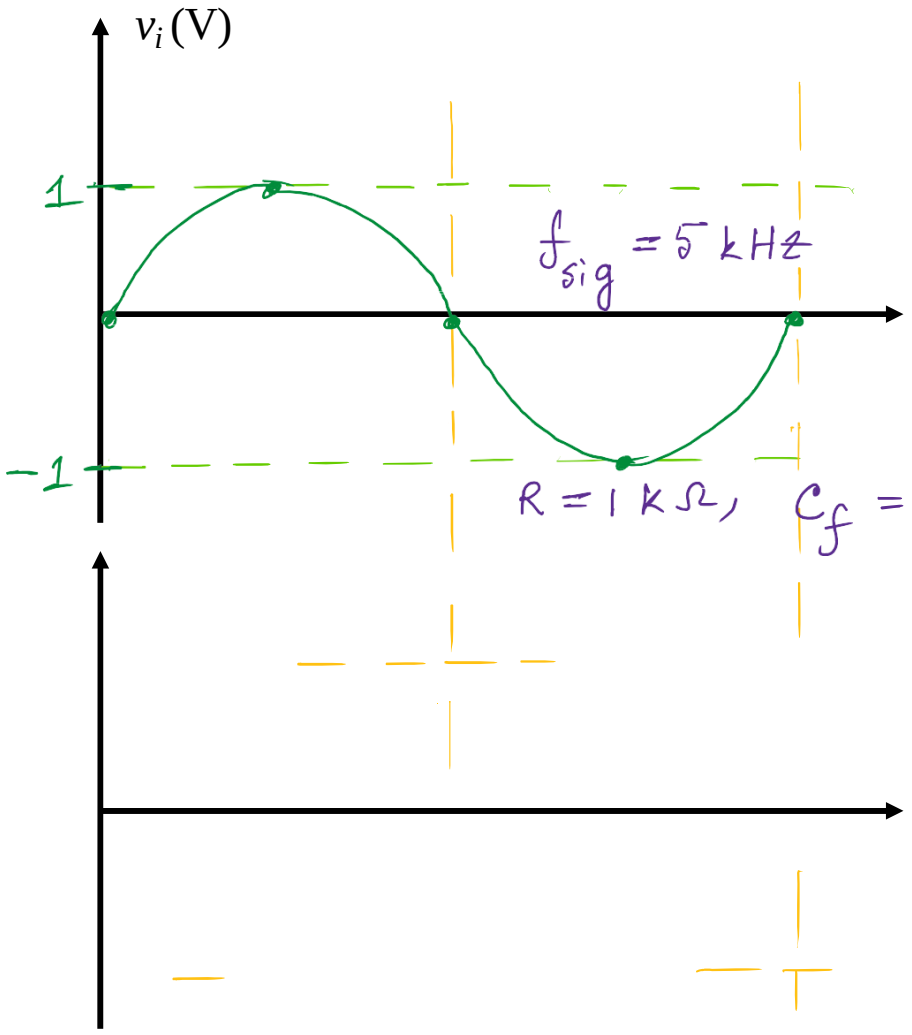
$$v_o = -\frac{1}{RC}\int_0^t v_i(\tau)\,d\tau$$

OR

$$V_{out}(t) = -\frac{1}{R_i C}\int V_{in}(t)\,dt$$

Basic Integrator





Example:

v_i is sin(xt) V.

v_0=?

Basic Integrator

Example:

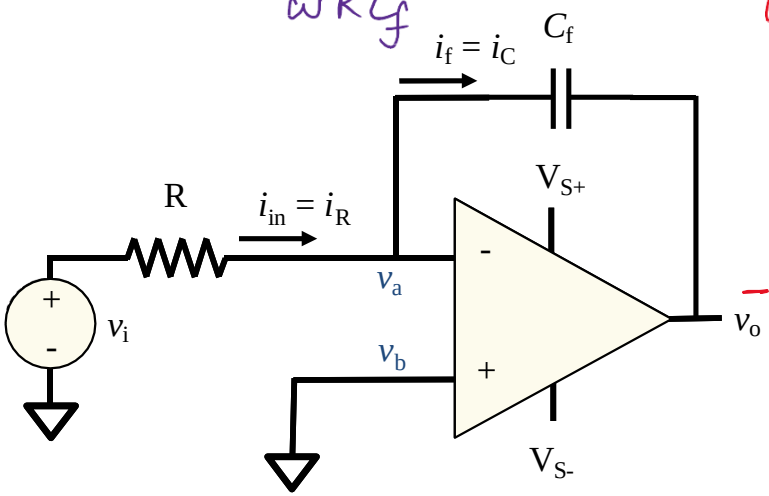
v_i is sin(xt) V.

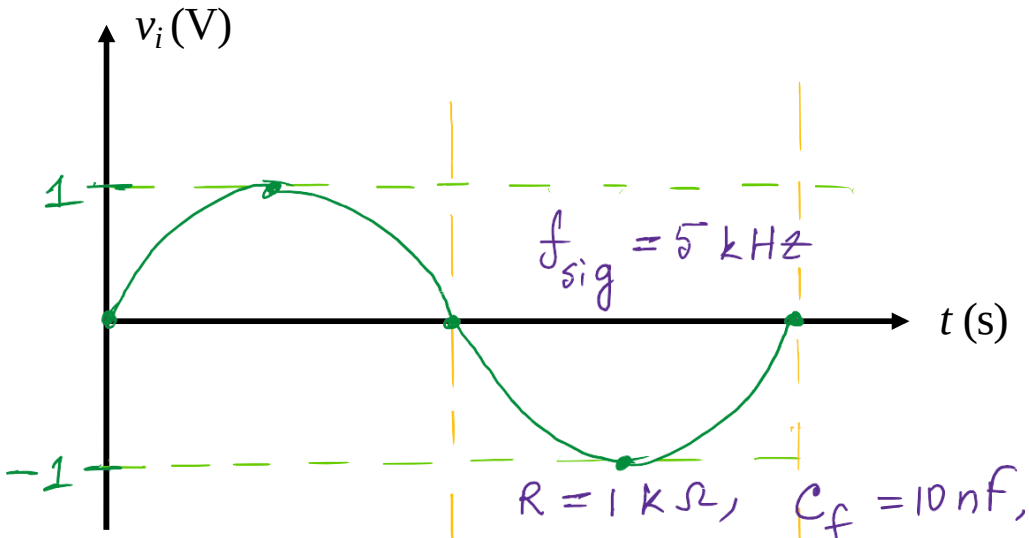
$v_o(0\,s) = 0\ V$

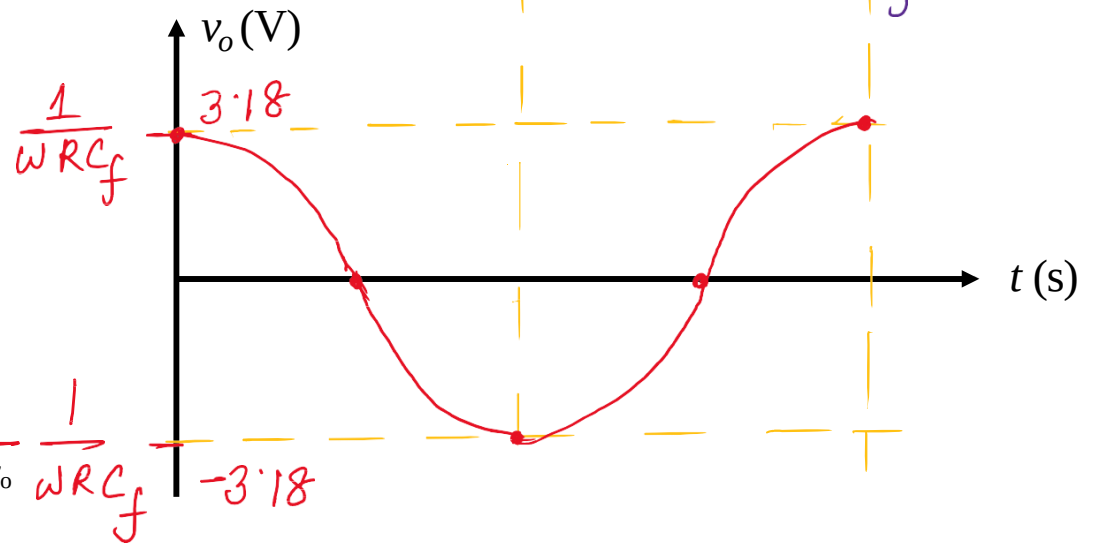
$$v_o(t) = -\frac{1}{RC_f}\int \sin(\omega t)\cdot dt$$

$$= -\frac{1}{RC_f}\left(-\frac{\cos\omega t}{\omega}\right)$$

$$= +\frac{1}{\omega RC_f}\cos\omega t$$







$$v_o = -\frac{1}{RC_f}\int v_i(t).dt$$

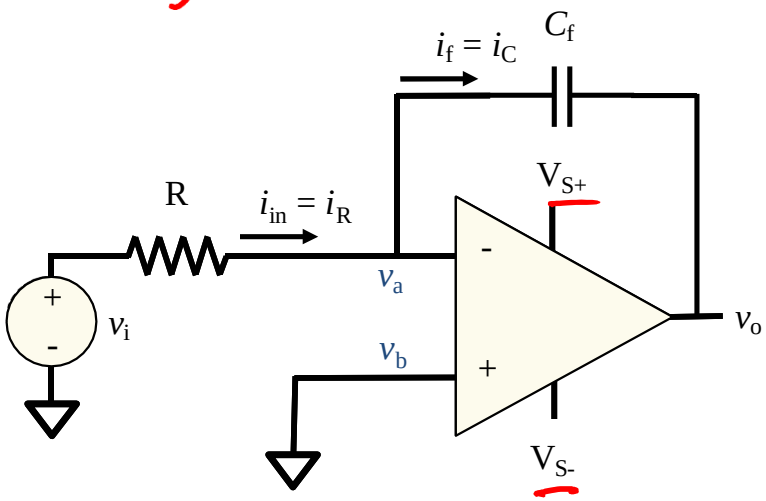
Basic Integrator

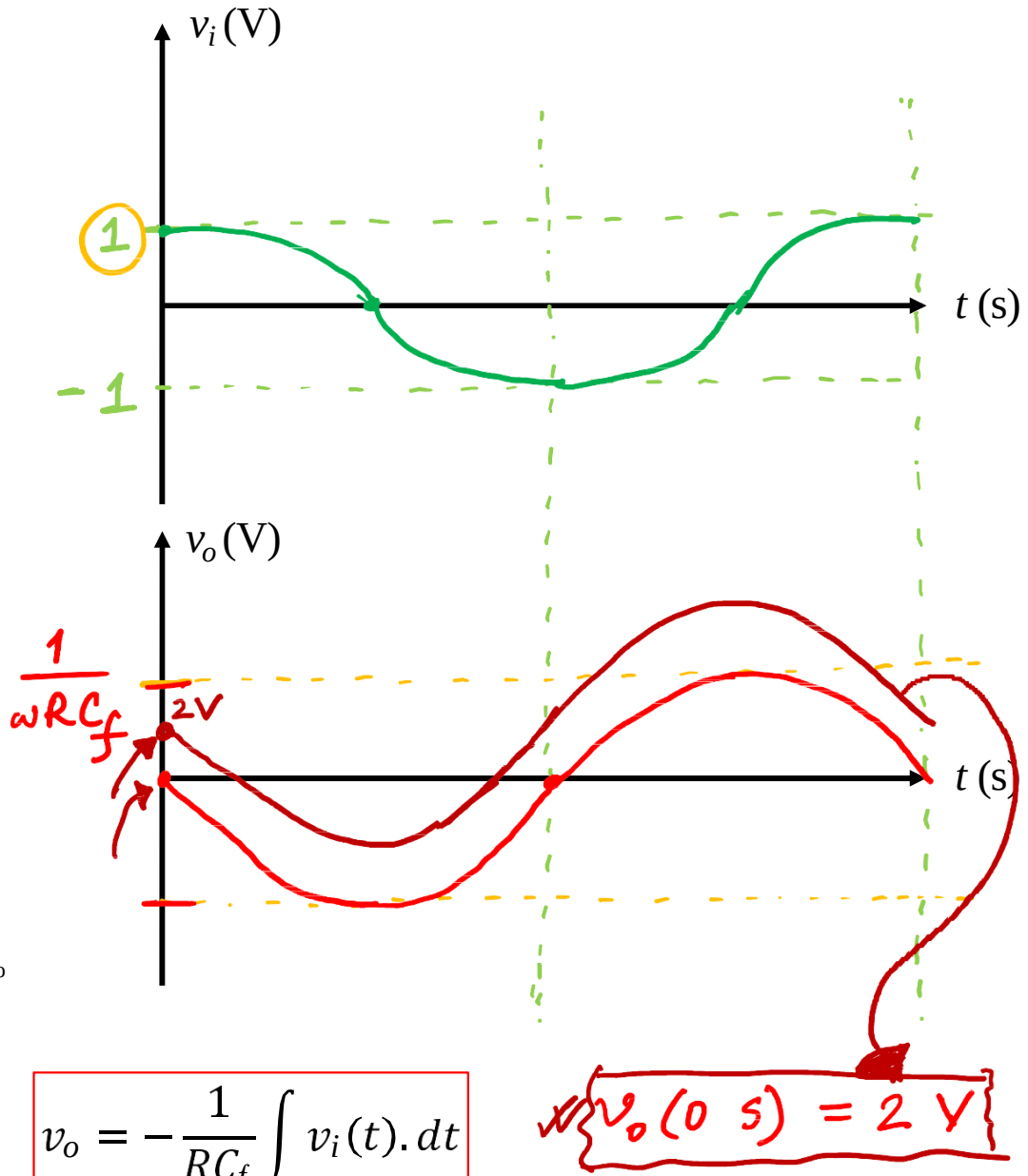
Example 3:

v_i is cos(xt) V.

$$v_o = -\frac{1}{RC_f}\int \cos(\omega t)\cdot dt$$

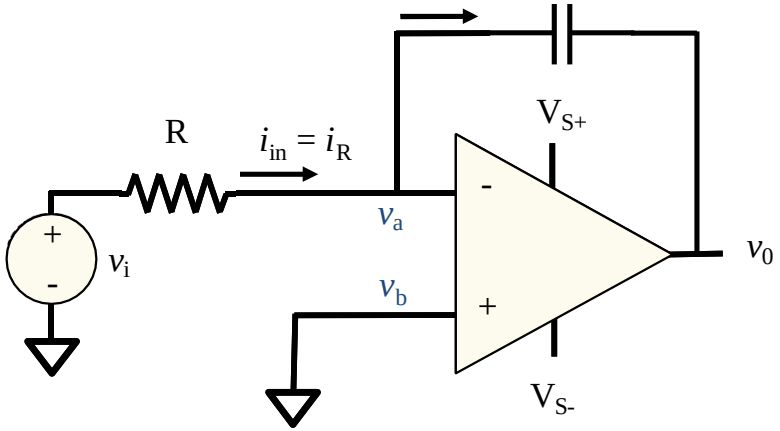
$$= \left(-\frac{1}{RC_f\omega}\right) \sin\omega t$$

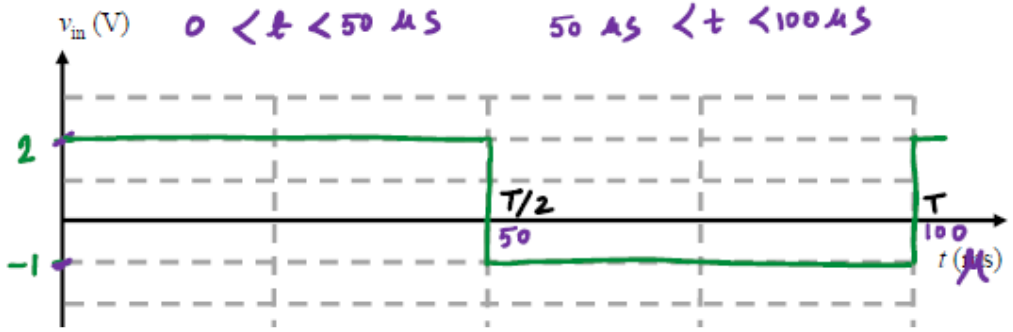




$$v_o = -\frac{1}{RC_f}\int v_i(t).dt$$

Basic Integrator





Example:

v_i is a square wave given in the figure.

v_0=?

Solution:

$C_f = 10\ nF$, $R = 1\ k\Omega$

$f = 10\ kHz$, $T = 100\ \mu s$

$$v_0(t) = -\frac{1}{RC_f}\int 2 \cdot dt, \quad 0 < t < 50\ \mu s$$

$$v_0(t) = \underbrace{\left(-\frac{2}{RC_f}\right)}_{m_1} \cdot t$$

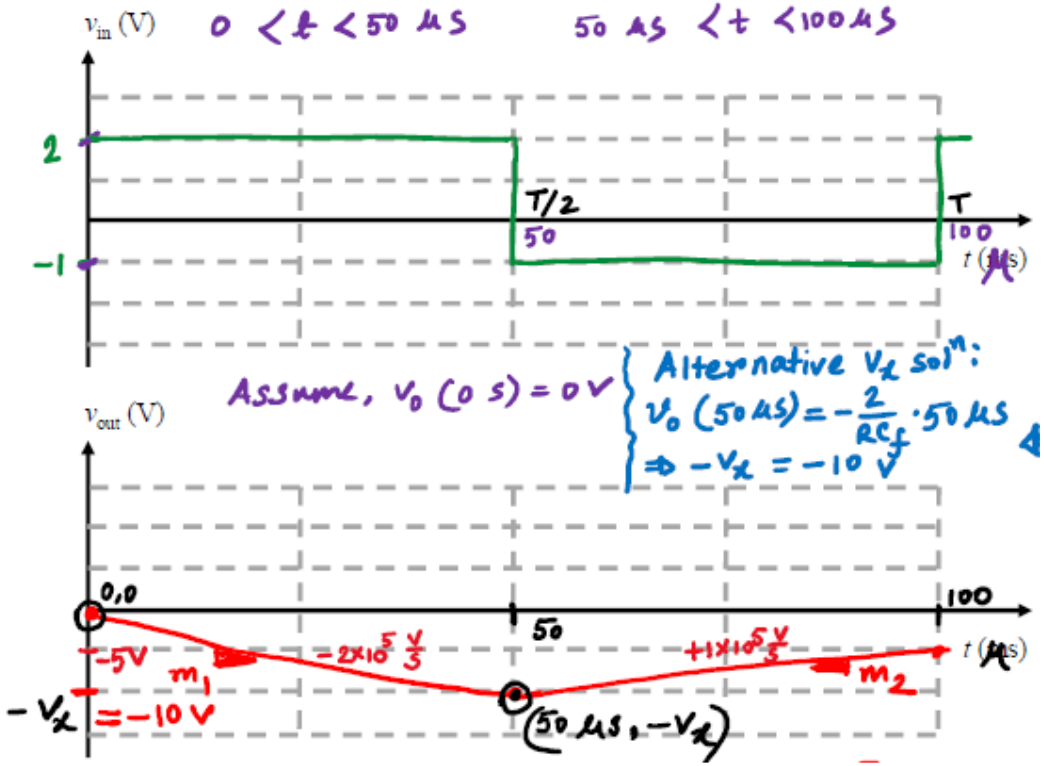
$$v_0(t) = -\frac{1}{RC_f}\int (-1) \cdot dt, \quad 50\ \mu s < t < 100\ \mu s$$

$$v_0(t) = \left(+\frac{1}{RC_f}\right) t$$

$$m_1 = -\frac{2}{RC_f} = -\frac{2}{1\ k\Omega \times 10\ nF} = -2 \times 10^5$$

$$\frac{-V_x - 0V}{50\ \mu s - 0} = -\frac{V_x}{50 \times 10^{-6}}$$

$$m_2 = +\frac{1}{RC_f} = +1 \times 10^5$$



$m_1 = -\frac{2}{RC_f} = -\frac{2}{1\,k\Omega \times 10\,nF} = -2\times10^5$

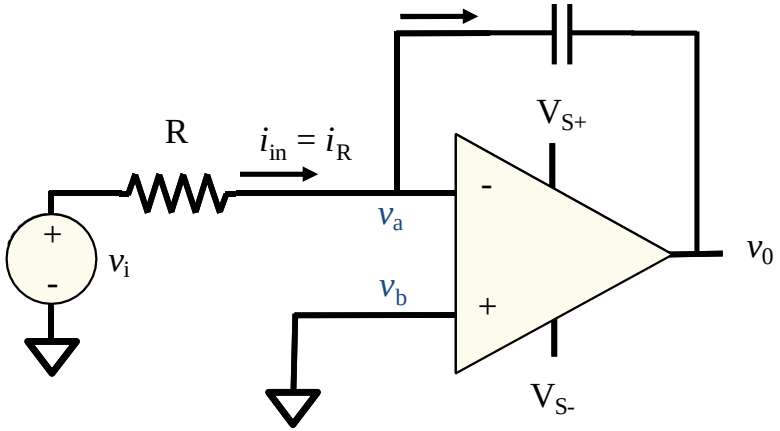
$\frac{-V_x - 0V}{50\,\mu s - 0} = -\frac{V_x}{50\times10^{-6}} = -2\times10^5$

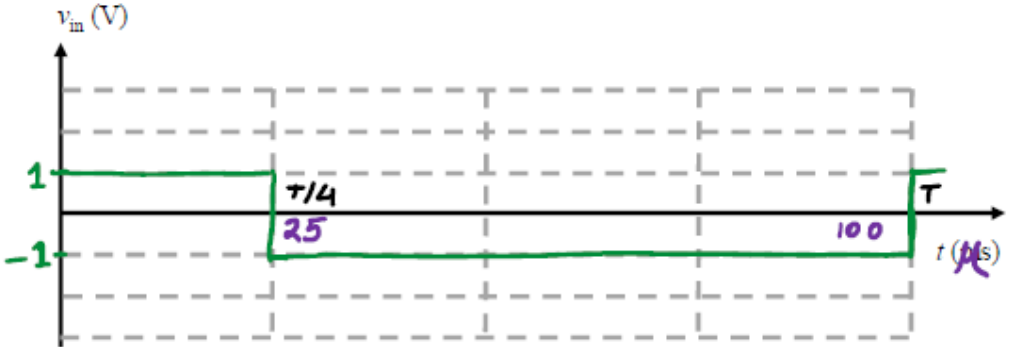
$V_x = 10V$

$m_2 = +\frac{1}{RC_f} = +1\times10^5$

$V_{x_2} = 5V$

Basic Integrator





Example:

v_i is a square wave given in the figure.

v_0=?

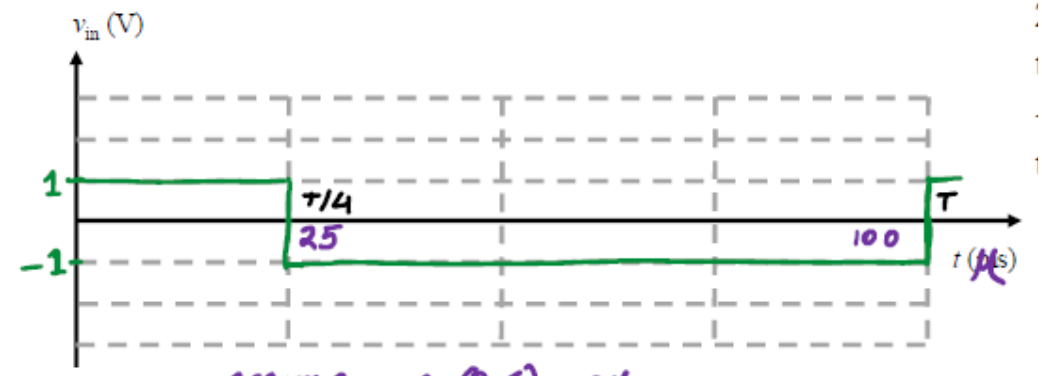
$v_0(t) = -\frac{1}{RC_f} \cdot t$, $0 < t < 25\,\mu s$

$v_0(t) = +\frac{1}{RC_f} \cdot t$, $25\,\mu s < t < 100\,\mu s$

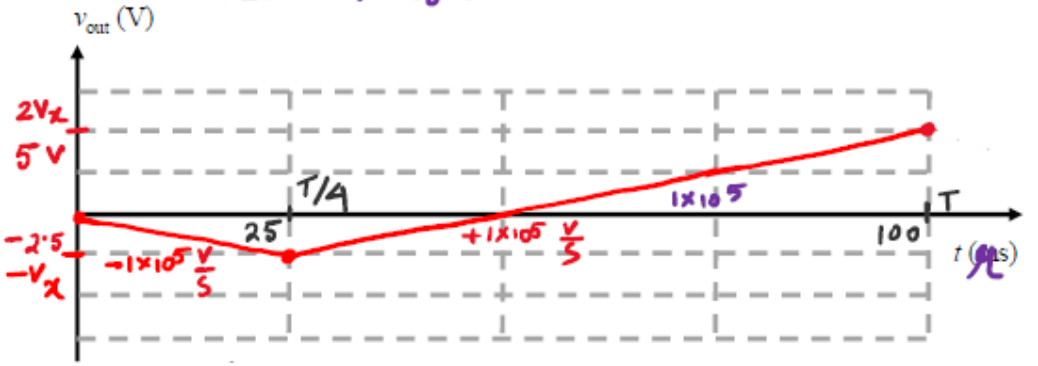
Calculation of $-V_x$:

$$v_0(25\,\mu s) = -\frac{1}{RC_f} \cdot (25\,\mu s) = -\frac{1}{1\,k\Omega \times 10\,nF} \cdot 25\,\mu s$$

$\Rightarrow$ $-V_x = -2.5\ V$ $\quad\therefore\quad$ $2V_x = +5\ V$



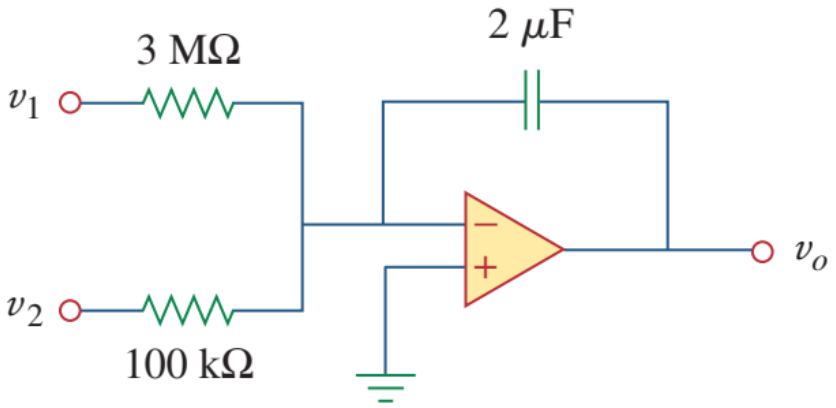
Assume, $v_o(0^-) = 0\,V$



Practice Problem

If $v_1 = 10 \cos 2t$ mV and $v_2 = 0.5t$ mV, find v_o in the op amp circuit

Assume that the voltage across the capacitor is initially zero.



Solution:

This is a summing integrator, and

$$v_o = -\frac{1}{R_1 C}\int v_1\,dt - \frac{1}{R_2 C}\int v_2\,dt$$

$$v_o = -\frac{1}{R_1 C}\int v_1\,dt - \frac{1}{R_2 C}\int v_2\,dt$$

$$= -\frac{1}{3\times 10^6 \times 2\times 10^{-6}}\int_0^t 10\cos(2\tau)d\tau$$

$$-\frac{1}{100\times 10^3 \times 2\times 10^{-6}}\int_0^t 0.5\tau d\tau$$

$$= -\frac{1}{6}\frac{10}{2}\sin 2t - \frac{1}{0.2}\frac{0.5t^2}{2} = -0.833\sin 2t - 1.25t^2\ \text{mV}$$

Practice Problem

Using a single op amp, a capacitor, and resistors of 100 kΩ or less, design a circuit to implement

$$v_0 = -50\int_0^t v_i(t)dt$$

Assume $v_o = 0$ at $t = 0$.

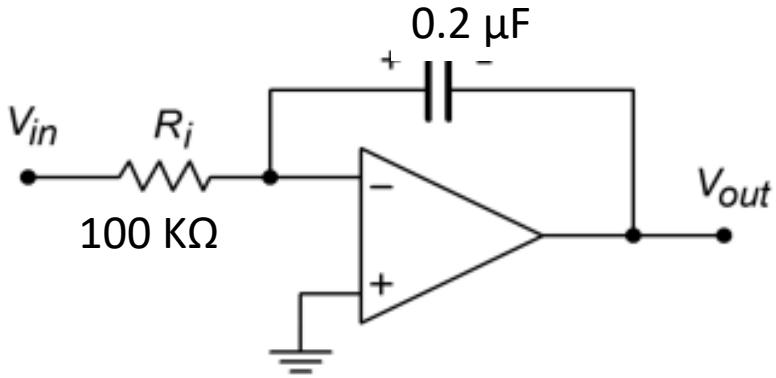
Solution:

$$V_{out}(t) = -\frac{1}{R_i C}\int V_{in}(t)\,dt$$

One possibility is as follows:

$$\frac{1}{RC} = 50$$

Let R = 100 kΩ, $C = \dfrac{1}{50\text{x}100\text{x}10^3} = 0.2\mu\text{F}$



Practice Problem

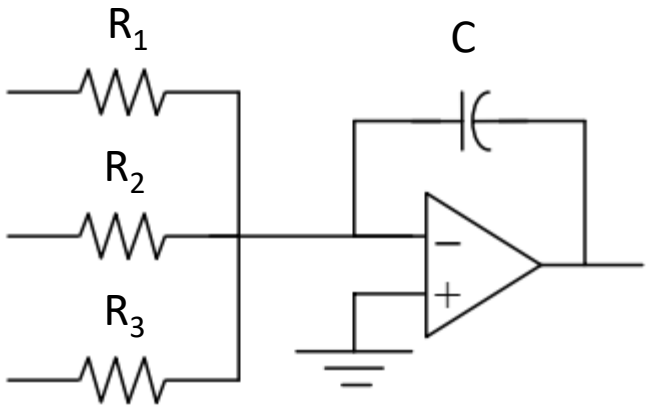
Show how you would use a single op amp to generate

$$v_0 = -\int (v_1 + 4v_2 + 10v_3)\,dt$$

If the integrating capacitor is $C = 2\ \mu$F, obtain other component values.

Solution:

By combining a summer with an integrator, we have the circuit below:



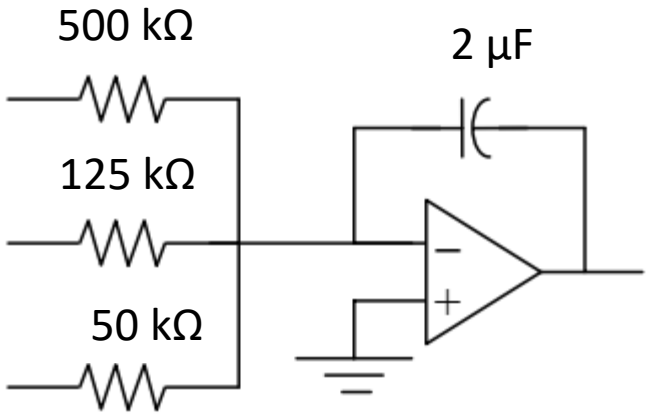
$$v_o = -\frac{1}{R_1 C}\int v_1 dt - \frac{1}{R_2 C}\int v_2 dt - \frac{1}{R_2 C}\int v_2 dt$$

For the given problem, C = **2μF**,

$R_1C = 1 \longrightarrow R_1 = 1/(C) = 10^6/(2) =$ **500 kΩ**

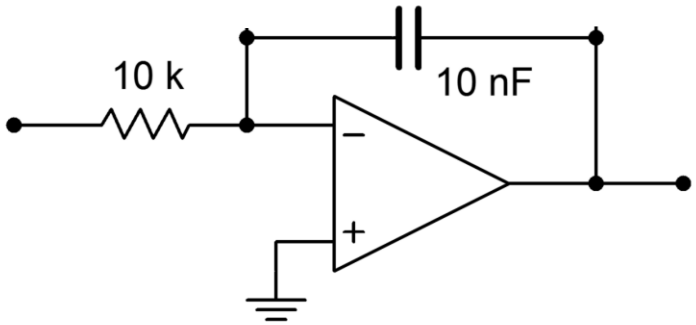
$R_2C = 1/(4) \longrightarrow R_2 = 1/(4C) = 500k\Omega/(4) =$ **125 kΩ**

$R_3C = 1/(10) \longrightarrow R_3 = 1/(10C) =$ **50 kΩ**



Practice Problem

Using the circuit give below, determine the output if the input is a 1 V peak sine wave at 5 kHz.



Solution:

The input signal as a function of time,

$$V_{in}(t) = 1 \sin 2\pi 5000 t$$

The output signal as a function of time,

$$V_{out}(t) = -\frac{1}{R_i C} \int V_{in}(t)\, dt$$

The output signal as a function of time,

$$V_{out}(t) = -\frac{1}{R_i C}\int V_{in}(t)\,dt$$

$$V_{out}(t) = -\frac{1}{10\,\mathrm{k}\times 10\,\mathrm{nF}}\int 1\sin 2\pi 5000\,t\,dt$$

$$V_{out}(t) = -10^4\int 1\sin 2\pi 5000\,t\,dt$$

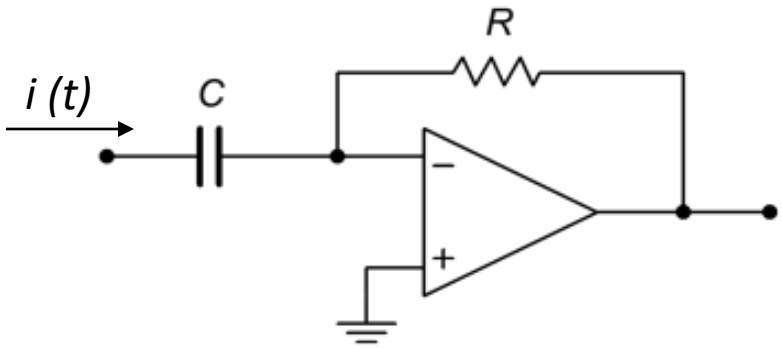
$$V_{out}(t) = -10^4\frac{1}{2\pi 5000}\int 2\pi 5000\sin 2\pi 5000\,t\,dt$$

$$V_{out}(t) = -0.318\left(-\cos 2\pi 5000\,t\right)$$

$$V_{out}(t) = 0.318\cos 2\pi 5000\,t$$

Differentiator

A **differentiator** is an op amp circuit whose output is proportional to the rate of change of the input signal.



$$i(t) = C\frac{dv(t)}{dt}$$

$$V_{out} = -V_{Rf} \qquad V_{Rf} = i\,R_f$$

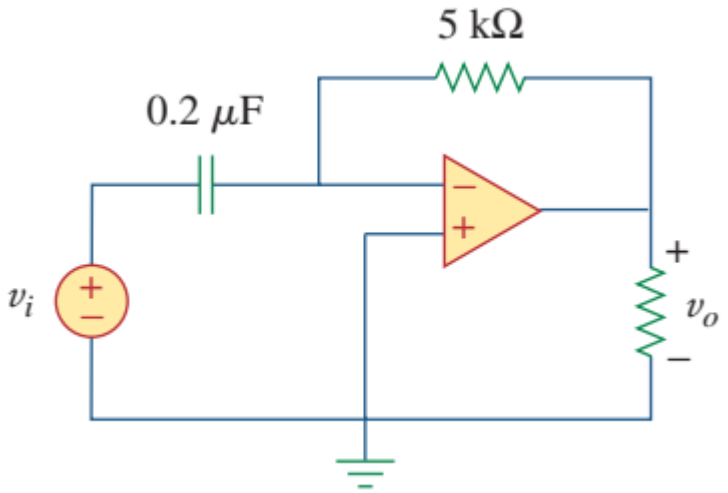
$$V_{out}(t) = -R_f C\frac{dv(t)}{dt}$$

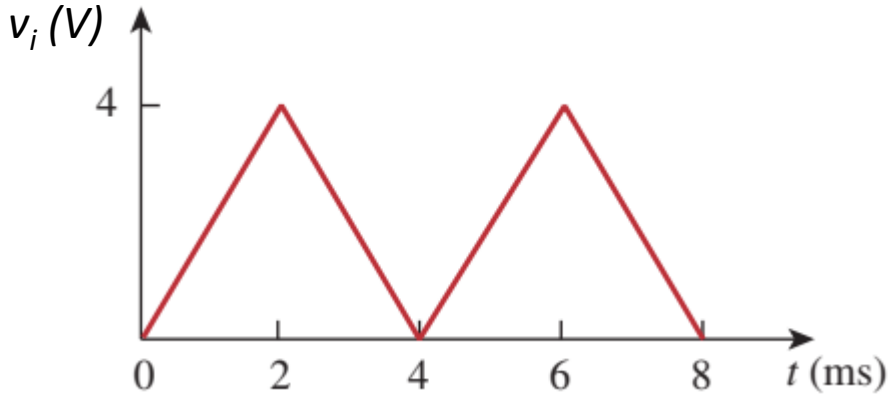
OR

$$v_o = -RC\frac{dv_i}{dt}$$

Practice Problem

Sketch the output voltage for the circuit, given the input voltage.
Take $v_0 = 0$ at $t = 0$.





Solution:

This is a differentiator with

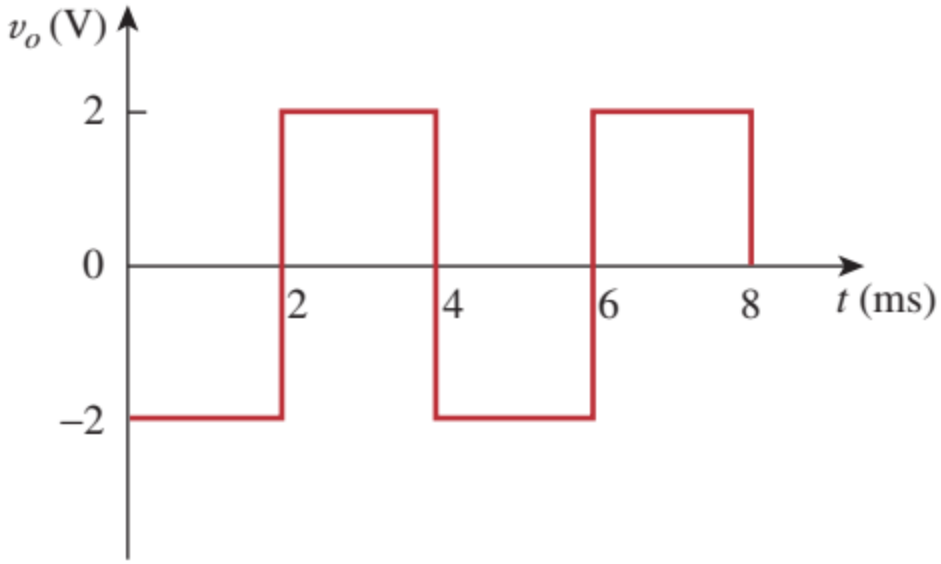
$$RC = 5 \times 10^3 \times 0.2 \times 10^{-6} = 10^{-3}\ \text{s}$$

For $0 < t < 4$ ms, we can express the input voltage

$$v_i = \begin{cases} 2000t & 0 < t < 2\ \text{ms} \\ 8 - 2000t & 2 < t < 4\ \text{ms} \end{cases}$$

This is repeated for $4 < t < 8$ ms.

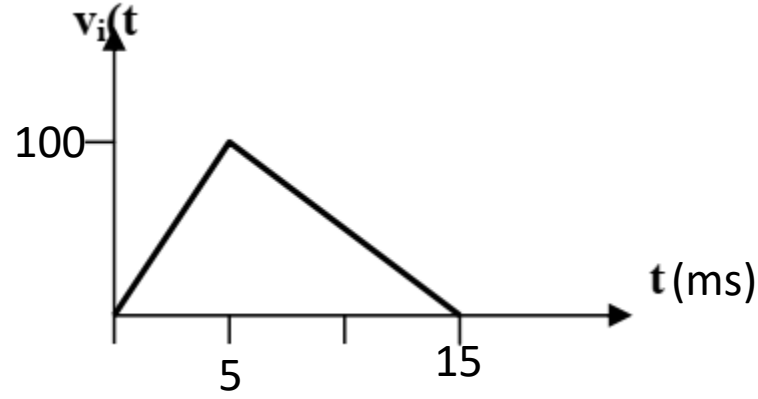
the output is $$v_o = -RC\frac{dv_i}{dt} = \begin{cases} -2\text{ V} & 0 < t < 2\text{ ms} \\ 2\text{ V} & 2 < t < 4\text{ ms} \end{cases}$$



Practice Problem

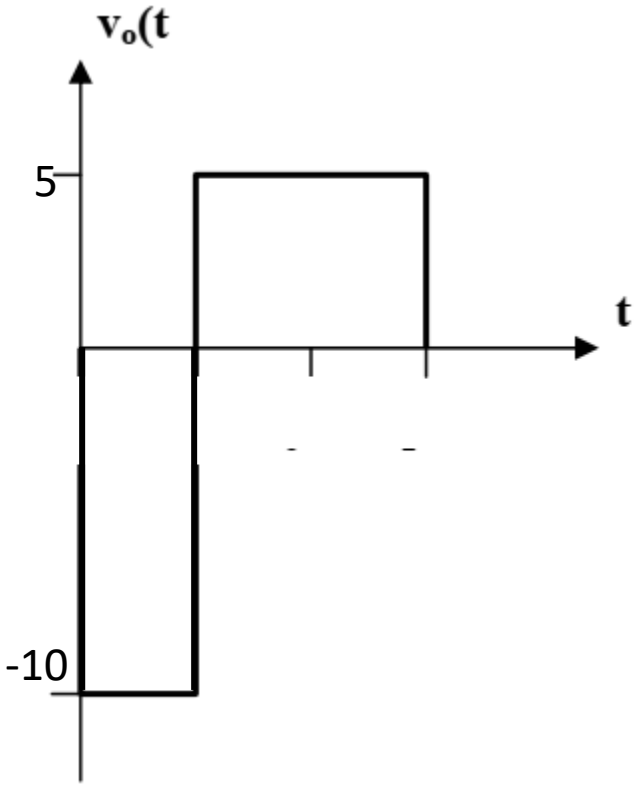
A voltage waveform has the following characteristics: a positive slope of 20 V/s for 5 ms followed by a negative slope of 10 V/s for 10 ms. If the waveform is applied to a differentiator with $\boldsymbol{R}$ = 50 kΩ, $\boldsymbol{C}$ = 10 μF, sketch the output voltage waveform.

Solution:



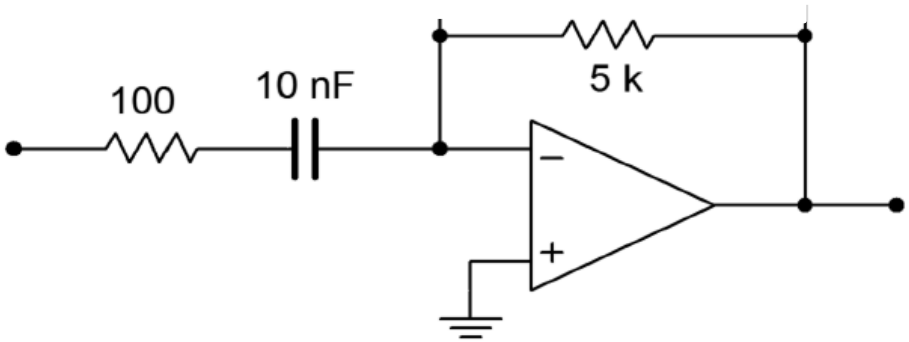
$$v_o = -RC\frac{dv_i}{dt},\quad RC = 50 \times 10^3 \times 10 \times 10^{-6} = 0.5$$

$$v_o = -0.5\frac{dv_i}{dt} = \begin{cases} -10, & 0 < t < 5 \\ 5, & 5 < t < 15 \end{cases}$$



Practice Problem

Sketch the output waveform for the circuit if the input is a 3 volt peak triangle wave at 4 kHz.



Solution:

The triangle wave may be broken into a positive-going portion and a negative-going portion. In either case, the total voltage change will be 6 V in one half-cycle. The period of the waveform is

$$T = \frac{1}{4\,\text{kHz}}$$

$$T = 250\,\mu\text{s}$$

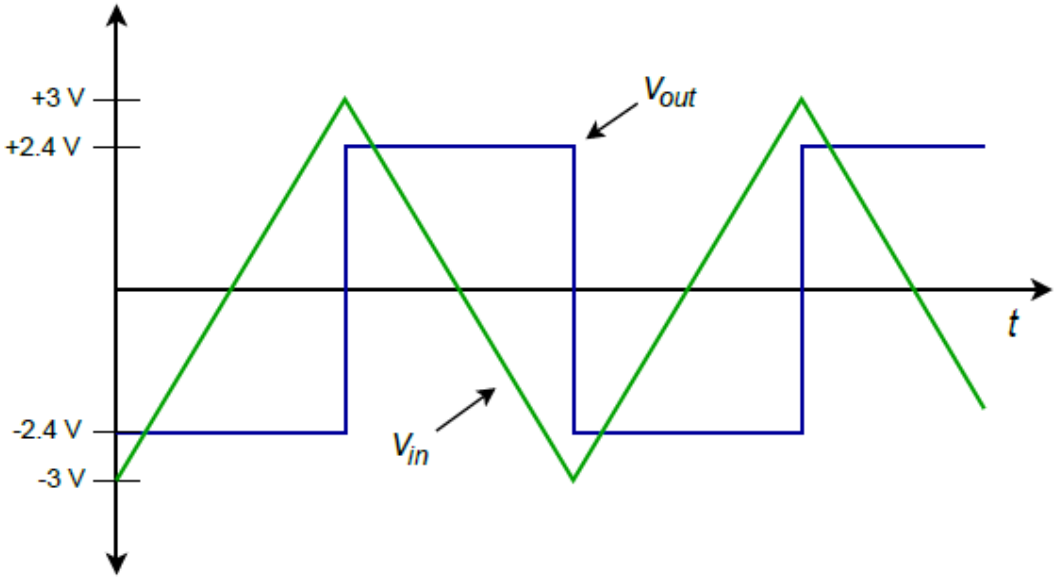
Therefore, for the positive going portion, a 6 volt change will be seen in 125 μs (−6 V in 125 μs for the negative going portion). The slope is

$$Slope = \frac{6\,\mathrm{V}}{125\,\mu\mathrm{s}}$$
$$Slope = 48000\,\mathrm{V/s}$$

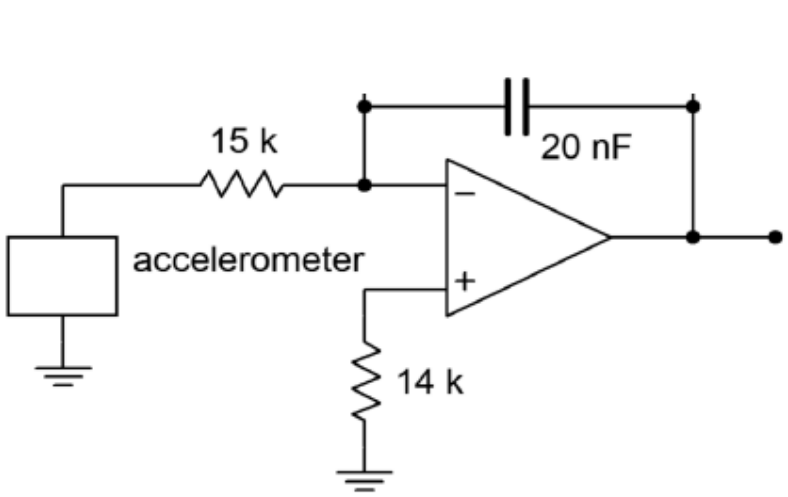
$$V_{in}(t) = 48000\,t$$

$$V_{out}(t) = -R_f C \frac{dV_{in}(t)}{dt}$$
$$V_{out}(t) = -5\,\mathrm{k} \times 10\,\mathrm{nF} \frac{d\,48000\,t}{dt}$$
$$V_{out}(t) = -2.4\,\mathrm{V}$$

During t = 0 through t = 125 µs, the output is −2.4 V. Differentiation of the second half of the wave is similar, but produces a positive output, +2.4 V. The result is a 4 kHz square wave that is 2.4 V peak. The resulting waveform is shown in Figure



Practice Problem



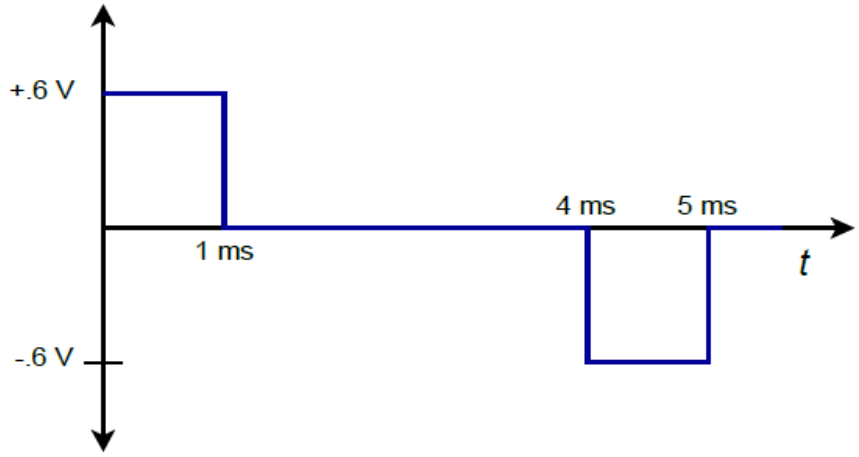


Figure shows an integrator connected to an accelerometer. If the accelerometer produces the voltage shown in Figure, determine the shape of the velocity curve.

Solution:

For the positive pulse (t = 0 to t =1 ms),

$$V_{out}(t) = -\frac{1}{R_i C}\int V_{in}(t)\,dt$$

$$V_{out}(t) = -\frac{1}{15\,\text{k}\times 20\,\text{nF}}\int_0^{10^{-3}} .6\,dt$$

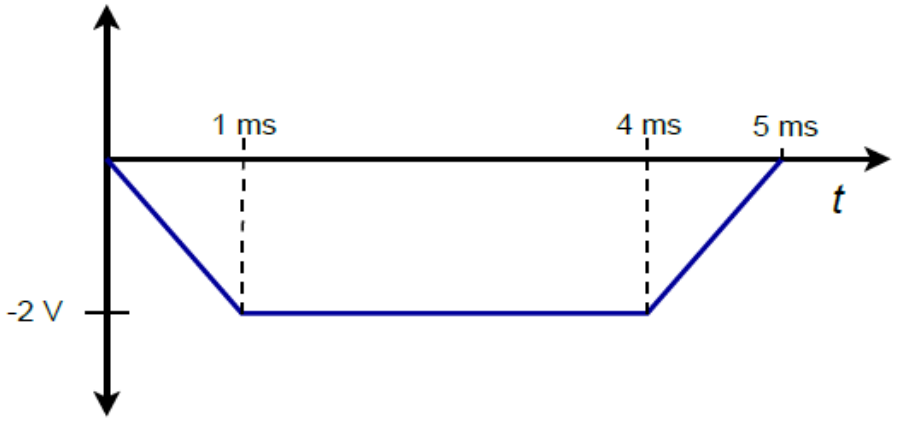
$$V_{out}(t) = -3333\times 0.6\times t\Big|_{t=0}^{t=10^{-3}}$$

$$V_{out} = -2000\times 10^{-3}$$

$$V_{out} = -2\,\text{V}$$

For the time period between 1 ms and 4 ms, the input signal is zero, and thus the change in output potential will be zero.

For the period between 4 ms and 5 ms, a positive going ramp will be produced. Because only the polarity is changed relative to the first pulse, we can quickly find that the change is +2 V.



Analog Electronics II: Op-amp & 555 Timer

Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

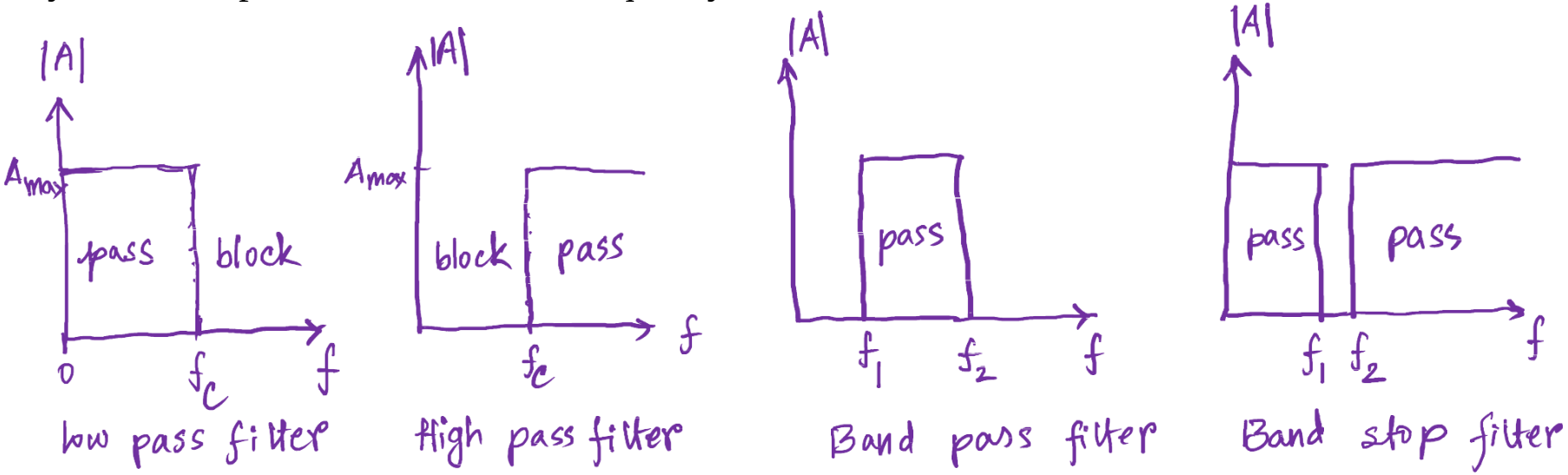
Lecture # 9

Frequency

The **frequency response** of a circuit is the variation in its behavior with change in signal frequency.

The sinusoidal steady-state frequency responses of circuits are of significance in many applications, especially in communications and control systems.

A specific application is in electric filters that block out or eliminate signals with unwanted frequencies and pass signals of the desired frequencies. Filters are used in radio, TV, and telephone systems to separate one broadcast frequency from another.



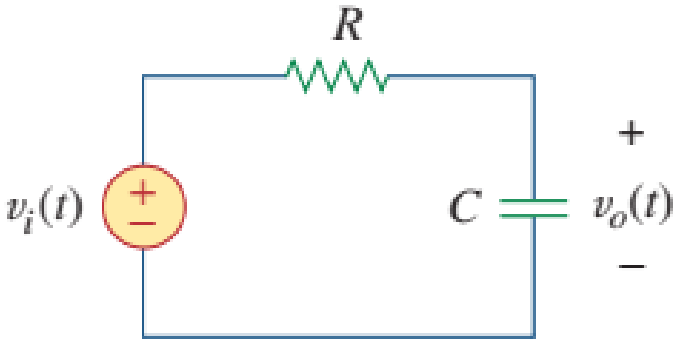
Source: Alexander-Sadiku Chapter 14

Active Filters vs Passive Filters

Basis for Comparison	Active Filter	Passive Filter
Composed of	Active components like op-amp, transistor etc.	Passive components like resistor, inductor and capacitor etc.
Cost	High	Comparatively low.
Circuit complexity	More complex	Less complex than active filter.
Weight	Low	Comparatively bulkier due to presence of inductors.
Q factor	High	Very low in comparison to active filters.
External power supply	Required	Not required
Sensitivity	More sensitive	Comparatively less sensitive.

Passive Filters: Lowpass Filter

A typical lowpass filter is formed when the output of an *RC* circuit taken off the capacitor



The transfer function $$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{1/j\omega C}{R + 1/j\omega C}$$

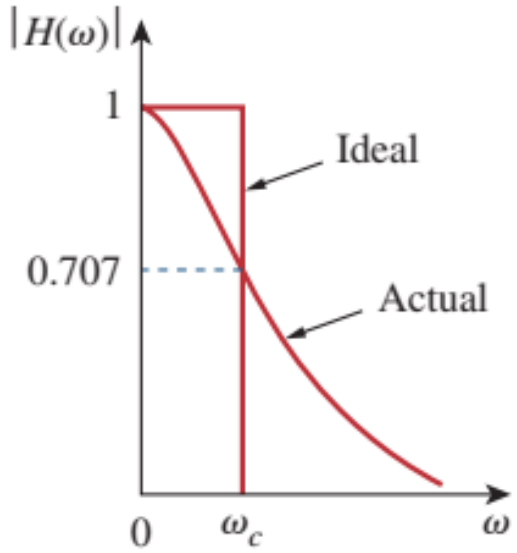
$$\mathbf{H}(\omega) = \frac{1}{1 + j\omega RC}$$

Note that $\mathbf{H}(0) = 1, \mathbf{H}(\infty) = 0.$

the *cutoff frequency* ω_c, is obtained by setting the magnitude of $\mathbf{H}(\omega)$ equal to $1/\sqrt{2}$,

$$H(\omega_c) = \frac{1}{\sqrt{1 + \omega_c^2 R^2 C^2}} = \frac{1}{\sqrt{2}}$$

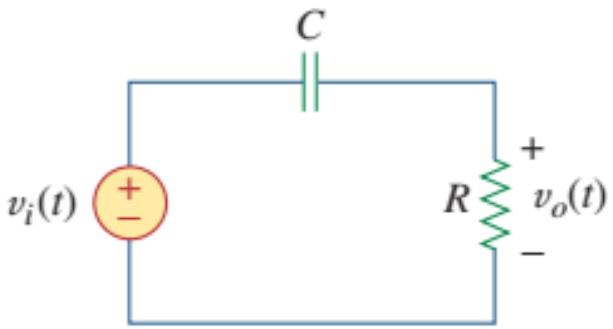
$$\omega_c = \frac{1}{RC}$$



A lowpass filter is designed to pass only frequencies from dc up to the cutoff frequency ω_c.

Passive Filters: Highpass Filter

A highpass filter is formed when the output of an *RC* circuit is taken off the resistor

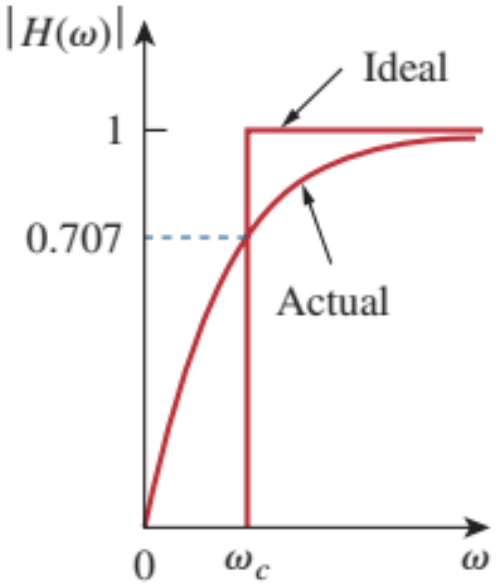


The transfer function is

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R}{R + 1/j\omega C}$$

$$\mathbf{H}(\omega) = \frac{j\omega RC}{1 + j\omega RC}$$

Note that $\mathbf{H}(0) = 0$, $\mathbf{H}(\infty) = 1$



the corner or cutoff frequency is

$$\omega_c = \frac{1}{RC}$$

A highpass filter is designed to pass all frequencies above its cutoff frequency ω_c.

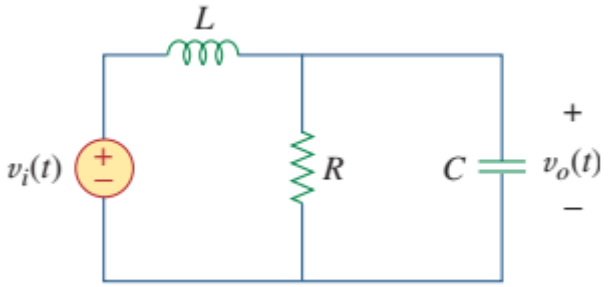
Passive Filters:

A lowpass filter can also be formed when the output of an *RL* circuit is taken off the resistor. Of course, there are many other circuits for lowpass filters.

A highpass filter can also be formed when the output of an *RL* circuit is taken off the inductor.

Example Problem 14.10

Determine what type of filter is shown in Fig. 14.39. Calculate the corner or cutoff frequency. Take $R = 2\ \text{k}\Omega$, $L = 2\ \text{H}$, and $C = 2\ \mu\text{F}$.



Solution: The transfer function is

$$\mathbf{H}(s) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R \parallel 1/sC}{sL + R \parallel 1/sC}, \qquad s = j\omega \qquad \textbf{(14.10.1)}$$

But

$$R \parallel \frac{1}{sC} = \frac{R/sC}{R + 1/sC} = \frac{R}{1 + sRC}$$

Substituting this into Eq. (14.10.1) gives

$$\mathbf{H}(s) = \frac{R/(1 + sRC)}{sL + R/(1 + sRC)} = \frac{R}{s^2RLC + sL + R}, \qquad s = j\omega$$

or

$$\mathbf{H}(\omega) = \frac{R}{-\omega^2RLC + j\omega L + R} \tag{14.10.2}$$

Since $\mathbf{H}(0) = 1$ and $\mathbf{H}(\infty) = 0$, we conclude from Table 14.5 that the circuit in Fig. 14.39 is a second-order lowpass filter. The magnitude of $\mathbf{H}$ is

$$H = \frac{R}{\sqrt{(R - \omega^2RLC)^2 + \omega^2L^2}} \tag{14.10.3}$$

The corner frequency is the same as the half-power frequency, i.e., where $\mathbf{H}$ is reduced by a factor of $1/\sqrt{2}$. Since the dc value of $H(\omega)$ is 1, at the corner frequency, Eq. (14.10.3) becomes after squaring

$$H^2 = \frac{1}{2} = \frac{R^2}{(R - \omega_c^2RLC)^2 + \omega_c^2L^2}$$

or

$$2 = (1 - \omega_c^2LC)^2 + \left(\frac{\omega_cL}{R}\right)^2$$

Substituting the values of R, L, and C, we obtain

$$2 = (1 - \omega_c^2\, 4 \times 10^{-6})^2 + (\omega_c\, 10^{-3})^2$$

Assuming that ω_c is in krad/s,

$$2 = (1 - 4\omega_c^2)^2 + \omega_c^2 \qquad \text{or} \qquad 16\omega_c^4 - 7\omega_c^2 - 1 = 0$$

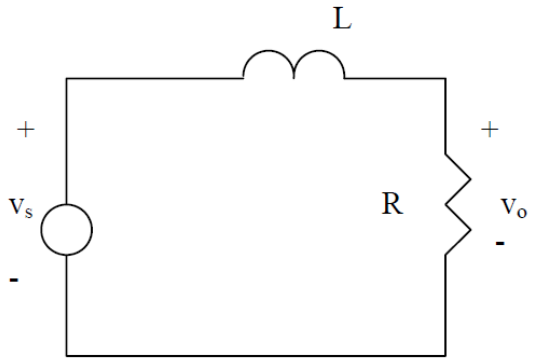
Solving the quadratic equation in ω_c^2, we get $\omega_c^2 = 0.5509$ and -0.1134. Since ω_c is real,

$$\omega_c = 0.742 \text{ krad/s} = 742 \text{ rad/s}$$

Practice Problem

Design an *RL* lowpass filter that uses a 40-mH coil and has a cutoff frequency of 5 kHz.

Solution:



The lowpass RL filter

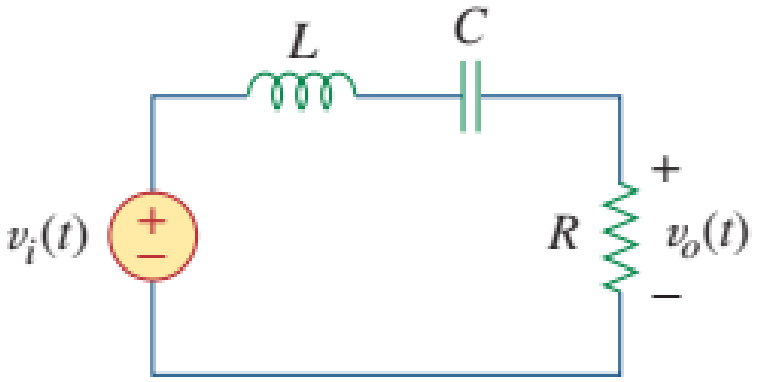
$$H = \frac{V_o}{V_s} = \frac{R}{R + j\omega L} = \frac{1}{1 + j\omega L / R}$$

$$\omega_c = \frac{R}{L} = 2\pi f_c$$

$$R = 2\pi f_c L = 2\pi x 5 x 10^3 \, x 40 x 10^{-3} = \underline{1.256 k\Omega}$$

Passive Filters: Bandpass Filter

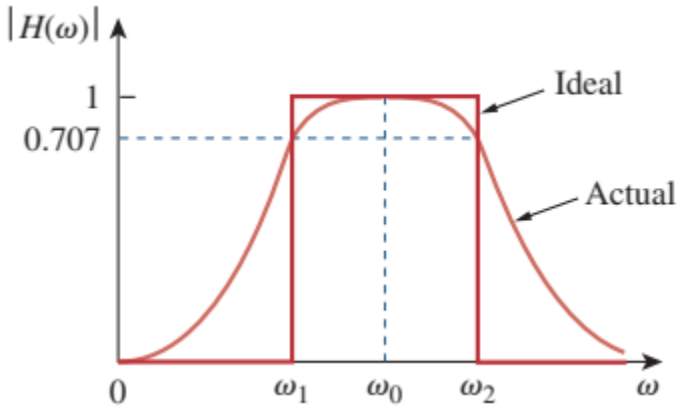
The *RLC* series resonant circuit provides a bandpass filter when the output is taken off the resistor



The transfer function is

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{R}{R + j(\omega L - 1/\omega C)}$$

We observe that $\mathbf{H}(0) = 0$, $\mathbf{H}(\infty) = 0$.

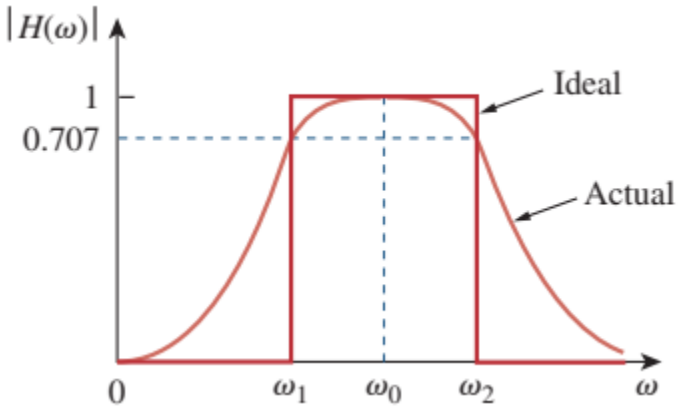


The bandpass filter passes a band of frequencies ($\omega_1 < \omega < \omega_2$) centered on ω_0, the center frequency, which is given by

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

A **bandpass filter** is designed to pass all frequencies within a band of frequencies, $\omega_1 < \omega < \omega_2$.

Bandpass Filter



ω_1 and ω_2 are called the *half-power frequencies.*

The half-power frequencies are obtained by setting input impedance Z equal to $\sqrt{2}R$,

$$\mathbf{Z} = \mathbf{H}(\omega) = \frac{\mathbf{V}_s}{\mathbf{I}} = R + j\omega L + \frac{1}{j\omega C}$$

$$\mathbf{Z} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \sqrt{2}R$$

Solving for ω, we obtain

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

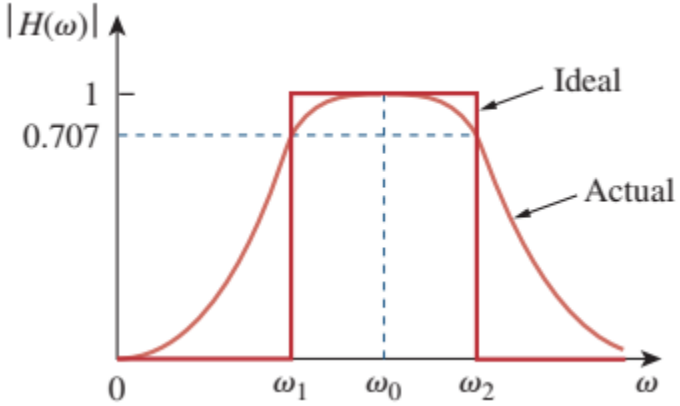
$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

the resonant frequency. $\omega_0 = \sqrt{\omega_1 \omega_2}$

Bandpass Filter

the *bandwidth* B, which is defined as the difference between the two half-power frequencies,

$$B = \omega_2 - \omega_1$$



The "sharpness" of the resonance in a resonant circuit is measured quantitatively by the *quality factor* Q.

$$Q = 2\pi \frac{\text{Peak energy stored in the circuit}}{\text{Energy dissipated by the circuit in one period at resonance}}$$

$$Q = 2\pi \frac{\frac{1}{2}LI^2}{\frac{1}{2}I^2R(1/f_0)} = \frac{2\pi f_0 L}{R} \qquad Q = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 CR}$$

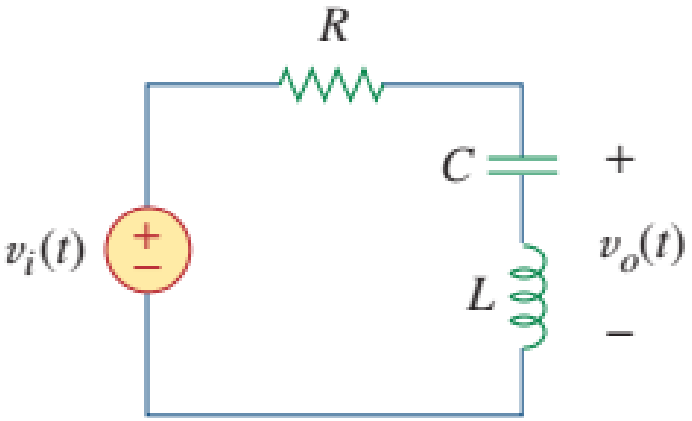
The relationship between the bandwidth B and the quality factor Q

$$B = \frac{R}{L} = \frac{\omega_0}{Q}$$

or $B = \omega_0^2 CR$.

Passive Filters: Bandstop Filter

A filter that prevents a band of frequencies between two designated values (ω_1 and ω_2) from passing is variably known as a *bandstop, band-reject*, or *notch* filter. A bandstop filter is formed when the output *RLC* series resonant circuit is taken off the *LC* series combination



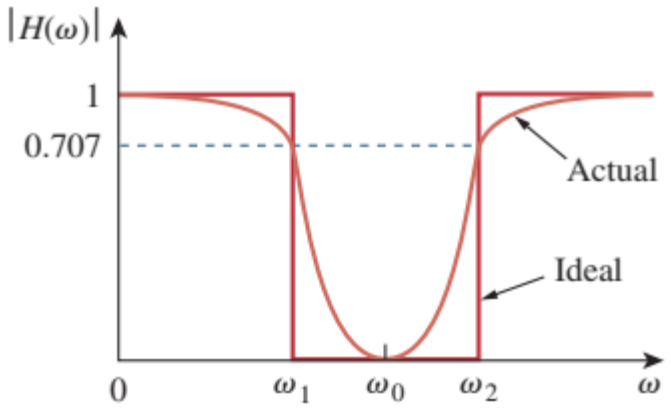
The transfer function is

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \frac{j(\omega L - 1/\omega C)}{R + j(\omega L - 1/\omega C)}$$

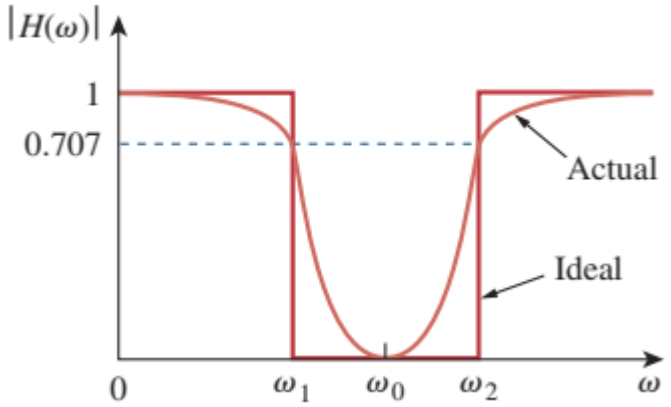
Notice that $\mathbf{H}(0) = 1$, $\mathbf{H}(\infty) = 1$.

Again, the center frequency is given by

$$\omega_0 = \frac{1}{\sqrt{LC}}$$



Bandstop Filter



ω_1 and ω_2 are called the *half-power frequencies.*

The half-power frequencies are obtained by setting input impedance Z equal to $\sqrt{2}R$,

$$\mathbf{Z} = \mathbf{H}(\omega) = \frac{\mathbf{V}_s}{\mathbf{I}} = R + j\omega L + \frac{1}{j\omega C}$$

$$\mathbf{Z} = R + j\left(\omega L - \frac{1}{\omega C}\right)$$

$$\sqrt{R^2 + \left(\omega L - \frac{1}{\omega C}\right)^2} = \sqrt{2}R$$

Solving for ω, we obtain

$$\omega_1 = -\frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

$$\omega_2 = \frac{R}{2L} + \sqrt{\left(\frac{R}{2L}\right)^2 + \frac{1}{LC}}$$

the resonant frequency. $\omega_0 = \sqrt{\omega_1 \omega_2}$

the *bandwidth* B, which is defined as the difference between the two half-power frequencies,

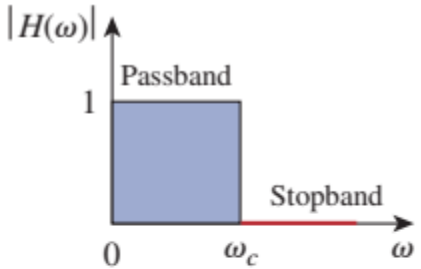
$$B = \omega_2 - \omega_1$$

A **bandstop filter** is designed to stop or eliminate all frequencies within a band of frequencies, $\omega_1 < \omega < \omega_2$.

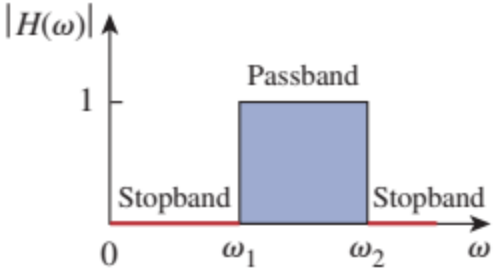
TABLE 14.5

Summary of the characteristics of ideal filters.

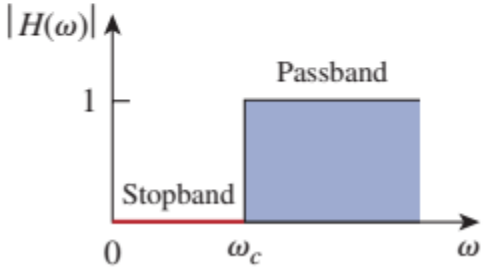
Type of Filter	$H(0)$	$H(\infty)$	$H(\omega_c)$ or $H(\omega_0)$
Lowpass	1	0	$1/\sqrt{2}$
Highpass	0	1	$1/\sqrt{2}$
Bandpass	0	0	1
Bandstop	1	1	0



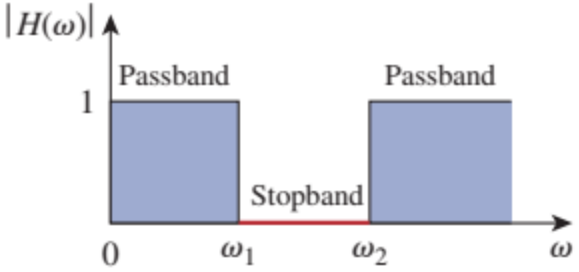
(a)



(c)



(b)



(d)

Example Problem 14.11

If the bandstop filter in Fig. 14.37 is to reject a 200-Hz sinusoid while passing other frequencies, calculate the values of L and C. Take $R = 150\ \Omega$ and the bandwidth as 100 Hz.

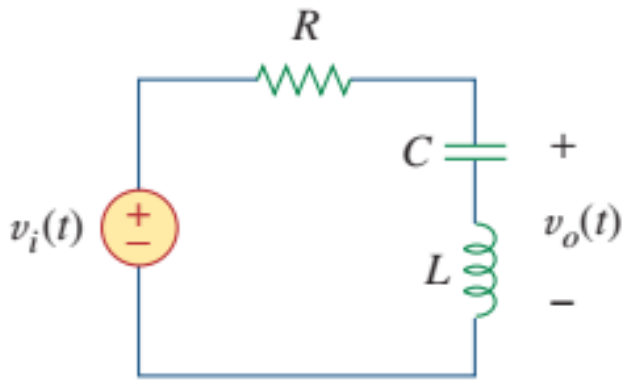


Figure 14.37
A bandstop filter.

Solution:

We use the formulas for a series resonant circuit in Section 14.5.

$$B = 2\pi(100) = 200\pi \text{ rad/s}$$

But

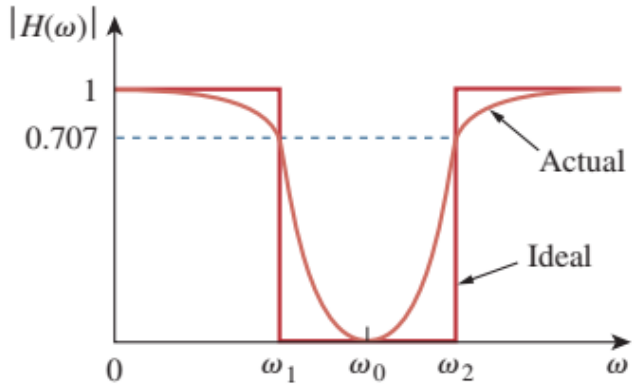
$$B = \frac{R}{L} \quad \Rightarrow \quad L = \frac{R}{B} = \frac{150}{200\pi} = 0.2387 \text{ H}$$

Rejection of the 200-Hz sinusoid means that f_0 is 200 Hz,

$$\omega_0 = 2\pi f_0 = 2\pi(200) = 400\pi$$

Since $\omega_0 = 1/\sqrt{LC}$,

$$C = \frac{1}{\omega_0^2 L} = \frac{1}{(400\pi)^2(0.2387)} = 2.653\ \mu\text{F}$$



Analog Electronics II: Op-amp & 555 Timer

Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

Lecture # 10

Unit 3: Filters Reference

1. Operarional Amplifiers and Linear Integrated Circuits by Robert Coughlin- Chapter-11

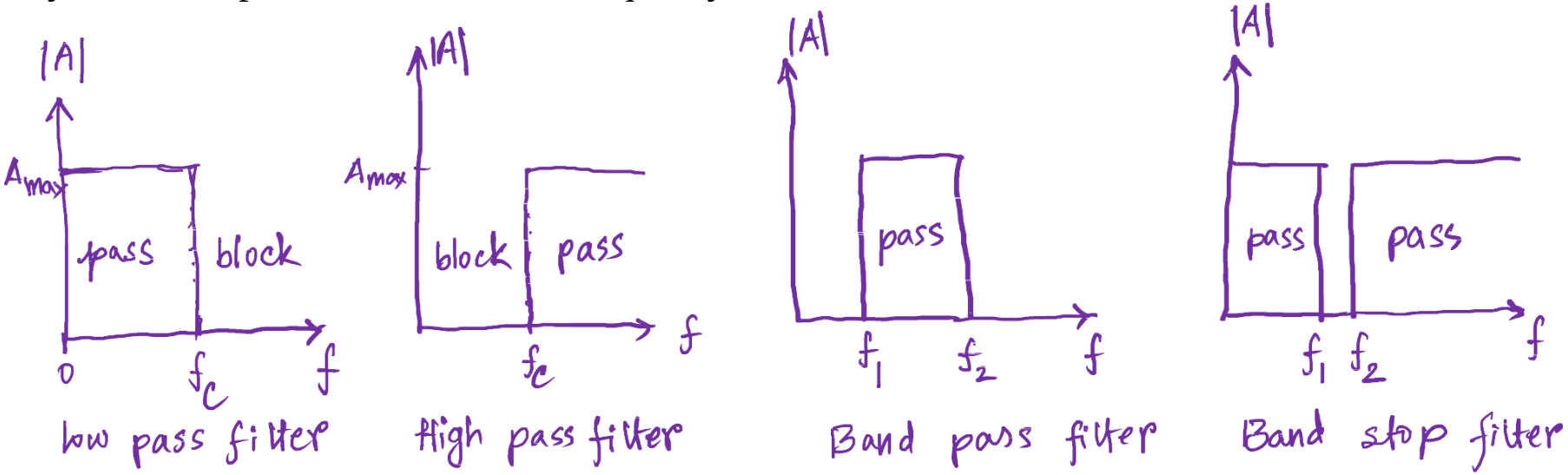
2. Fundamentals of Electric Circuits by Alexander Sadiku- Chapter-14

Frequency

The **frequency response** of a circuit is the variation in its behavior with change in signal frequency.

The sinusoidal steady-state frequency responses of circuits are of significance in many applications, especially in communications and control systems.

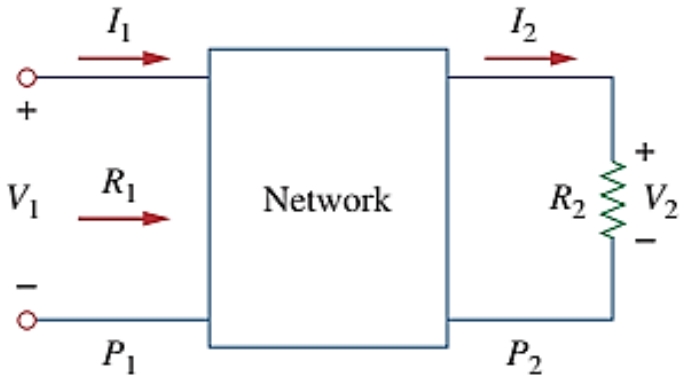
A specific application is in electric filters that block out or eliminate signals with unwanted frequencies and pass signals of the desired frequencies. Filters are used in radio, TV, and telephone systems to separate one broadcast frequency from another.



Source: Alexander-Sadiku Chapter 14

Decibel

Alternatively, the gain G can be expressed in terms of voltage or current ratio.



$$G_{\text{dB}} = 10 \log_{10} \frac{P_2}{P_1} = 10 \log_{10} \frac{V_2^2/R_2}{V_1^2/R_1}$$

$$= 10 \log_{10} \left(\frac{V_2}{V_1} \right)^2 + 10 \log_{10} \frac{R_1}{R_2}$$

$$G_{\text{dB}} = 20 \log_{10} \frac{V_2}{V_1} - 10 \log_{10} \frac{R_2}{R_1}$$

For the case when $R_2 = R_1$, $\quad G_{\text{dB}} = 20 \log_{10} \dfrac{V_2}{V_1}$

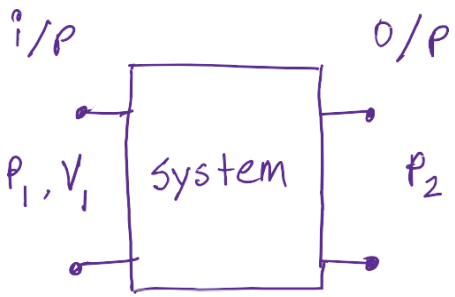
if $P_1 = I_1^2 R_1$ and $P_2 = I_2^2 R_2$,

for $R_1 = R_2$,

$$G_{\text{dB}} = 20 \log_{10} \frac{I_2}{I_1}$$

Source: Alexander-Sadiku Chapter 14

Decibel



P_1 = input power

P_2 = output ”

V_1 = input voltage

V_2 = output ”

The *decibel* (dB) provides us with a unit of less magnitude. It is 1/10th of a bel and is given by

$$G_{\text{dB}} = 10 \log_{10} \frac{P_2}{P_1} \tag{14.5}$$

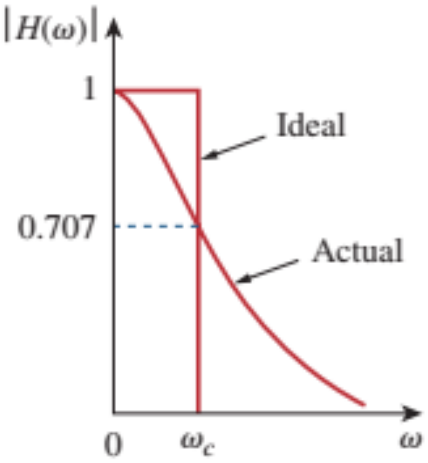
When $P_1 = P_2$, there is no change in power and the gain is 0 dB. If $P_2 = 2P_1$, the gain is

$$G_{\text{dB}} = 10 \log_{10} 2 \simeq 3 \text{ dB} \tag{14.6}$$

and when $P_2 = 0.5P_1$, the gain is

$$G_{\text{dB}} = 10 \log_{10} 0.5 \simeq -3 \text{ dB} \tag{14.7}$$

The half-power frequency, which is equivalent to the corner frequency on the Bode plots in the context of filters is usually known as the cutoff frequency is obtained by setting the magnitude of equal to $1/\sqrt{2}$

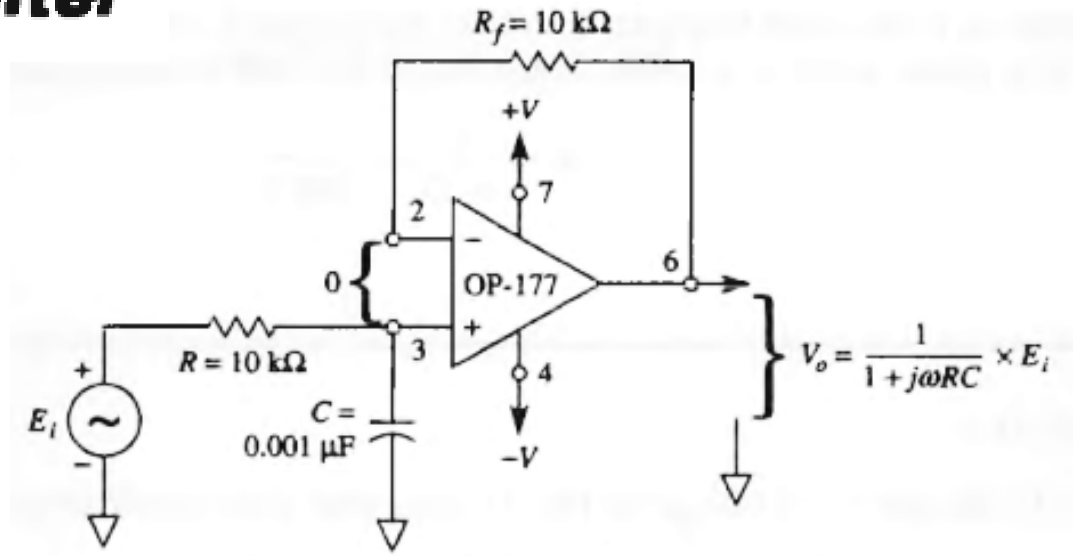


Source: Alexander-Sadiku Chapter 14

Active Filters: First-Order Lowpass Filter

BUTTERWORTH FILTER

– 20-dB/Decade Filter



(a) Low-pass filter for a roll-off of −20 dB/decade.

Designing the Filter

$$V_o = \frac{1/j\omega C}{R + 1/j\omega C} \times E_i$$

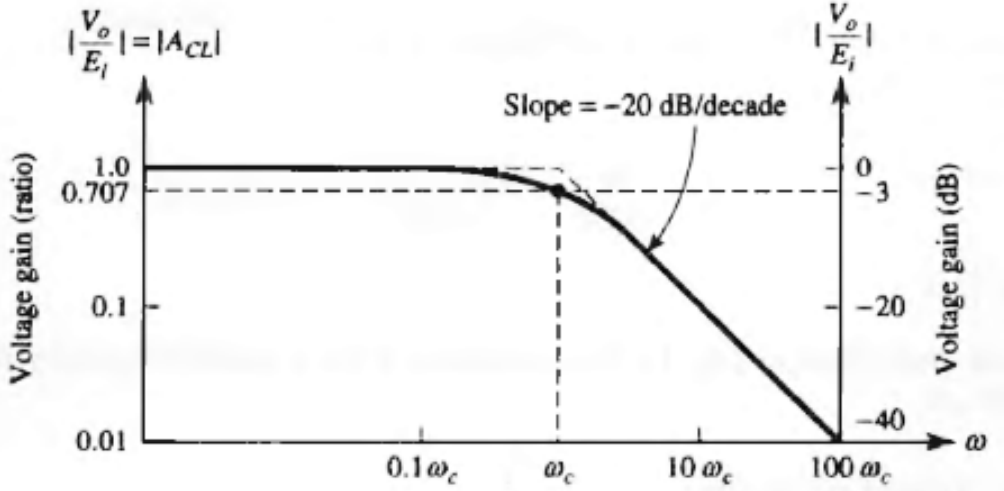
$$A_{CL} = \frac{V_o}{E_i} = \frac{1}{1 + j\omega RC}$$

$$\omega_c = \frac{1}{RC} = 2\pi f_c$$

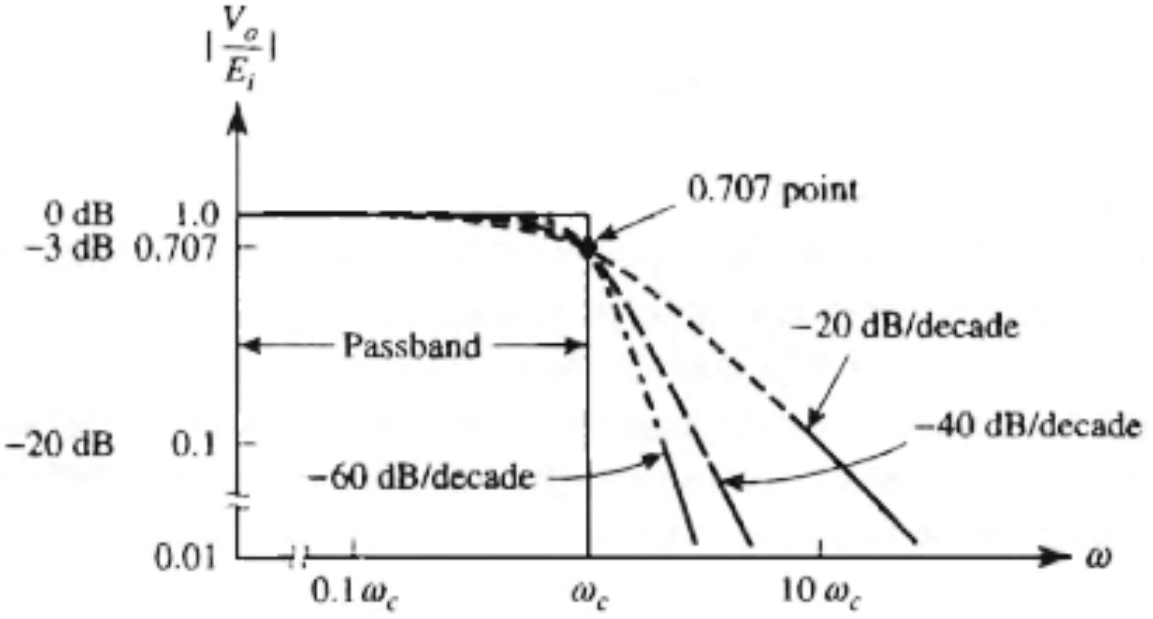
Active Low Pass Filter Frequency Response

$$\text{dB} = 20 \log_{10} \frac{Vo}{E_i}$$

BASIC LOW-PASS FILTER



BUTTERWORTH FILTER



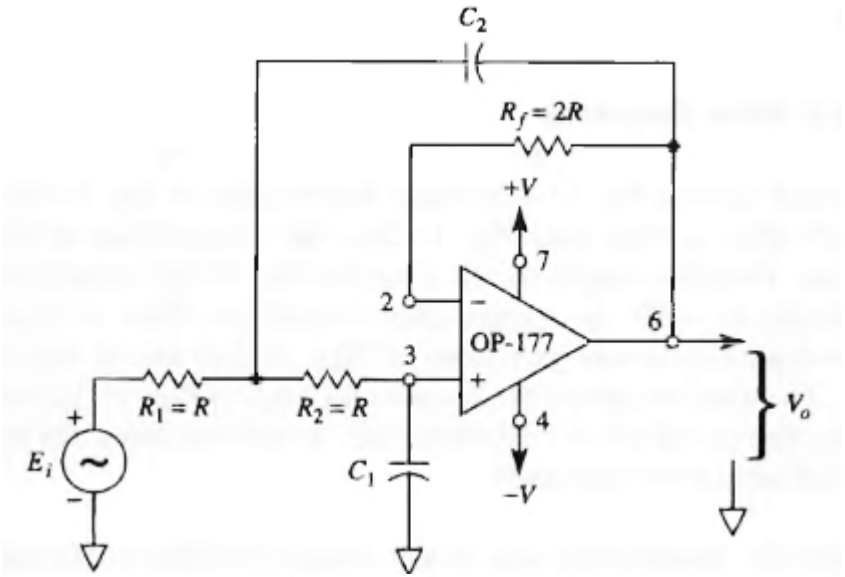
— 40-DB/DECADE LOW-PASS BUTTERWORTH FILTER

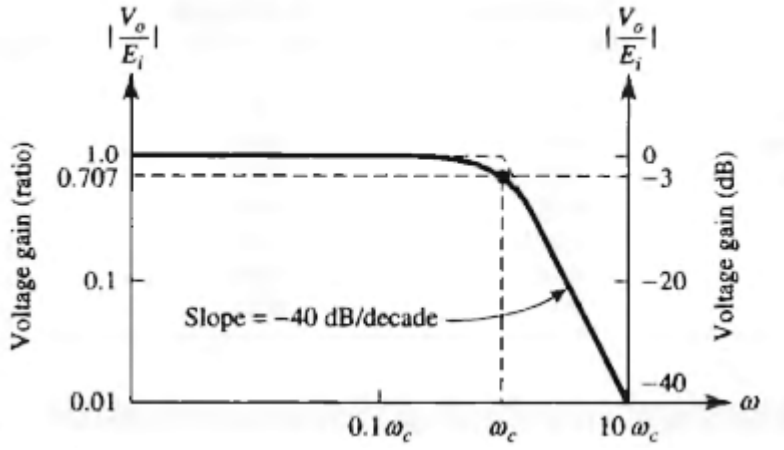
Design procedure

1. Choose the cutoff frequency, ω_c or f_c.
2. Pick C_1; choose a convenient value between 100 pF and 0.1 μF.
3. Make $C_2 = 2C_1$.
4. Calculate

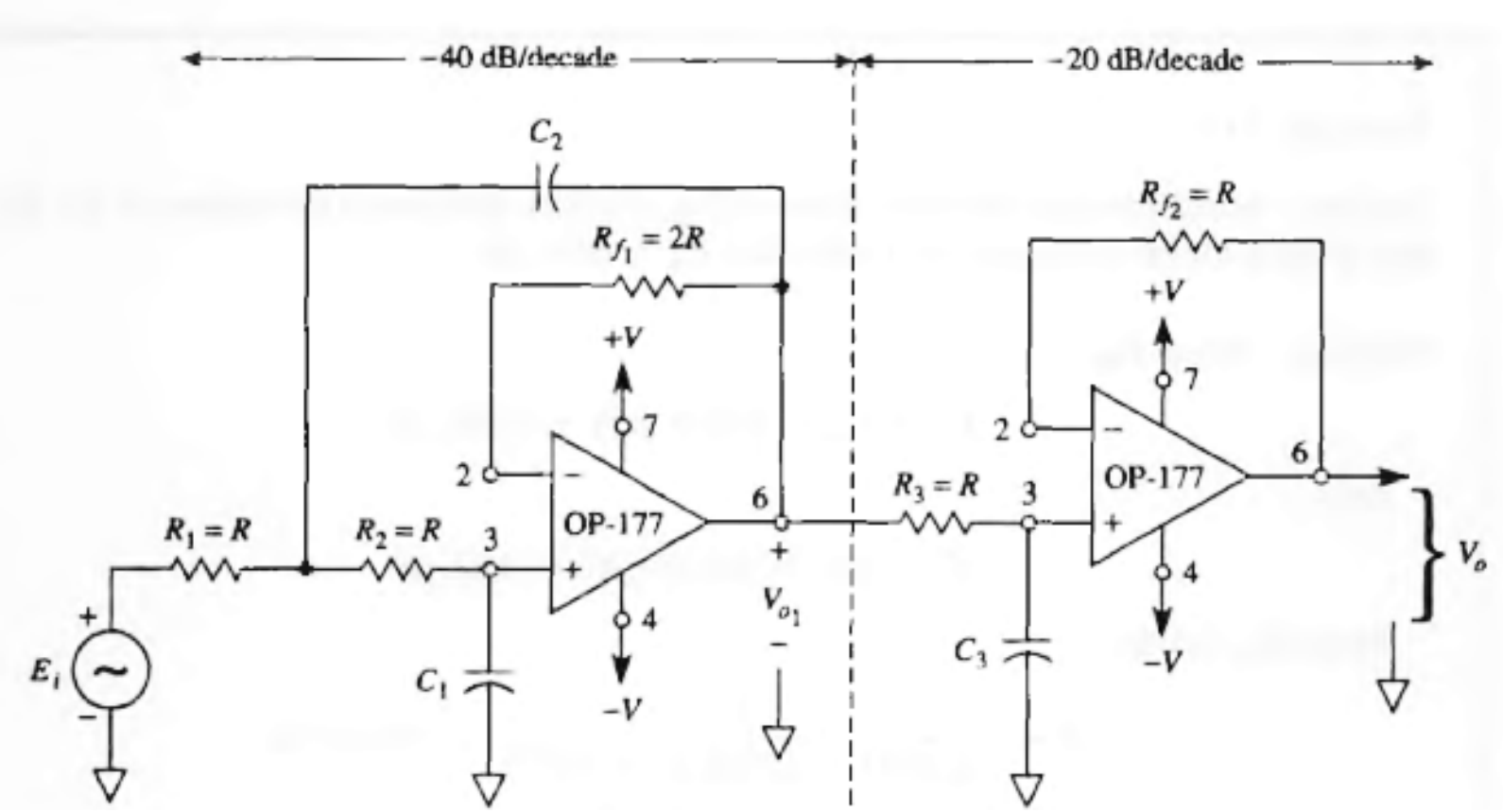
$$R = \frac{0.707}{\omega_c C_1}$$

5. Choose $R_f = 2R$.





– 60-DB/DECADE LOW-PASS BUTTERWORTH FILTER



— 60-DB/DECADE LOW-PASS BUTTERWORTH FILTER

Design procedure

1. Choose the cutoff frequency, ω_c or f_c.
2. Pick C_3; choose a convenient value between 0.001 and 0.1 μF.
3. Make

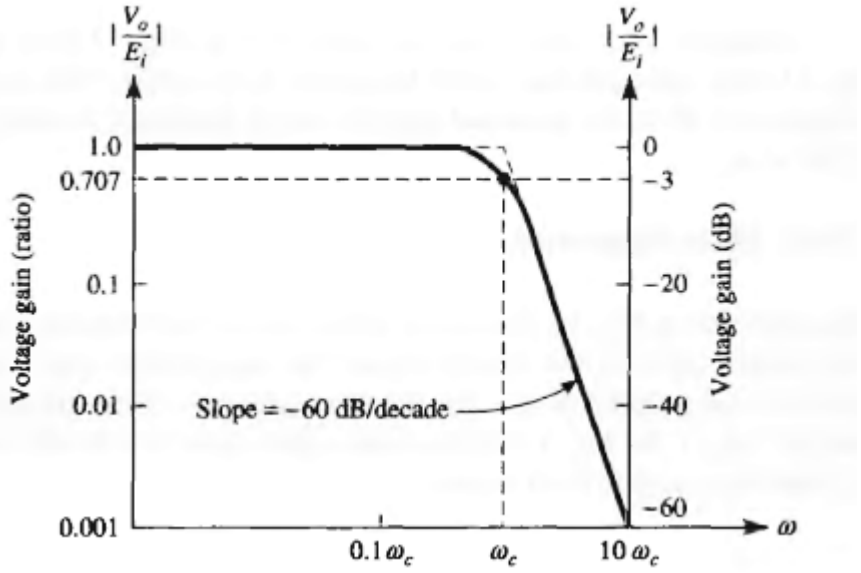
$$C_1 = \tfrac{1}{2}C_3 \qquad \text{and} \qquad C_2 = 2C_3$$

4. Calculate

$$R = \frac{1}{\omega_c C_3}$$

5. Make $R_1 = R_2 = R_3 = R$.
6. $R_{f_1} = 2R$ and $R_{f_2} = R$.

For best results the value of R should be between 10 and 100 kΩ.



$|A_{CL}|$ FOR THE LOW-PASS FILTERS

ω	−20 dB/decade; Fig. 11-2(a)	−40 dB/decade; Fig. 11-4(a)	−60 dB/decade; Fig. 11-5(a)
$0.1\omega_c$	1.0	1.0	1.0
$0.25\omega_c$	0.97	0.998	0.999
$0.5\omega_c$	0.89	0.97	0.992
ω_c	0.707	0.707	0.707
$2\omega_c$	0.445	0.24	0.124
$4\omega_c$	0.25	0.053	0.022
$10\omega_c$	0.1	0.01	0.001

Example 11-4

Determine R_1 and R_2 in Fig. 11-4(a) for a cutoff frequency of 1 kHz. Let $C_1 = 0.01\ \mu F$.

Solution Pick $C_2 = 2C_1 = 2(0.01\ \mu F) = 0.02\ \mu F$. Select $R_1 = R_2 = R$ from Eq. (11.3):

$$R = \frac{0.707}{(6.28)(1 \times 10^3)(0.01 \times 10^{-6})} = 11{,}258\ \Omega$$

and

$$R_f = 2(11{,}258\ \Omega) = 22{,}516\ \Omega$$

Example 11-5

For the -60-dB/decade low-pass filter of Fig. 11-5(a), determine the values of C_1, C_2, and R for a cutoff frequency of 1 kHz. Let $C_3 = 0.01\ \mu F$.

Solution From Eq. (11-5),

$$C_1 = \tfrac{1}{2}C_3 = \tfrac{1}{2}(0.01\ \mu F) = 0.005\ \mu F$$

and

$$C_2 = 2C_3 = 2(0.01\ \mu F) = 0.02\ \mu F$$

From Eq. (11-6),

$$R = \frac{1}{(6.28)(1 \times 10^3)(0.01 \times 10^{-6})} = 15{,}915\ \Omega$$

Example Problem 14.12

Design a lowpass active filter with a dc gain of 4 and a corner frequency of 500 Hz.

Solution:

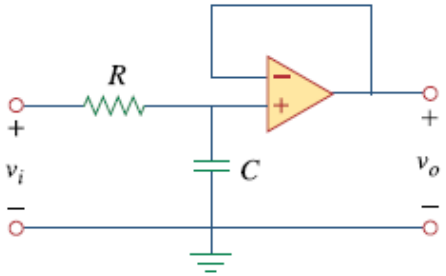
The corner frequency is

$$\omega_c = 2\pi f_c = 2\pi(500) = \frac{1}{R_f C_f}$$

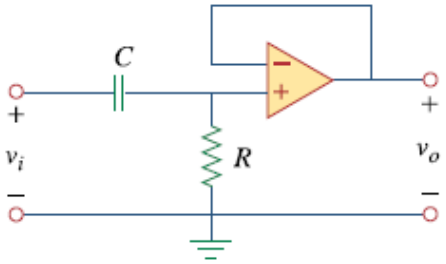
If we select $C_f = 0.2\ \mu\text{F}$ $\qquad R_f = \dfrac{1}{2\pi(500)0.2 \times 10^{-6}} = 1.59\ \text{k}\Omega$

Practice Problem

Find the transfer function for each of the active filters



(a)



(b)

Solution: (a) $\mathbf{V}_+ = \dfrac{1/j\omega C}{R + 1/j\omega C}\mathbf{V}_i, \qquad \mathbf{V}_- = \mathbf{V}_o$

Since $\mathbf{V}_+ = \mathbf{V}_-$,

$$\frac{1}{1 + j\omega RC}\mathbf{V}_i = \mathbf{V}_o$$

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \underline{\frac{\mathbf{1}}{\mathbf{1 + j\omega RC}}}$$

(b) $\mathbf{V}_+ = \dfrac{R}{R + 1/j\omega C}\mathbf{V}_i, \qquad \mathbf{V}_- = \mathbf{V}_o$

Since $\mathbf{V}_+ = \mathbf{V}_-$,

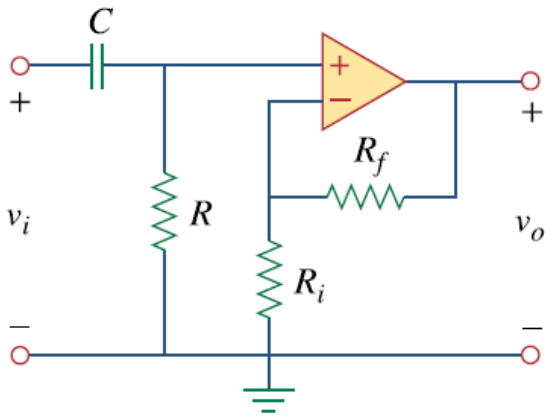
$$\frac{j\omega RC}{1 + j\omega RC}\mathbf{V}_i = \mathbf{V}_o$$

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \underline{\frac{\mathbf{j\omega RC}}{\mathbf{1 + j\omega RC}}}$$

Practice Problem

A highpass filter is shown in Fig. 14.92. Show that the transfer function is

$$\mathbf{H}(\omega) = \left(1 + \frac{R_f}{R_i}\right) \frac{j\omega RC}{1 + j\omega RC}$$



Solution:

$$\mathbf{V}_+ = \frac{\mathrm{R}}{\mathrm{R} + 1/\mathrm{j}\omega\mathrm{C}}\mathbf{V}_\mathrm{i} = \frac{\mathrm{j}\omega\mathrm{RC}}{1 + \mathrm{j}\omega\mathrm{RC}}\mathbf{V}_\mathrm{i}$$

$$\mathbf{V}_- = \frac{\mathrm{R_i}}{\mathrm{R_i} + \mathrm{R_f}}\mathbf{V}_\mathrm{o}$$

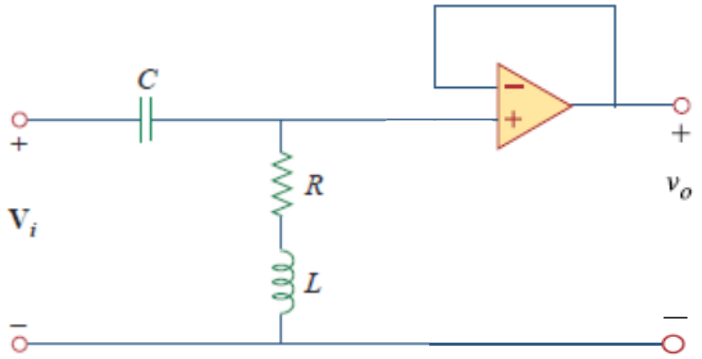
Since $\mathbf{V}_+ = \mathbf{V}_-$,

$$\frac{\mathrm{R_i}}{\mathrm{R_i} + \mathrm{R_f}}\mathbf{V}_\mathrm{o} = \frac{\mathrm{j}\omega\mathrm{RC}}{1 + \mathrm{j}\omega\mathrm{RC}}\mathbf{V}_\mathrm{i}$$

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_\mathrm{o}}{\mathbf{V}_\mathrm{i}} = \underline{\left(1 + \frac{\mathbf{R_f}}{\mathbf{R_i}}\right)\left(\frac{\mathbf{j}\omega\mathbf{RC}}{\mathbf{1} + \mathbf{j}\omega\mathbf{RC}}\right)}$$

Practice Problem

Find the transfer function $H(\omega)= V_0/V_i$ of the circuit below. What kind of filter it is and why? Sketch the approximate frequency response.



Solution:

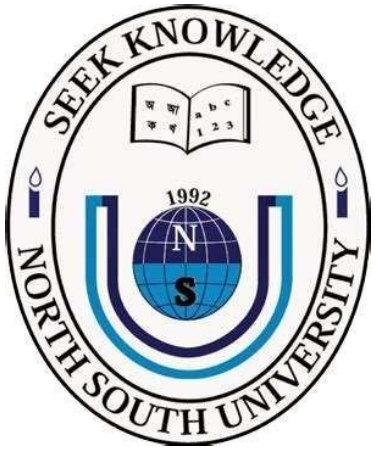
Practice Problem

Design a low-pass filter for a biomedical signal conditioning circuit used to measure ECG signals. The filter must effectively remove high-frequency noise (e.g., muscle artifacts and power line interference) while preserving the ECG waveform.

Design Requirements:
Desired cutoff frequency, fc = 150 Hz
Roll-off rate: –60 dB/decade
Passband gain: 1 (unity gain)
Input Resistance: at least 5kΩ

Solution:

Analog Electronics II: Op-amp & 555 Timer



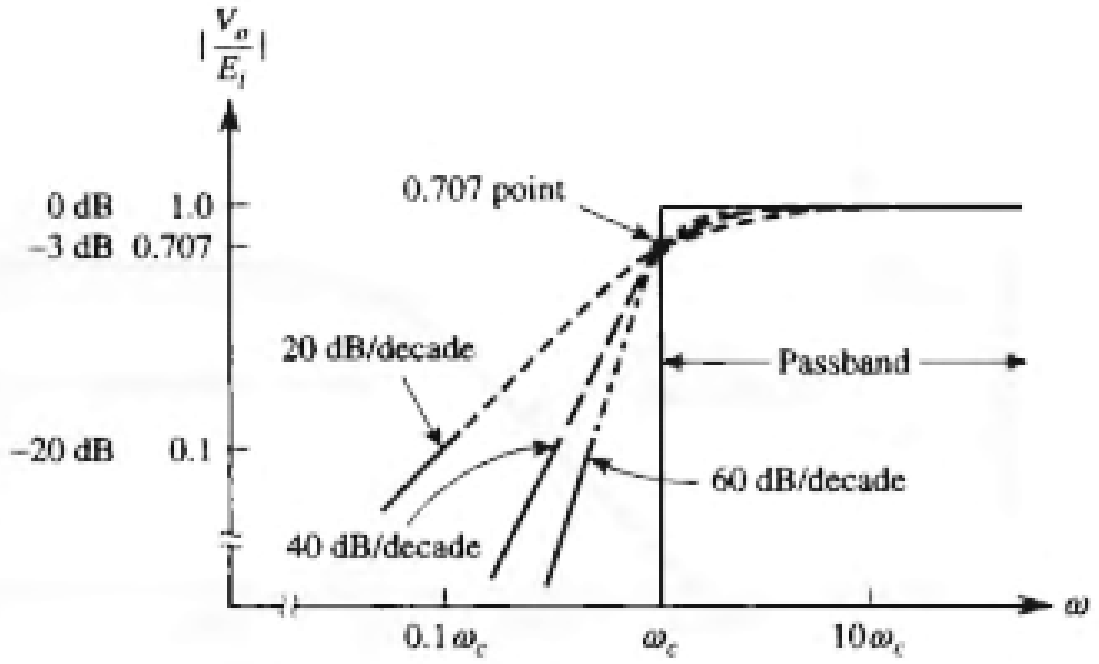
Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

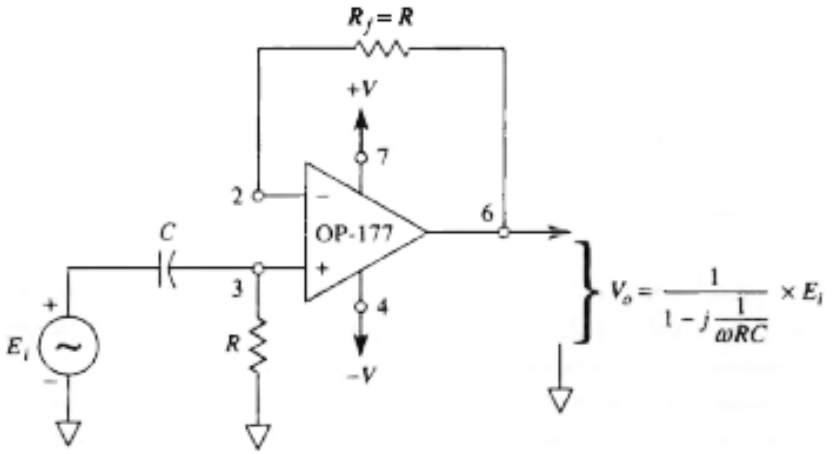
Lecture # 11

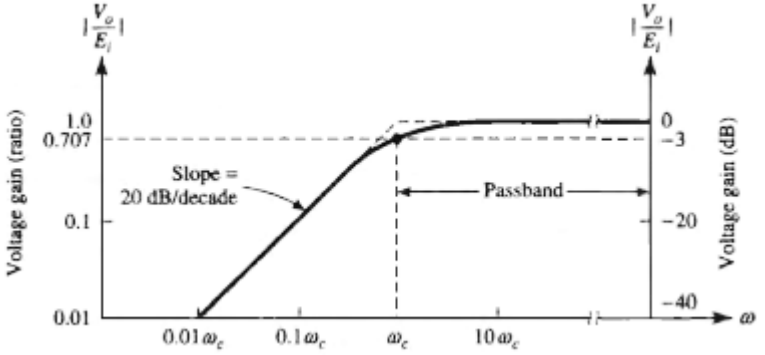
Active Filters: First-Order Highpass Filter

HIGH-PASS BUTTERWORTH FILTERS



20-dB/Decade Filter

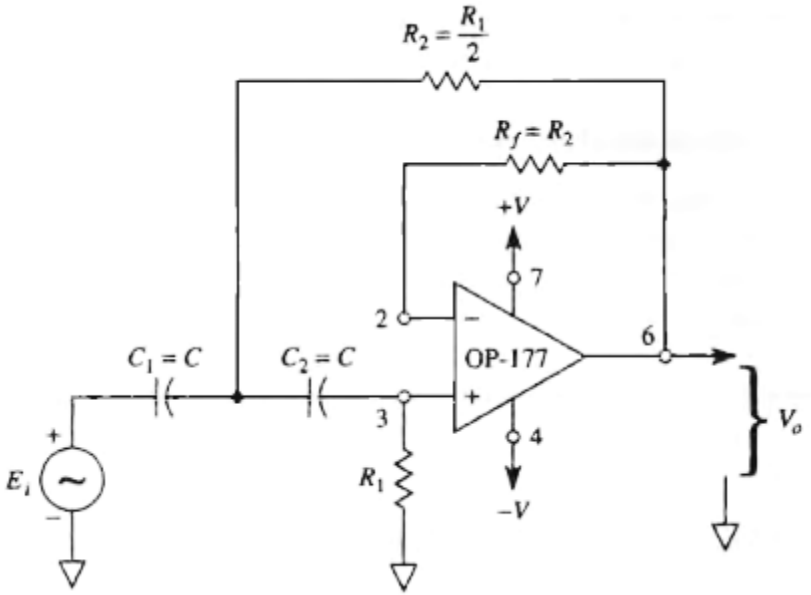


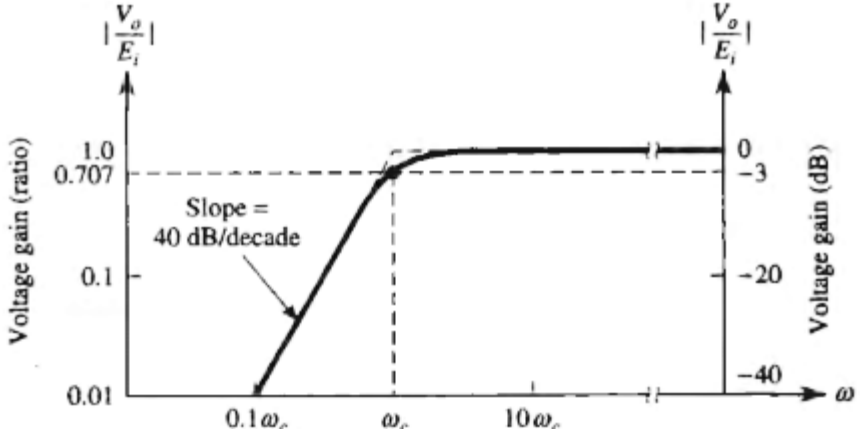


Design procedure for 20-dB/decade high-pass

1. Choose the cutoff frequency, ω_c or f_c.
2. Choose a convenient value of C, usually between 0.001 and 0.1 μF.
3. Calculate R from $\omega_c = \frac{1}{RC} = 2\pi f_c$
4. Choose $R_f = R$.

40-dB/Decade Filter





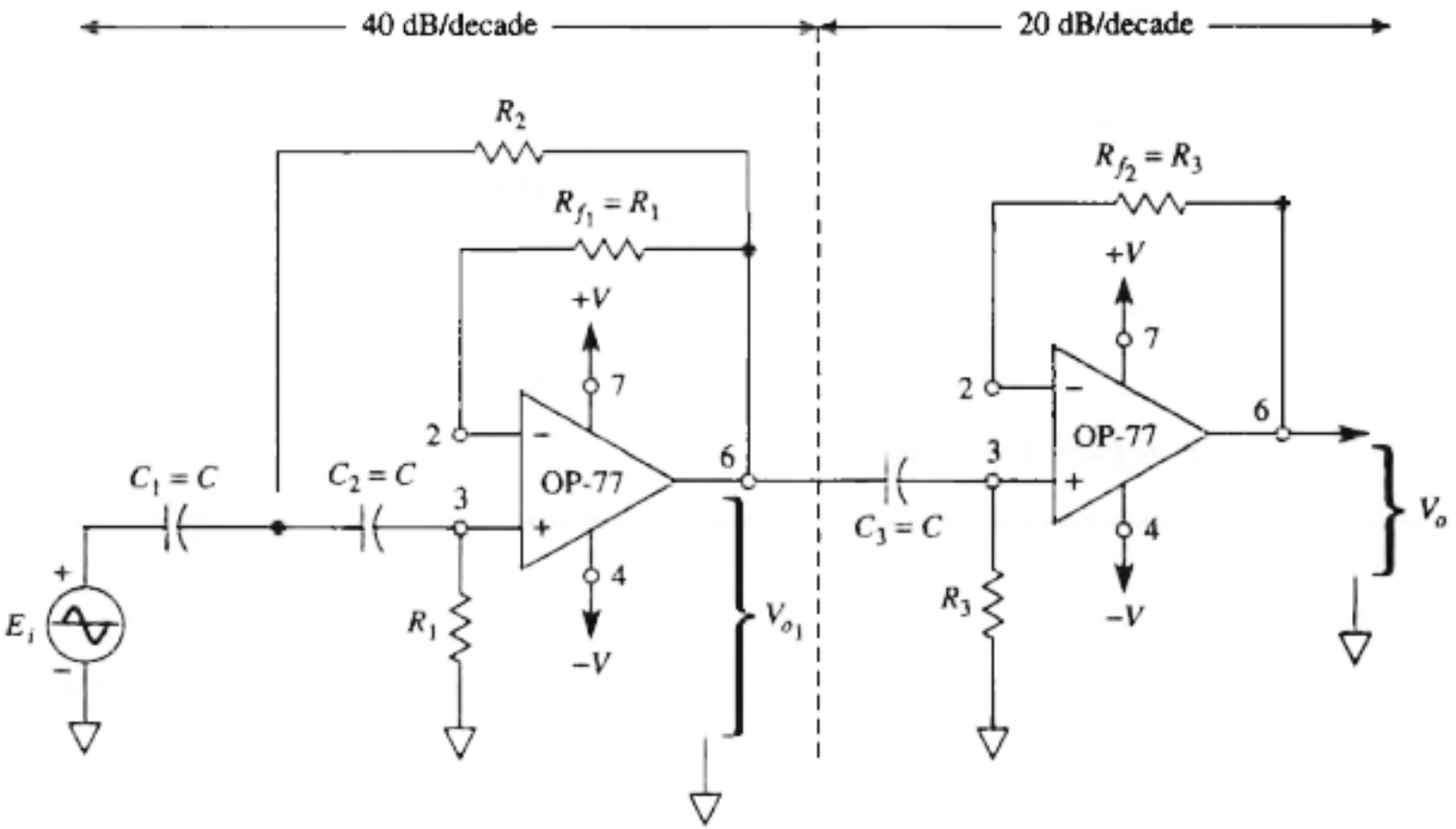
Design procedure for 40-dB/decade high-pass

1. Choose a cutoff frequency, ω_c or f_c.
2. Let $C_1 = C_2 = C$ and choose a convenient value.
3. Calculate R_1 from

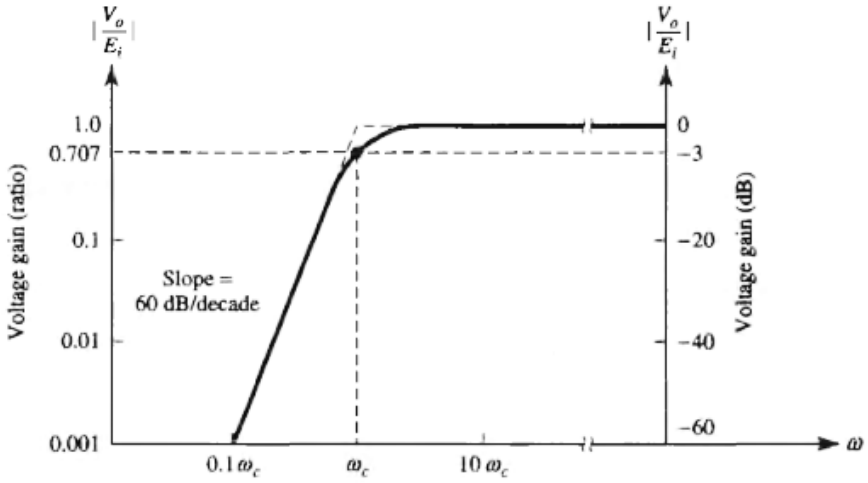
$$R_1 = \frac{1.414}{\omega_c C}$$

4. Select $R_2 = \frac{1}{2}R_1$
5. To minimize dc offset, let $R_f = R_1$.

60-dB/Decade Filter



60-dB/Decade Filter



Design procedure for 60-dB/decade high-pass

1. Choose the cutoff frequency, ω_c or f_c.
2. Let $C_1 = C_2 = C_3 = C$ and choose a convenient value between 100 pF and 0.1 μF.
3. Calculate R_3 from

$$R_3 = \frac{1}{\omega_c C}$$

4. Select

$$R_1 = 2R_3$$

5. Select

$$R_2 = \tfrac{1}{2}R_3$$

6. To minimize dc offset current, let $R_{f_1} = R_1$ and $R_{f_2} = R_3$.

COMPARISON OF $|A_{CL}|$

ω	20 dB/decade; Fig. 11-7(a)	40 dB/decade; Fig. 11-8(a)	60 dB/decade; Fig. 11-9(a)
$0.1\omega_c$	0.1	0.01	0.001
$0.25\omega_c$	0.25	0.053	0.022
$0.5\omega_c$	0.445	0.24	0.124
ω_c	0.707	0.707	0.707
$2\omega_c$	0.89	0.97	0.992
$4\omega_c$	0.97	0.998	0.999
$10\omega_c$	1.0	1.0	1.0

Example 11-10

For Fig. 11-9(a), let $C_1 = C_2 = C_3 = C = 0.1\ \mu$F. Determine (a) R_3, (b) R_1, and (c) R_2 for $\omega_c = 1$ krad/s. ($f_c = 159$ Hz.)

Solution (a) By Eq. (11-11),

$$R_3 = \frac{1}{(1 \times 10^3)(0.1 \times 10^{-6})} = 10\ \text{k}\Omega$$

(b) $R_1 = 2R_3 = 2(10\ \text{k}\Omega) = 20\ \text{k}\Omega$.
(c) $R_2 = \frac{1}{2}R_3 = \frac{1}{2}(10\ \text{k}\Omega) = 5\ \text{k}\Omega$.

Example 11-11

Determine (a) R_3, (b) R_1, and (c) R_2 in Fig. 11-9(a) for a cutoff frequency of 60 kHz. Let $C_1 = C_2 = C_3 = C = 220$ pF.

Solution (a) From Eq. (11-11),

$$R_3 = \frac{1}{(6.28)(60 \times 10^3)(220 \times 10^{-12})} = 12\ \text{k}\Omega$$

(b) $R_1 = 2R_3 = 2(12\ \text{k}\Omega) = 24\ \text{k}\Omega$.
(c) $R_2 = \frac{1}{2}R_3 = \frac{1}{2}(12\ \text{k}\Omega) = 6\ \text{k}\Omega$.

If desired, the 20-dB/decade section can come before the 40-dB/decade section, because the op amps provide isolation and do not load one another.

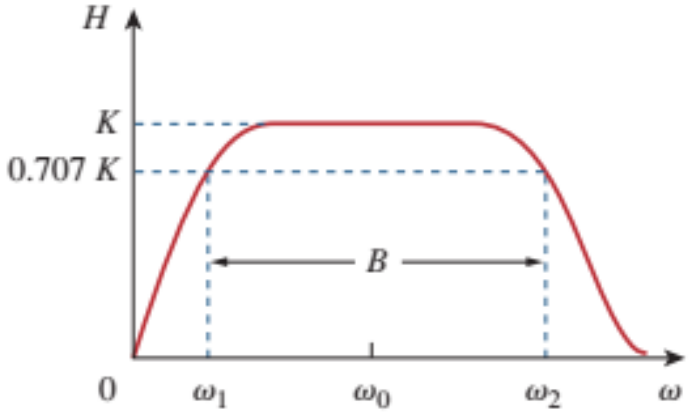
Practice Problem

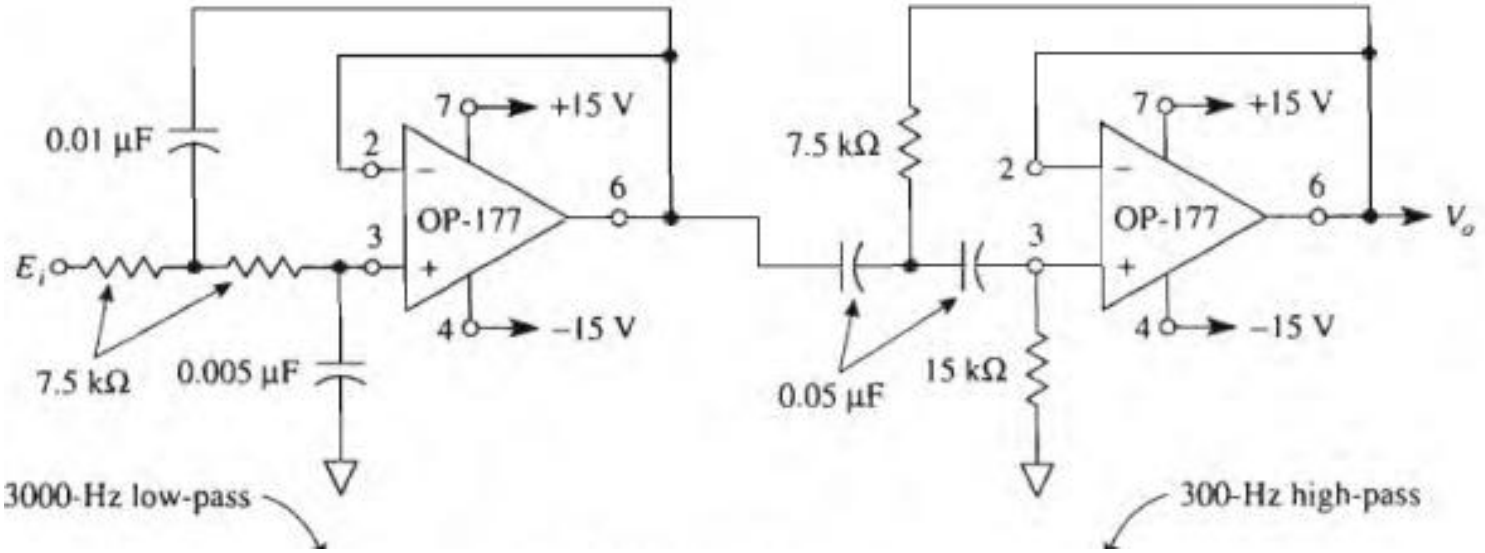
An Electrocardiogram (ECG or EKG) is a medical test that records the electrical activity of the heart over time. It helps diagnose heart conditions by analyzing the heart's rhythm, electrical conduction, and abnormalities in heart function.
ECG electrodes placed on the skin detect electrical activity from the heart. The raw signal contains useful heart rhythms (0.5 Hz – 100 Hz) but also high-frequency noise (>100 Hz) from muscle activity (EMG) and electrical interference (60 Hz power lines).
Design a Butterworth high pass filter with 40 dB/decade slope on the frequency response. The minimum value of the resistances should be 1 kΩ.

Solution:

Bandpass Filter

Wideband Filter Circuit





Bandpass Filter

$$\mathbf{H}(\omega) = \frac{\mathbf{V}_o}{\mathbf{V}_i} = \left(-\frac{1}{1 + j\omega C_1 R}\right)\left(-\frac{j\omega C_2 R}{1 + j\omega C_2 R}\right)\left(-\frac{R_f}{R_i}\right)$$

$$= -\frac{R_f}{R_i}\frac{1}{1 + j\omega C_1 R}\frac{j\omega C_2 R}{1 + j\omega C_2 R}$$

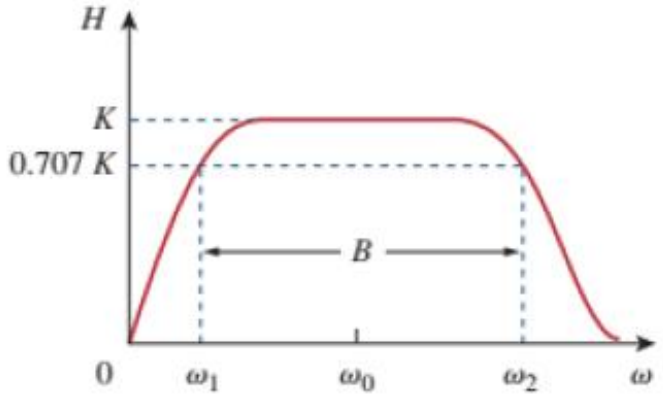
The lowpass section sets the upper corner frequency $\omega_2 = \dfrac{1}{RC_1}$

the highpass section sets the lower corner frequency $\omega_1 = \dfrac{1}{RC_2}$

the center frequency $\omega_0 = \sqrt{\omega_1 \omega_2}$

bandwidth $B = \omega_2 - \omega_1$

quality factor $Q = \dfrac{\omega_0}{B}$



Bandpass Filter

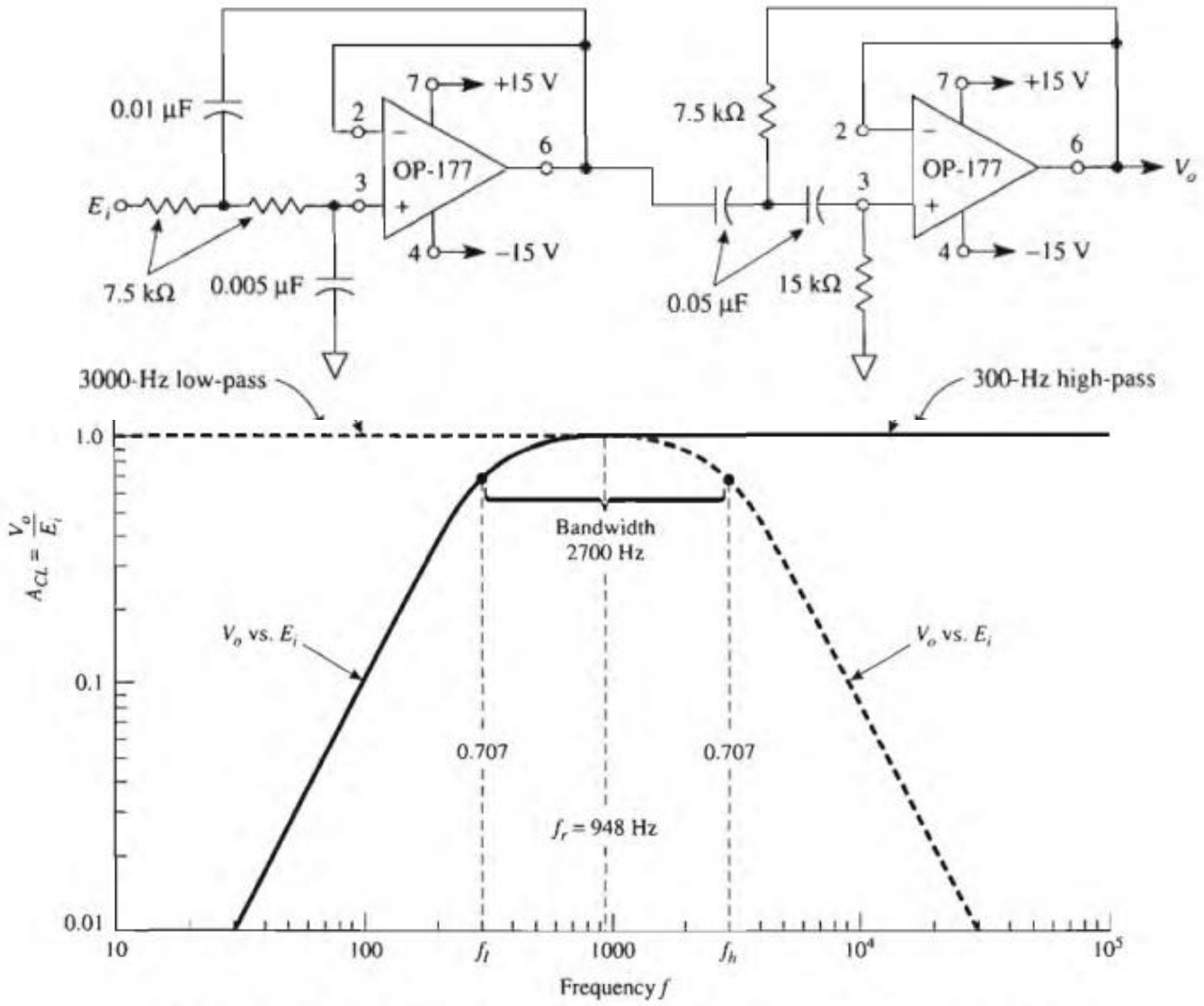
$$\mathbf{H}(\omega) = -\frac{R_f}{R_i}\frac{j\omega/\omega_1}{(1 + j\omega/\omega_1)(1 + j\omega/\omega_2)} = -\frac{R_f}{R_i}\frac{j\omega\omega_2}{(\omega_1 + j\omega)(\omega_2 + j\omega)}$$

At the center frequency $\omega_0 = \sqrt{\omega_1\omega_2}$, the magnitude of the transfer function is

$$|\mathbf{H}(\omega_0)| = \left|\frac{R_f}{R_i}\frac{j\omega_0\omega_2}{(\omega_1 + j\omega_0)(\omega_2 + j\omega_0)}\right| = \frac{R_f}{R_i}\frac{\omega_2}{\omega_1 + \omega_2}$$

Thus, the passband gain is $K = \dfrac{R_f}{R_i}\dfrac{\omega_2}{\omega_1 + \omega_2}$

Wideband Filter



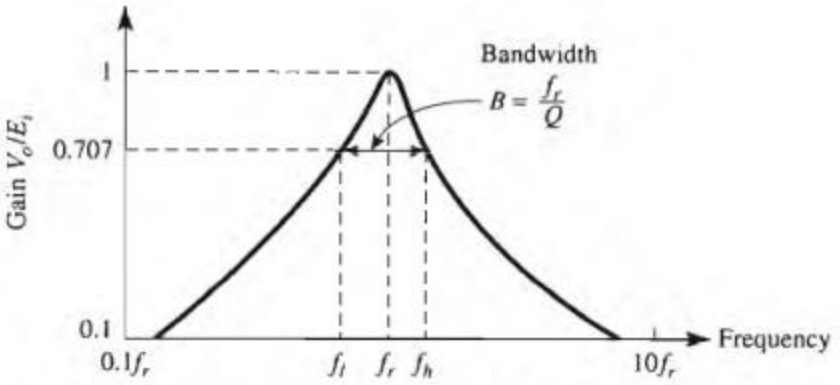
Practice Problem

For a music production, a particular frequency range 250 Hz- 3000 Hz needs to be filtered in order to eliminate noise and other interfering sounds. Design an active bandpass filter.
Design Requirements:

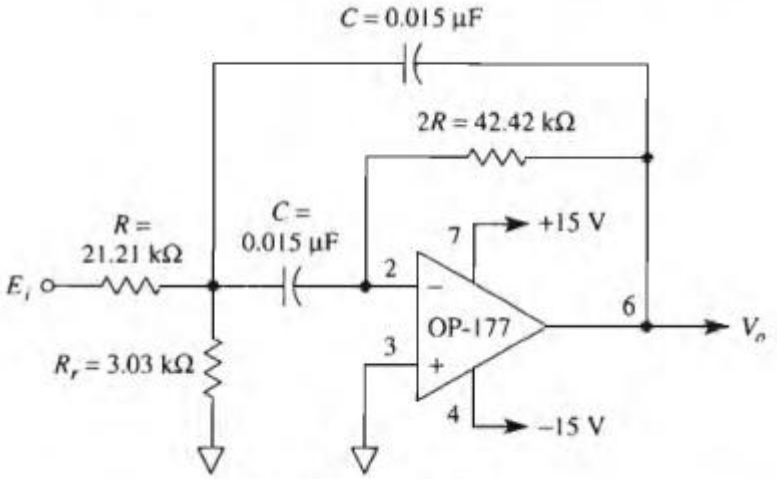
Passband frequencies: between 250 Hz and 3,000 Hz
Passband Gain = 10
Input impedance R= 10 kΩ.
Draw the circuit diagram and find out the required values of corresponding circuit components.

Solution:

NARROWBAND BANDPASS FILTERS



(a) Typical frequency response curve of a narrowband filter.



(b) Narrowband filter.

FIGURE 11-12 Narrow bandpass filter circuit and its frequency response for the component values shown; $f_r = 100$ Hz, $B = 500$ Hz, $Q = 2$, $f_l = 780$ Hz, and $f_h = 1280$ Hz. (a) Typical frequency response of a bandpass filter; (b) narrow bandpass filter circuit.

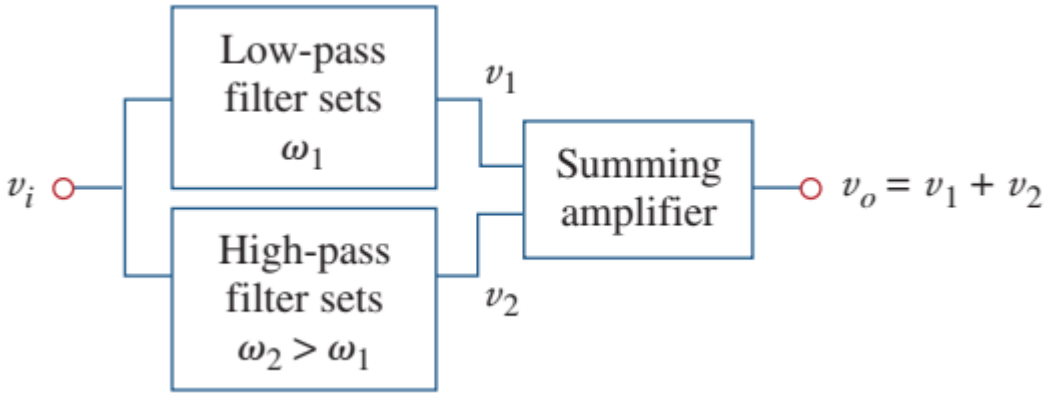
Design procedure

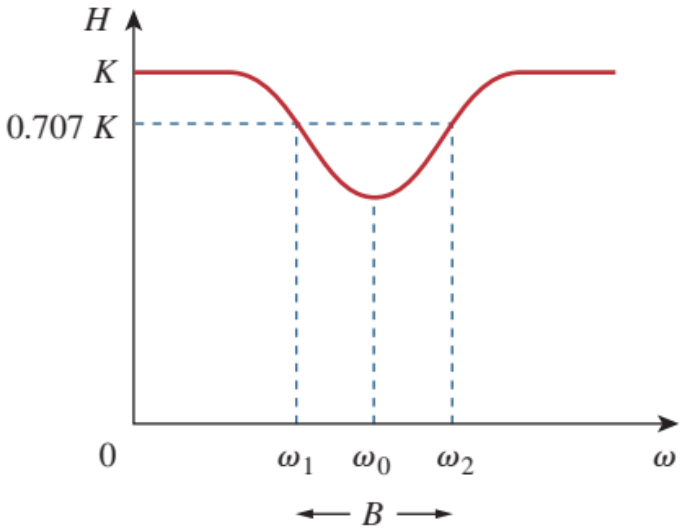
$$B = \frac{0.1591}{RC} \qquad R_r = \frac{R}{2Q^2 - 1}$$

$$B = \frac{f_r}{Q} \qquad f_r = \frac{0.1125}{RC}\sqrt{1 + \frac{R}{R_r}}$$

Active Filters: Bandreject (or Notch) Filter

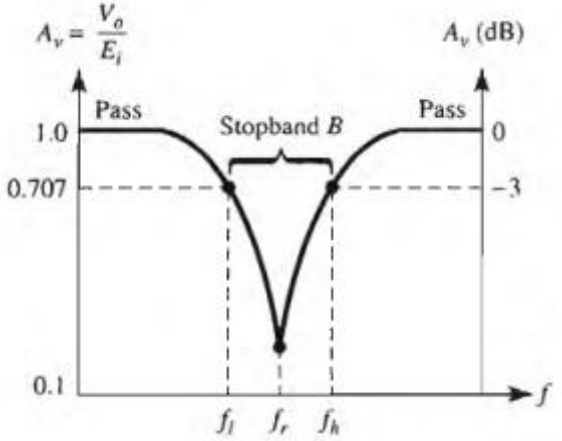
Wideband Filter Circuit





NOTCH FILTERS

NARROWBAND BANDREJECT FILTER



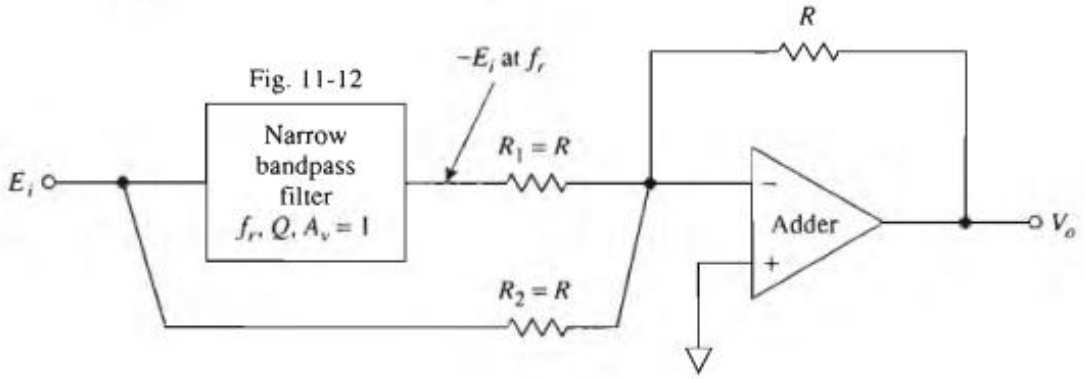


FIGURE 11-14 A notch filter is made by a circuit that subtracts the output of a bandpass filter from the original signal.

Procedure to Make a Notch Filter

1. Make a bandpass filter that has the same resonant frequency, bandwidth, and consequently Q as the notch filter.
2. Connect the inverting adder of Fig. 11-15 by selecting equal resistors for R. Usually, $R = 10\ \text{k}\Omega$. (A practical fine-tuning procedure is presented in the next section.)

NOTCH FILTERS

NARROWBAND BANDREJECT FILTER

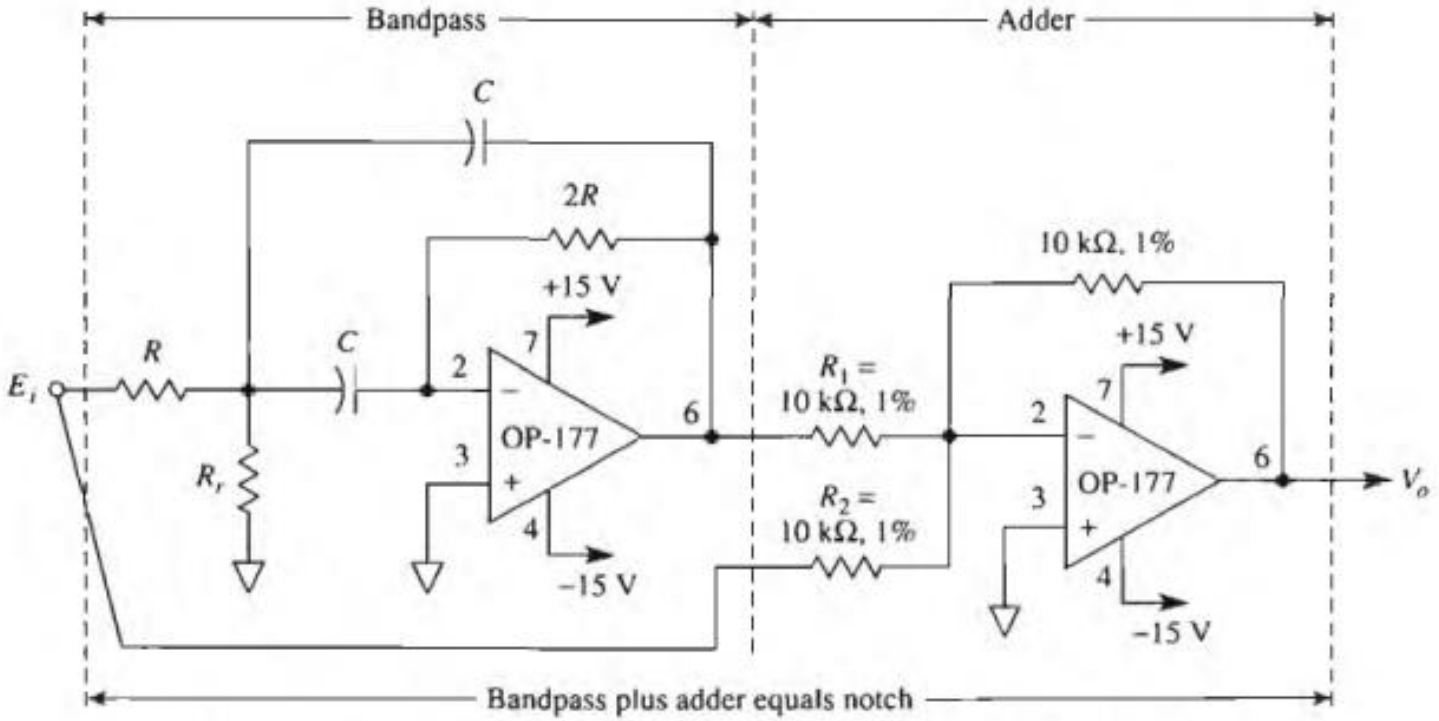


FIGURE 11-15 This two-op-amp notch filter is made from a bandpass filter plus an inverting adder. If $C = 0.33\ \mu F$, $R = 40.2\ k\Omega$, and $R_f = 201\ \Omega$, the notch frequency will be 120 Hz and reject a bandwidth of 12 Hz.

Practice Problem

Design a **biomedical signal conditioning circuit** for an **EEG amplifier** that picks up unwanted **50 Hz power-line interference**.
To remove this narrowband interference without affecting nearby signal frequencies, you decide to implement a **narrowband band-reject filter** using a **band-pass–plus–subtractor method**.

Design Specifications:
- Interference (resonant) frequency, f_0=50 Hz
- Desired quality factor, Q=25
- Passband gain of the final output = 1 (unity)
- Supply: ±15 V op-amp system
- Available passive components: capacitors (100 pF – 0.1 µF)

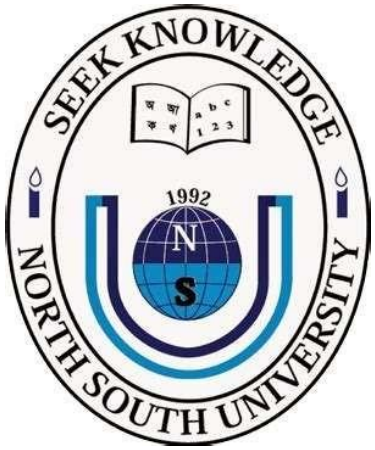
Solution:

References:

1. Butterworth Filters:
Low-Pass, High-Pass, Wideband/Narrowband Bandpass, Wideband/Narrowband Bandreject Filters.

-Operational Amplifiers and Linear Integrated Circuits- by Robert F. Coughlin & Frederick F. Driscoll, **Chapter 11**

Analog Electronics II: Op-amp & 555 Timer

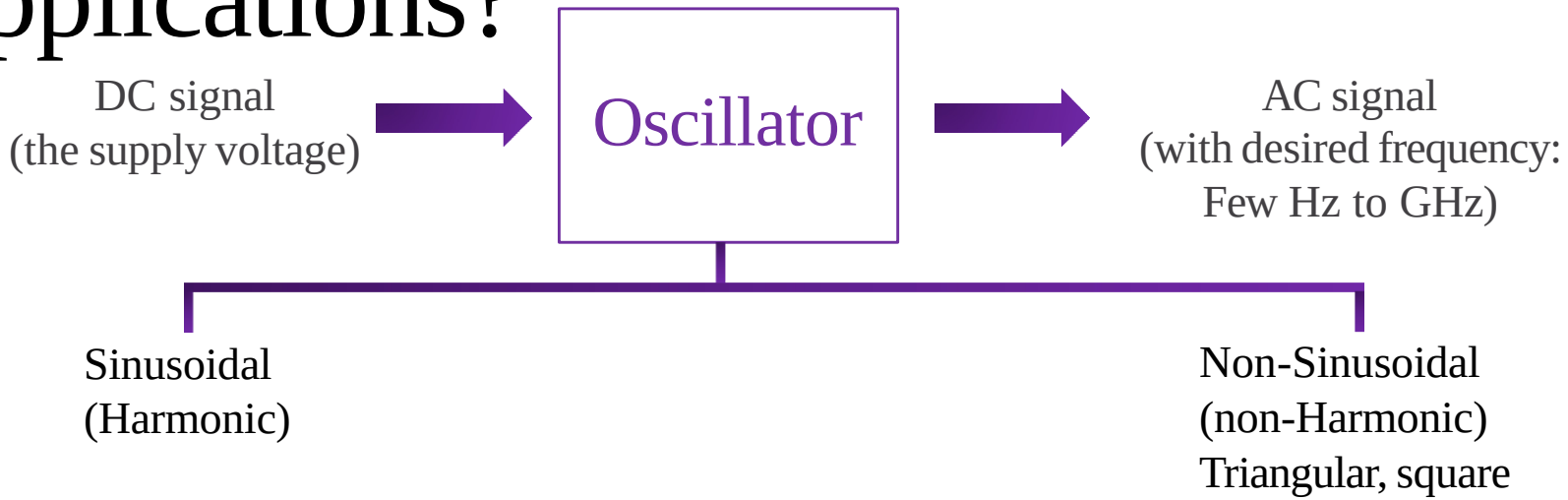


Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

Lecture # 13

Oscillator – What is it? Applications?



Applications:

Oscillators are used for generating:

- Clock signals
- Carrier frequency
- In signal generator

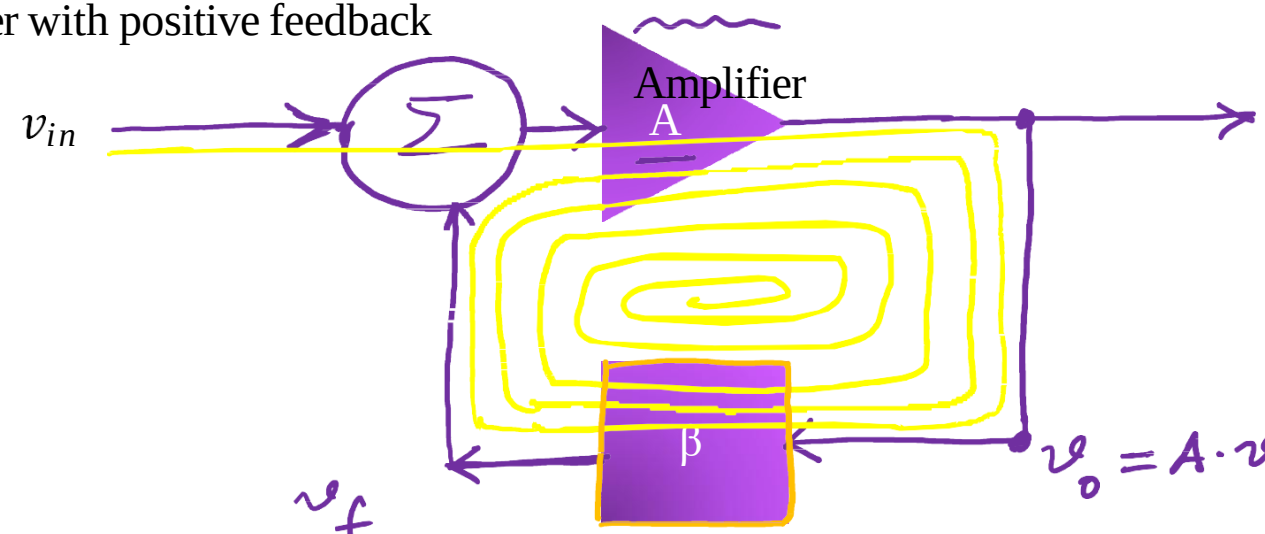
Oscillator – block diagram

$A = \text{gain}$

$A > 1$

$A = \frac{v_o}{v_{in}}$

Amplifier with positive feedback



$\beta = \frac{v_f}{v_o}$

Positive feedback path /
Regenerative feedback path /
Frequency selector Circuit /
Resonant Circuit

$v_f = \beta \cdot v_o$

$= \beta (A v_{in})$

$v_f = (A\beta) \cdot v_{in}$

$A\beta = \frac{v_f}{v_{in}}$

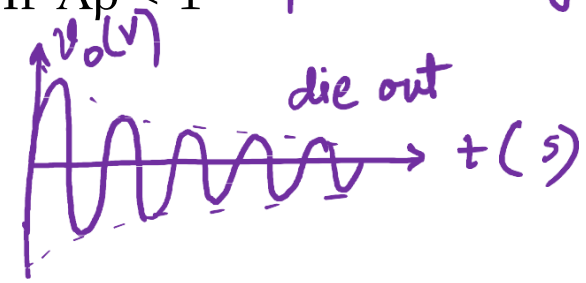
loop gain

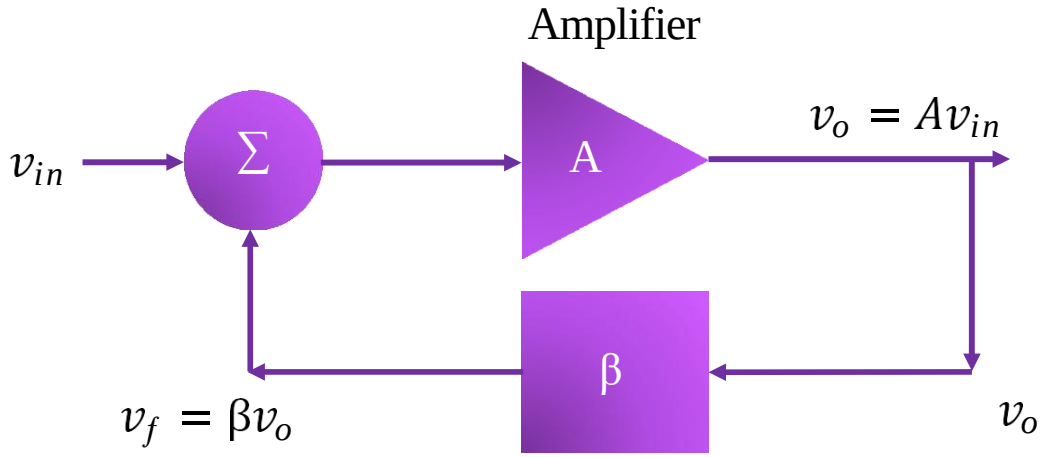
β = feedback fraction

$\beta < 1$

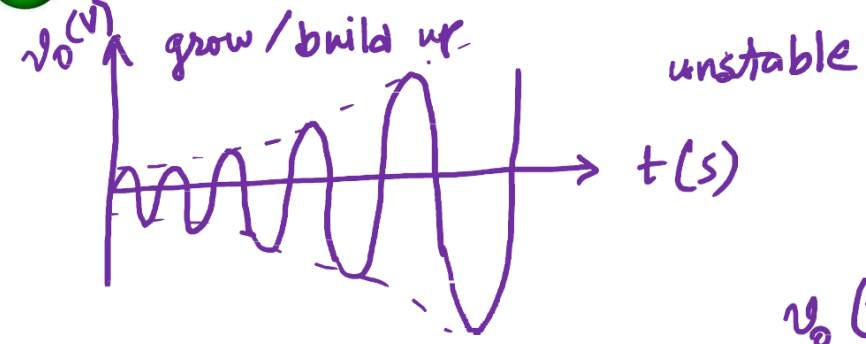
Loop gain

- If $A\beta < 1$ — β dominating

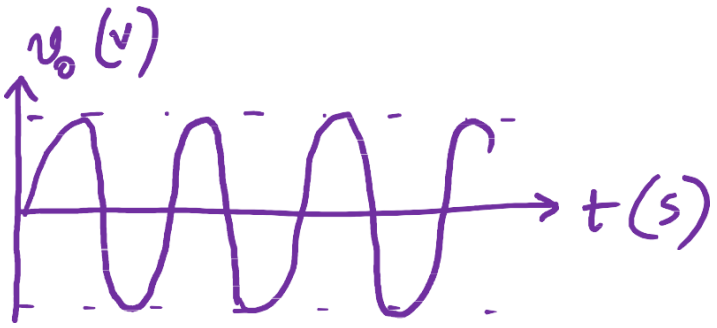




- If $A\beta > 1$ — A dominating

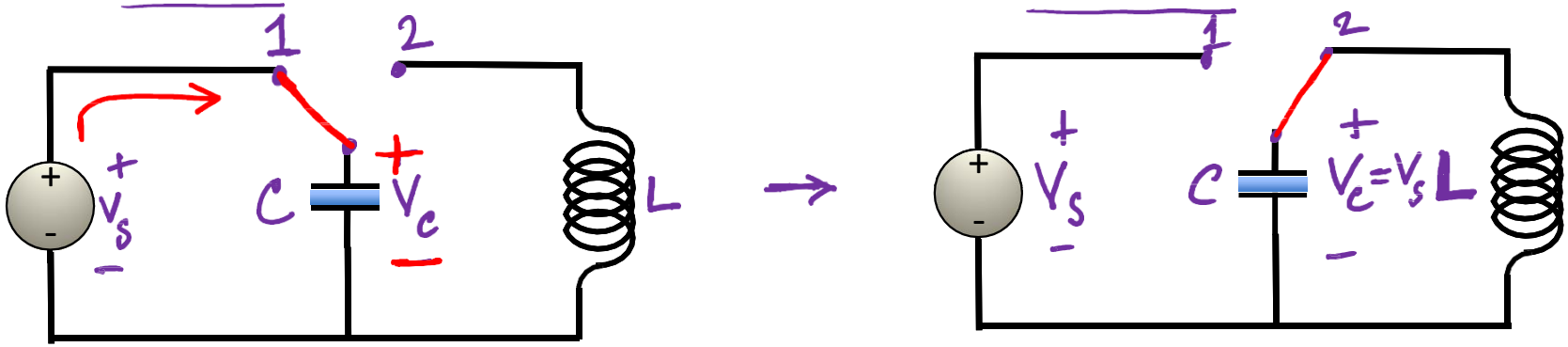


- If $A\beta = 1$ (for sustained oscⁿ)

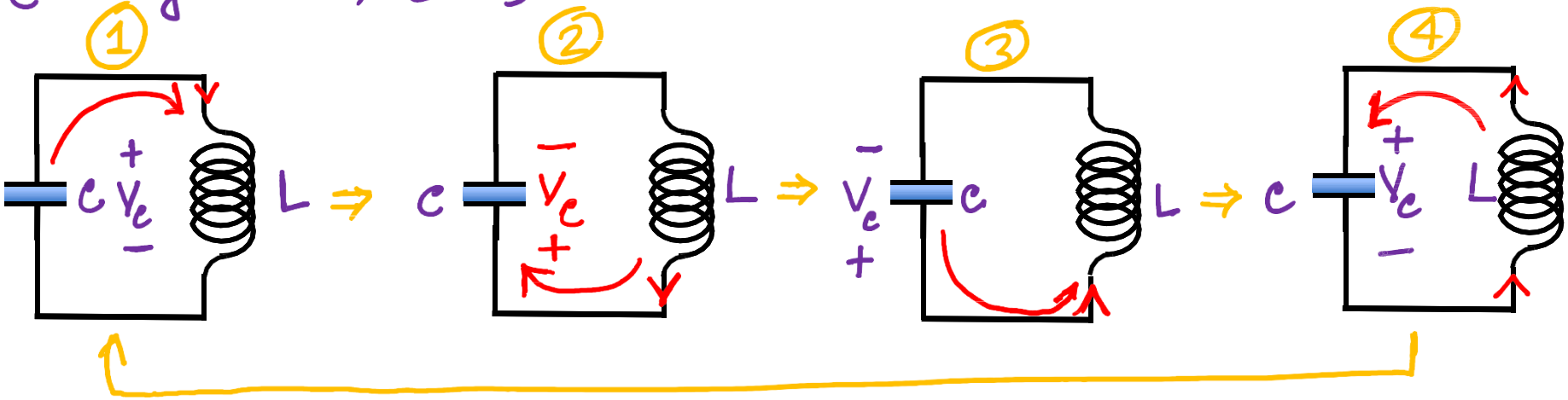


LC tank circuit

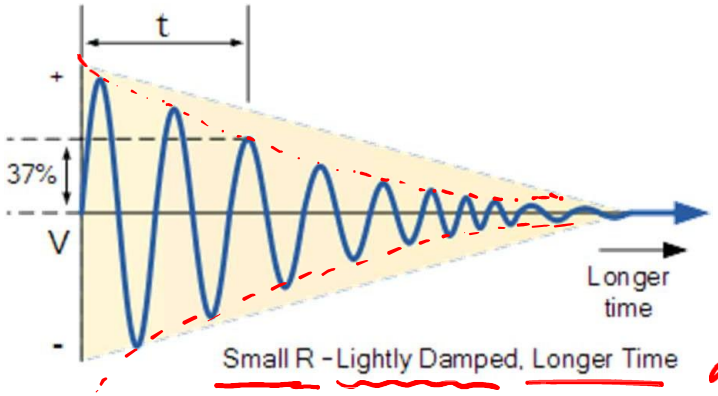
- ✓ The capacitor stores energy in the form of an electrostatic field and which produces a potential (*static voltage*) across its plates.
- ✓ The inductive coil stores its energy in the form of an electromagnetic field.

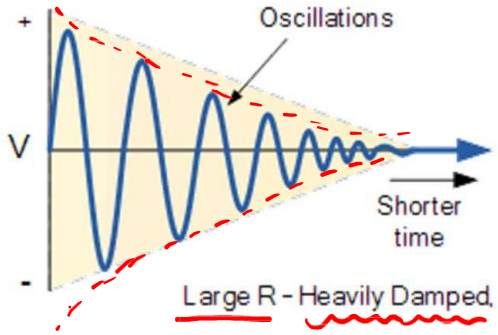


@ steady state, $V_c = V_s$



LC tank resonance





$X_L = 2\pi f L$

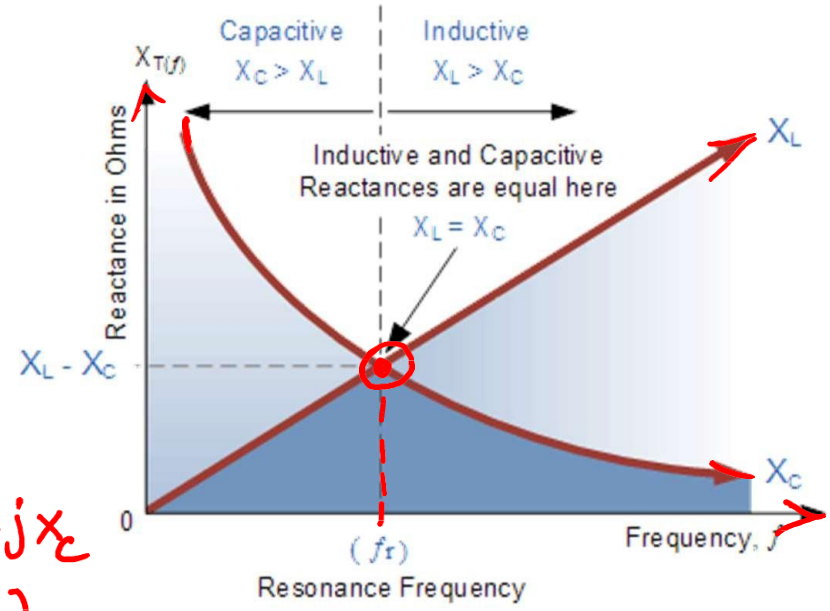
$X_C = \frac{1}{2\pi f C}$

$Z = R + jX_L - jX_C$

$Z = R$ (real)

v_o & v_{in} in phase

@ $f = f_r$



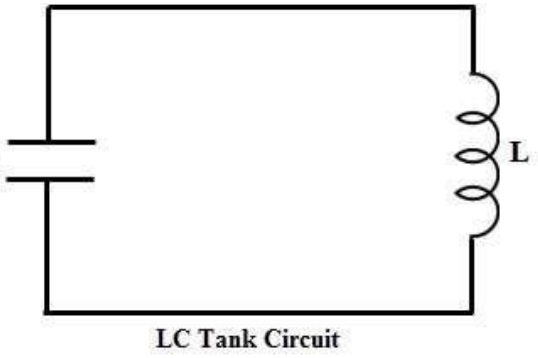
@ $f = f_r$

$X_L = X_C$

$\Rightarrow 2\pi f_r L = \frac{1}{2\pi f_r C}$

$\Rightarrow f_r^2 = \frac{1}{(2\pi)^2 LC}$

$$f_r = \frac{1}{2\pi\sqrt{LC}}$$



Further reading:

ELECTRONICS HUB

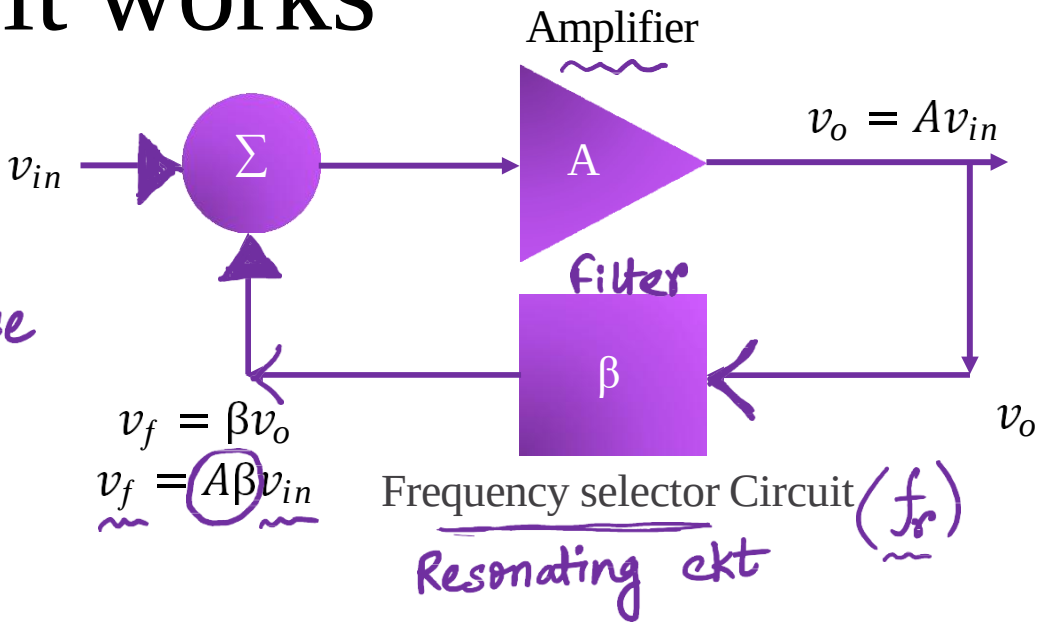
ElectronicsTutorials

Oscillator – how it works

- If the phase shift introduced by the amplifier and feedback circuit is 0 deg,

v_f & v_{in} ⇒ in the same phase

$\angle A\beta = 0^\circ$



- After the first cycle, if v_{in} is removed,

v_f works as the i/p

- Whether we will have sustained oscillations or not,

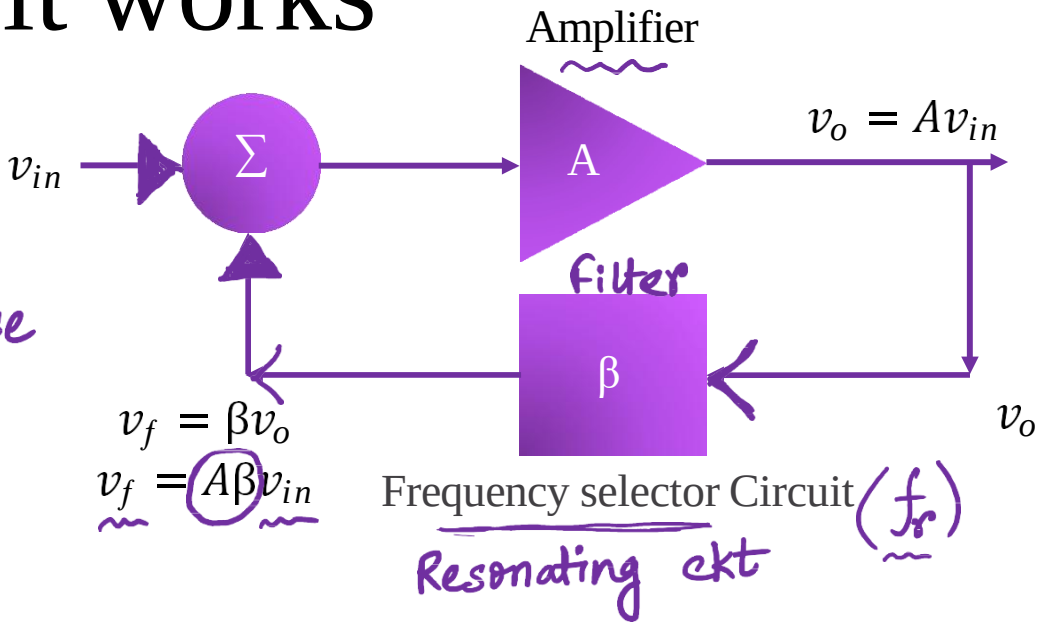
$A\beta \begin{cases} > 1 \\ < 1 \\ = 1 \end{cases}$

Oscillator – how it works

- If the phase shift introduced by the amplifier and feedback circuit is 0 deg,

v_f & v_{in} ⇒ in the same phase

$\angle A\beta = 0^\circ$



- After the first cycle, if v_{in} is removed,

v_f works as the i/p

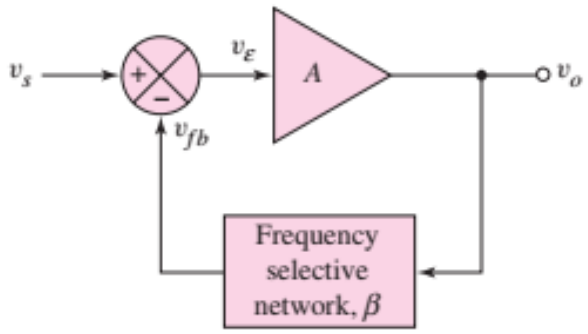
- Whether we will have sustained oscillations or not,

$A\beta \begin{cases} > 1 \\ < 1 \\ = 1 \end{cases}$

Sinusoidal Oscilator

Chapter 15.2 from Microelectronic Circuit Analysis and Design by Donald A Neaman 4th Ed.

The basic oscillator consists of an amplifier and a frequency-selective network connected in a feedback loop.



Block diagram of the fundamental feedback circuit

For the circuit shown, the ideal closed-loop transfer function is given by

$$A_f(s) = \frac{A(s)}{1 + A(s)\beta(s)}$$

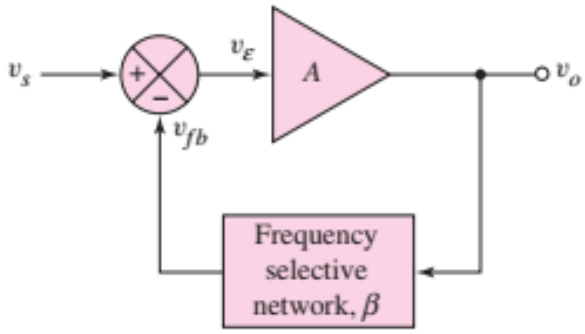
The feedback transfer function β is the principal portion of the loop gain that is dependent on frequency.

Oscillators Basics

The loop gain of the feedback circui $T(s) = A(s)\beta(s)$

If T(s) = −1, the closed-loop transfer function goes to infinity, which means that the circuit can have a finite output for a zero input signal.

Assume that T(s) ≈ −1 so that positive feedback exists over a particular frequency range.

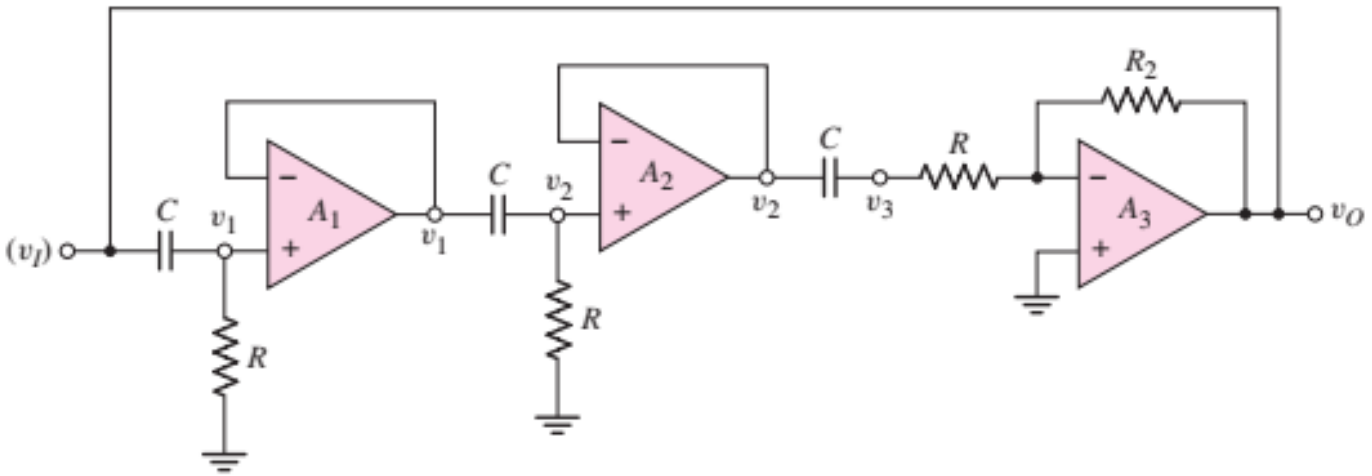


The condition for oscillation is that, at a specific frequency, we have

$$T(j\omega_o) = A(j\omega_o)\beta(j\omega_o) = -1$$

The condition that $T(j\omega_o) = -1$ is called the **Barkhausen criterion.**

Phase-Shift Oscillator

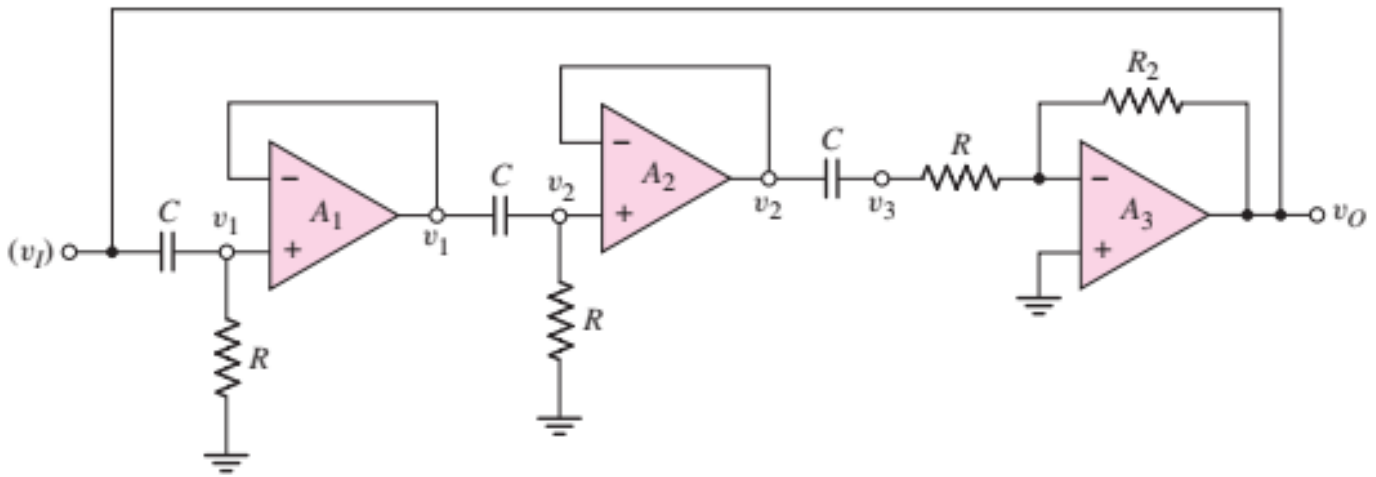


The basic amplifier of the circuit is the op-amp A3, which is connected as an inverting amplifier with its output connected to a three-stage RC filter. The voltage followers in the circuit eliminate loading effects between each RC filter stage.

The transfer function of the first RC network is

$$v_1 = \left(\frac{sRC}{1+sRC}\right)(v_I)$$

Phase-Shift Oscillator



$$\frac{v_3}{(v_I)} = \left(\frac{sRC}{1+sRC}\right)^3 = \beta(s)$$ where $\beta(s)$ is the feedback transfer function.

The amplifier gain $A(s)$ $$A(s) = \left|\frac{v_O}{v_3}\right| = \frac{R_2}{R}$$

The loop gain is then

$$T(s) = A(s)\beta(s) = \left(\frac{R_2}{R}\right)\left(\frac{sRC}{1+sRC}\right)^3$$

$$T(s) = A(s)\beta(s) = \left(\frac{R_2}{R}\right)\left(\frac{sRC}{1+sRC}\right)^3$$

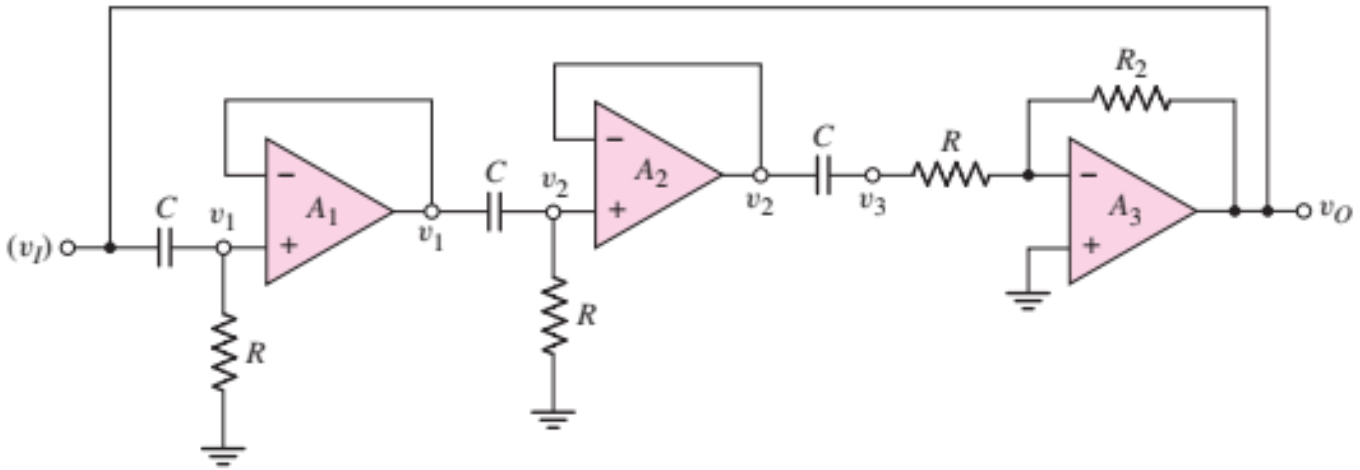
$$T(j\omega) = \left(\frac{R_2}{R}\right)\frac{(j\omega RC)^3}{(1+j\omega RC)^3}$$

$$= -\left(\frac{R_2}{R}\right)\frac{(j\omega RC)(\omega RC)^2}{[1-3\omega^2R^2C^2]+j\omega RC[3-\omega^2R^2C^2]}$$

The condition for oscillation is that |T(jωo)| = 1 and the phase of T(jωo) must be 180 degrees.

To satisfy the condition T(jωo) = −1, the imaginary component of the Equation must equal zero. Since the numerator is purely imaginary, the denominator must become purely imaginary,

$$[1-3\omega_o^2R^2C^2] = 0 \qquad \omega_o = \frac{1}{\sqrt{3}RC}$$



$$\omega_o = \frac{1}{\sqrt{3}RC}$$

where ω_o is the oscillation frequency. At this frequency,

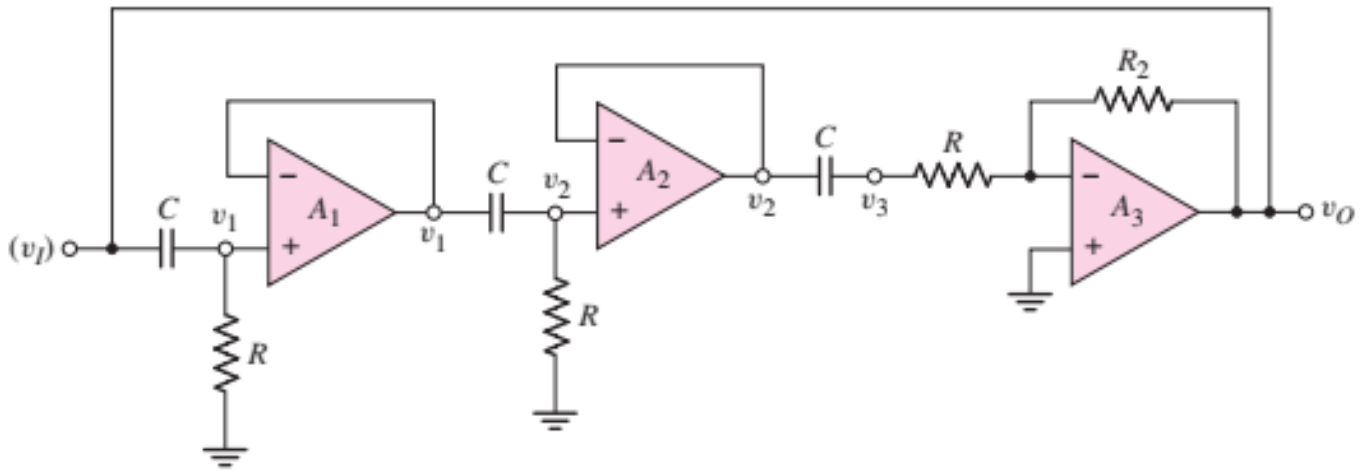
$$T(j\omega_o) = -\left(\frac{R_2}{R}\right)\frac{(j/\sqrt{3})(1/3)}{0 + (j/\sqrt{3})[3 - (1/3)]} = -\left(\frac{R_2}{R}\right)\left(\frac{1}{8}\right)$$

the condition $T(j\omega_o) = -1$ is satisfied when $\dfrac{R_2}{R} = 8$

Example Problem 15.4

Determine the oscillation frequency and required amplifier gain for a phase-shift oscillator.

Consider the phase-shift oscillator in the Figure with parameters C = 0.1 μF and R = 1 kΩ.



Solution:

The oscillation frequency is $$f_o = \frac{1}{2\pi\sqrt{3}RC} = \frac{1}{2\pi\sqrt{3}(10^3)(0.1 \times 10^{-6})} = 919\,\text{Hz}$$

The minimum amplifier gain magnitude is 8; therefore, the minimum value of R_2 is 8 kΩ

Practice Problem 15.24

Consider the phase shift oscillator in Figure 15.15. (a) For $R = 20\,\text{k}\Omega$ and $C = 0.001\,\mu\text{F}$, determine the frequency of oscillation. What is the required value of R_2? (b) Design the circuit such that the frequency of oscillation is $f_o = 25\,\text{kHz}$. Let $R = 20\,\text{k}\Omega$.

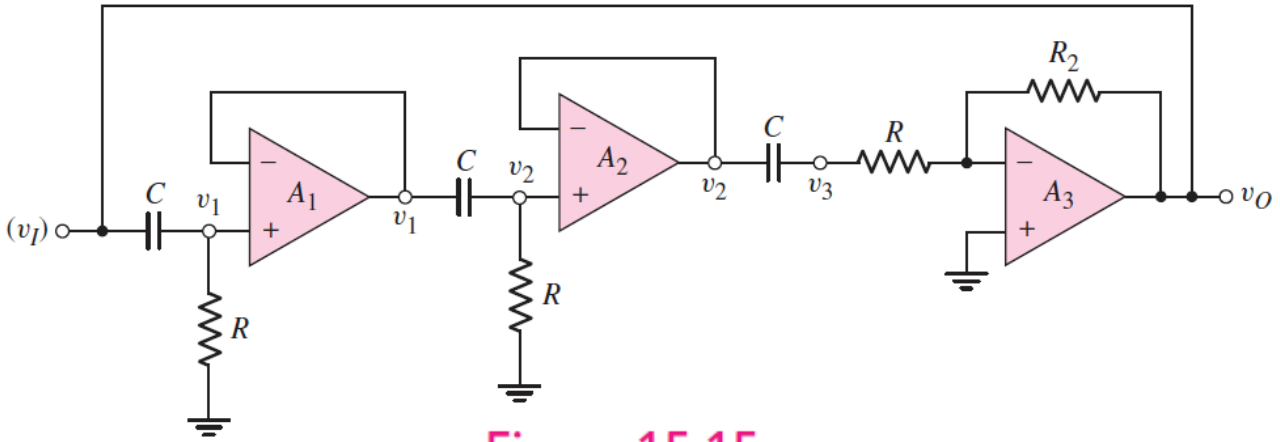


Figure 15.15

Solution:

$$\text{(a)}\quad f_o = \frac{1}{2\pi\sqrt{3}RC} = \frac{1}{2\pi\sqrt{3}\left(20\times10^{3}\right)\left(0.001\times10^{-6}\right)}$$

$$f_o = 4.59\,\text{kHz},\quad R_2 = 8R = 160\,\text{k}\Omega$$

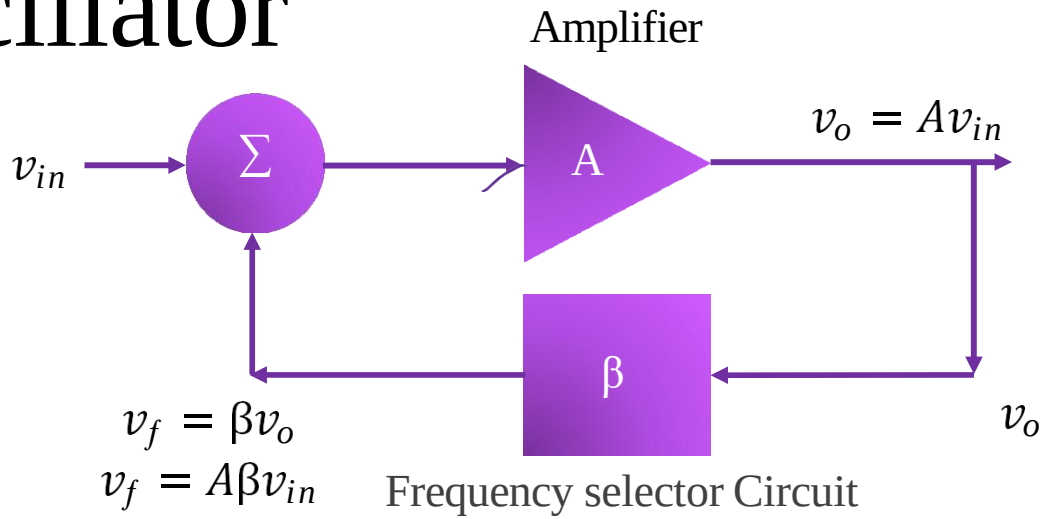
$$\text{(b)}\quad C = \frac{1}{2\pi\sqrt{3}Rf_o} = \frac{1}{2\pi\sqrt{3}\left(20\times10^{3}\right)\left(25\times10^{3}\right)}$$

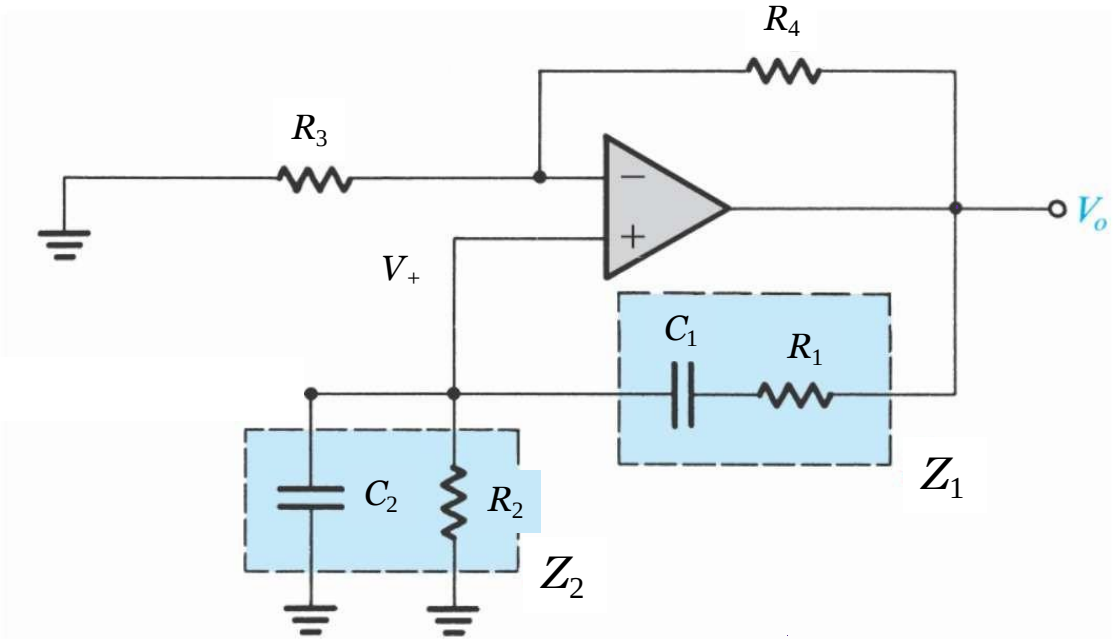
$$C = 184\,\text{pF},\quad R_2 = 160\,\text{k}\Omega$$

Wein Bridge Oscillator

- ✓ Harmonic oscillator

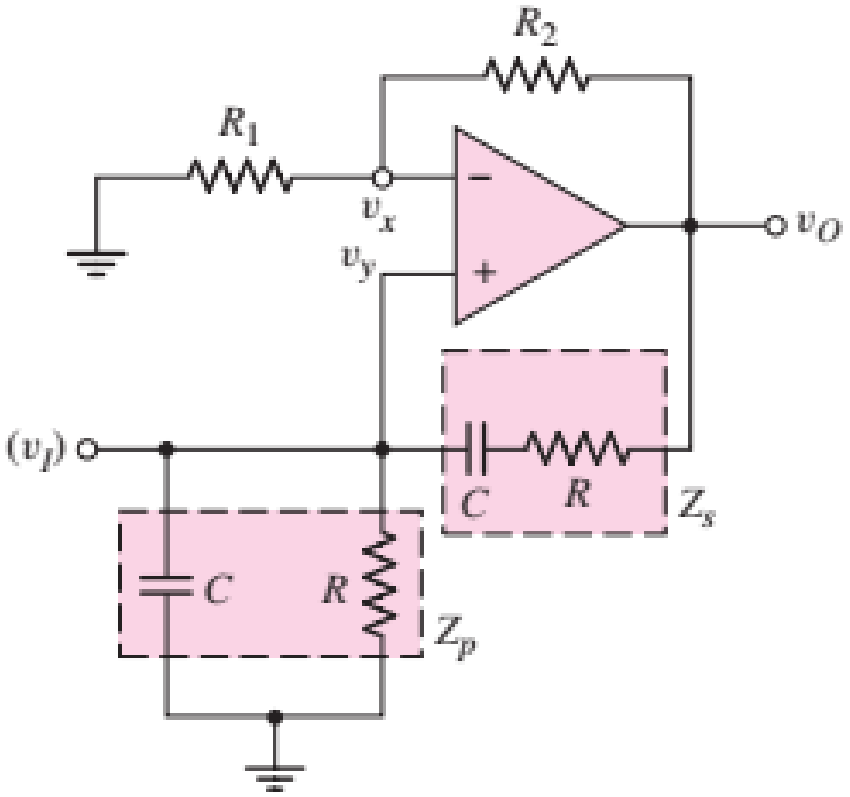
  (generates sinusoidal waveform)
- ✓ Feedback network: RC
- ✓ In audio frequency range





Wien-Bridge Oscillator

The circuit consists of an op-amp connected in a noninverting configuration and two RC networks connected as the frequency-selecting feedback circuit.

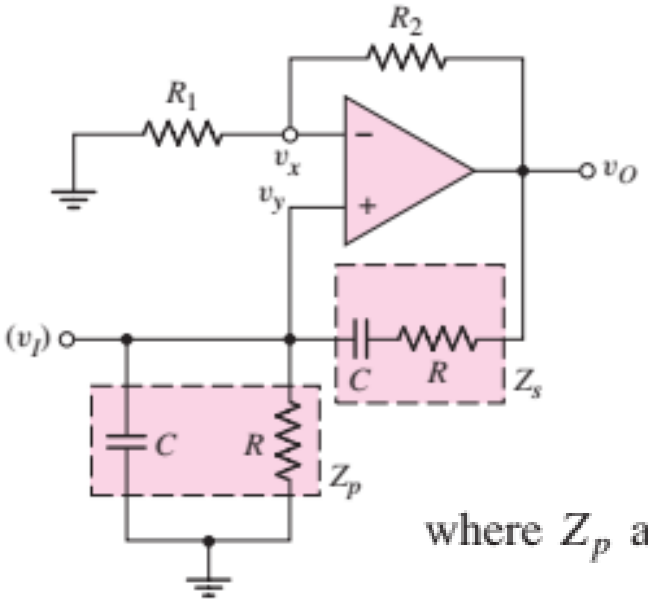


Since the noninverting amplifier introduces zero phase shift, the frequency-selective feedback circuit must also introduce zero phase shift to create the positive feedback condition.

The loop gain is the product of the amplifier gain and the feedback transfer function,

$$T(s) = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{Z_p}{Z_p + Z_s}\right)$$

Wien-Bridge Oscillator



$$T(s) = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{Z_p}{Z_p + Z_s}\right)$$

where Z_p and Z_s are the parallel and series RC network impedances.

$$Z_p = \frac{R}{1 + sRC} \qquad Z_s = \frac{1 + sRC}{sC}$$

Combining Equations $$T(s) = \left(1 + \frac{R_2}{R_1}\right)\left[\frac{1}{3 + sRC + (1/sRC)}\right]$$

Since this circuit has no explicit negative feedback

the condition for oscillation is $T(j\omega_o) = 1$

Calculation of *T(s)= A(s) β(s)*

$$\beta(s) = \left(\frac{Z_p}{Z_p + Z_s}\right) = \frac{\frac{R}{1+sRC}}{\frac{sRC+1}{sC} + \frac{R}{1+sRC}}$$

$$\beta(s) = \frac{\frac{R}{1+sRC} \cdot (1+sRC) \cdot sC}{\frac{sRC+1}{sC} \cdot (1+sRC) \cdot sC \; + \; \frac{R}{1+sRC} \cdot (1+sRC) \cdot sC}$$

$$\beta(s) = \frac{RsC}{(sRC+1)(1+sRC) \quad + \quad RsC} = \frac{RsC}{(sRC+1)^2 + RsC}$$

$$= \frac{RsC}{s^2R^2C^2 + 2sRC + 1 \; + \; RsC}$$

$$= \frac{sRC}{s^2R^2C^2 + 3sRC + 1} = \frac{1}{sRC + 3 + \frac{1}{sRC}}$$

the condition for oscillation is

$$T(j\omega_o) = 1 = \left(1 + \frac{R_2}{R_1}\right)\left[\frac{1}{3 + j\omega_o RC + (1/j\omega_o RC)}\right]$$

Since T(jωo) must be real, the imaginary component of the above Equation must be zero;

$$j\omega_o RC + \frac{1}{j\omega_o RC} = 0 \qquad \omega_o = \frac{1}{RC}$$

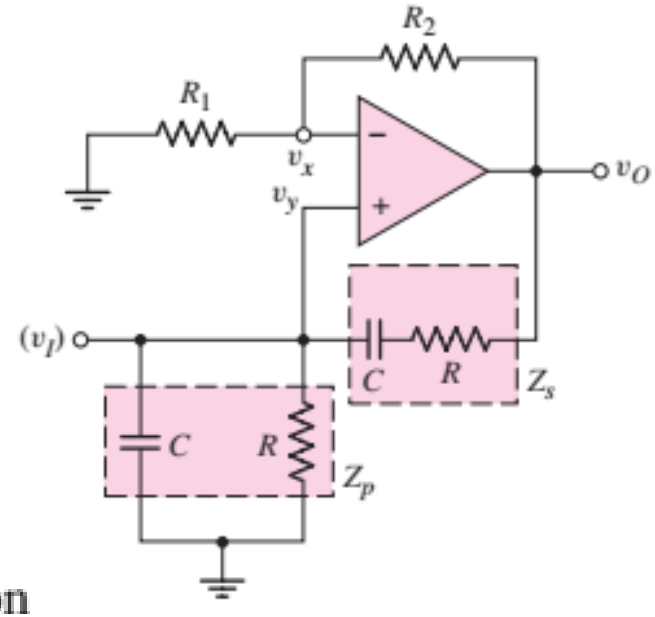
The magnitude condition is then

$$1 = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{1}{3}\right) \qquad \frac{R_2}{R_1} = 2$$

Example Problem 15.5

Design a Wien-bridge circuit to oscillate at a specified frequency.

Specifications: Design the Wien-bridge oscillator shown in the Figure to oscillate at f_o = 20 kHz.

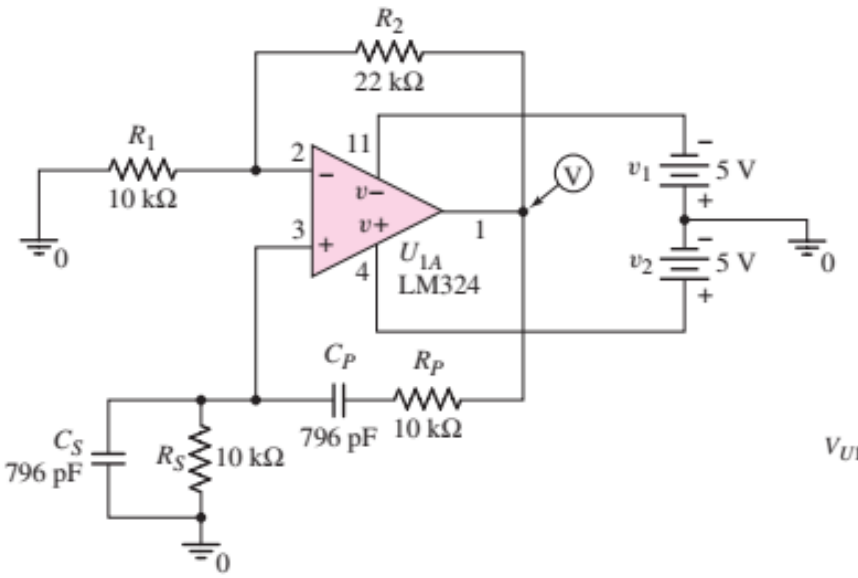


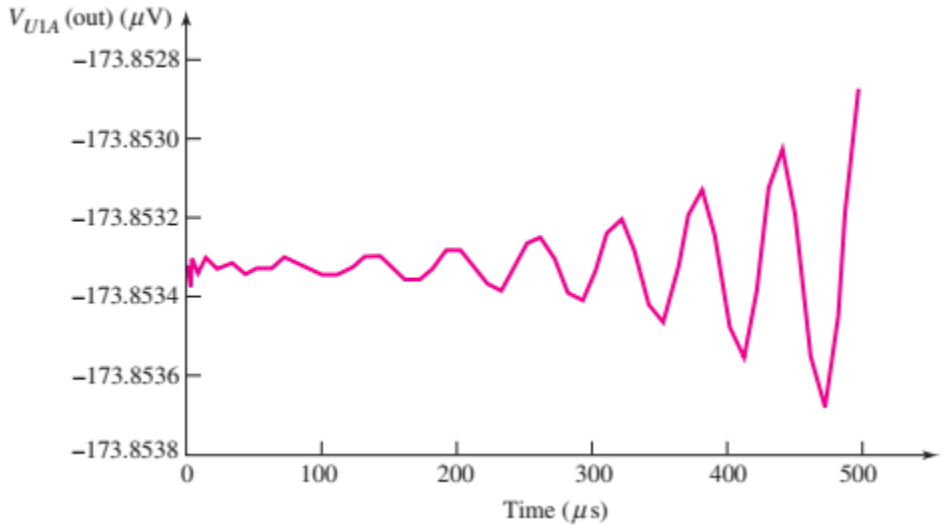
Solution: The oscillation frequency given by Equation

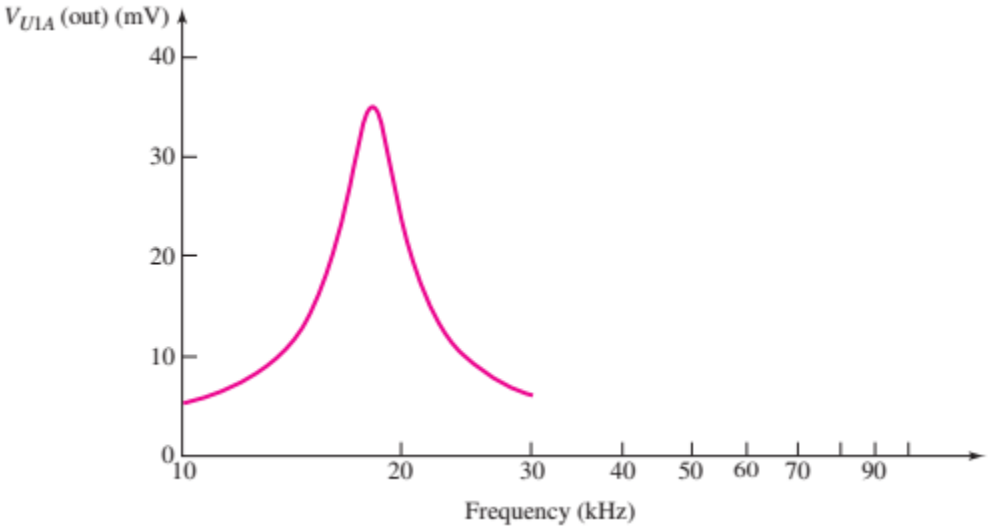
$$RC = \frac{1}{2\pi f_o} = \frac{1}{2\pi(20 \times 10^3)} = 7.96 \times 10^{-6}$$

A 10 kΩ resistor and 796 pF capacitor satisfy this requirement. Since the amplifier resistor ratio must be $R_2/R_1 = 2$, we could, for example, have $R_2 = 20\,\text{k}\Omega$ and $R_1 = 10\,\text{k}\Omega$, which would satisfy the requirement.

Simulation of Problem 15.5



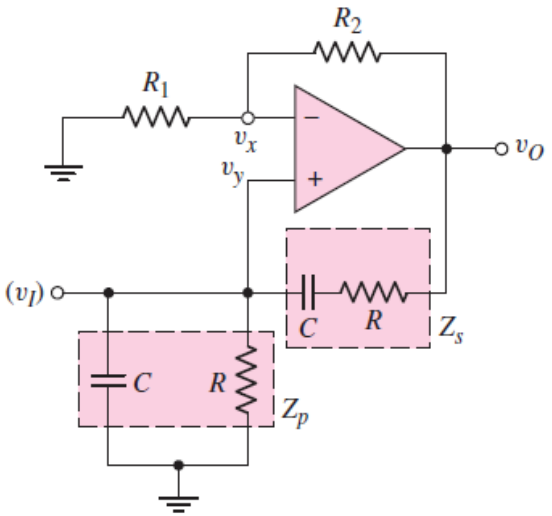




Practice Problem 15.34

Design the Wein-bridge oscillator to oscillate at $f_o = 35\,\text{kHz}$. Choose appropriate component values.

Solution:



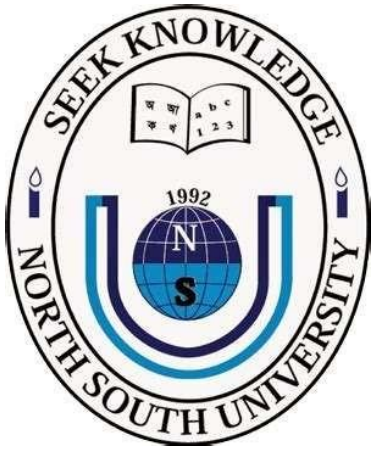
$$f_o = \frac{1}{2\pi RC}$$

$$RC = \frac{1}{2\pi f_o} = \frac{1}{2\pi\left(35\times10^3\right)} = 4.547\times10^{-6}$$

Let $C = 0.001\,\mu\text{F}$, then $R = 4.55\,\text{k}\Omega$

Set $\dfrac{R_2}{R_1} = 2$

Analog Electronics II: Op-amp & 555 Timer



Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

Lecture # 14

OSCILLATORS

Reference:

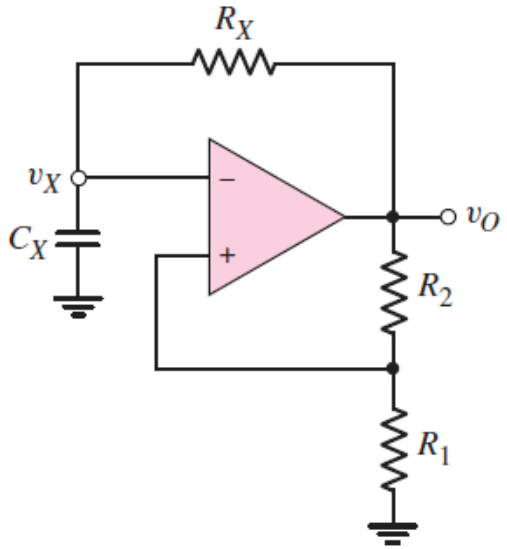
1. Lecture Slides
2. Microelectronic Circuit Analysis and Design, by Donald Neaman, Chapter 15 (15.2)
3. Operational Amplifiers and Linear Integrated Circuits, by Robert Coughlin Chapter Chapter 6 (6.1-6.4)

NONSINUSOIDAL OSCILLATORS

Chapter 15.4. from Microelectronic Circuit Analysis and Design by Donald A Neaman 4th Ed.

Many applications, especially digital electronic systems, use a nonsinusoidal squarewave oscillator to provide a clock signal for the system.
In other applications, a single pulse of known height and width is used to initiate a particular set of functions.

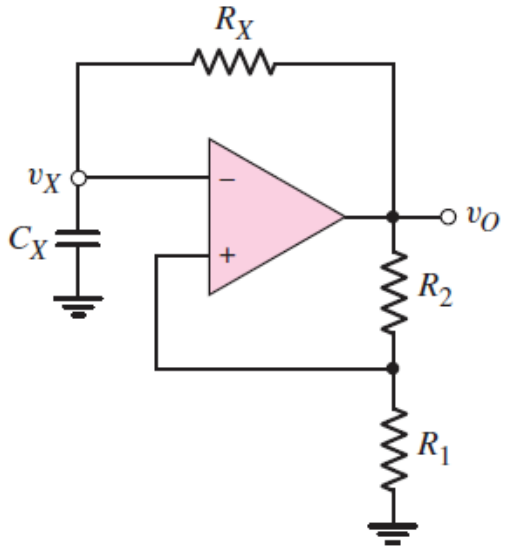
Schmitt Trigger Oscillator



The Schmitt trigger can be used in an oscillator circuit to generate a square-wave output signal.

Schmitt Trigger Oscillator

This circuit has no stable states. It is therefore called an **astable multivibrator.**



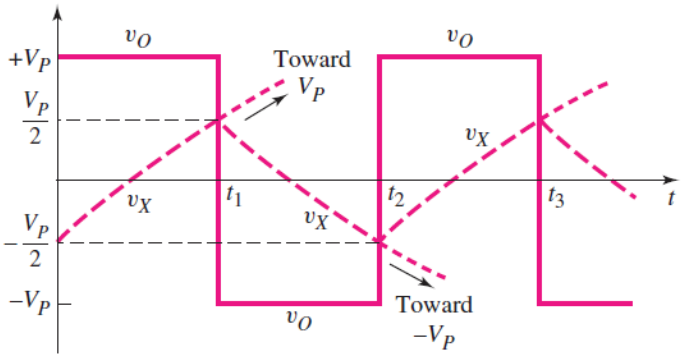
The high saturated output denoted by $V_H = V_P$
The low saturated output denoted by $V_L = -V_P$

$$R_1 = R_2 \equiv R$$

If v_O is low, or $v_O = -V_P$, then $v+ = -(1/2)V_P$.

When v_X drops just slightly below v+, the output switches high so that $v_O = +V_P$ and $v+ = +(1/2)V_P$

The R_XC_X network sees a positive step-increase in voltage, so capacitor C_X begins to charge and voltage v_X starts to increase toward a final value of V_P .



Schmitt Trigger Oscillator

The general equation for the voltage across a capacitor in an RC network is

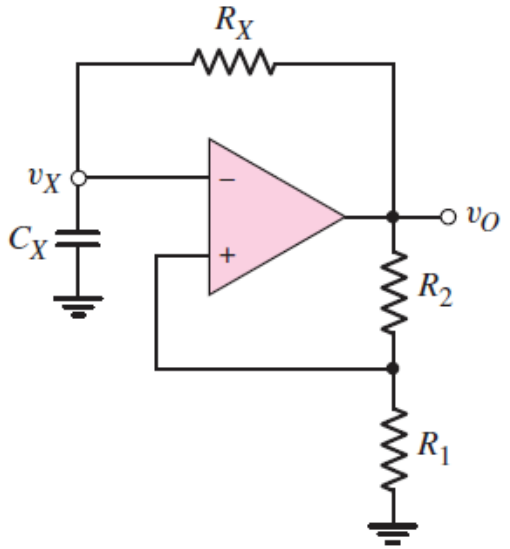
$$v_X = v_{\text{Final}} + (v_{\text{Initial}} - v_{\text{Final}})\, e^{-t/\tau}$$

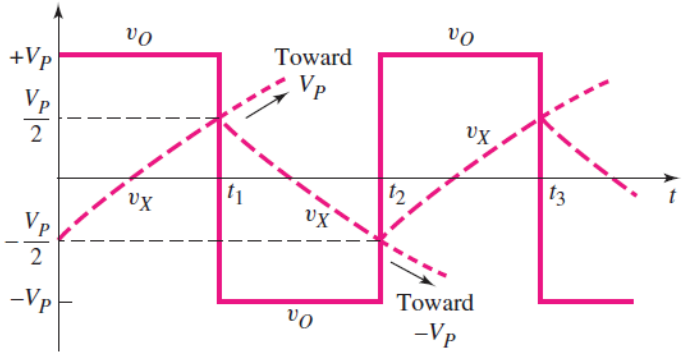
$$v_X = V_P + \left(-\frac{V_P}{2} - V_P\right) e^{-t/\tau_x}$$

$$v_X = V_P - \frac{3V_P}{2} e^{-t/\tau_x}$$

where $\tau_x = R_X C_X$

$$R_1 = R_2 \equiv R$$

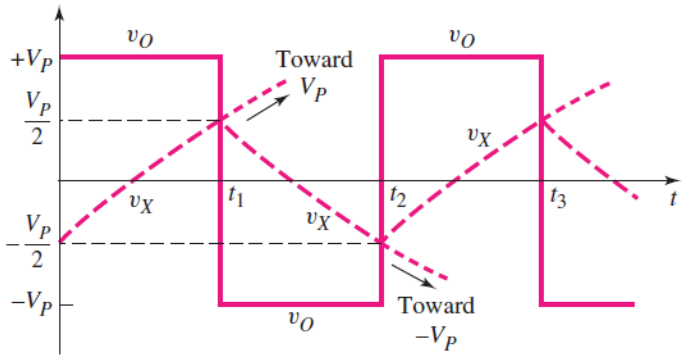




Voltage v_X increases exponentially with time toward a final voltage V_P. However, when v_X becomes just slightly greater than $v_+ = +(1/2)V_P$, the output switches to its low state of $v_O = -V_P$ and $v_+ = -(1/2)V_P$.

Schmitt Trigger Oscillator

The $R_X C_X$ network sees a negative step change in voltage, so capacitor C_X now begins to discharge and voltage v_X starts to decrease toward a final value of $-V_P$.



$$v_X = -V_P + \left[+\frac{V_P}{2} - (-V_P)\right] e^{-(t-t_1)/\tau_x}$$

$$v_X = -V_P + \frac{3V_P}{2} e^{-(t-t_1)/\tau_x}$$

Time t_1 can be found by setting $t = t_1$ when $v_X = V_P/2$,

$$\frac{V_P}{2} = V_P - \frac{3V_P}{2} e^{-t_1/\tau_x}$$

$$t_1 = \tau_x \ln 3 = 1.1 R_X C_X$$

$$T = 2.2 R_X C_X \qquad f = \frac{1}{T} = \frac{1}{2.2 R_X C_X}$$

Example Problem 15.8

Objective: Design a Schmitt trigger oscillator for a specified frequency.

Specifications: Assume that an ideal comparator is available. Use standard-valued resistors and capacitors in the final design.

Consider the oscillator in Figure 15.35. Design the circuit to oscillate at $f_o = 1$ kHz.

Solution: Using Equation (15.88), we can write

$$R_X C_X = \frac{1}{2.2 f_o} = \frac{1}{2.2(10^3)} = 4.55 \times 10^{-4}$$

If $C_X = 0.1\,\mu$F, then $R_X = 4.55\,\text{k}\Omega$.

Multivibrator

Multivibrator is an electronic circuit used to implement 2-state devices, such as oscillator, timer, flip flop, etc.

2-state: 2 voltage levels V_{high} and V_{low}
In digital electronics, Logic 1 and Logic 0

Multivibrators can be made with op-amp / 555 timer / transistor.

Based on the number of stable states, multivibrators can be categorized into 3 types

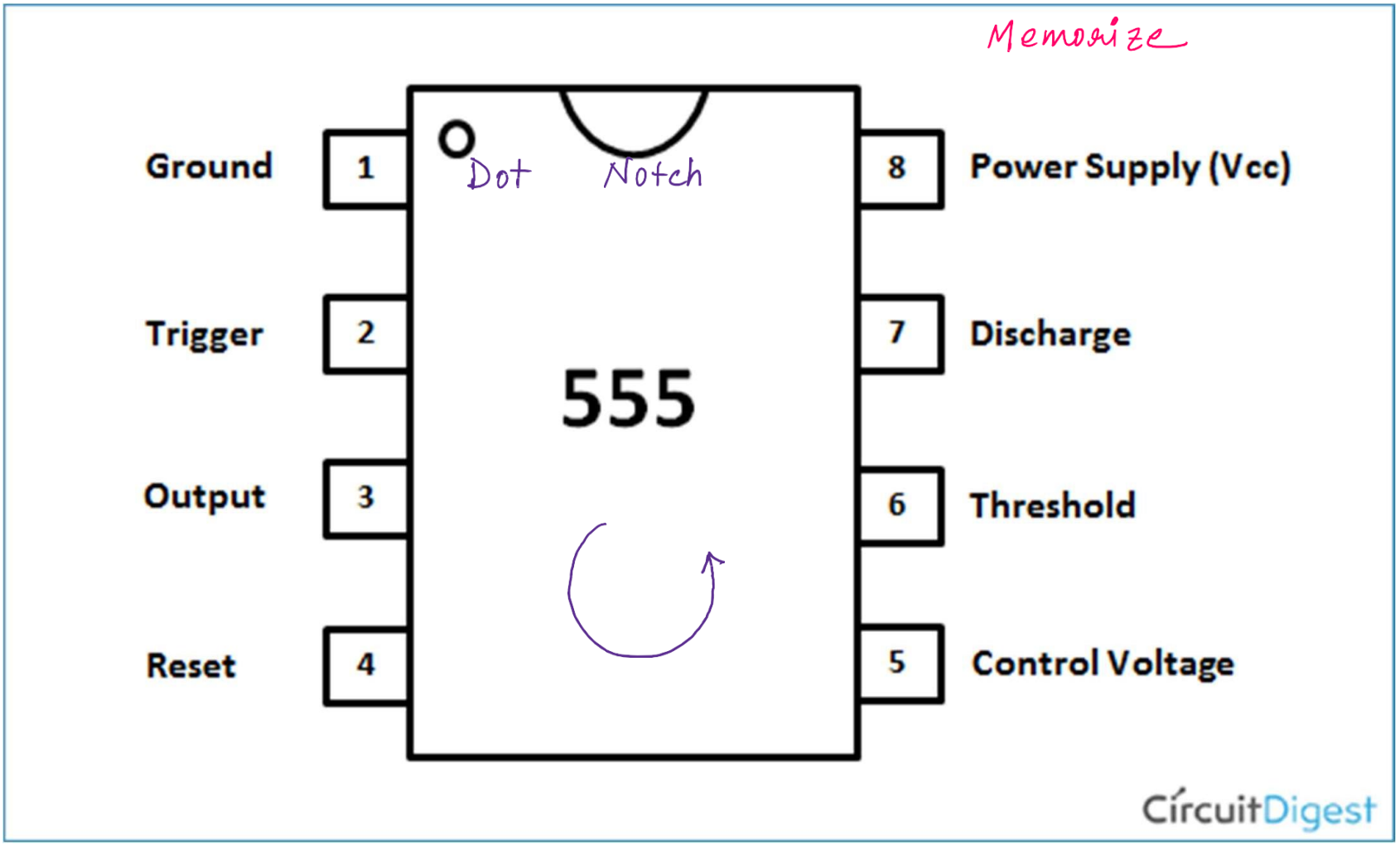
555 Timer

Popular in timing related applications

- ✓ Tone and alarm generation
- ✓ Frequency division
- ✓ Provide timing delays
- ✓ As relaxation oscillator

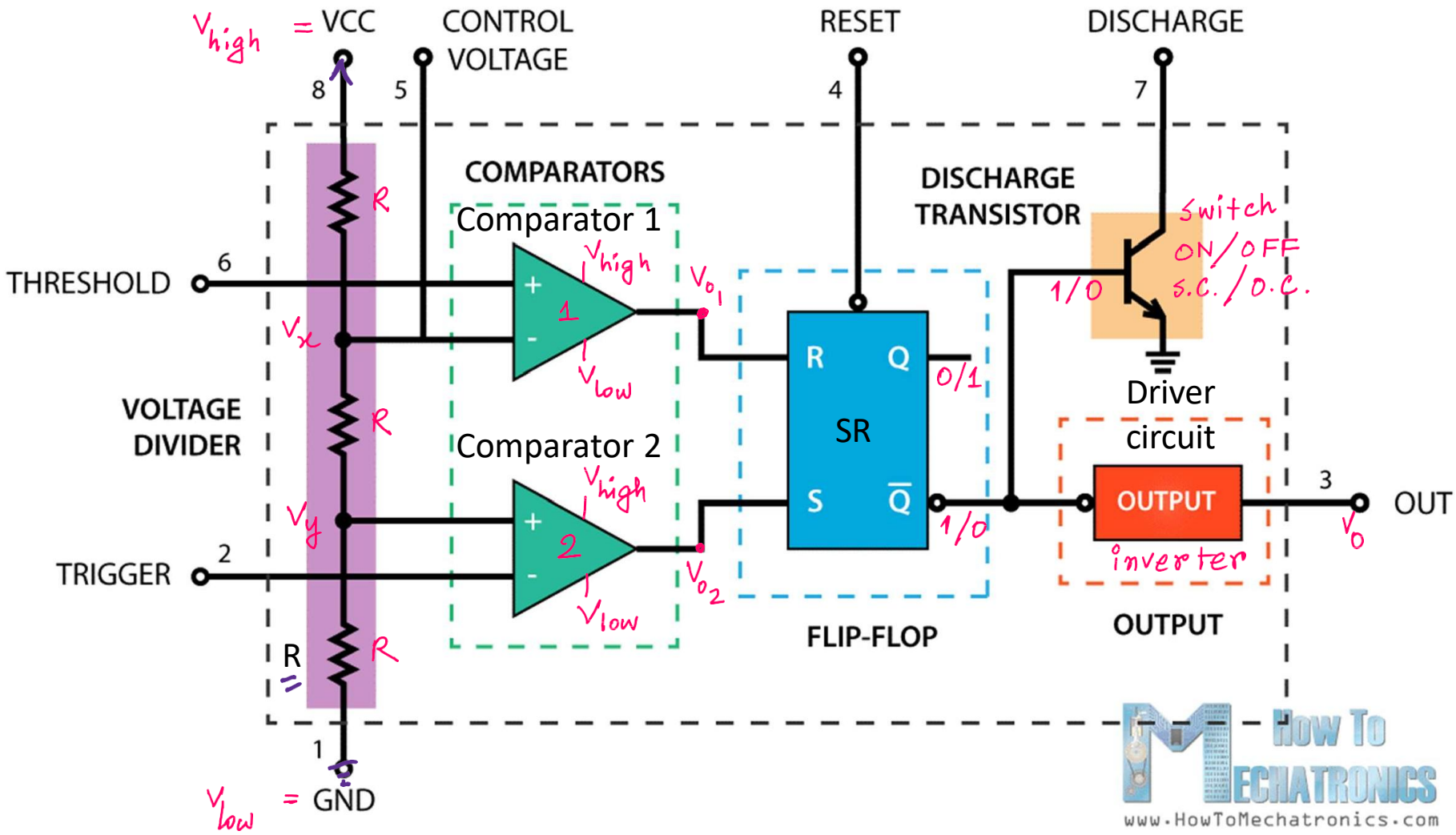
555 Timer – Pin diagram

Dual-in line packaging



8 pin DIP Package

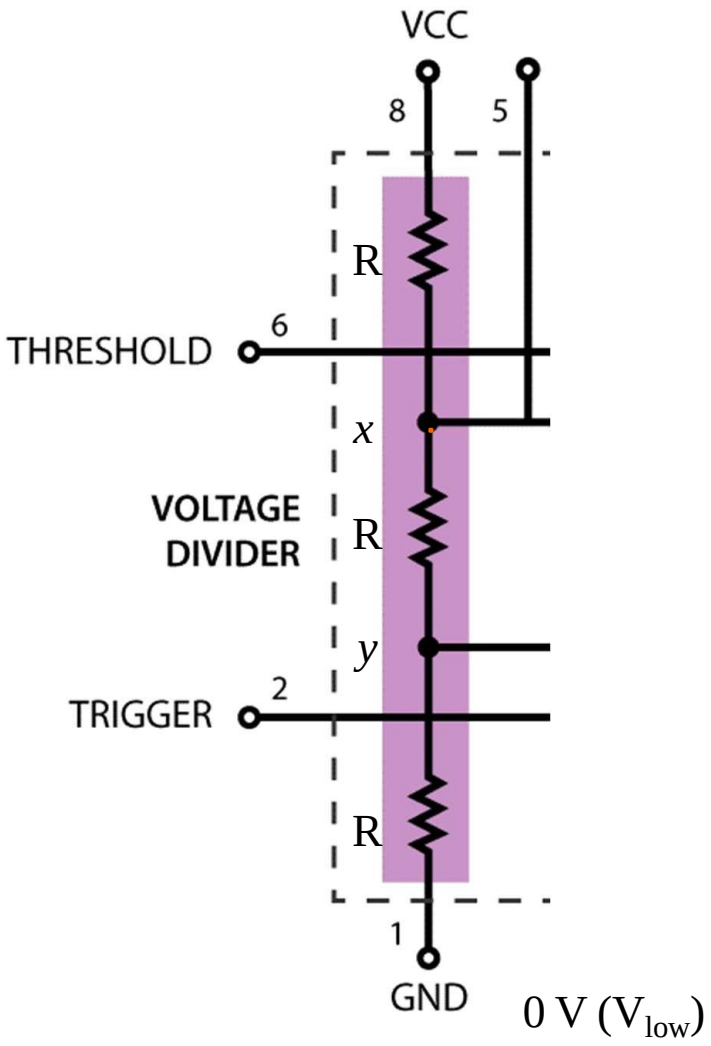
555 Timer – Block diagram



How To MECHATRONICS
www.HowToMechatronics.com

555 Timer – Voltage divider network

Supply voltage pin: 4.5 to 15 V (V_{high})



0 V (V_{low})

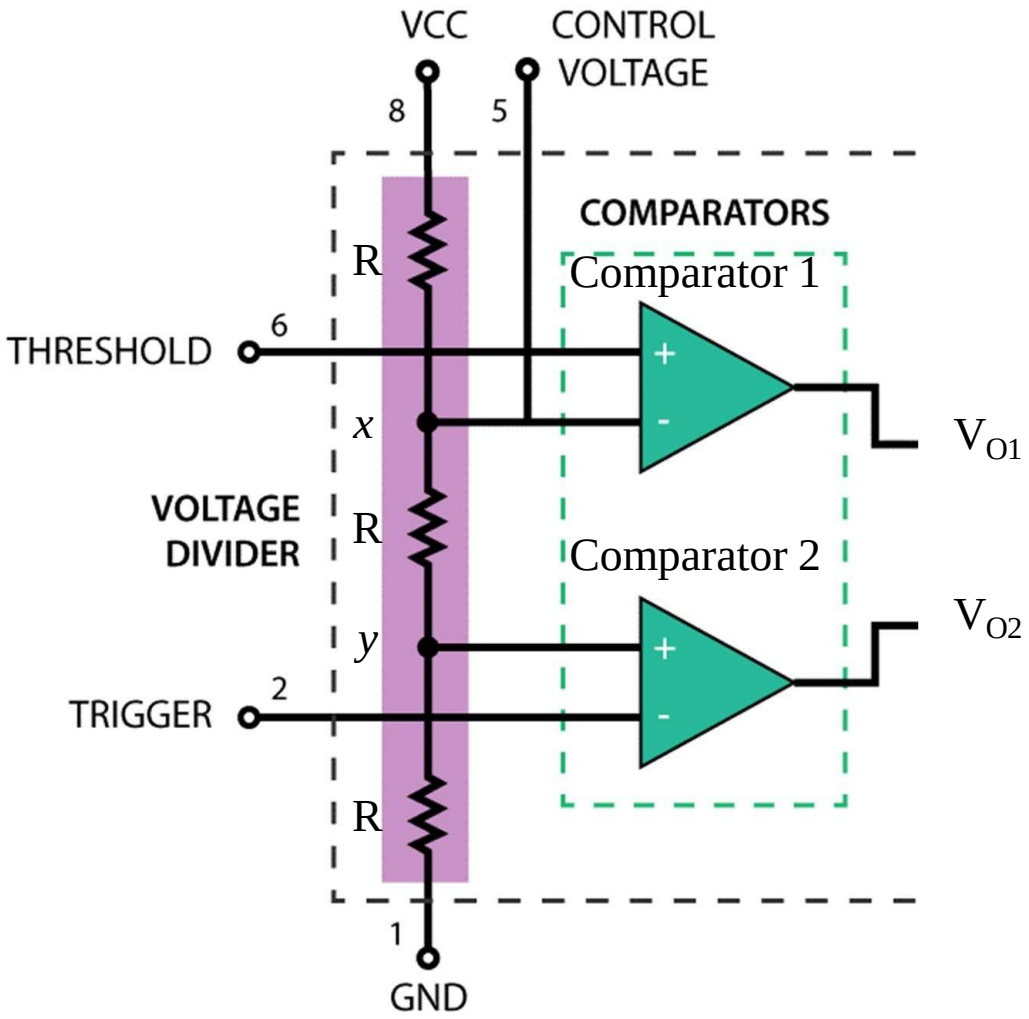
Applying voltage divider rule:

$$V_x = \frac{2R}{3R} V_{CC} = \frac{2}{3} V_{CC}$$

$$V_y = \frac{R}{3R} V_{CC} = \frac{1}{3} V_{CC}$$

555 Timer – comparators

Pin 6 and pin 2 can change the timings of the output signal



Pin 6, V_x: input of comparator 1

$V_{threshold} > V_x$: $V_{O1} = V_{high}$

$V_+ > V_-$

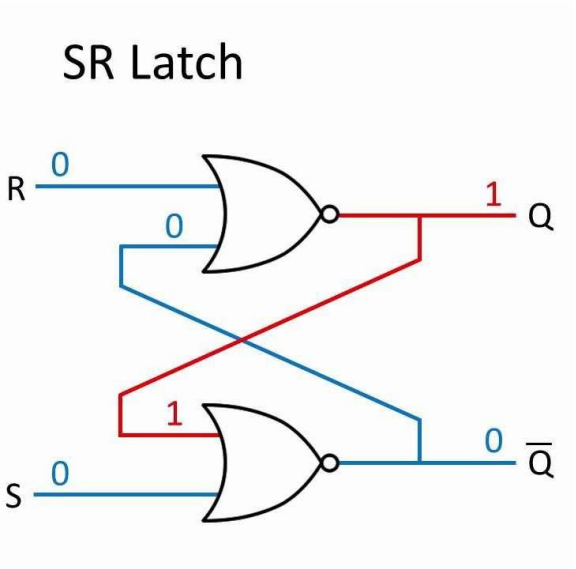
Pin 2, V_y: input of comparator 2

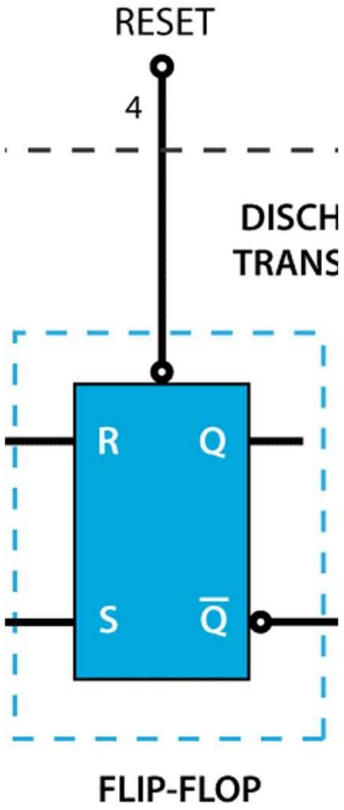
$V_{trigger} < V_y$: $V_{O2} = V_{high}$

$V_- < V_+$

SR Flip-flop

S	R	Q(n)	Q(n+1)
0	0	0/1	0/1
0	1	0/1	0 (RESET)
1	0	0/1	1 (SET)
1	1	0/1	× (INVALID)



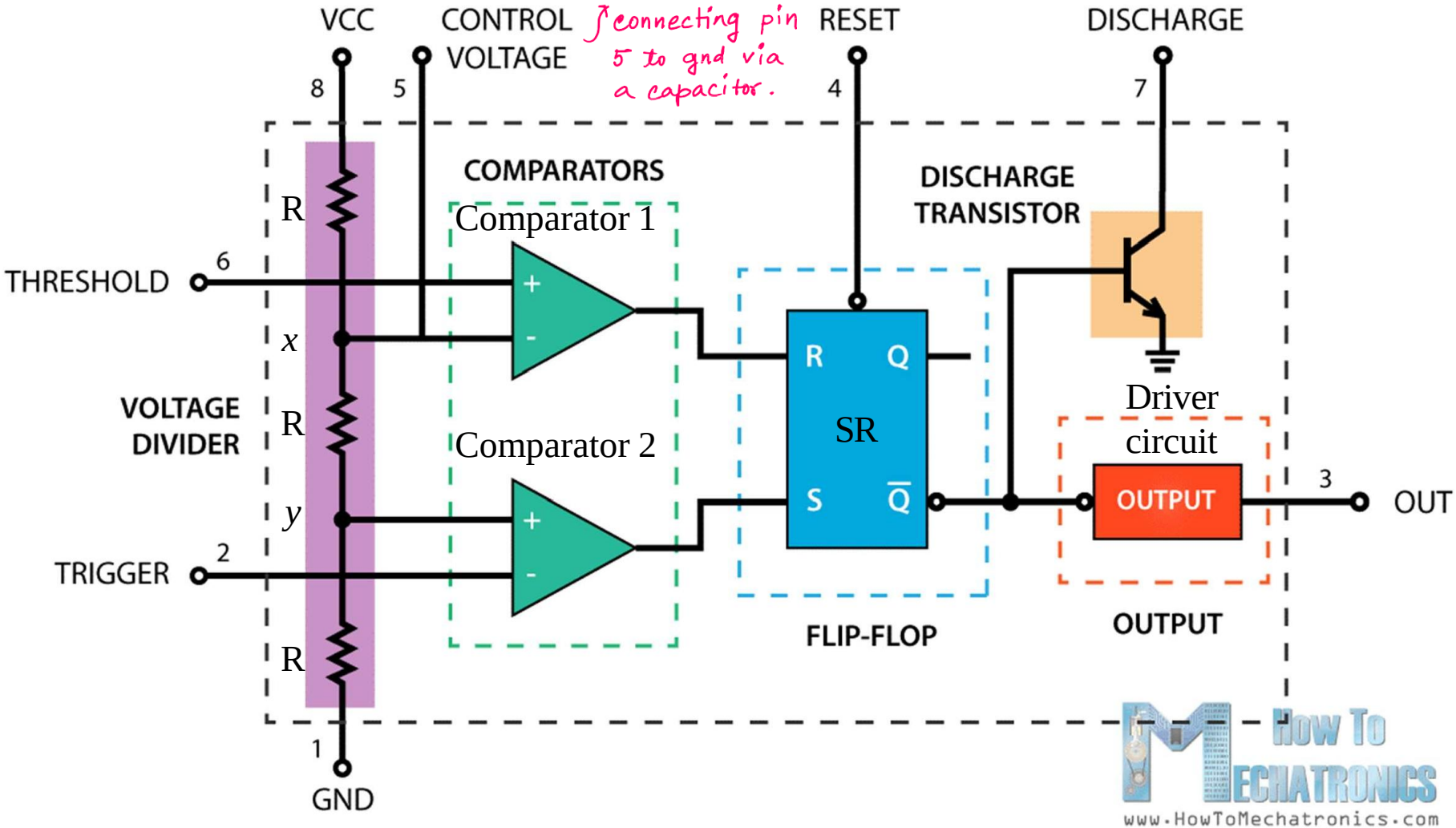


555 Timer – Block diagram

External R and C are connected between pin 6 (threshold) and 7(discharge) between pins 8 (Vcc) and 1 (Gnd)

Externally other V_{ref} can be applied to comparator 1, it can change the o/p signal timing, too

We will deactivate control voltage option by connecting pin 5 to gnd via a capacitor.

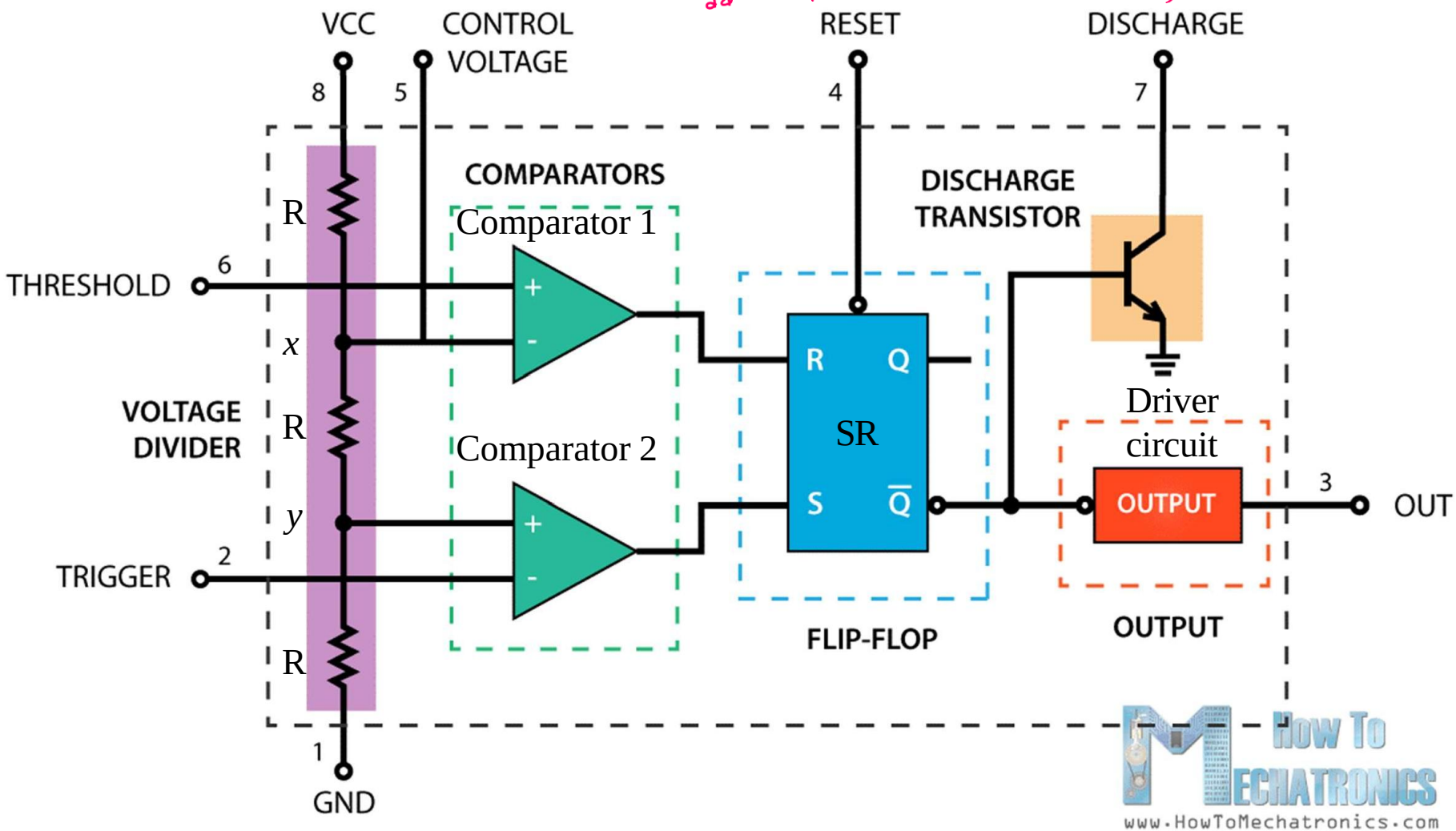


How To MECHATRONICS
www.HowToMechatronics.com

555 Timer – Block diagram

Reset pin: Logic 0 (Active low pin)
It resets the flip-flop and o/p of flip-flop

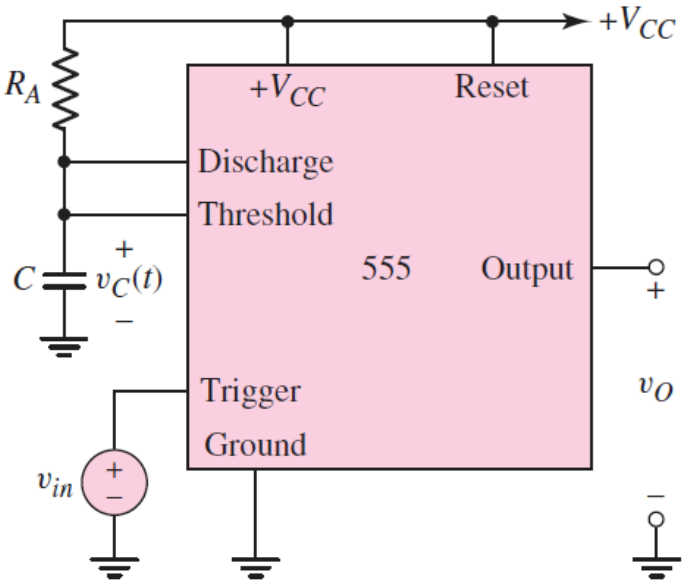
We will deactivate reset pin by applying V_{dd} at pin 4 (∵ active low pin).



How To MECHATRONICS
www.HowToMechatronics.com

Monostable Multivibrator

A monostable multivibrator, also called a one-shot, operates by charging a timing capacitor with a current set by an external resistance.

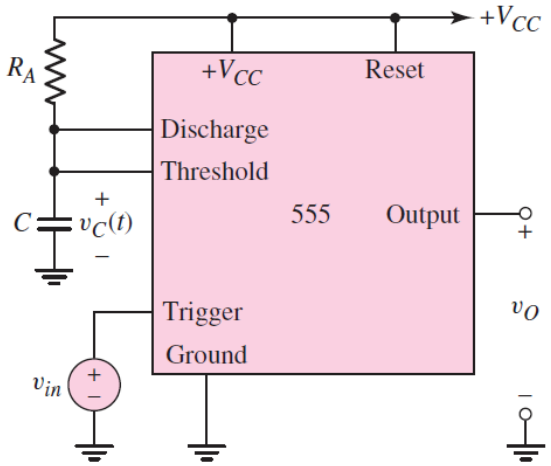


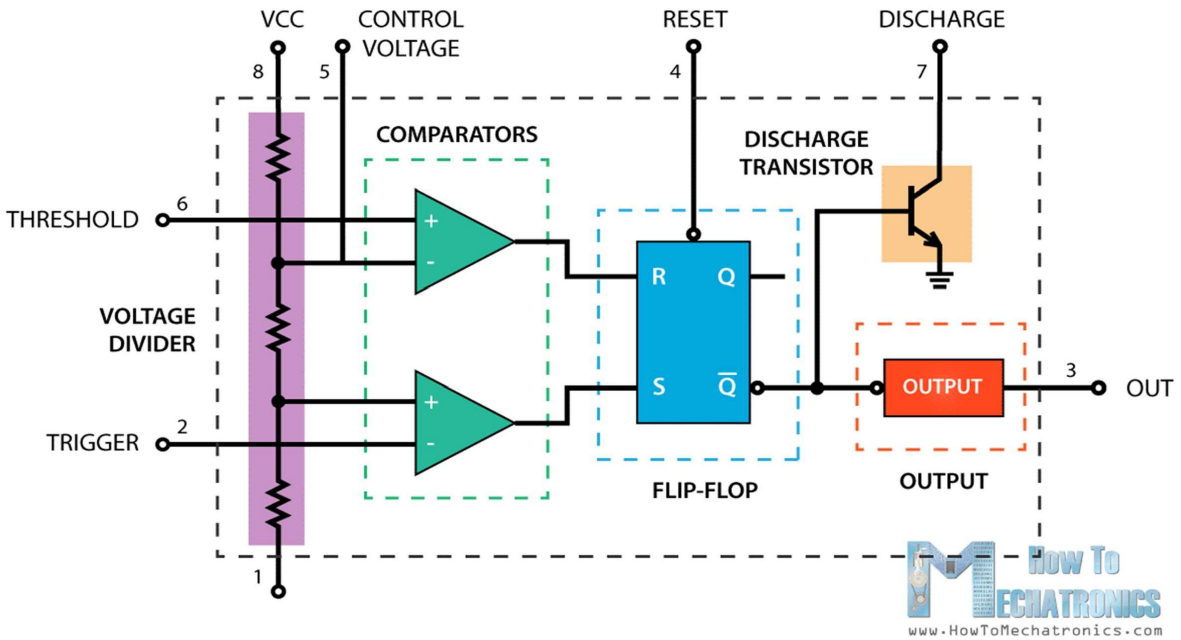
The 555 circuit connected as a monostable multivibrator

With a high voltage V_+ applied to the trigger input, the trigger comparator output is low, the flip-flop output .Q is high.

The discharge transistor is turned on, and the timing capacitor C is discharged to nearly ground potential. The output of the 555 circuit is then low,

Monostable Multivibrator

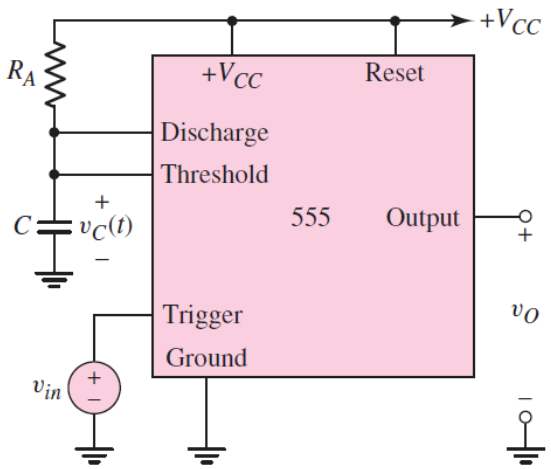


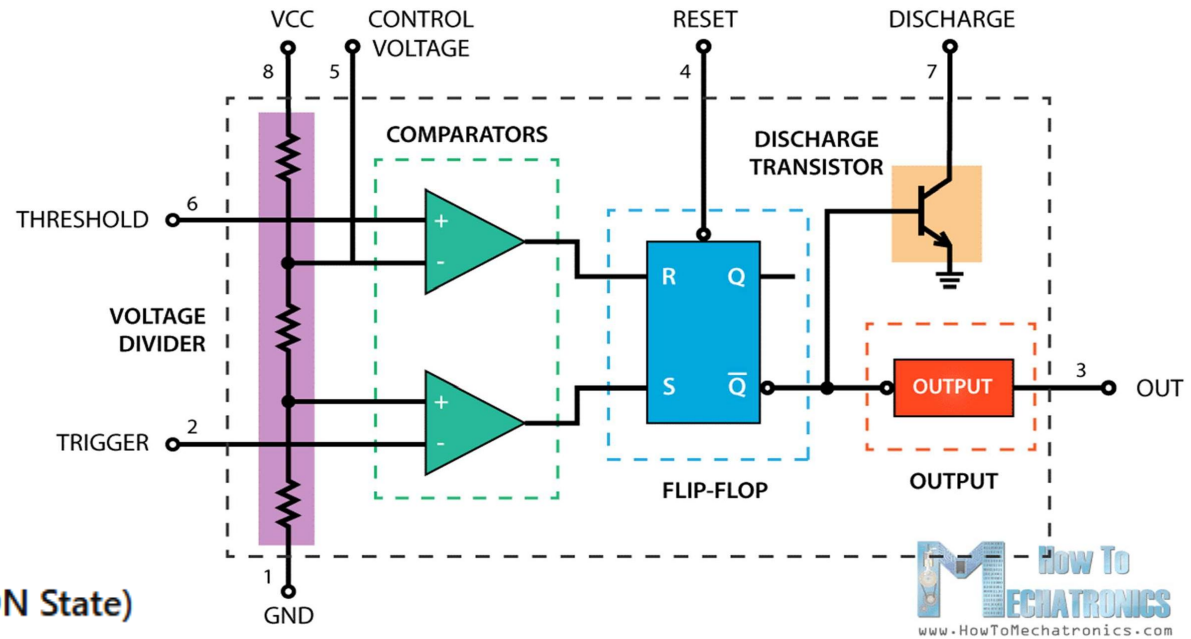


Step 1: Negative Trigger Signal

- A **negative pulse** (voltage drops below **1/3 Vcc**) is applied to the **Trigger input (pin 2)**.
- The **Trigger Comparator** detects this and **sets the SR Flip-Flop**.
- This makes the **output HIGH (Q = 1)**.
- At the same time, the **Discharge Transistor turns OFF**, allowing the capacitor to **start charging** through resistor R.

Monostable Multivibrator





Step 2: Capacitor Charging (Unstable ON State)

- The capacitor starts charging from 0V towards Vcc via the resistor.
- Voltage across the capacitor $V_C(t) = V_{cc}(1 - e^{-t/RC})$.
- As long as $V_C < 2/3Vcc$, nothing changes.

Step 3: Threshold Reached (Return to Stable State)

- Once V_C reaches **2/3 Vcc**, the **Threshold Comparator** triggers.
- It **resets the SR Flip-Flop**.
- The **output goes LOW** (Q = 0).
- The **Discharge Transistor turns ON**, discharging the capacitor quickly to 0V.

Back to Initial State Now the system is **back to the stable state** (OFF), waiting for another trigger.

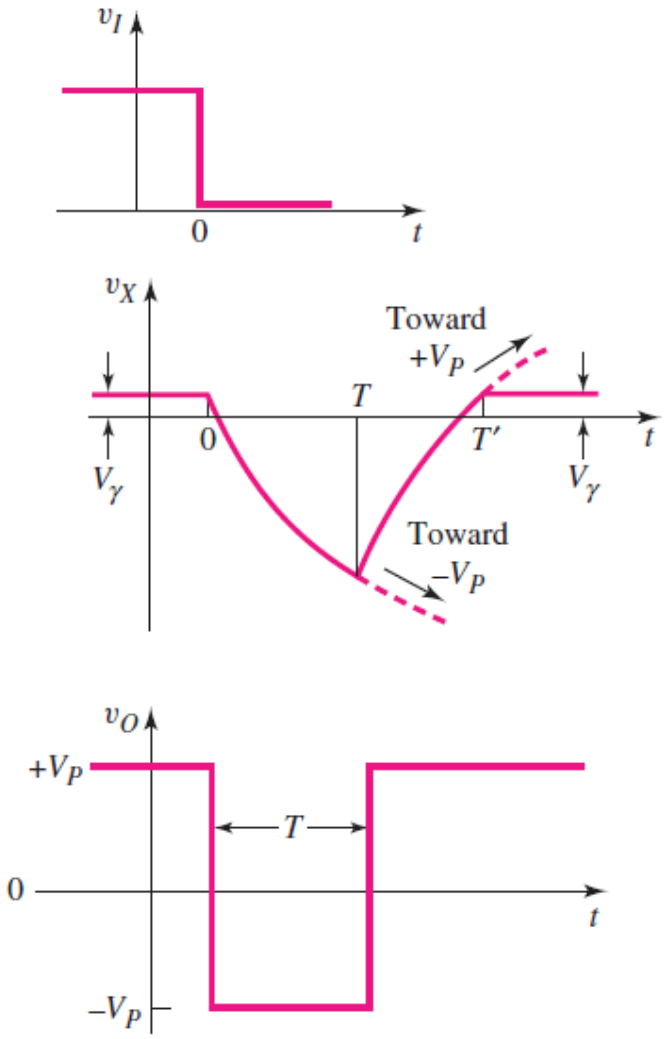
Monostable Multivibrator

When a negative-going pulse is applied to the trigger input, the output of the trigger comparator goes high when the trigger pulse drops below $(1/3)V_+$.
Then output Q goes low, which means that the output of the 555 goes high, and the discharge transistor turns off.

The output of the 555 remains high even if the trigger pulse returns
to its initial high value, because the reset input to the flip-flop is still low.

The capacitor voltage is given by

$$v(t) = V^+(1 - e^{-t/RC})$$



Monostable Multivibrator

The width of the output pulse is determined by setting v(t) = (2/3)V+ and t = T

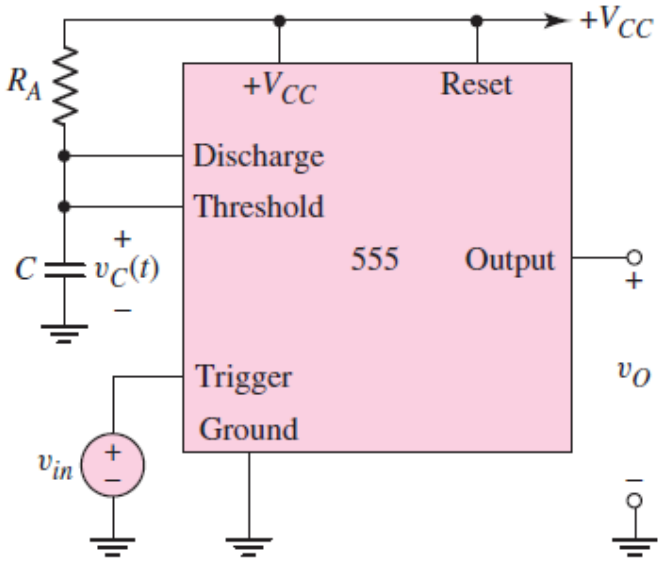
$$\left(\frac{2}{3}\right)V^{+} = V^{+}\left(1 - e^{-T/RC}\right)$$

$$T = RC\,\ln(3) = 1.1\,RC$$

Example Problem 15.10

Objective: Design a 555 IC as a monostable multivibrator to produce a specified output pulse width.

Specifications: The circuit with the configuration shown in Figure 15.40 is to be designed to produce an output pulse width of 100 μs.



Solution:

Let $C = 15\,\text{nF}$.

$$R = \frac{T}{1.1C} = \frac{100 \times 10^{-6}}{(1.1)(15 \times 10^{-9})} \Rightarrow 6.06\,\text{k}\Omega$$

Astable Multivibrator (Free running) - Review

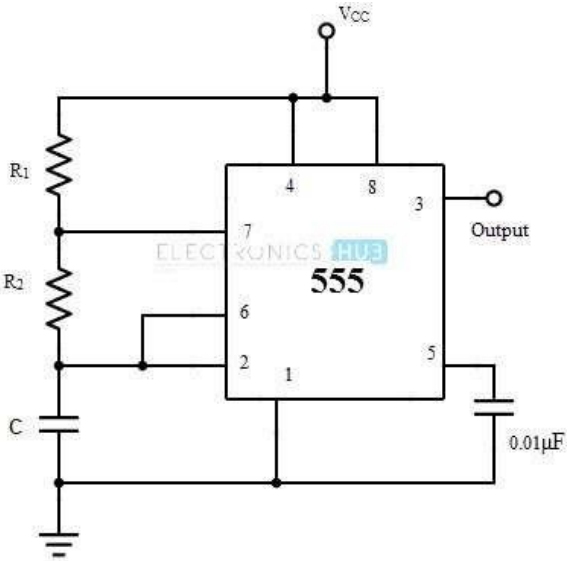
Output is not stable in any of the two states.

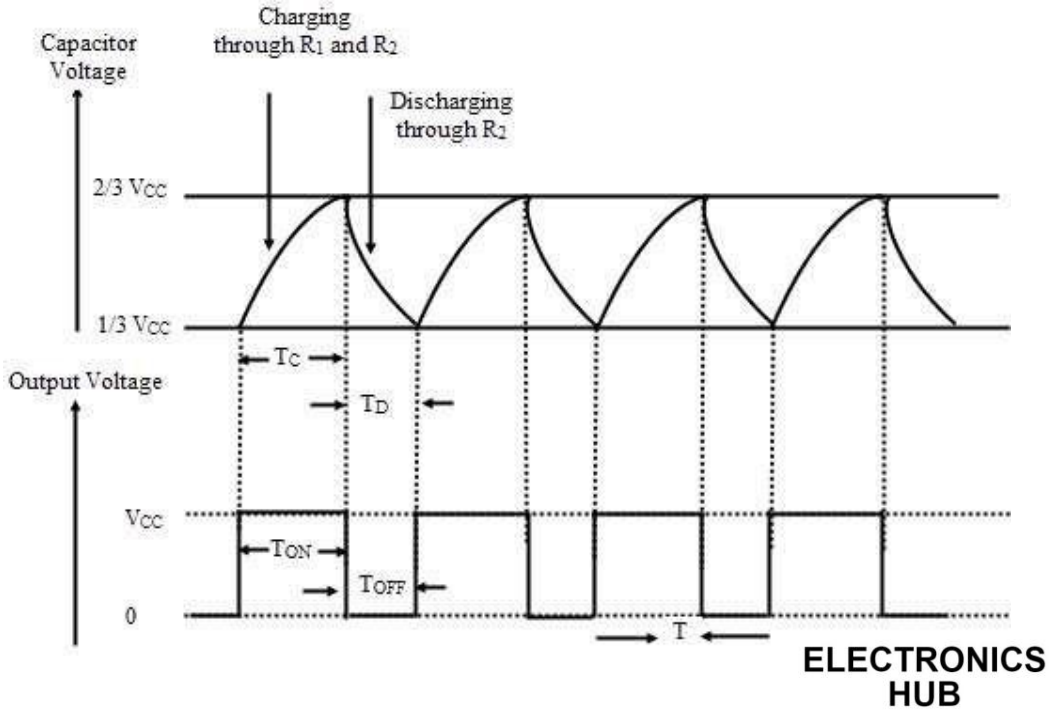
Output continuously change between the 2 states.

There is no input trigger

T_{high} and T_{low} depend on externally connected R and C

Application: relaxation oscillator





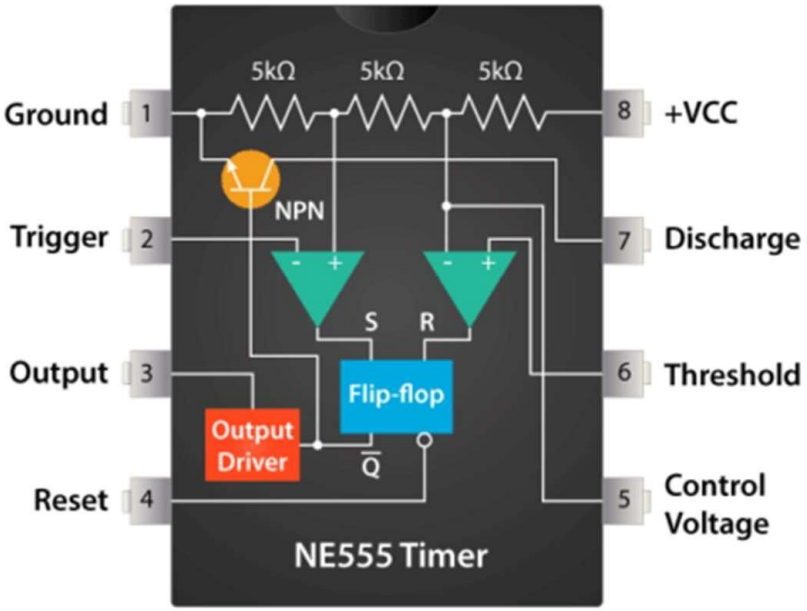
Astable Multivibrator

Reset (4) and control (5) pins are not used.

Reset pin (4) is connected to $V_{high} = V_{cc}$ (∵ active low).

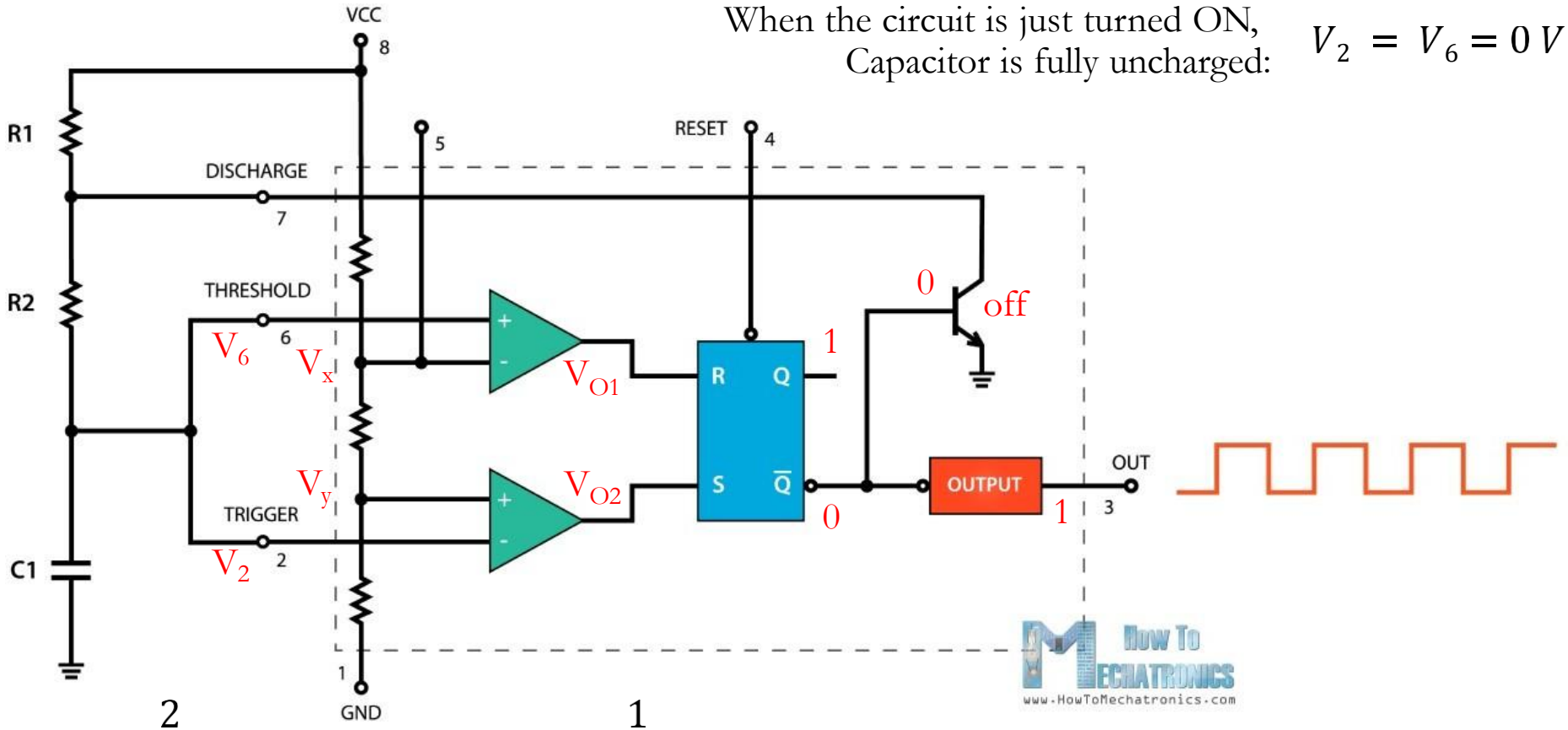
Control pin (5) is connected to $V_{low} = 0$ V via a capacitor.

Threshold (6) and trigger (2) pins are connected (∵ no input trigger).



Astable Multivibrator

When the circuit is just turned ON,
Capacitor is fully uncharged: $V_2 = V_6 = 0\ V$



$$V_x = \frac{2}{3}V_{CC} \qquad V_y = \frac{1}{3}V_{CC}$$

$$V_x > V_2 \qquad V_y > V_6$$

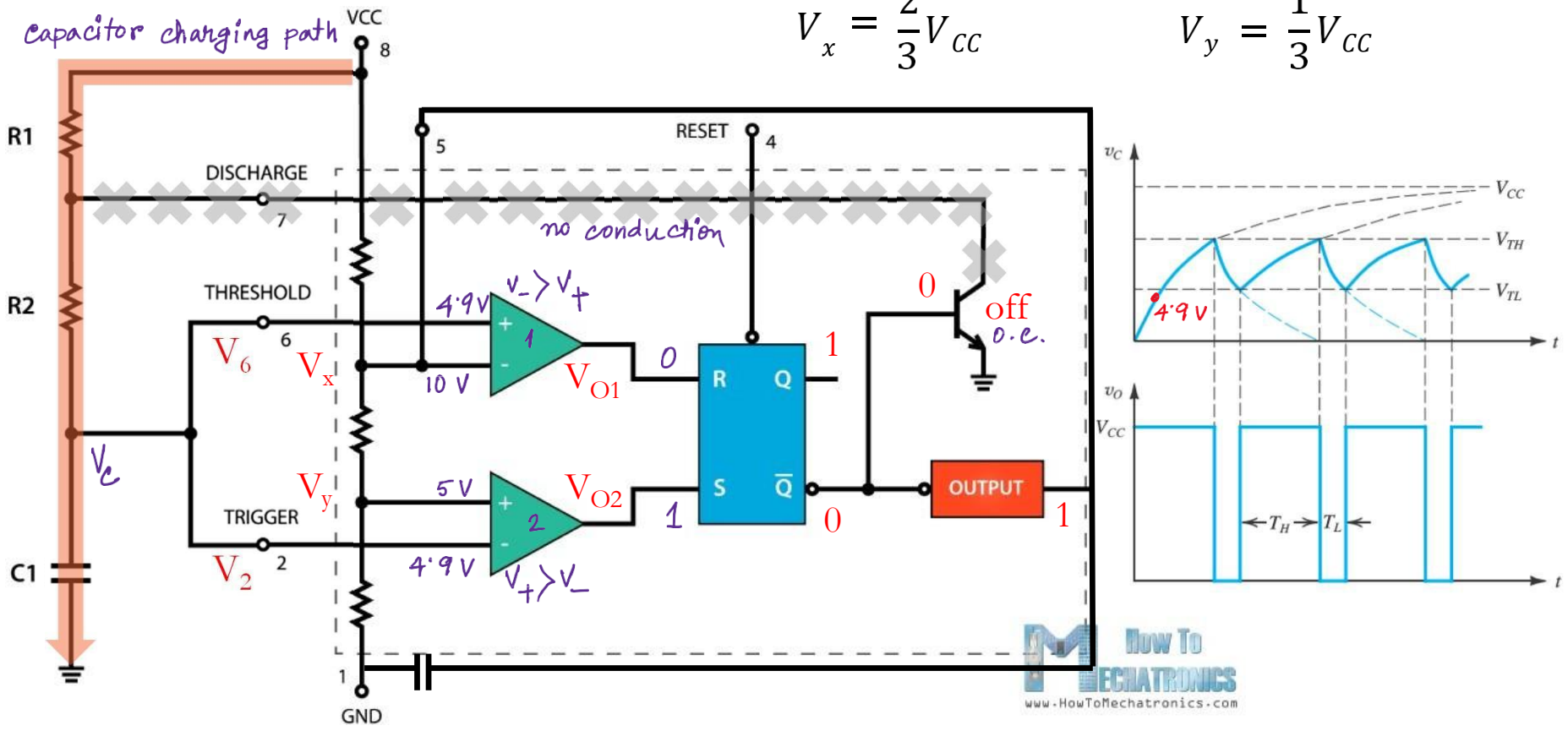
$$V_{O1} = V_{low} \qquad V_{O2} = V_{high}$$

$$R = 0 \qquad S = 1$$

How To MECHATRONICS

Astable Multivibrator

$$V_x = \frac{2}{3}V_{CC} \qquad V_y = \frac{1}{3}V_{CC}$$



When $Q = 1: Q' = 0$

Transistor is OFF

$V_o = V_{high}$

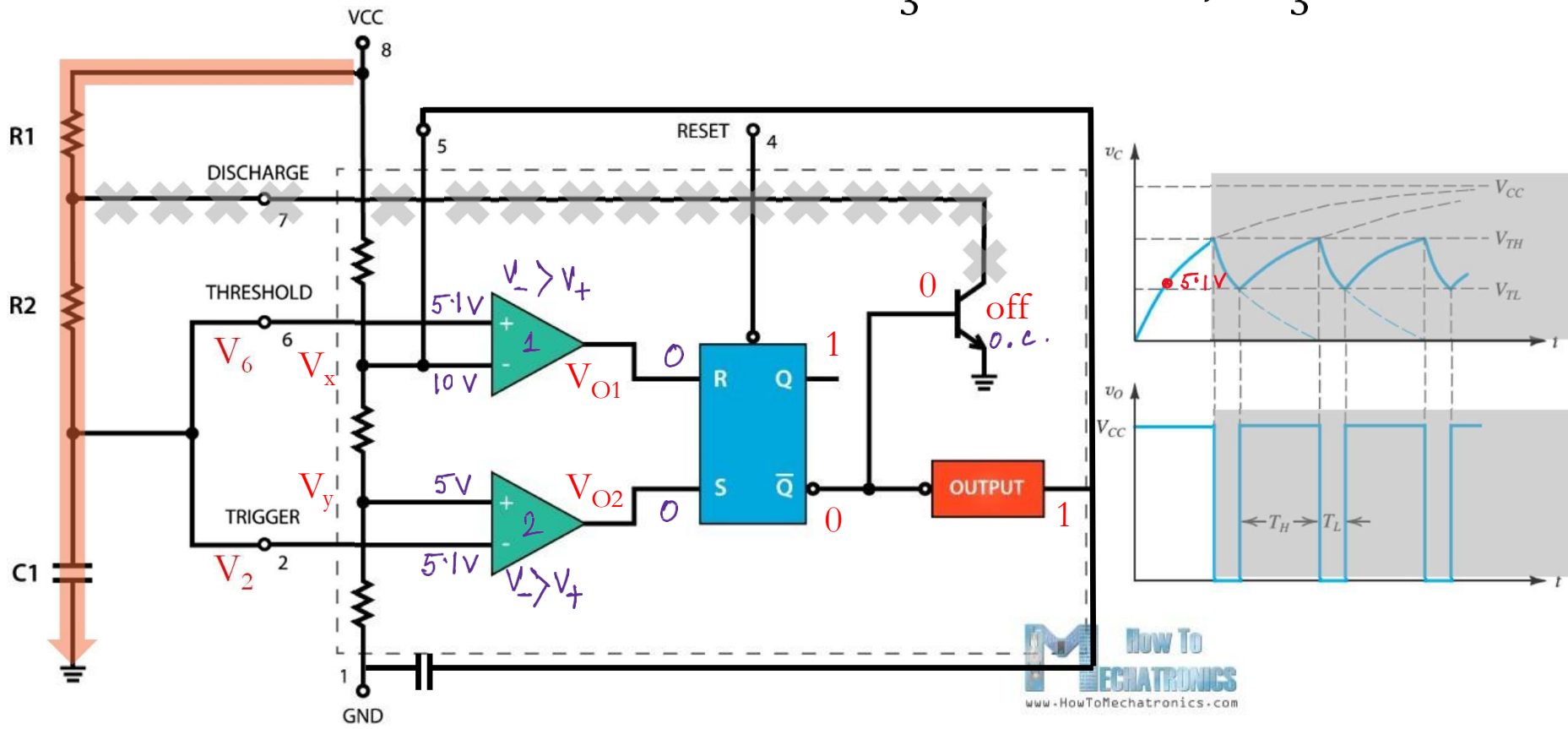
Capacitor will charge through R_1 and R_2

$V_2 = V_6$ will gradually increase

How To MECHATRONICS

Astable Multivibrator

$V_x = \frac{2}{3}V_{CC}$ $V_y = \frac{1}{3}V_{CC}$



When $V_2 > Vy$:

$V_{O2} = V_{low}$

$R = 0$ $S = 0$ **Memory state**

C_1 will continue charging as long as $V_c < V_x$.

$Q = 1 : Q' = 0$

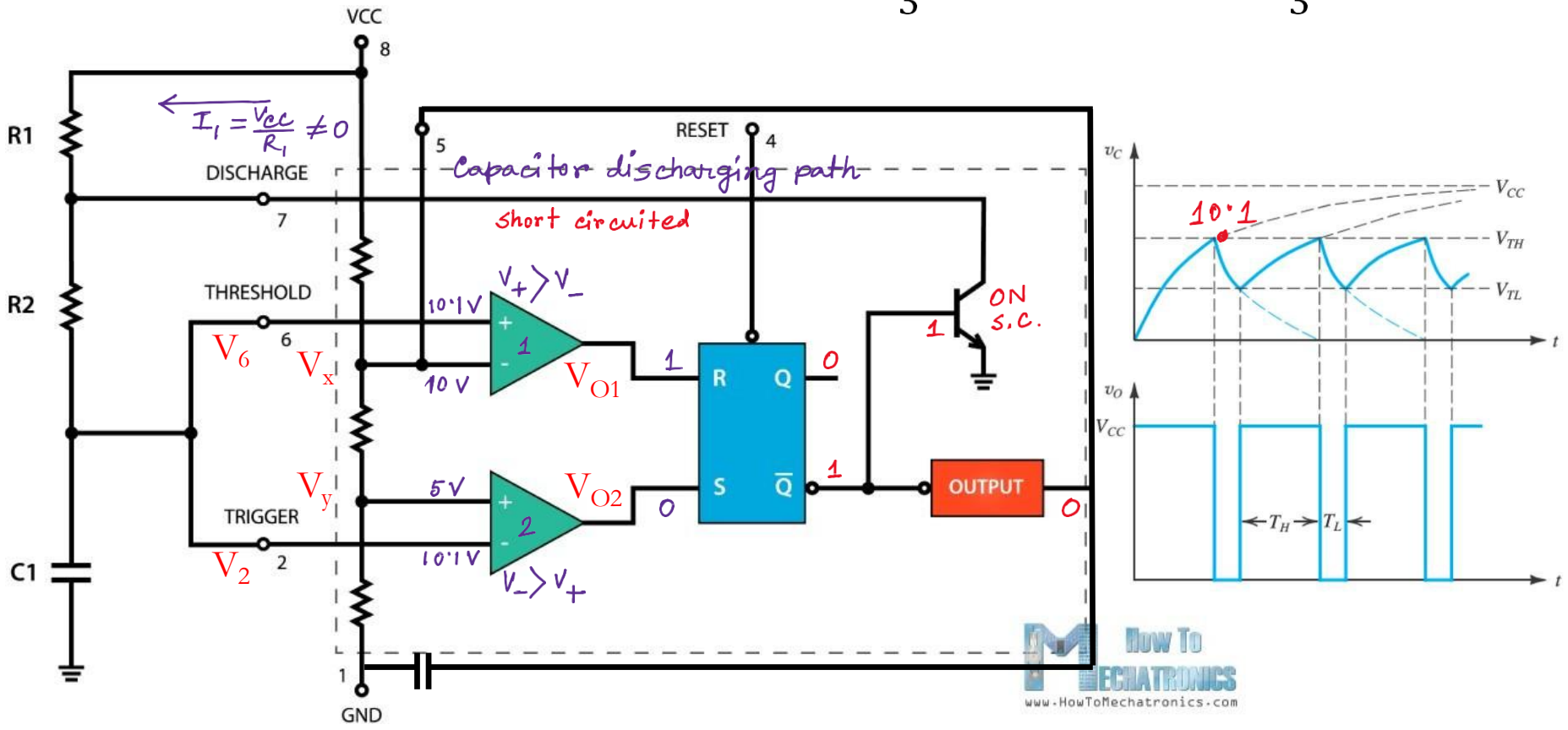
Transistor is OFF

$V_o = V_{high}$

How To MECHATRONICS

Astable Multivibrator

$V_x = \frac{2}{3}V_{CC}$ $V_y = \frac{1}{3}V_{CC}$



When $V_6 > Vx$:

$V_{O1} = V_{high}$

∴ As soon as $V_c > V_x$, C_1 interrupts charging and instead starts discharging.

$R = 1$ $S = 0$ **Reset state** $Q = 0: Q' = 1$

Capacitor will discharge through R_2

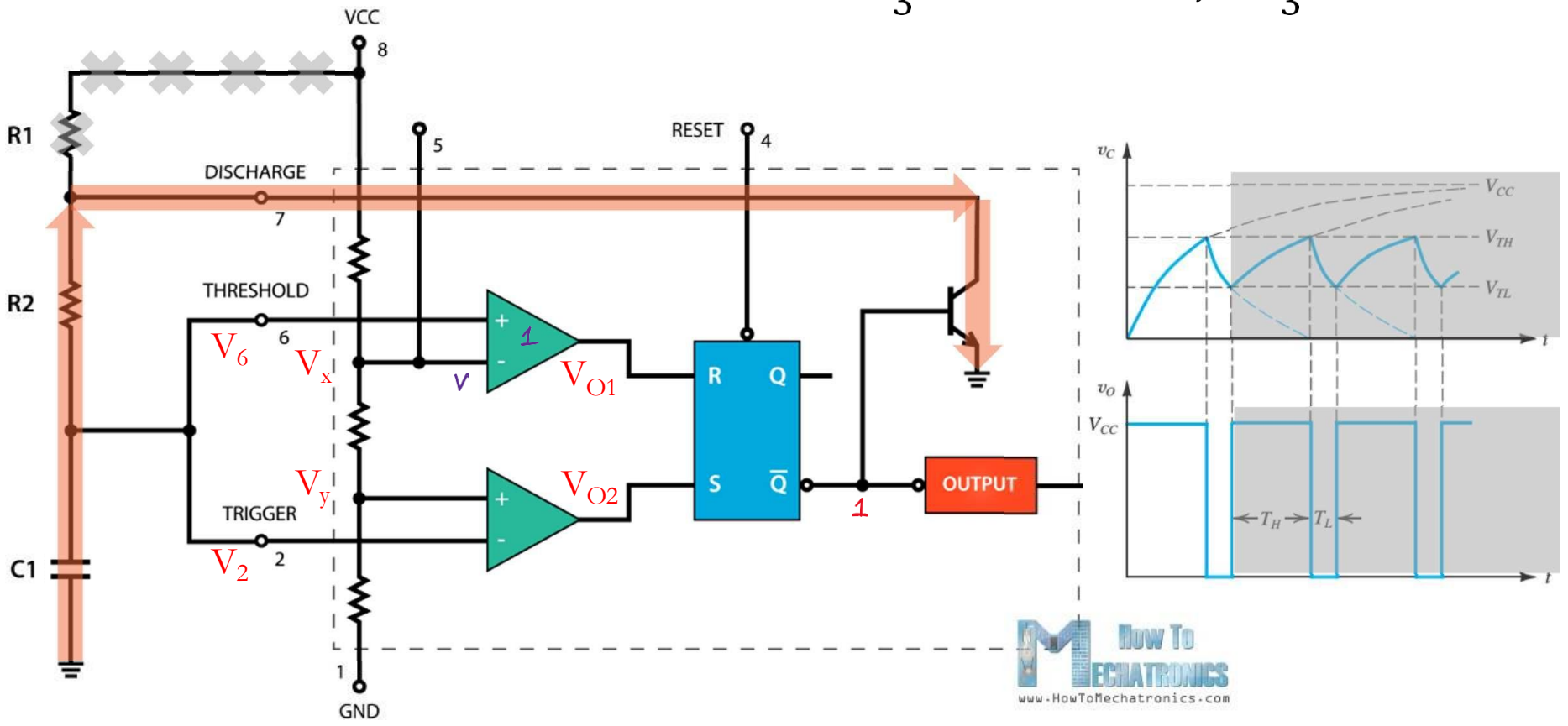
$V^2 = V^6$ will gradually decrease

Transistor is ON

$V_o = V_{low}$

Astable Multivibrator

$V_x = \frac{2}{3}V_{CC}$ $V_y = \frac{1}{3}V_{CC}$



When $V_6 < Vx$:

$V_{O1} = V_{low}$

$R = 0$ $S = 0$ **Memory state** $Q = 0: Q' = 1$

Transistor is ON

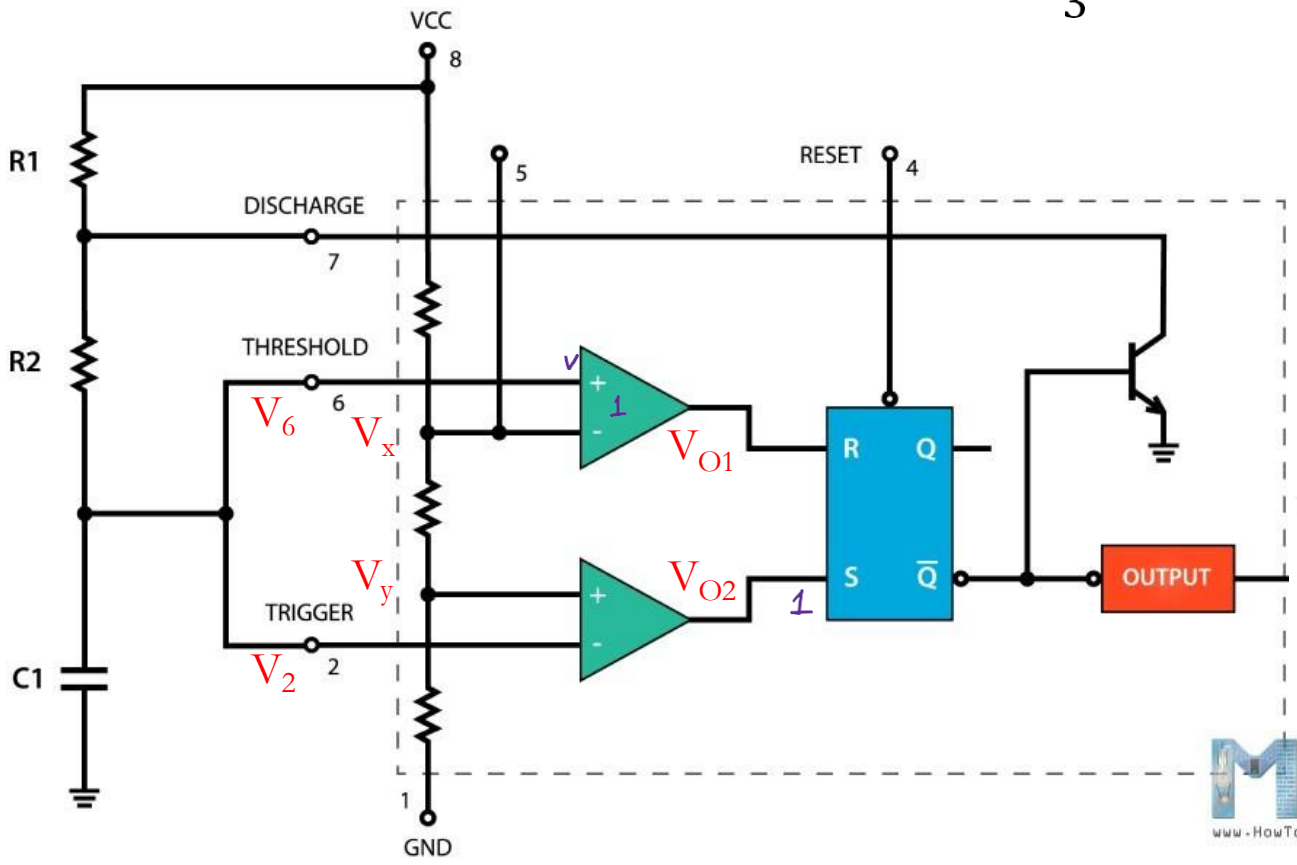
Capacitor will discharge through R_2

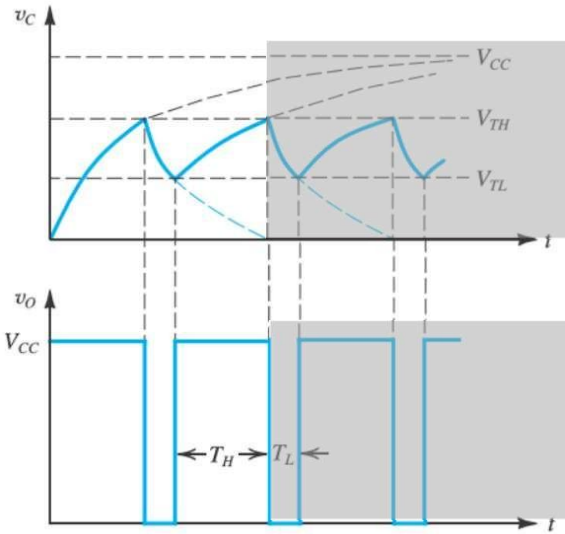
$V_2 = V_6$ will gradually decrease

$V_o = V_{low}$

Astable Multivibrator

$V_x = \frac{2}{3}V_{CC}$ $V_y = \frac{1}{3}V_{CC}$





When $V_2 < Vy$:

$V_{O2} = V_{high}$

$R = 0$ $S = 1$ **Set state**

$Q = 1: Q' = 0$

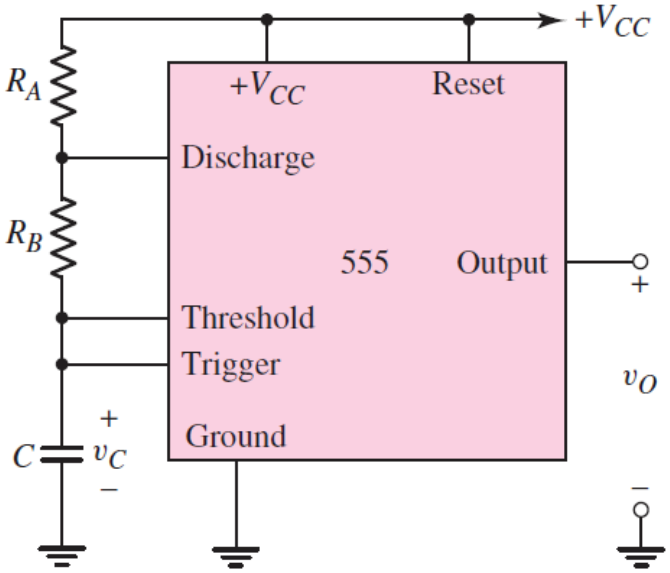
Transistor is OFF

$V_o = V_{high}$

Capacitor will charge through R_1 and R_2

$V_2 = V_6$ will gradually increase

Astable Multivibrator



Astable multivibrator 555 circuit

In the astable mode, the timing capacitor C charges through $R_A = R_B$ until $v(t)$ reaches $(2/3)V_+$. The threshold comparator output then goes high, forcing the flip-flop output Q to go high

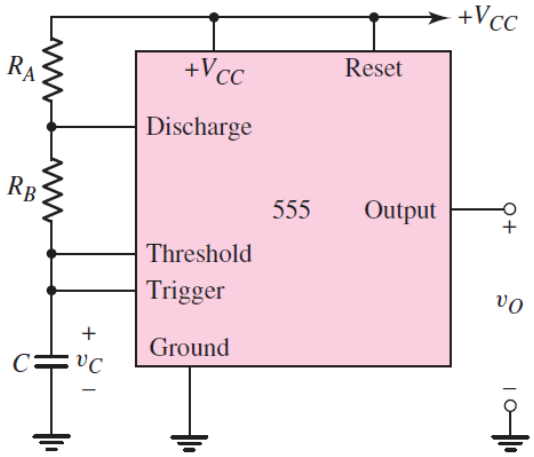
The discharge transistor turns on, and the timing capacitor C discharges through R_B and the discharge transistor. The capacitor voltage decreases until it reaches (1/3)V+, at which point the trigger comparator switches states and sends Q low.

When v(t) reaches the threshold level of (2/3)V+, the cycle repeats itself.

Astable Multivibrator

When the timing capacitor is charging, during the time 0 < t < TC

$$v(t) = \frac{1}{3}V^+ + \frac{2}{3}V^+(1 - e^{-t/\tau_A})$$ where $\tau_A = (R_A + R_B)C$.



At time $t = T_C$ the capacitor voltage reaches the threshold level

$$v(T_C) = \frac{2}{3}V^+ = \frac{1}{3}V^+ + \frac{2}{3}V^+(1 - e^{-T_C/\tau_A})$$

$$T_C = \tau_A \ln(2) = 0.693(R_A + R_B)C$$

When the timing capacitor is discharging, during the time $0 < t < T_D$

$$v(t') = \frac{2}{3}V^+ e^{-t'/\tau_B}$$ where $\tau_B = R_B C$.

Astable Multivibrator

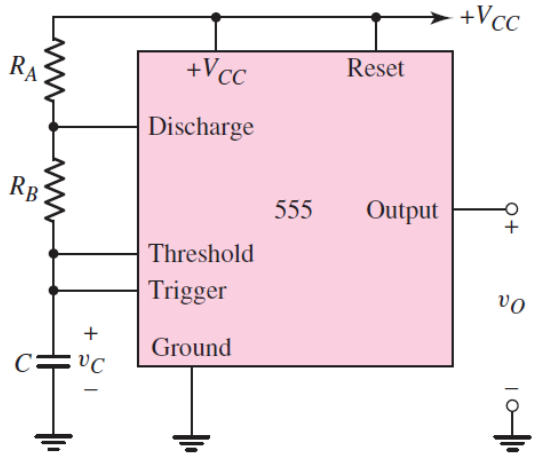
At time $t = T_D$, the capacitor voltage reaches the trigger level

$$v(T_D) = \frac{1}{3}V^+ = \frac{2}{3}V^+ e^{-T_D/\tau_B}$$

$$T_D = \tau_B \ln(2) = 0.693 R_B C$$

$$f = \frac{1}{T} = \frac{1}{T_C + T_D} = \frac{1}{0.693(R_A + 2R_B)C}$$

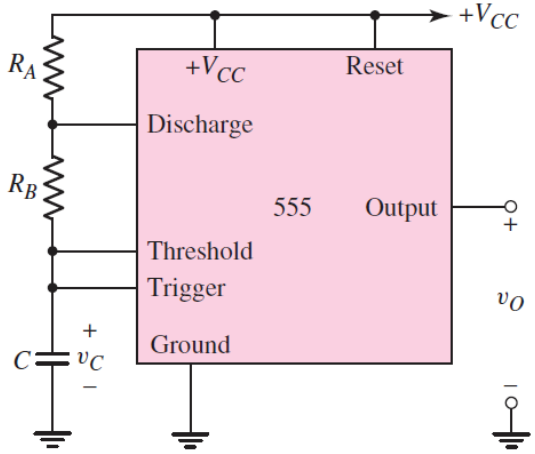
$$\text{Duty cycle} = \frac{T_C}{T} \times 100\% = \frac{R_A + R_B}{R_A + 2R_B} \times 100\%$$



Example Problem 15.8

Objective: Design the 555 IC as an astable multivibrator for a specified frequency and duty cycle.

Specifications: The circuit with the configuration in Figure 15.41 is to be designed such that the frequency is 50 kHz and the duty cycle is 75 percent.



Solution: The frequency of oscillation,

$$f = \frac{1}{0.693(R_A + 2R_B)C}$$

Therefore,

$$R_A + 2R_B = \frac{1}{(0.693)fC} = \frac{1}{(0.693)(50 \times 10^3)(1 \times 10^{-9})} \Rightarrow 28.9\,\text{k}\Omega$$

$$\text{Duty cycle} = 0.75 = \frac{R_A + R_B}{R_A + 2R_B}$$

Example Problem 15.11

$$0.75 = \frac{R_A + R_B}{R_A + 2R_B}$$

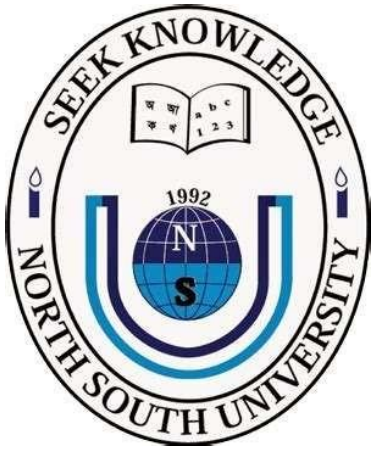
which yields

$$R_A = 2R_B$$

Combining Equations of R_A and R_B

$$R_A = 14.5\,\text{k}\Omega \qquad \text{and} \qquad R_B = 7.23\,\text{k}\Omega$$

Analog Electronics II: Op-amp & 555 Timer



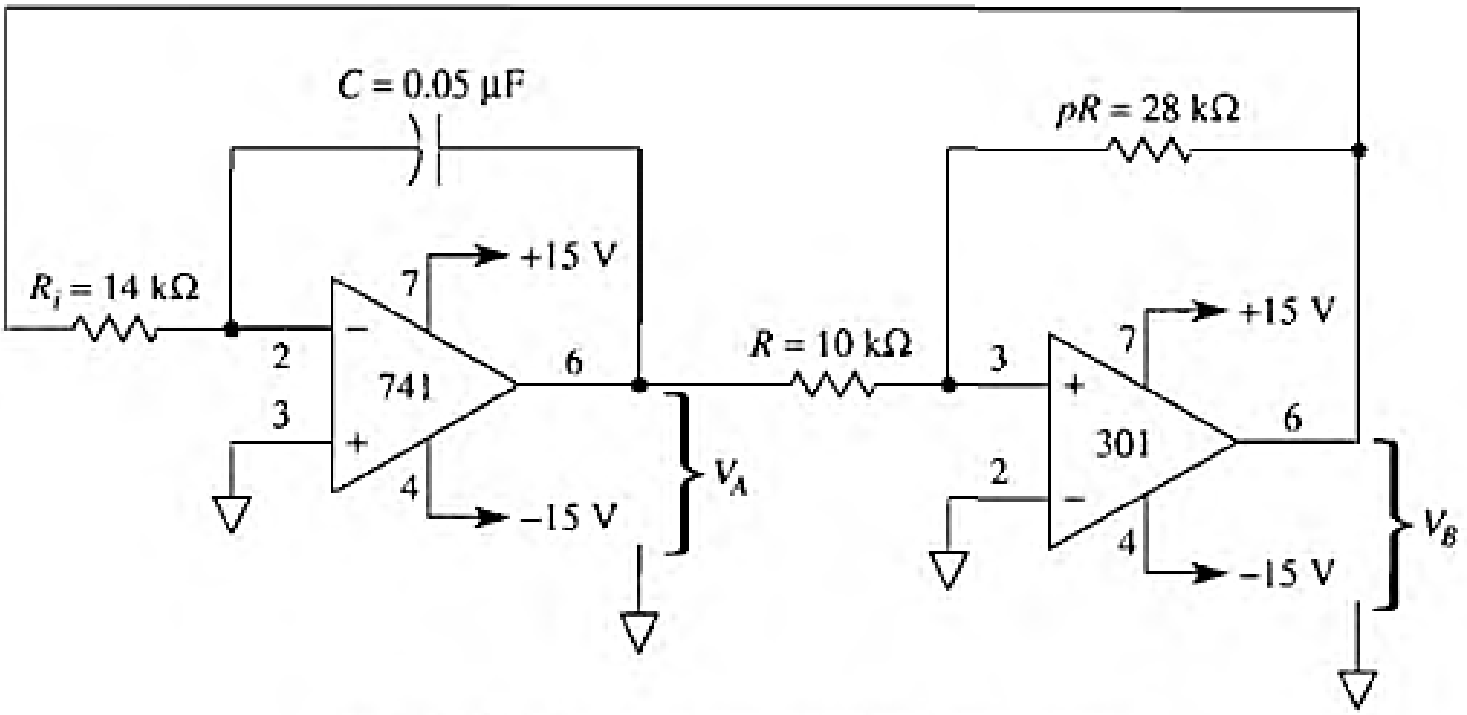
Course instructor:
Dr. Mohsin Sajjad
Assistant Professor
North South University

EEE 311

Lecture # 15

Triangle Wave Generators

The bipolar triangle-wave generator circuit

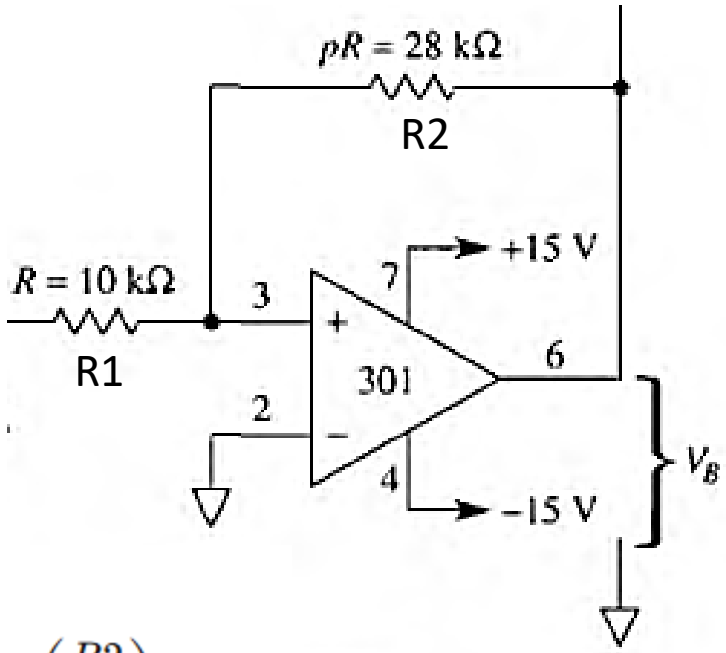


(a) The 741 integrator circuit and 301 comparator circuit are wired to make a triangle-wave generator.

Threshold Voltages:

$$V_+ = \frac{R2}{R1+R2}V_{out} + \frac{R1}{R1+R2}V_{in}$$

$$V_{in} = -\frac{R2}{R1}V_{out}$$



- When output = **+ V_{sat}** , the threshold is:

$$V_{TH-} = -\frac{R2}{R1}(+V_{sat}) = -\left(\frac{R2}{R1}\right)V_{sat}$$

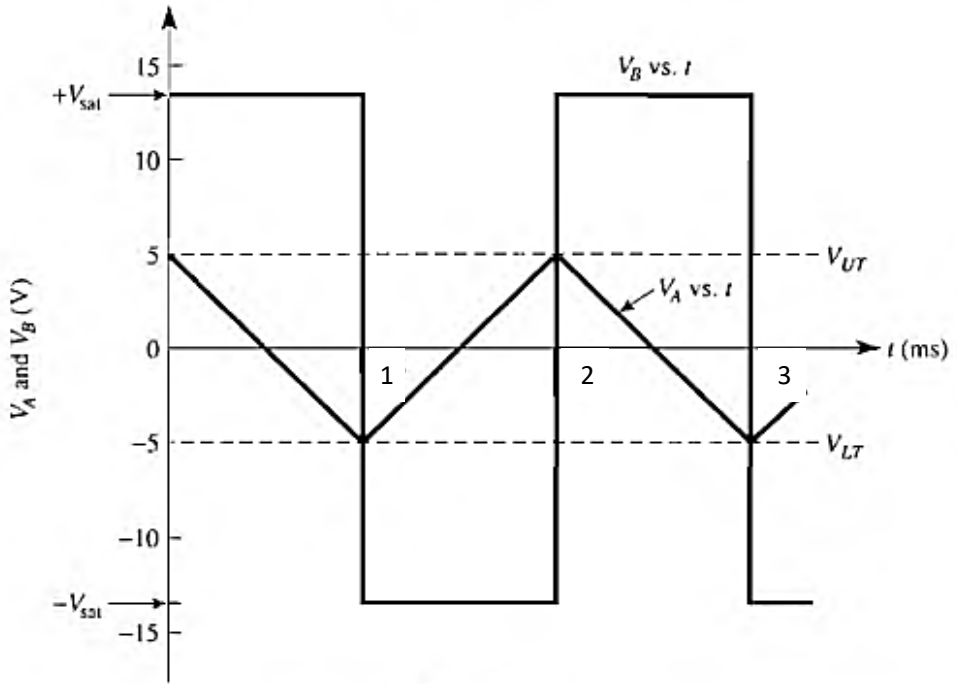
- When output = **- V_{sat}** , the threshold is:

$$V_{TH+} = -\frac{R2}{R1}(-V_{sat}) = +\left(\frac{R2}{R1}\right)V_{sat}$$

- If Vin rises **above** $+\left(\frac{R2}{R1}\right)V_{sat}$, output goes **HIGH**
- If Vin falls **below** $-\left(\frac{R2}{R1}\right)V_{sat}$, output goes **LOW**

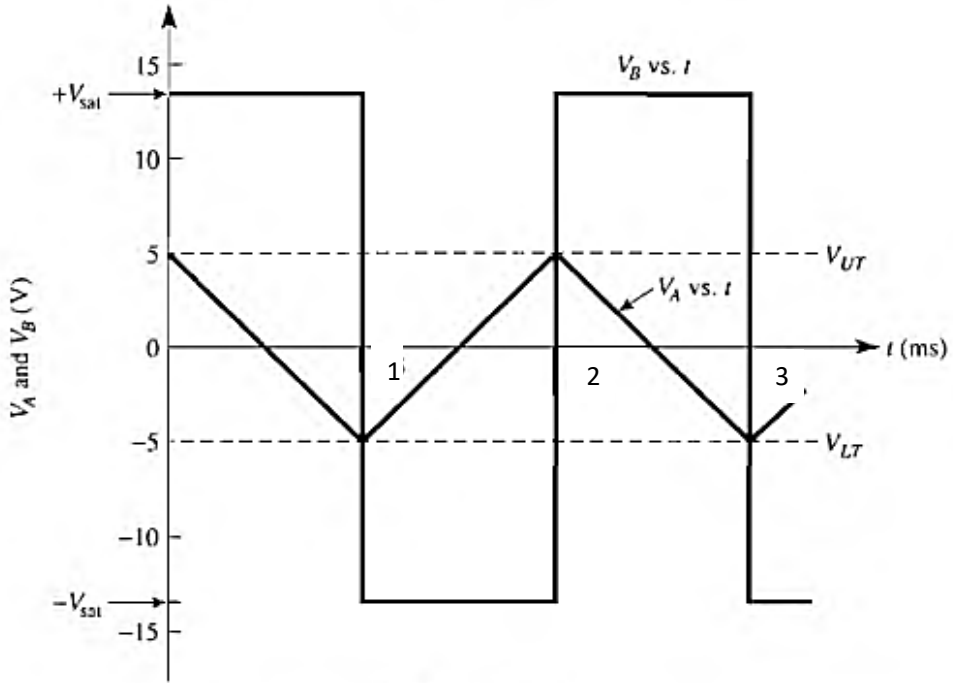
In between these, the output remains in its previous state → **hysteresis**.

Triangle Wave Generators



To understand circuit operation, refer to time interval 0 to 1 ms in Fig. 6-6. Assume that V_B is high at $+V_{sat}$. This forces a constant current (V_{sat}/R_1) through C (left to right) to drive V_A negative from V_{UT} to V_{LT}. When V_A reaches V_{LT}, pin 3 of the 301 goes negative, and V_B snaps to $-V_{sat}$, and $t = 1$ ms.

Triangle Wave Generators



$$V_{UT} = -\frac{-V_{sat}}{p}$$

$$V_{LT} = -\frac{+V_{sat}}{p}$$

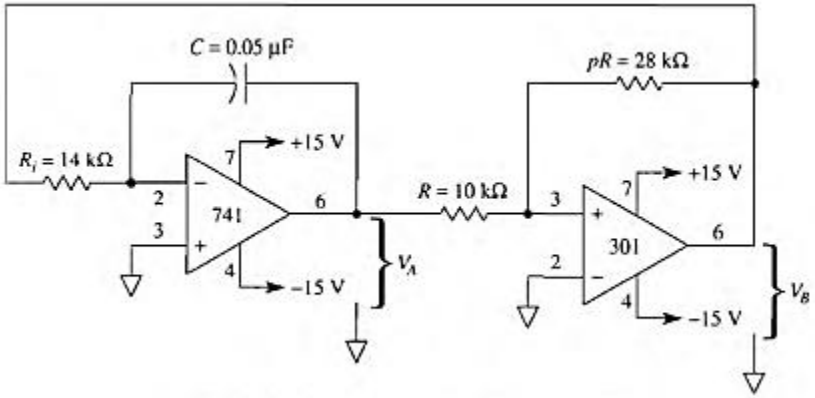
$$p = \frac{pR}{R}$$

$$f = \frac{p}{4R_fC}$$

When V_B is at $-V_{sat}$, it forces a constant current (right to left) through C to drive V_A positive from V_{LT} toward V_{UT} (see the time interval 1 to 2 ms). When V_A reaches V_{UT} at $t = 2$ ms, pin 3 of the 301 goes positive and V_B snaps to $+V_{sat}$. This initiates the next cycle of oscillation.

Example 6-6

A triangle-wave generator oscillates at a frequency of 1000 Hz with peak values of approximately +5 V. Calculate the required values for pR, R_i, and C in Fig. 6-6.



Solution

$$p = -\frac{-V_{sat}}{V_{UT}} = -\frac{-13.8\text{ V}}{5\text{ V}} = +2.76 \approx 2.8$$

Choose $R = 10\text{ k}\Omega$. Then from Eq. (6-5c) we solve for pR as

$$pR = 2.8(10\text{ k}\Omega) = 28\text{ k}\Omega$$

Next we select R_i and C. Begin by making a *trial* choice for $C = 0.05\ \mu\text{F}$. Then calculate a value for R_i to see if R_i is greater than 10 kΩ. From Eq. (6-6),

$$R_i = \frac{p}{4fC} = \frac{2.8}{4(1000\text{ Hz})(0.05\ \mu\text{F})} = 14\text{ k}\Omega$$